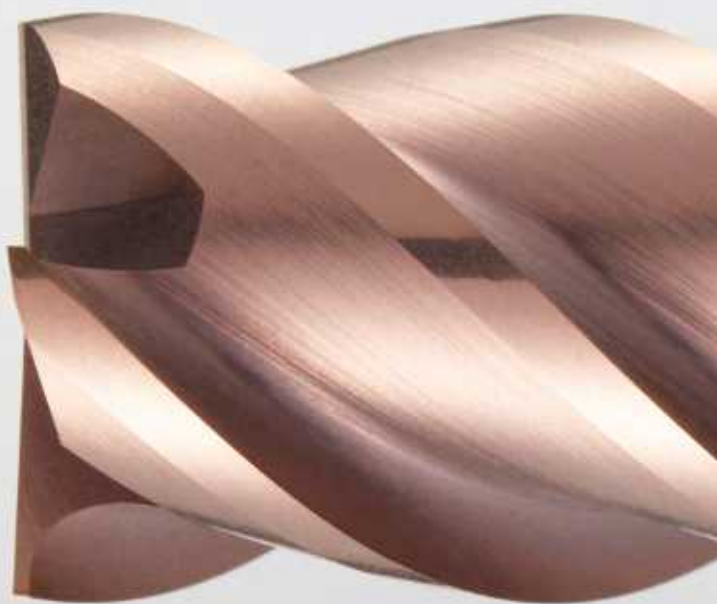




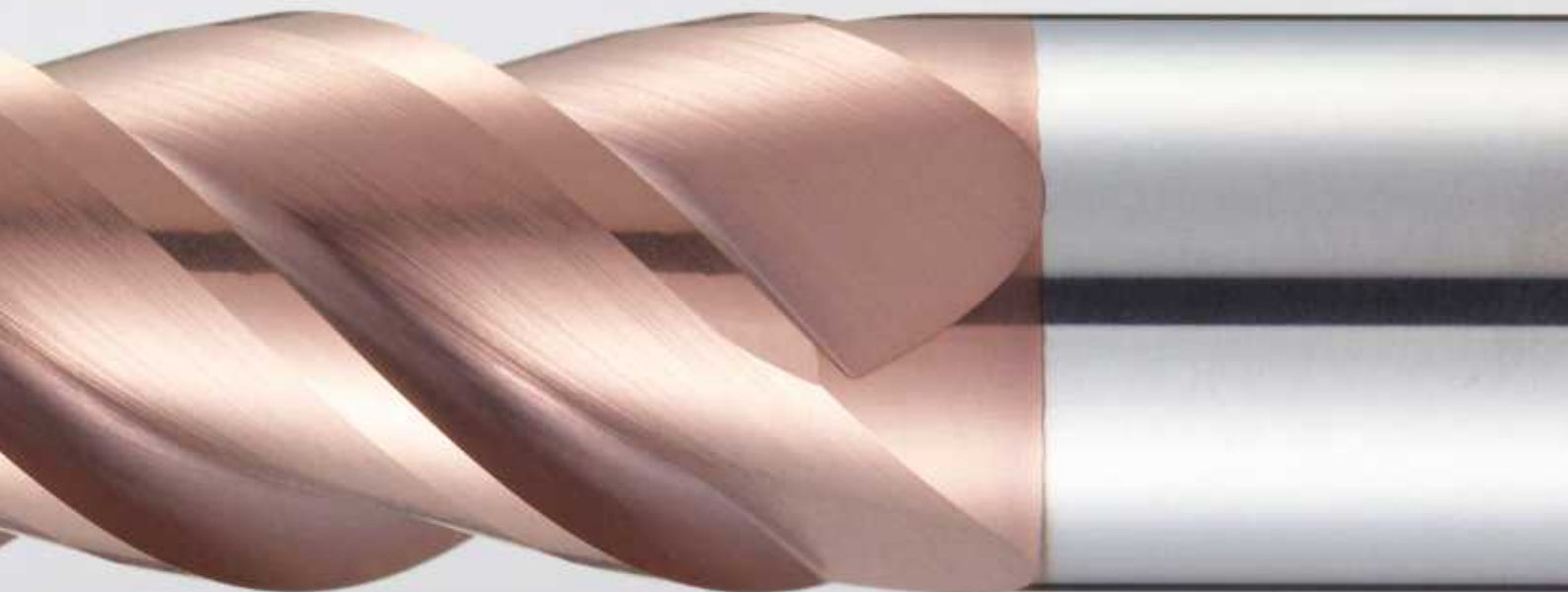
ENDMILLS

Korloy Endmills, with New technology and our technical know-how, are the best for increasing productivity and machinability

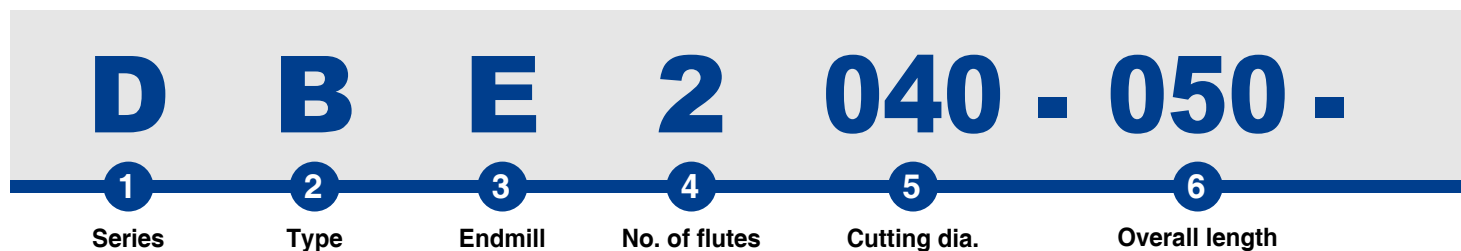


Technical information for ENDMILLS

B3	Code system
B5	KORLOY Endmill
B11	Endmill selection guide
B13	Super Endmill for HRSA [for difficult-to-cut materials HRSA]
B17	Super Endmill for Ti [for difficult-to-cut materials Ti]
B22	The Mirror Endmill [for high precision mold manufacture]
B24	H-Star Endmill [for high hardness steel(HrC50~63)]
B92	U-Star Endmill [for medium hardness steel(HrC30~50)]
B175	G-Star Endmill [for low hardness steel(HrC10~30)]
B200	R+ Endmill [for high efficient roughing]
B207	S-Star Endmill [for stainless steel]
B219	A-Star Endmill [for aluminum]
B230	D Endmill [for graphite]
B239	Composite Router Endmill [for composite materials]
B243	T Endmill [for dental prostheses]
B262	C-Max [for copper based electrode]
B266	PCD Endmill [for non-ferrous metals]
B267	Special Order Form for Endmills



B Code System



1 Series
D B E 2 040 - 050 - R T - V N S

Z, IP, ZP : Endmill for general usage **FM** : High feed Endmill
P : High speed/ hardness Endmill **SSEA, AP** : Aluminum Endmill
C : Copper, Copper alloy Endmill **SP** : Stainless Endmill
D : Graphite, Non-Ferrous Endmill **CC** : Composite Router Endmill
V : Variable Endmill **T** : Dental Endmill

5 Cutting dia
D B E 2 040 - 050 - R T - V N S

Cutting dia

Notation	DC(mm)
040	Ø4.0
060	Ø6.0
080	Ø8.0
100	Ø10.0

2 Type
D B E 2 040 - 050 - R T - V N S

Flat type Ball type Radius type

F **B** **R**

3 Endmill
D B E 2 040 - 050 - R T - V N S

4 No. of flutes
D B E 2 040 - 050 - R T - V N S

2 Flutes 3 Flutes 4 Flutes 6 Flutes

2 **3** **4** **6**

6 Overall length
D B E 2 040 - 050 - R T - V N S

Overall length

Notation	LF(mm)
050	50
080	80
100	100

※ The above code system is not applied for SSEA (Aluminum Endmill) and ZSE (Brazing Endmill)

R02 T00 - V05 N12 S06

7

Corner radius

8

Taper angle

9

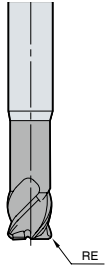
Flute length

10

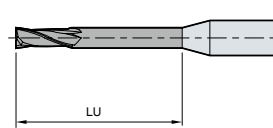
Neck length

11

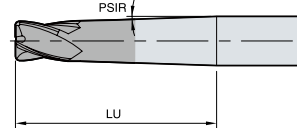
Shank diameter

7**Corner radius**DBE2040-050-**R**T-VNS

Corner radius	
Notation	RE(mm)
R02	r 0.2
R05	r 0.5
R10	r 1.0
R15	r 1.5

10**Neck length**DBE2040-050-RT-V**N**S

Long Neck



Taper Long Neck

LU (mm) : Neck Length

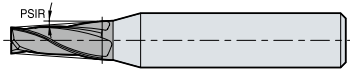
PSIR(BHTA) : Taper Angle

Long	
Notation	LU(mm)
N05	5
N08	8
N10	10
N12	12

Taper long neck	
Notation	LU + PSIR(mm)
N0510	5+1°
N0815	8+1.5°
N1020	10+2°
N1225	12+2.5°

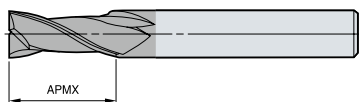
8**Taper angle**

DBE2040-050-RT-VNS



PSIR(BHTA) : Taper Angle

Taper Angle	
Notation	PSIR(BHTA)
T10	1°
T15	1.5°
T20	2°

9**Flute length**DBE2040-050-RT-V**N**S

Taper length	
Notation	APMX(mm)
V05	5
V10	10
V15	15

11**Shank diameter**DBE2040-050-RT-VNS**S**

Shank diameter	
Notation	DCON-MS
S06	Ø6
S08	Ø8
S10	Ø10
S12	Ø12
S16	Ø16

※ This code system is also for special endmills

Type	Used	Shape	Designation	Figure	Coated	No. of flute	Size (DC)		Workpiece						page
							Min	Max	P	M	K	N	S	H	
									Steel	Stainless steel	Cast iron	Non-ferrous metal	Heat resistant alloy Titanium alloy	Hardened steel	
Super Endmill	HRSA	Flat	SFES		SL	4	3.0	20.0					◎		B14
		Radius	SRES		SL	4	3.0	20.0					◎		B15
	Ti/STS	Flat	SFET		UL	4	3.0	20.0	◎				◎		B18
		Radius	SRET		UL	4	3.0	20.0	◎				◎		B19
		Ball	SBET		UL	2	4.0	12.0	◎				◎		B20
			SBET		UL	4	4.0	12.0	◎				◎		B21
The Mirror Endmill	High precision mold manufacture	Ball	PCD-MBE		-	-	0.3	2.0	◎					◎	B22
			cBN-MBE		-	2	0.4	2.0						◎	B23
		Radius	cBN-MRE		-	2	0.4	2.0						◎	B23
H-Star Endmill	High speed, High hardness	Flat	ESE702		AlTiN	2	0.1	20.0	○				○	◎	B26
			ESE712		AlTiN	2	1.0	6.0	○				○	◎	B27
			ESE704		AlTiN	4	1.0	20.0	○				○	◎	B28
			ESE714		AlTiN	4	1.0	12.0	○				○	◎	B29
			ESE724(6)		AlTiN	4/6	1.0	12.0	○				○	◎	B30
			ESE744		AlTiN	4	1.0	12.0	○				○	◎	B31
			ESE716		AlTiN	6	6.0	20.0	○				○	◎	B32
			ESRE712		AlTiN	2	0.1	12.0	○				○	◎	B33
			ESRE714		AlTiN	4	0.5	12.0	○				○	◎	B36
			ESXE704		AlTiN	4	1.0	12.0	○				○	◎	B38
			ESXE714		AlTiN	4	2.0	12.0	○				○	◎	B38
			ESLNS20		AlTiN	2	0.1	5.0	○				○	◎	B39
			ESLNS40		AlTiN	4	1.0	5.0	○				○	◎	B43
		Radius	ESR702		AlTiN	2	1.0	12.0	○				○	◎	B45
			ESR732		AlTiN	2	1.0	12.0	○				○	◎	B48
			ESR704		AlTiN	4	1.0	12.0	○				○	◎	B49
			ESR714		AlTiN	4	3.0	12.0	○				○	◎	B51
			ESR724		AlTiN	4	6.0	12.0	○				○	◎	B51
			ESR734		AlTiN	4	1.0	12.0	○				○	◎	B52
			ESR706		AlTiN	6	6.0	12.0	○				○	◎	B53

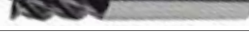
◎ : Excellent ○ : Good

Type	Used	Shape	Designation	Figure	Coated	No. of flute	Size (DC)		Workpiece						page
							Min	Max	P	M	K	N	S	H	
									Steel	Stainless steel	Cast iron	Non-ferrous metal	Heat resistant alloy, Titanium alloy	Hardened steel	
H-Star Endmill	High speed, High hardness	Radius	ESR736		AlTiN	6	6.0	12.0	○				○	◎	B53
			ESRR712		AlTiN	2	0.2	16.0	○				○	◎	B54
			ESRR714		AlTiN	4	0.5	20.0	○				○	◎	B60
			ESXR704		AlTiN	4	2.0	12.0	○				○	◎	B67
			ESLNR20		AlTiN	2	0.2	3.0	○				○	◎	B68
			ESTNR20		AlTiN	2	0.2	3.0	○				○	◎	B72
		High feed	ESPM4		AlTiN	4	3.0	12.0	○				○	◎	B74
		Ball	ESB702		AlTiN	2	0.1	12.0	○				○	◎	B75
			ESB712		AlTiN	2	1.0	12.0	○				○	◎	B76
			ESB703		AlTiN	3	2.0	12.0	○				○	◎	B77
			ESB734		AlTiN	4	2.0	10.0	○				○	◎	B77
			ESRB712		AlTiN	2	0.1	12.0	○				○	◎	B78
			ESLNB20		AlTiN	2	0.1	5.0	○				○	◎	B83
			ESTNB20		AlTiN	2	0.2	10.0	○				○	◎	B87
			ESTNB30		AlTiN	3	2.0	5.0	○				○	◎	B91
U-Star Endmill	General	Flat	UE502		AlCrN	2	0.1	25.0	◎	○	○			○	B94
			UE512		AlCrN	2	0.1	12.0	◎	○	○			○	B96
			UE522		AlCrN	2	1.0	25.0	◎	○	○			○	B99
			UXE502		AlCrN	2	0.1	20.0	◎	○	○			○	B101
			UE504H		AlCrN	4	1.0	20.0	◎	○	○			○	B103
			UE514		AlCrN	4	1.0	12.0	◎	○	○			○	B104
			UE524		AlCrN	4	1.0	25.0	◎	○	○			○	B106
			ULE504		AlCrN	4	3.0	16.0	◎	○	○			○	B108
			UE504		AlCrN	4	0.8	25.0	◎	○	○			○	B109
			UXE504		AlCrN	4	1.0	20.0	◎	○	○			○	B110
			UE506		AlCrN	6	6.0	20.0	◎	○	○			○	B111
			UTE502		AlCrN	2	0.3	10.0	◎	○	○			○	B112
			UTE504		AlCrN	4	0.8	10.0	◎	○	○			○	B114

◎ : Excellent ○ : Good

Type	Used	Shape	Designation	Figure	Coated	No. of flute	Size (DC)		Workpiece						page
							Min	Max	P	M	K	N	S	H	
									Steel	Stainless steel	Cast iron	Non-ferrous metal	Heat resistant alloy Titanium alloy	Hardened steel	
U-Star Endmill	General	Radius	UR502		AlCrN	2	0.2	20.0	◎	○	○			○	B117
			UR512		AlCrN	2	0.2	20.0	◎	○	○			○	B120
			UR542		AlCrN	2	0.2	4.0	◎	○	○			○	B126
			UR504		AlCrN	4	3.0	20.0	◎	○	○			○	B132
			UR544		AlCrN	4	1.0	4.0	◎	○	○			○	B133
			UXR504		AlCrN	4	1.0	20.0	◎	○	○			○	B137
			UXR514		AlCrN	4	1.0	20.0	◎	○	○			○	B140
			UR506		AlCrN	6	6.0	20.0	◎	○	○			○	B145
			UDR503		AlCrN	3	6.0	20.0	◎	○	○			○	B146
			USPM4		AlCrN	4	1.0	20.0	◎	○	○			○	B147
			UTR504		AlCrN	4	0.8	2.5	◎	○	○			○	B148
		Ball	UB502		AlCrN	2	1.0	25.0	◎	○	○			○	B152
			UB502P		AlCrN	2	0.1	12.0	◎	○	○			○	B154
			UB512		AlCrN	2	0.1	12.0	◎	○	○			○	B155
			UB512S6		AlCrN	2	0.5	2.0	◎	○	○			○	B159
			UB532		AlCrN	2	3.0	12.0	◎	○	○			○	B160
			UB542		AlCrN	2	0.1	12.0	◎	○	○			○	B161
			USB502		AlCrN	2	3.0	20.0	◎	○	○			○	B168
			UB503		AlCrN	3	1.0	12.0	◎	○	○			○	B169
			UB504		AlCrN	4	1.0	12.0	◎	○	○			○	B169
			UTB502		AlCrN	2	0.3	2.0	◎	○	○			○	B170
		Roughing	UF50		AlCrN	3~5	3.0	25.0	◎	○	○			○	B171
			UF51		AlCrN	3~5	3.0	25.0	◎	○	○			○	B172
			UF51H		AlCrN	3~5	3.0	25.0	◎	○	○			○	B173
G-Star Endmill	General	Ball	DB312		AlTiN	2	1.0	20.0	◎	○	◎		○		B177
			DB342		AlTiN	2	1.0	12.0	◎	○	◎		○		B178
		Flat	TX202		AlTiN	2	1.0	20.0	◎	○	◎		○		B179
			TX204		AlTiN	4	1.0	20.0	◎	○	◎		○		B180

◎ : Excellent ○ : Good

Type	Used	Shape	Designation	Figure	Coated	No. of flute	Size (DC)		Workpiece						page
							Min	Max	P	M	K	N	S	H	
									Steel	Stainless steel	Cast iron	Non-ferrous metal	Heat resistant alloy, Titanium alloy	Hardened steel	
G-Star Endmill	General	Flat	TX222		AlTiN	2	3.0	20.0							B181
			TX224		AlTiN	4	3.0	20.0							B182
			TX302		AlTiN	2	1.0	20.0							B183
			TX304		AlTiN	4	1.0	20.0							B184
			TX304H		AlTiN	4	3.0	20.0							B185
		Ball	TXB202		AlTiN	2	1.0	20.0							B186
			TXB204		AlTiN	4	2.0	20.0							B187
			TXB222		AlTiN	2	3.0	20.0							B188
			TXB232		AlTiN	2	3.0	20.0							B189
			TXB302		AlTiN	2	1.0	20.0							B190
		Flat	TXB304		AlTiN	4	1.0	20.0							B191
			ZE302P		AlTiN	2	1.0	20.0							B192
			ZE304P		AlTiN	4	1.0	20.0							B193
			ZE322		AlTiN	2	3.0	20.0							B194
		Radius	ZE324		AlTiN	4	3.0	20.0							B195
			ZR304H		AlTiN	4	3.0	12.0							B196
		Radius	ZR322		AlTiN	2	3.0	12.0							B197
			ZR324		AlTiN	4	3.0	12.0							B198
			ZR324H		AlTiN	4	6.0	12.0							B199
R+ Endmill	Aluminum	Roughing	RPAE		Carbide, Non	3	6.0	25.0							B202
	General		RPE-FP-H		Carbide, TiAlN	4	5.0	20.0							B202
			RPLE-FP-H		Carbide, TiAlN	4	5.0	20.0							B203
			RPE-XG		Carbide, TiAlN	4	5.0	20.0							B203
			RPE-FP-L		Carbide, TiAlN	4	5.0	20.0							B204
			RPE-RG		Carbide, TiAlCrN	4	5.0	20.0							B204
			RPE-RG		HSS, TiAlN	4	6.0	20.0							B205
			RPE-FF		HSS, TiAlN	4	6.0	20.0							B205
			RPE-FP		HSS, TiAlN	4	6.0	20.0							B206
			RPE-RG		HSS, TiCN HSS, TiN	4	6.0	50.0							B206

◎ : Excellent ○ : Good

Type	Used	Shape	Designation	Figure	Coated	No. of flute	Size (DC)		Workpiece						page
							Min	Max	P	M	K	N	S	H	
									Steel	Stainless steel	Cast Iron	Non-ferrous metal	Heat resistant alloy, Titanium alloy	Hardened steel	
S-Star Endmill	STS	Flat	SPFE2000		AlCrN	2	1.0	20.0	○	◎			○		B208
			SPFE3000		AlCrN	3	1.0	20.0	○	◎			○		B209
			SPFE4000		AlCrN	4	1.0	20.0	○	◎			○		B210
			SPFE6000		AlCrN	6	6.0	20.0	○	◎			○		B211
		Radius	SPRE4000		AlCrN	4	1.0	20.0	○	◎			○		B212
			SPRE5000		AlCrN	5	6.0	20.0	○	◎			○		B214
			SPRE7000		AlCrN	7	6.0	20.0	○	◎			○		B215
		Ball	SPBE2000		AlCrN	2	1.0	12.0	○	◎			○		B216
			SPBE4000		AlCrN	4	3.0	20.0	○	◎			○		B217
		Roughing	SPXE3000 SPXE4000 SPXE5000		AlCrN	3~5	3.0	20.0	○	◎			○		B218
A-Star Endmill	Aluminum	Flat	APFE		Non	2	2.5	20.0	○			◎			B220
					Non	3	2.5	20.0	○			◎			B220
		Middle Flat	APMFE		Non	2	3.0	20.0	○			◎			B221
					Non	3	3.0	20.0	○			◎			B221
		Long Flat	APLFE		Non	2	3.0	20.0	○			◎			B222
					Non	3	3.0	20.0	○			◎			B222
		Ball	APBE		Non	2	1.0	12.0	○			◎			B223
		Short Flat	AFE		Non	3	1.0	20.0	○			◎			B224
		Flat			Non	3	1.0	20.0	○			◎			B225
		Long Flat			Non	3	1.0	20.0	○			◎			B226
		Roughing	APRE		Non	3	4.0	25.0	○			◎			B228
		Wave Roughing	RPAE		Carbide, Non	3	6.0	25.0				◎			B229
D Endmill	Graphite, Ceramics	Flat	DFE		Diamond	2	1.0	12.0				◎			B231
					Diamond	4	2.0	12.0				◎			B232
		Radius	DRE		Diamond	2	0.5	3.0				◎			B233
					Diamond	4	2.0	12.0				◎			B234
		Ball	DBE		Diamond	2	0.6	12.0				◎			B236
					Diamond	4	2.0	12.0				◎			B238

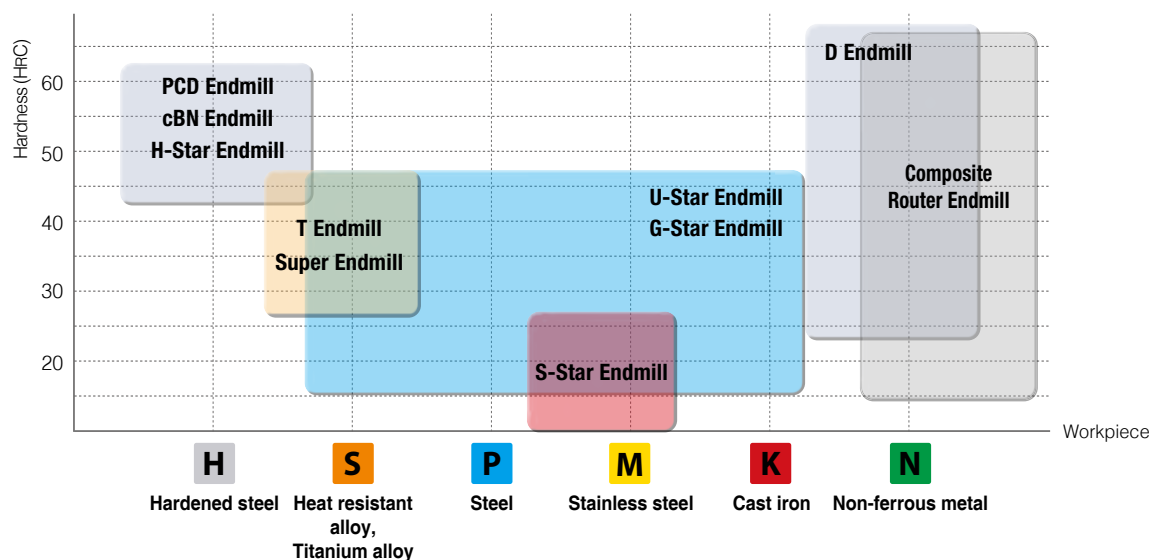
◎ : Excellent ○ : Good

Type	Used	Shape	Designation	Figure	Coated	No. of flute	Size (DC)		Workpiece						page
							Min	Max	P	M	K	N	S	H	
									Steel	Stainless steel	Cast iron	Non-ferrous metal	Heat resistant alloy, Titanium alloy	Hardened steel	
Composite Router Endmill	Composite CFRP, GFRP	Flat	CCDR		Diamond	4	6.0	8.0				◎			B240
					Diamond	6	10.0	12.0				◎			B240
			CCHR		Diamond	4	6.0	8.0				◎			B240
					Diamond	6	10.0	12.0				◎			B240
			CCR		Diamond	2	4.0	12.0				◎			B241
			CCLR		Diamond	4	4.0	12.0				◎			B241
			CCRR		Diamond	6	6.0	8.0				◎			B242
					Diamond	8	10.0	12.0				◎			B242
T Endmill	Dental, Zirconia	Ball	TZBE		Diamond	2	0.6	3.0				◎			B243
	Dental, Metal		TTBE		Diamond	2	0.6	3.0					◎		B243
	Dental, Wax		TWBE		-	2	0.6	3.0				◎			B243
C-Max	Cooper, Cooper alloy	Flat	CFE2000		CrN	2	1	12	○			◎			B262
		Long Flat	CFNE2000		CrN	2	0.5	4	○			◎			B263
		Ball	CBE2000		CrN	2	1	12	○			◎			B264
		Long Ball	CBNE2000		CrN	2	0.5	4	○			◎			B264
		Radius	CRE2000		CrN	2	2	12	○			◎			B265
		Long Radius	CRNE2000		CrN	2	1	4	○			◎			B265
PCD Endmill	Non-ferrous, High speed	Flat	PDE		-	1	4.6	6.0				◎			B266
					-	2	6.0	12.0				◎			B266

◎ : Excellent ○ : Good

B Endmill Selection Guide

Application range



Tool selection guideline by functions

● 1st recommended ◐ 2nd recommended ○ Not recommended

Type	No. of flute							
		Precis finishing	Finishing	Roughing	Slotting	Plunging	Copying	Trochoidal milling
Flat/ Radius	2 flutes	○	○	◐	●	●	○	○
	3 flutes	○	◐	◐	●	◐	○	○
	4 flutes	●	●	●	●	○	○	●
	6 flutes or over	●	●	○	○	○	○	●
Ball	2 flutes	○	○	○	●	○	●	○
	4 flutes	○	○	○	◐	○	●	○

- It is recommended to choose the shortest length tool in every application as possible.
- Stable machining actualizes long tool life and enhanced surface finish.

Line-up and features

Workpiece	Used	Product name	Type	No. of tooth	Diameter (Range)	Picture	Features
						No. of standard items	
H	High hardness (~HRC65)	The Mirror Endmill (PCD)		-	0.3~2	 6 Items	- Optimal surface finish by PCD ball Endmill with no edge - Nano-level surface finish due to its ultra-fine Endmill
	High hardness (~HRC65)	The Mirror Endmill (cBN)		2	0.4~2	 9 Items	- Higher productivity and surface finish in high speed cutting - Stable tool life and surface from high precision Endmill
	High hardness (HRC50~63)	H-Star Endmill	  	2~6	1.0~20	 3,007 Items	- Suitable for high speed machining of hardened workpieces (HRC50~63) - Improved initial chipping resistance and good wear resistance with optimized edge treatment for high hardness steel cutting
P K	Hardness (HRC30~50)	U-Star Endmill	  	2~6	0.1~25	 4,585 Items	Suitable for machining medium hardness workpieces (HRC30~50) made of alloy steel, carbon steel, die steel, etc.
	General (~HRC30)	G-Star Endmill	  	2~4	1.0~20	 456 Items	- For general machining with high performance and high quality - For various workpiece machining (carbon steel, alloy steel, cast iron, pre-hardened, etc.)
M	Stainless steel	S-Star Endmill	   	2~7	1.0~20	 218 Items	- Optimal performance in stainless machining - Enhanced oxidation resistance
S	HRSA	Super Endmill	 	4	3.0~20	 162 Items	- Endmill for HRSA machining - Optimal for machining of Ni based HRSA such as Inconel, Hastelloy, Waspaloy, etc.
	Titanium	Super Endmill	 	2/4	1.0~20	 64 Items	- Endmill for titanium and stainless steel cutting - Longer tool life : high toughness substrate and high lubrication coating layer
N	Non-ferrous metal, Aluminum	A-Star Endmill	  	2~3	1.0~20	 187 Items	- Effective chip evacuation in high feed machining with U-shape - Double relief angle (stronger cutting edge)
	Composite materials	Composite Router Endmill		2~8	4.0~12	 44 Items	- Router for composite material machining - High performance due to Nano-Crystalline dia-coating
	Graphite, Ceramics	D Endmill	  	2~4	0.6~12	 280 Items	- Longer tool life due to high hardness dia-coating - Applying one-pass grinding and good surface finish
	Dental, metal, wax, Zirconia	T Endmill		2	0.6~3	 214 Items	- Endmill for machining materials for dental prosthesis, Zirconia, Titanium, Co-Cr, Wax, PMMA, etc. - Applicable to dental milling machine and various materials for steeping flutes
	Copper, Copper alloy	C-Max Endmill	  	2	0.5~12	 94 Items	- Application of K-Silver Coating(wear resistance, chipping resistance) - Optimal for copper and non-ferrous metal machining
	For general machining with special function	Roughing		2~4	5.0~25	 204 Items	Endmill with a shape minimizing cutting load for roughing

B Technical Information for Super Endmill for HRSA

Endmill for Ni series HRSA machining (Inconel, Hasteloy, Waspaloy and etc.)

Super Endmill for HRSA

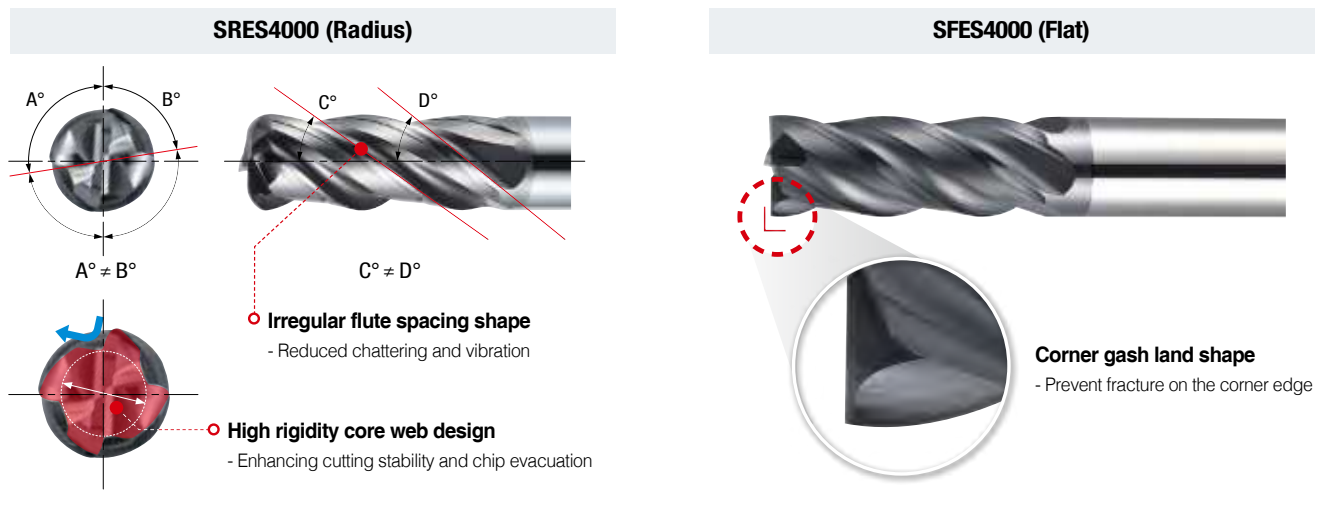
- Super Endmill For HRSA increases cutting performance and cutting stability applying positive rake angle and irregular flute spacing. Also, the new coating layer with high hardness controls fracture in cutting edge and ensures long tool life for HRSA machining by its increased wear resistance.

Code system

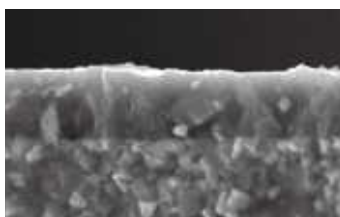
S	RE	S	4	120	-	080	-	R30
Super Endmill	Type	Workpiece	No. of flute	Tool diameter		Overall length		Corner R
	FE: Flat RE: Radius BE: Ball	S: Super alloy • Inconel718 • Waspaloy • Hastelloy T: Titanium/STS	4: 4 flutes	120: Ø12.0mm		080: 80mm		R30: 3.0mm

Features

- **Aerospace and generation industries:** Exclusive Endmill series for HRSA workpieces engine, turbine parts and etc.
- **Sharp cutting edge:** Reducing cutting load and suppression of work hardening
- **Longer tool life:** High toughness substrate and new grade with high wear resistance applied



Grade features



SL Coated (Super Lubricating coating)

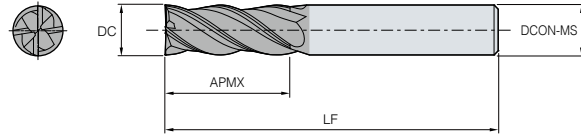
- **Applying high lubrication coating and special surface treatment technology**
- Increased welding resistance, chipping resistance and cutting stability by surface treatment technology

SFES4000

Flat



DC	Tolerance
Ø3 - Ø6	0.00 ~ -0.015
Ø8 - Ø20	0.00 ~ -0.020



(mm)

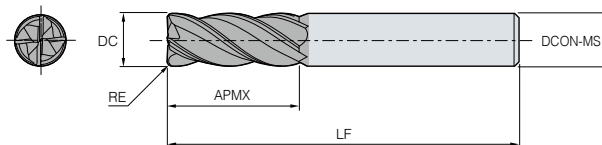
Designation	DC	DCON-MS	APMX	LF
SFES 4030-050	3	6	8	50
4040-050	4	6	10	50
4050-060	5	6	15	60
4060-060	6	6	15	60
4080-070	8	8	20	70
4100-075	10	10	25	75
4120-080	12	12	30	80
4140-090	14	14	35	90
4160-100	16	16	42	100
4200-100	20	20	48	100

SRES4000

Radius



DC	Tolerance
Ø3 ~ Ø6	0.00 ~ -0.015
Ø8 ~ Ø20	0.00 ~ -0.020



(mm)

Designation	DC	DCON-MS	APMX	LF	RE
SRES 4030-055-R02	3	6	8	55	0.2
4030-055-R03	3	6	8	55	0.3
4030-055-R05	3	6	8	55	0.5
4040-055-R02	4	6	10	55	0.2
4040-055-R03	4	6	10	55	0.3
4040-055-R05	4	6	10	55	0.5
4040-070-R02	4	6	10	70	0.2
4040-070-R03	4	6	10	70	0.3
4040-070-R05	4	6	10	70	0.5
4050-055-R02	5	6	15	55	0.2
4050-055-R03	5	6	15	55	0.3
4050-055-R05	5	6	15	55	0.5
4050-090-R02	5	6	15	90	0.2
4050-090-R03	5	6	15	90	0.3
4050-090-R05	5	6	15	90	0.5
4060-060-R03	6	6	15	60	0.3
4060-060-R05	6	6	15	60	0.5
4060-060-R08	6	6	15	60	0.8
4060-060-R10	6	6	15	60	1
4060-060-R15	6	6	15	60	1.5
4060-060-R20	6	6	15	60	2
4060-090-R03	6	6	15	90	0.3
4060-090-R05	6	6	15	90	0.5
4060-090-R08	6	6	15	90	0.8
4060-090-R10	6	6	15	90	1
4060-090-R15	6	6	15	90	1.5
4060-090-R20	6	6	15	90	2
4080-070-R03	8	8	20	70	0.3
4080-070-R05	8	8	20	70	0.5
4080-070-R08	8	8	20	70	0.8
4080-070-R10	8	8	20	70	1
4080-070-R15	8	8	20	70	1.5
4080-070-R20	8	8	20	70	2
4080-070-R25	8	8	20	70	2.5
4080-070-R30	8	8	20	70	3
4080-100-R03	8	8	20	100	0.3
4080-100-R05	8	8	20	100	0.5
4080-100-R08	8	8	20	100	0.8
4080-100-R10	8	8	20	100	1
4080-100-R15	8	8	20	100	1.5
4080-100-R20	8	8	20	100	2
4080-100-R25	8	8	20	100	2.5

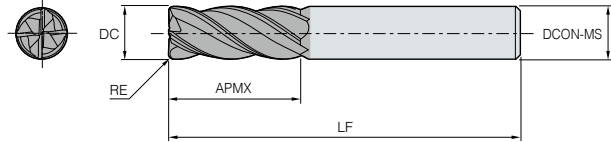
Designation	DC	DCON-MS	APMX	LF	RE
SRES 4080-100-R30	8	8	20	100	3
4100-075-R03	10	10	25	75	0.3
4100-075-R05	10	10	25	75	0.5
4100-075-R08	10	10	25	75	0.8
4100-075-R10	10	10	25	75	1
4100-075-R15	10	10	25	75	1.5
4100-075-R20	10	10	25	75	2
4100-075-R25	10	10	25	75	2.5
4100-075-R30	10	10	25	75	3
4100-100-R03	10	10	25	100	0.3
4100-100-R05	10	10	25	100	0.5
4100-100-R08	10	10	25	100	0.8
4100-100-R10	10	10	25	100	1
4100-100-R15	10	10	25	100	1.5
4100-100-R20	10	10	25	100	2
4100-100-R25	10	10	25	100	2.5
4100-100-R30	10	10	25	100	3
4120-080-R05	12	12	30	80	0.5
4120-080-R08	12	12	30	80	0.8
4120-080-R10	12	12	30	80	1
4120-080-R15	12	12	30	80	1.5
4120-080-R20	12	12	30	80	2
4120-080-R25	12	12	30	80	2.5
4120-080-R30	12	12	30	80	3
4120-080-R35	12	12	30	80	3.5
4120-080-R40	12	12	30	80	4
4120-110-R05	12	12	30	110	0.5
4120-110-R08	12	12	30	110	0.8
4120-110-R10	12	12	30	110	1
4120-110-R15	12	12	30	110	1.5
4120-110-R20	12	12	30	110	2
4120-110-R25	12	12	30	110	2.5
4120-110-R30	12	12	30	110	3
4120-110-R35	12	12	30	110	3.5
4120-110-R40	12	12	30	110	4
4140-090-R05	14	14	35	90	0.5
4140-090-R08	14	14	35	90	0.8
4140-090-R10	14	14	35	90	1
4140-090-R15	14	14	35	90	1.5
4140-090-R20	14	14	35	90	2
4140-090-R30	14	14	35	90	3
4140-150-R05	14	14	35	150	0.5

SRES4000

Radius



DC	Tolerance
Ø3 - Ø6	0.00 - -0.015
Ø8 - Ø20	0.00 - -0.020



Designation	DC	DCON-MS	APMX	LF	RE
SRES 4140-150-R08	14	14	35	150	0.8
4140-150-R10	14	14	35	150	1
4140-150-R15	14	14	35	150	1.5
4140-150-R20	14	14	35	150	2
4140-150-R30	14	14	35	150	3
4160-100-R05	16	16	42	100	0.5
4160-100-R08	16	16	42	100	0.8
4160-100-R10	16	16	42	100	1
4160-100-R15	16	16	42	100	1.5
4160-100-R20	16	16	42	100	2
4160-100-R25	16	16	42	100	2.5
4160-100-R30	16	16	42	100	3
4160-100-R35	16	16	42	100	3.5
4160-100-R40	16	16	42	100	4
4160-100-R50	16	16	42	100	5
4160-100-R60	16	16	42	100	6
4160-150-R05	16	16	42	150	0.5
4160-150-R08	16	16	42	150	0.8
4160-150-R10	16	16	42	150	1
4160-150-R15	16	16	42	150	1.5
4160-150-R20	16	16	42	150	2
4160-150-R25	16	16	42	150	2.5
4160-150-R30	16	16	42	150	3
4160-150-R35	16	16	42	150	3.5
4160-150-R40	16	16	42	150	4
4160-150-R50	16	16	42	150	5
4160-150-R60	16	16	42	150	6
4180-100-R05	18	20	45	100	0.5
4180-100-R08	18	20	45	100	0.8
4180-100-R10	18	20	45	100	1

Designation	DC	DCON-MS	APMX	LF	RE
SRES 4180-100-R15	18	20	45	100	1.5
4180-100-R20	18	20	45	100	2
4180-100-R30	18	20	45	100	3
4180-150-R05	18	20	45	150	0.5
4180-150-R08	18	20	45	150	0.8
4180-150-R10	18	20	45	150	1
4180-150-R15	18	20	45	150	1.5
4180-150-R20	18	20	45	150	2
4180-150-R30	18	20	45	150	3
4200-100-R05	20	20	48	100	0.5
4200-100-R10	20	20	48	100	1
4200-100-R15	20	20	48	100	1.5
4200-100-R20	20	20	48	100	2
4200-100-R25	20	20	48	100	2.5
4200-100-R30	20	20	48	100	3
4200-100-R35	20	20	48	100	3.5
4200-100-R40	20	20	48	100	4
4200-100-R50	20	20	48	100	5
4200-100-R60	20	20	48	100	6
4200-150-R05	20	20	48	150	0.5
4200-150-R10	20	20	48	150	1
4200-150-R15	20	20	48	150	1.5
4200-150-R20	20	20	48	150	2
4200-150-R25	20	20	48	150	2.5
4200-150-R30	20	20	48	150	3
4200-150-R35	20	20	48	150	3.5
4200-150-R40	20	20	48	150	4
4200-150-R50	20	20	48	150	5
4200-150-R60	20	20	48	150	6

B Technical Information for Super Endmill For Ti

Endmills series for difficult-to-cut materials

Super Endmill for Ti

• With its optimal edge structure for Titanium machining and enlarged chip pocket in flutes design, Super Endmill For Ti reduces cutting load and cutting heat and it improves chip evacuation. In addition, applying high toughness substrate and high lubrication coating layer minimizing irregular tool fracture and welding ensure maximized tool life.

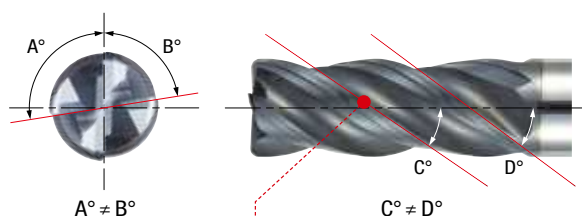
Code system

S	RE	T	4	120	-	080	-	R30
Super Endmill	Type	Workpiece	No. of flute	Tool diameter		Overall length		Corner R
	FE: Flat RE: Radius BE: Ball	S: Super alloy • Inconel718 • Waspaloy • Hastelloy T: Titanium/STS	4: 4 flutes	120: Ø12.0mm		080: 80mm		R30: 3.0mm

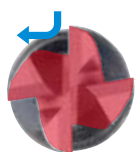
Features

- Endmill for titanium and stainless steel cutting
- **Longer tool life** : high toughness substrate and high lubrication coating layer

SFET (Flat) / SRET (Radius)

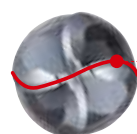
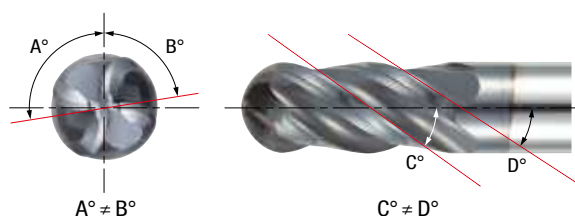


• **Irregular flute spacing shape**
- Reduced chattering and vibration



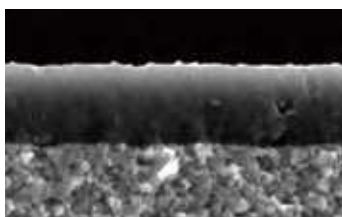
• **Large chip pocket and streamlined flute design**
- Good chip evacuation

SBET (Ball)



• **S-curve cutting edge**
- Reduced cutting load

Grade features



UL Coated (Ultra Lubricating coating)

- Enhanced chip control and welding resistance by exclusive lubrication coating technology
- High chipping resistance substrate

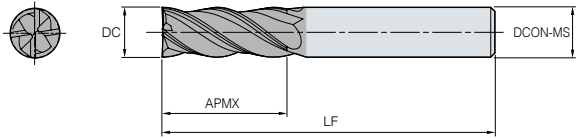
SFET4000

Flat



DC
Ø3 - Ø6
Ø8 - Ø20

Tolerance
0.00 ~ -0.015
0.00 ~ -0.020



(mm)

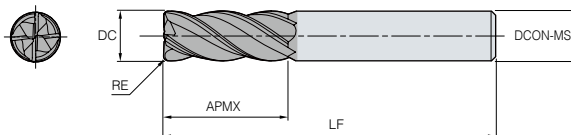
Designation		DC	DCON-MS	APMX	LF
SFET	4030-050	3	6	8	50
	4040-050	4	6	10	50
	4050-060	5	6	15	60
	4060-060	6	6	15	60
	4080-070	8	8	20	70
	4100-075	10	10	25	75
	4120-080	12	12	30	80
	4160-100	16	16	42	100
	4200-100	20	20	48	100

SRET4000

Radius



DC	Tolerance
Ø3 ~ Ø6	0.00 ~ -0.015
Ø8 ~ Ø20	0.00 ~ -0.020



(mm)

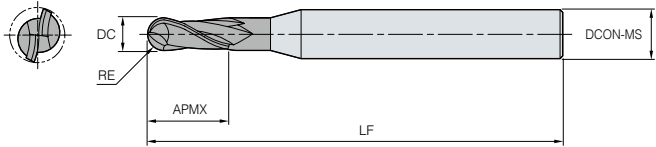
Designation		DC	DCON-MS	APMX	LF	RE
SRET	4030-050-R02	3	6	8	50	0.2
	4030-050-R05	3	6	8	50	0.5
	4040-050-R02	4	6	10	50	0.2
	4040-050-R05	4	6	10	50	0.5
	4050-060-R02	5	6	15	60	0.2
	4050-060-R05	5	6	15	60	0.5
	4050-060-R10	5	6	15	60	1
	4060-060-R03	6	6	15	60	0.3
	4060-060-R05	6	6	15	60	0.5
	4060-060-R10	6	6	15	60	1
	4080-070-R03	8	6	20	70	0.3
	4080-070-R05	8	8	20	70	0.5
	4080-070-R10	8	8	20	70	1
	4100-075-R03	10	8	25	75	0.3
	4100-075-R05	10	8	25	75	0.5
	4100-075-R10	10	10	25	75	1
	4100-075-R15	10	10	25	75	1.5
	4100-075-R20	10	10	25	75	2
	4120-080-R05	12	12	30	80	0.5
	4120-080-R10	12	12	30	80	1
	4120-080-R15	12	16	30	80	1.5
	4120-080-R20	12	16	30	80	2
	4120-080-R25	12	16	30	80	2.5
	4120-080-R30	12	16	30	80	3
	4160-100-R05	16	20	42	100	0.5
	4160-100-R10	16	20	42	100	1
	4200-100-R05	20	20	48	100	0.5
	4200-100-R10	20	20	48	100	1

SBET2000

Ball



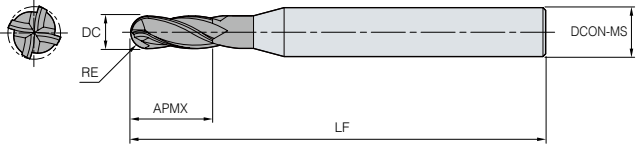
					DC Ø4 - Ø6 Ø8 - Ø12	Tolerance 0.00 ~ -0.015 0.00 ~ -0.020
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Designation		DC	DCON-MS	APMX	LF	RE
SBET	2040-050	4	6	8	50	2
	2040-070	4	6	8	70	2
	2050-060	5	6	12	60	2.5
	2050-080	5	6	12	80	2.5
	2060-060	6	6	12	60	3
	2060-090	6	6	12	90	3
	2080-070	8	8	16	70	4
	2080-100	8	8	16	100	4
	2100-075	10	10	20	75	5
	2100-100	10	10	20	100	5
	2120-080	12	12	25	80	6
	2120-100	12	12	25	100	6

SBET4000

Ball



						(mm)
Designation		DC	DCON-MS	APMX	LF	RE
SBET	4040-050	4	6	8	50	2
	4040-070	4	6	8	70	2
	4050-060	5	6	12	60	2.5
	4050-080	5	6	12	80	2.5
	4060-060	6	6	12	60	3
	4060-090	6	6	12	90	3
	4080-070	8	8	16	70	4
	4080-100	8	8	16	100	4
	4100-075	10	10	20	75	5
	4100-100	10	10	20	100	5
	4120-080	12	12	25	80	6
	4120-100	12	12	25	100	6

High precision mold manufacture solution

The Mirror Endmill

- Suitable for high hardness (higher than HRC60) die mold super precision cutting
- Enhanced wear resistance from applying the optimal grade for PCD, cBN

PCD Endmill

For polishing of high precision workpiece and high hardness mold

- Optimal surface finish by PCD ball Endmill with no edge
- Nano-level surface finish due to its ultra-fine Endmill
- Enhanced wear resistance from applying the optimal grade for PCD



cBN Endmill (Ball)

For ultra-fine and mirror-like workpiece and mold with over HRC60 machining

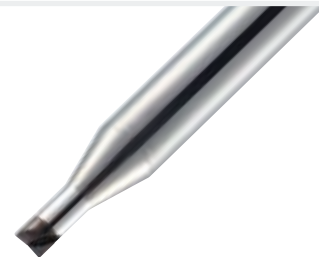
- Higher productivity and surface finish in high speed cutting
- Enhanced wear resistance due to the optimal cBN grade
- Longer tool life by shape with strong cutting edge
- Stable tool life and surface from high precision Endmill



cBN Endmill (Radius)

For medium cutting of high precision workpiece and mold machining above HRC60

- Higher productivity in high speed machining
- Better wear resistance of tool due to applying the optimal grade for cBN
- Good surface finish through connecting smooth cutting edge and body
- Long tool life from strong cutting edge



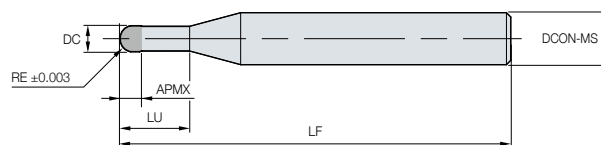
PCD-MBE

Ball



h3
shank

DC	Tolerance
Ø0.3 ~ Ø2	0 ~ -0.008mm



(mm)

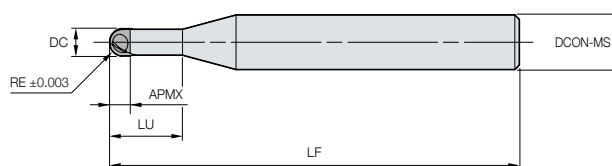
	Designation	DC	DCON-MS	APMX	LU	LF	RE
PCD-MBE	0003-048-N007S04	0.3	4	0.15	0.75	48	0.15
	0004-048-N010S04	0.4	4	0.2	1	48	0.2
	0006-048-N015S04	0.6	4	0.3	1.5	48	0.3
	0010-048-N025S04	1	4	0.5	2.5	48	0.5
	0015-048-N040S04	1.5	4	0.75	4	48	0.75
	0020-048-N050S04	2	4	1	5	48	1

cBN-MBE2000

Ball



DC	Tolerance
Ø0.4 - Ø2	0 - -0.008mm



(mm)

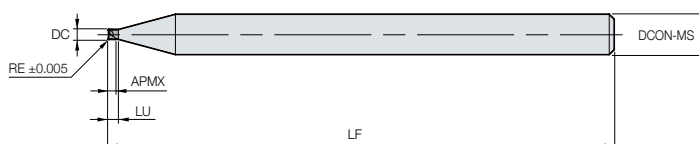
Designation	DC	DCON-MS	APMX	LU	LF	RE
cBN-MBE 2004-050-N005S04	0.4	4	0.33	0.5	50	0.2
2004-050-N010S04	0.4	4	0.33	1	50	0.2
2005-050-N010S04	0.5	4	0.38	1	50	0.25
2006-050-N015S04	0.6	4	0.5	1.5	50	0.3
2008-050-N020S04	0.8	4	0.6	2	50	0.4
2010-050-N025S04	1	4	0.7	2.5	50	0.5
2012-050-N030S04	1.2	4	0.8	3	50	0.6
2015-050-N040S04	1.5	4	1	4	50	0.75
2020-050-N050S04	2	4	1.2	5	50	1

cBN-MRE2000

Radius



DC	Tolerance
Ø0.4 - Ø2	0 - -0.008mm



(mm)

Designation	DC	DCON-MS	APMX	LU	LF	RE
cBN-MRE 2004-050-R005-N005S04	0.4	4	0.24	0.5	50	0.05
2004-050-R005-N010S04	0.4	4	0.24	1	50	0.05
2004-050-R010-N005S04	0.4	4	0.24	0.5	50	0.1
2004-050-R010-N010S04	0.4	4	0.24	1	50	0.1
2005-050-R005-N005S04	0.5	4	0.3	0.5	50	0.05
2005-050-R005-N010S04	0.5	4	0.3	1	50	0.05
2005-050-R010-N005S04	0.5	4	0.3	0.5	50	0.1
2005-050-R010-N010S04	0.5	4	0.3	1	50	0.1
2010-050-R010-N010S04	1	4	0.7	1	50	0.1
2010-050-R010-N020S04	1	4	0.7	2	50	0.1
2010-050-R010-N030S04	1	4	0.7	3	50	0.1
2010-050-R010-N040S04	1	4	0.7	5	50	0.1
2010-050-R020-N010S04	1	4	0.7	1	50	0.2
2010-050-R020-N020S04	1	4	0.7	2	50	0.2
2015-050-R010-N030S04	1.5	4	0.85	3	50	0.1
2015-050-R010-N045S04	1.5	4	0.85	4.5	50	0.1
2015-050-R010-N075S04	1.5	4	0.85	7.5	50	0.1
2015-050-R020-N045S04	1.5	4	0.85	4.5	50	0.2
2020-050-R010-N040S04	2	4	0.85	4	50	0.1
2020-050-R010-N060S04	2	4	0.85	6	50	0.1
2020-050-R010-N100S04	2	4	0.85	10	50	0.1
2020-050-R020-N040S04	2	4	0.85	4	50	0.2
2020-050-R020-N060S04	2	4	0.85	6	50	0.2
2020-050-R020-N100S04	2	4	0.85	10	50	0.2

※ In case longer flute length than 0.85mm with Ø1.5~ 2.0 is necessary, it could be order made by 1.2mm.

For high hardness (HRC50~63)

H-Star Endmill

- **High hardness coating layer** - Ensuring stable cutting from high Si content, increased wear resistance and frictional heat resistance due to applying a new AlTiSiN series coating layer
- **High hardness substrate** - Containing ultra-fine WC + Co 9% and expanded general application range by maximizing cutting edge feature
- **Edge treatment** - Increased chipping resistance in the beginning of high hardness steel cutting and enhanced wear resistance lead to stable cutting

Code system

ES	R	70	4	100	15	32	-	10
H-Star Endmill	Type	Length, Shank type	No. of flute	Tool dia.	Corner R	Effective		Shank dia.
	B: Ball E: Flat R: Radius XE: Flat (irregular flute spacing) XR: Radius (irregular flute spacing) PM: Power mill RB: Rib ball RE: Rib flat RR: Rib radius LNB: Long neck ball TNB: Taper neck ball LNS: Long neck flat LNR: Long neck radius	70: Neck 71: Straight, neck 72: Long shank neck 73: Long shank 74: Helix 35°	2: 2 flutes 3: 3 flutes 4: 4 flutes 6: 6 flutes	010: Ø1.0mm 060: Ø6.0mm 065: Ø6.5mm 100: Ø10.0mm	05: 0.5mm 15: 1.5mm 20: 2.0mm	10: 10mm 12: 12mm 32: 32mm		06: Ø6.0mm 10: Ø10.0mm 12: Ø12.0mm

Features

• Edge treatment



- Enhancing chipping resistance in the beginning of high hardness steel cutting
- Increased wear resistance and stable cutting performance

• High hardness substrate



- Ultra-fine WC + Co 9%
- Expanded general application range by maximizing cutting edge feature

• High hardness coating layer (PC305H)



- High Si content
- Enhanced wear resistance
- Stable cutting through frictional heat resistance increase



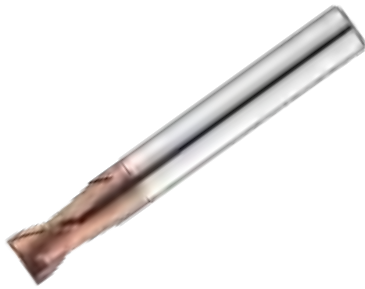
B Technical Information for H-Star Endmill

Line-up

Designation	Appearance	Type	Tolerance	Irregular flute spacing	page
ESE702		2 flutes neck type flat Endmill	Ø0.1 ~ 20.0		B26
ESE712		2 flutes flat Endmill	Ø1.0 ~ 12.0		B27
ESE704		4 flutes neck type flat Endmill	Ø1.0 ~ 20.0		B28
ESE714		4 flutes high helix flat Endmill	Ø1.0 ~ 12.0		B29
ESE724(6)		4&6 flutes neck type flat Endmill	Ø1.0 ~ 12.0		B30
ESE744		4 flutes high helix flat Endmill	Ø1.0 ~ 12.0		B31
ESE716		6 flutes high helix flat Endmill	Ø6.0 ~ 20.0		B32
ESRE712		2 flutes rib neck type flat Endmill	Ø0.1 ~ 12.0		B33~B35
ESRE714		4 flutes rib flat Endmill	Ø0.5 ~ 12.0		B36~B37
ESXE704		4 flutes neck type flat Endmill	Ø1.0 ~ 12.0	●	B38
ESXE714		4 flutes flat Endmill	Ø2.0 ~ 12.0	●	B38
ESLNS20		2 flutes long neck type flat Endmill	Ø0.1 ~ 5.0		B39~B42
ESLNS40		4 flutes long neck type flat Endmill	Ø1.0 ~ 5.0		B43~B44
ESR702		2 flutes neck type radius Endmill	Ø1.0 ~ 12.0		B45~B47
ESR732		2 flutes long shank radius Endmill	Ø1.0 ~ 12.0		B48
ESR704		4 flutes neck type radius Endmill	Ø1.0 ~ 12.0		B49~B50
ESR714		4 flutes radius Endmill	Ø3.0 ~ 12.0		B51
ESR724		4 flutes neck type radius Endmill	Ø6.0 ~ 12.0		B51
ESR734		4 flutes long shank radius Endmill	Ø1.0 ~ 12.0		B52
ESR706		6 flutes neck type radius Endmill	Ø6.0 ~ 12.0		B53
ESR736		6 flutes radius Endmill	Ø6.0 ~ 12.0		B53
ESRR712		2 flutes rib radius Endmill	Ø0.2 ~ 16.0		B54~B59
ESRR714		4 flutes rib radius Endmill	Ø0.5 ~ 20.0		B60~B66
ESXR704		4 flutes neck type radius Endmill	Ø1.0 ~ 12.0	●	B67
ESLNR20		2 flutes long radius Endmill	Ø0.2 ~ 3.0		B68~B71
ESTNR20		2 flutes tapered neck type radius Endmill	Ø0.2 ~ 3.0		B72~B73
ESPM4		4 flutes neck type radius Endmill	Ø3.0 ~ 12.0		B74
ESB702		2 flutes neck type ball Endmill	Ø0.1 ~ 12.0		B75
ESB712		2 flutes ball Endmill	Ø1.0 ~ 12.0		B76
ESB703		3 flutes neck type ball Endmill	Ø2.0 ~ 12.0		B77
ESB734		4 flutes 15° helix ball Endmill	Ø2.0 ~ 10.0		B78
ESRB712		2 flutes rib ball Endmill	Ø0.1 ~ 12.0		B79~B82
ESLNB20		2 flutes long neck type ball Endmill	Ø0.1 ~ 5.0		B83~B86
ESTNB20		2 flutes tapered neck type ball Endmill	Ø0.2 ~ 10.0		B87~B90
ESTNB30		3 flutes tapered neck type ball Endmill	Ø2.0 ~ 5.0		B91

ESE702

2 flutes neck type flat



CARBIDE

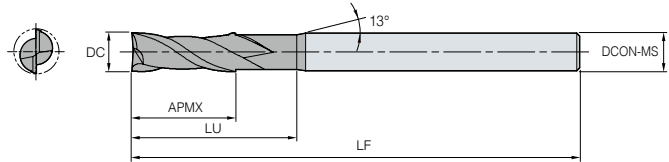
2

H-A
30°

Grade
PC305H

h5
shank

DC
Ø0.1 ~ Ø6
Ø8 ~ Ø20

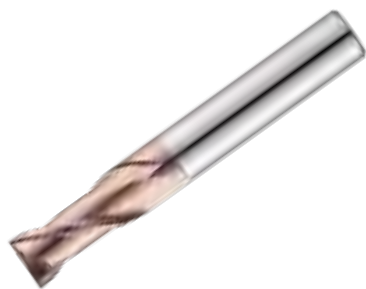
Tolerance
0.000 ~ -0.012
0.000 ~ -0.015


(mm)

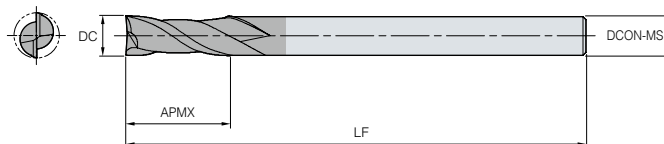
Designation	DC	DCON-MS	APMX	LU	LF
ESE 702001	0.1	4	0.2	-	40
702002	0.2	4	0.4	-	40
702003	0.3	4	0.5	-	40
702004	0.4	4	0.7	-	40
702005	0.5	4	1	-	40
702006	0.6	4	1.2	-	40
702007	0.7	4	1.4	-	40
702008	0.8	4	1.6	-	40
702009	0.9	4	2	-	40
702010	1	6	1.5	-	40
702010S4	1	4	1.5	-	40
702015	1.5	6	2.2	-	40
702020	2	6	3	6	40
702020S4	2	4	3	6	40
702025	2.5	6	4	6	40
702030	3	6	4	7	45
702035	3.5	6	6	9	45
702040	4	6	6	9	45
702045	4.5	6	6	10	45
702050	5	6	6	11	50
702060	6	6	7	14	50
702080	8	8	9	18	60
702100	10	10	12	25	75
702120	12	12	15	30	75
702160	16	16	18	38	90
702200	20	20	24	45	100

ESE712

2 flutes flat



DC	Tolerance
Ø1 ~ Ø6	0.000 ~ -0.012
Ø8 ~ Ø12	0.000 ~ -0.015

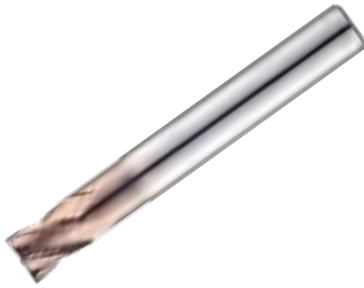


(mm)

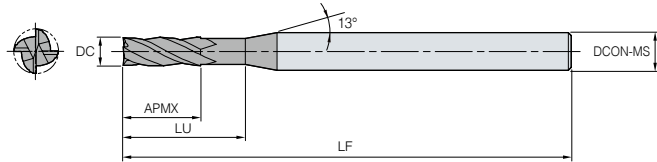
	Designation	DC	DCON-MS	APMX	LF
ESE	712010	1	6	3	40
	712010-02	1	6	2	40
	712010-02S4	1	4	2	40
	712010-04	1	6	4	40
	712012	1.2	6	3	40
	712015	1.5	6	4	40
	712015S4	1.5	4	4	40
	712015-06	1.5	6	6	40
	712015-08	1.5	6	8	40
	712020	2	6	5	40
	712020S4	2	4	5	40
	712020-08	2	6	8	40
	712020-10	2	6	10	50
	712025	2.5	6	6	40
	712025S4	2.5	4	6	40
	712030	3	6	8	45
	712030S4	3	4	8	45
	712030-10	3	6	10	50
	712030-12	3	6	12	50
	712035	3.5	6	10	45
	712040	4	6	10	45
	712040S4	4	4	10	45
	712040-12	4	6	12	50
	712040-16	4	6	16	60
	712045	4.5	6	11	45
	712050	5	6	13	50
	712055	5.5	6	13	50
	712060	6	6	13	50
	712060-15	6	6	15	60
	712065	6.5	8	16	60
	712070	7	8	18	60
	712080	8	8	19	60
	712100	10	10	22	70
	712100-25	10	10	25	70
	712120	12	12	26	75
	712120-30	12	12	30	75

ESE704

4 flutes neck type flat



CARBIDE	4	H-A 30°	Grade PC305H	h5 shank	DC Ø1 - Ø6 Ø8 - Ø20	Tolerance 0.000 ~ -0.012 0.000 ~ -0.015
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(mm)

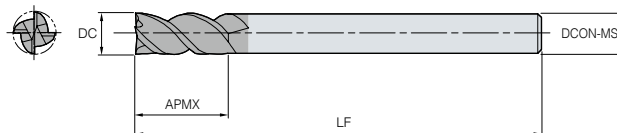
Designation	DC	DCON-MS	APMX	LU	LF
ESE 704010	1	6	1.5	-	40
704010S4	1	4	1.5	-	40
704015	1.5	6	2.2	-	40
704015S4	1.5	4	2.2	-	40
704020	2	6	3	6	40
704020S4	2	4	3	6	40
704025	2.5	6	4	6	40
704025S4	2.5	4	4	6	40
704030	3	6	4	7	45
704030S4	3	4	4	7	45
704035	3.5	6	5	9	45
704040	4	6	5	9	45
704040S4	4	4	5	9	45
704045	4.5	6	6	10	45
704050	5	6	6	11	50
704060	6	6	7	14	50
704080	8	8	9	18	60
704100	10	10	12	25	75
704120	12	12	15	30	75
704160	16	16	18	38	90
704200	20	20	24	45	100

ESE714

4 flutes high helix flat



DC	Tolerance
Ø1 ~ Ø6	0.000 ~ -0.012
Ø8 ~ Ø12	0.000 ~ -0.015



(mm)

Designation	DC	DCON-MS	APMX	LF
ESE 714010	1	6	2.5	40
714010S4	1	4	2.5	40
714012	1.2	6	3	40
714015	1.5	6	4	40
714015S4	1.5	4	4	40
714020	2	6	5	40
714020S4	2	4	5	40
714025	2.5	6	6	40
714025S4	2.5	4	6	40
714030	3	6	8	45
714030S4	3	4	8	45
714035	3.5	6	9	45
714040	4	6	10	45
714040S4	4	4	10	45
714050	5	6	13	50
714060	6	6	13	50
714060-15	6	6	15	60
714060-15L	6	6	15	90
714080	8	8	19	60
714080L	8	8	19	100
714100	10	10	22	70
714100-25	10	10	25	70
714100-25L	10	10	25	100
714120	12	12	26	75
714120-30	12	12	30	80
714120-30L	12	12	30	100

ESE724(6)

4&6 flutes neck type flat

CARBIDE

4

6

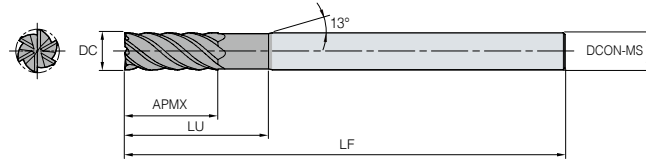
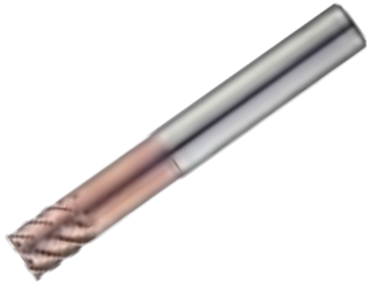
H-A
45°

H-A
60°

Grade
PC305H

h5
shank

DC	Tolerance
Ø1 - Ø6	0.000 ~ -0.012
Ø8 - Ø12	0.000 ~ -0.015



(mm)

Designation		DC	DCON-MS	APMX	LU	LF	NOF
ESE	724010	1	6	1.5	5	45	4
	724015	1.5	6	2.2	6	45	4
	724020	2	6	3	8	45	4
	724030	3	6	4	9	50	4
	724040	4	6	5	12	50	4
	724040S4L	4	4	5	12	75	4
	724050	5	6	6	15	50	4
	726060	6	6	7	20	60	6
	726080	8	8	9	25	70	6
	726100	10	10	12	32	75	6
	726120	12	12	15	38	80	6

ESE744

4 flutes flat



CARBIDE

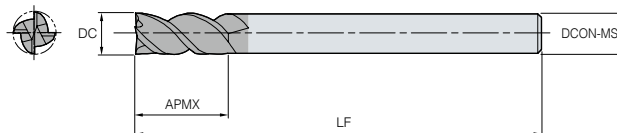
4

H-A
35°

Grade
PC305H

h5
shank

DC	Tolerance
Ø1 ~ Ø6	0.000 ~ -0.012
Ø8 ~ Ø12	0.000 ~ -0.015

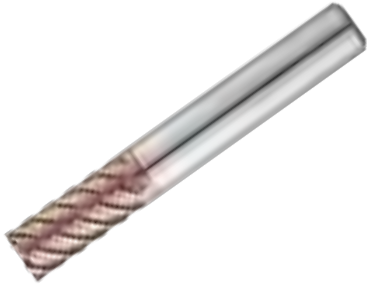


(mm)

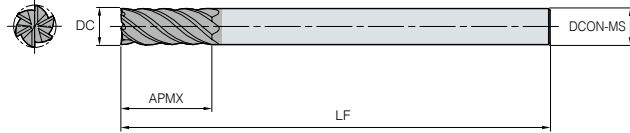
Designation		DC	DCON-MS	APMX	LF
ESE	744010S3	1	3	2.5	40
	744010S4	1	4	2.5	40
	744010	1	6	2.5	40
	744012S3	1.2	3	3	40
	744012S4	1.2	4	3	40
	744015S3	1.5	3	4	40
	744015S4	1.5	4	4	40
	744015	1.5	6	4	40
	744020S3	2	3	6	40
	744020S4	2	4	6	40
	744020	2	6	6	40
	744025S3	2.5	3	8	45
	744025S4	2.5	4	8	45
	744025	2.5	6	8	45
	744030S3	3	3	8	50
	744030S4	3	4	8	45
	744030	3	6	8	45
	744035	3.5	6	10	45
	744040S4	4	4	11	45
	744040	4	6	11	45
	744045	4.5	6	11	45
	744050	5	6	13	50
	744055	5.5	6	13	50
	744060	6	6	13	50
	744080	8	8	19	60
	744100	10	10	22	70
	744120	12	12	26	75

ESE716

6 flutes high helix flat



CARBIDE	6	H-A 50°	Grade PC305H	h5 shank	DC Ø6 Ø8 - Ø20	Tolerance 0.000 ~ -0.012 0.000 ~ -0.015
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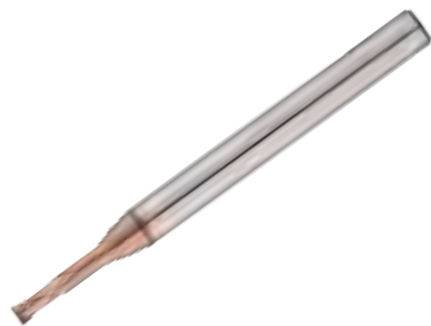


(mm)

Designation		DC	DCON-MS	APMX	LF
ESE	716060	6	6	13	50
	716080	8	8	18	60
	716100	10	10	22	70
	716120	12	12	26	75
	716160	16	16	35	90
	716200	20	20	44	100

ESRE712

2 flutes rib neck type flat



CARBIDE

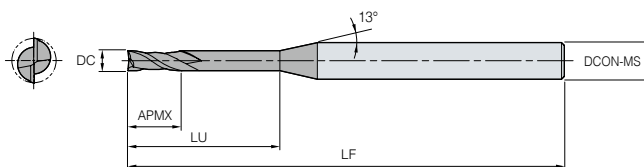
2

H-A
30°

Grade
PC305H

h5 shank

DC	Tolerance
Ø0.1 ~ Ø6	0.000 ~ -0.012
Ø8 ~ Ø12	0.000 ~ -0.015



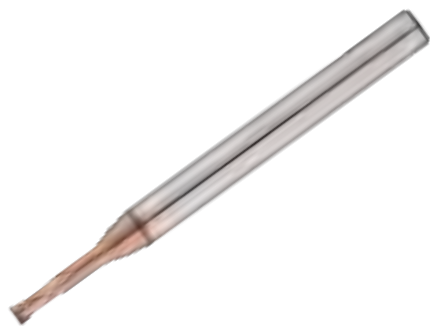
(mm)

	Designation	DC	DCON-MS	APMX	LU	LF
ESRE	712001003	0.1	4	0.15	0.3	40
	712001005	0.1	4	0.15	0.5	40
	71200101	0.1	4	0.15	1	40
	712002005	0.2	4	0.3	0.5	40
	712002015	0.2	4	0.3	1.5	40
	71200201	0.2	4	0.3	1	40
	71200202	0.2	4	0.3	2	40
	712003015	0.3	4	0.5	1.5	40
	71200301	0.3	4	0.5	1	40
	712003025	0.3	4	0.5	2.5	40
	71200302	0.3	4	0.5	2	40
	71200303	0.3	4	0.5	3	40
	71200304	0.3	4	0.5	4	40
	71200305	0.3	4	0.5	5	40
	712004015	0.4	4	0.6	1.5	40
	71200401	0.4	4	0.6	1	40
	712004025	0.4	4	0.6	2.5	40
	71200402	0.4	4	0.6	2	40
	71200403	0.4	4	0.6	3	40
	71200404	0.4	4	0.6	4	40
	71200405	0.4	4	0.6	5	40
	71200406	0.4	4	0.6	6	40
	71200408	0.4	4	0.6	8	40
	71200410	0.5	4	0.6	10	40
	712005015	0.5	4	0.7	1.5	45
	71200501	0.5	4	0.7	1	45
	712005025	0.5	4	0.7	2.5	45
	71200502	0.5	4	0.7	2	45
	71200503	0.5	4	0.7	3	45
	71200504	0.5	4	0.7	4	45
	71200505	0.5	4	0.7	5	45
	71200506	0.5	4	0.7	6	45
	71200508	0.5	4	0.7	8	45
	71200510	0.5	4	0.7	10	45
	71200512	0.5	4	0.7	12	45
	71200514	0.5	4	0.7	14	45
	71200516	0.5	4	0.7	16	45
	71200602	0.6	4	0.9	2	45

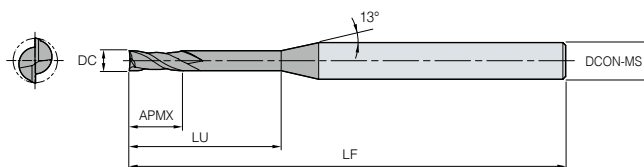
	Designation	DC	DCON-MS	APMX	LU	LF
ESRE	71200603	0.6	4	0.9	3	45
	71200604	0.6	4	0.9	4	45
	71200605	0.6	4	0.9	5	45
	71200606	0.6	4	0.9	6	45
	71200608	0.6	4	0.9	8	45
	71200610	0.6	4	0.9	10	45
	71200612	0.6	4	0.9	12	45
	71200614	0.6	4	0.9	14	45
	71200616	0.6	4	0.9	16	45
	71200702	0.7	4	1.2	2	45
	71200704	0.7	4	1.2	4	45
	71200706	0.7	4	1.2	6	45
	71200708	0.7	4	1.2	8	45
	71200710	0.7	4	1.2	10	45
	71200712	0.7	4	1.2	12	45
	71200802	0.8	4	1.2	2	45
	71200803	0.8	4	1.2	3	45
	71200804	0.8	4	1.2	4	45
	71200805	0.8	4	1.2	5	45
	71200806	0.8	4	1.2	6	45
	71200808	0.8	4	1.2	8	45
	71200810	0.8	4	1.2	10	45
	71200812	0.8	4	1.2	12	45
	71200814	0.8	4	1.2	14	45
	71200816	0.8	4	1.2	16	45
	71200820	0.8	4	1.2	20	50
	71200906	0.9	4	1.3	6	45
	71200908	0.9	4	1.3	8	45
	71200910	0.9	4	1.3	10	45
	71201002	1	4	1.5	2	50
	71201003	1	4	1.5	3	50
	71201004	1	4	1.5	4	50
	71201005	1	4	1.5	5	50
	71201006	1	4	1.5	6	50
	71201007	1	4	1.5	7	50
	71201008	1	4	1.5	8	50
	71201010	1	4	1.5	10	50
	71201012	1	4	1.5	12	50

ESRE712

2 flutes rib neck type flat



CARBIDE	2	H-A 30°	Grade PC305H	h5 shank	DC Ø0.1 ~ Ø6 Ø8 ~ Ø12	Tolerance 0.000 ~ -0.012 0.000 ~ -0.015
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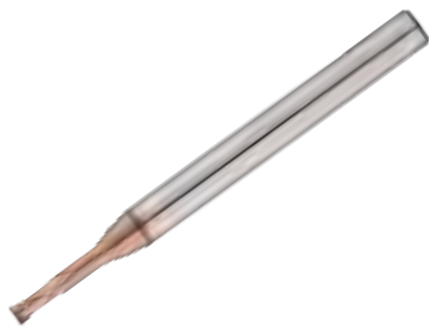


	Designation	DC	DCON-MS	APMX	LU	LF
ESRE	71201014	1	4	1.5	14	50
	71201016	1	4	1.5	16	50
	71201018	1	4	1.5	18	50
	71201020	1	4	1.5	20	50
	71201022	1	4	1.5	22	60
	71201026	1	4	1.5	26	60
	71201030	1	4	1.5	30	70
	71201040	1	4	1.5	40	80
	71201050	1	4	1.5	50	100
	71201204	1.2	4	1.8	4	50
	71201206	1.2	4	1.8	6	50
	71201208	1.2	4	1.8	8	50
	71201210	1.2	4	1.8	10	50
	71201212	1.2	4	1.8	12	50
	71201214	1.2	4	1.8	14	50
	71201216	1.2	4	1.8	16	50
	71201220	1.2	4	1.8	20	50
	71201226	1.2	4	1.8	26	60
	71201230	1.2	4	1.8	30	70
	71201406	1.4	4	2.1	6	50
	71201408	1.4	4	2.1	8	50
	71201410	1.4	4	2.1	10	50
	71201414	1.4	4	2.1	14	50
	71201416	1.4	4	2.1	16	50
	71201420	1.4	4	2.1	20	50
	71201504	1.5	4	2.3	4	50
	71201505	1.5	4	2.3	5	50
	71201506	1.5	4	2.3	6	50
	71201507	1.5	4	2.3	7	50
	71201508	1.5	4	2.3	8	50
	71201510	1.5	4	2.3	10	50
	71201512	1.5	4	2.3	12	50
	71201514	1.5	4	2.3	14	50
	71201516	1.5	4	2.3	16	50
	71201518	1.5	4	2.3	18	50
	71201520	1.5	4	2.3	20	50
	71201522	1.5	4	2.3	22	60
	71201526	1.5	4	2.3	26	60

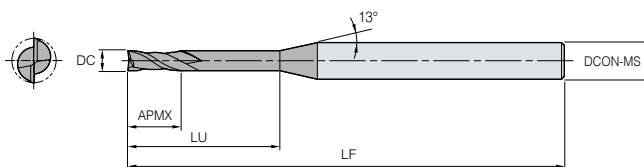
	Designation	DC	DCON-MS	APMX	LU	LF
ESRE	71201530	1.5	4	2.3	30	70
	71201608	1.6	4	2.3	8	50
	71201610	1.6	4	2.3	10	50
	71201612	1.6	4	2.3	12	50
	71201616	1.6	4	2.3	16	50
	71201620	1.6	4	2.3	20	50
	71201808	1.8	4	2.7	8	50
	71201810	1.8	4	2.7	10	50
	71201812	1.8	4	2.7	12	50
	71201816	1.8	4	2.7	16	50
	71201820	1.8	4	2.7	20	50
	71202006	2	4	3	6	50
	71202008	2	4	3	8	50
	71202010	2	4	3	10	50
	71202012	2	4	3	12	50
	71202014	2	4	3	14	50
	71202016	2	4	3	16	50
	71202018	2	4	3	18	50
	71202020	2	4	3	20	50
	71202022	2	4	3	22	60
	71202026	2	4	3	26	60
	71202030	2	4	3	30	70
	71202035	2	4	3	35	70
	71202040	2	4	3	40	80
	71202045	2	4	3	45	90
	71202050	2	4	3	50	100
	71202060	2	4	3	60	110
	71202508	2.5	4	4	8	50
	71202510	2.5	4	4	10	50
	71202512	2.5	4	4	12	50
	71202514	2.5	4	4	14	50
	71202516	2.5	4	4	16	50
	71202518	2.5	4	4	18	50
	71202520	2.5	4	4	20	50
	71202522	2.5	4	4	22	60
	71202526	2.5	4	4	26	60
	71202530	2.5	4	4	30	70
	71202535	2.5	4	4	35	70

ESRE712

2 flutes rib neck type flat



DC	Tolerance
Ø0.1 ~ Ø6	0.000 ~ -0.012
Ø8 ~ Ø12	0.000 ~ -0.015



(mm)

	Designation	DC	DCON-MS	APMX	LU	LF
ESRE	71202540	2.5	4	4	40	80
	71202545	2.5	4	4	45	90
	71202550	2.5	4	4	50	100
	71203006	3	6	4.5	6	50
	71203008	3	6	4.5	8	50
	71203010	3	6	4.5	10	50
	71203012	3	6	4.5	12	50
	71203014	3	6	4.5	14	60
	71203016	3	6	4.5	16	60
	71203018	3	6	4.5	18	60
	71203020	3	6	4.5	20	60
	71203022	3	6	4.5	22	65
	71203026	3	6	4.5	26	65
	71203030	3	6	4.5	30	70
	71203035	3	6	4.5	35	70
	71203040	3	6	4.5	40	80
	71203045	3	6	4.5	45	90
	71203050	3	6	4.5	50	100
	71203060	3	6	4.5	60	100
	71204008	4	6	6	8	50
	71204010	4	6	6	10	50
	71204012	4	6	6	12	50
	71204014	4	6	6	14	60
	71204016	4	6	6	16	60
	71204018	4	6	6	18	60
	71204020	4	6	6	20	60
	71204022	4	6	6	22	65
	71204026	4	6	6	26	65

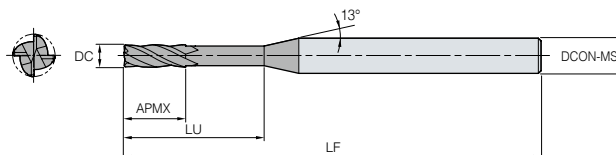
	Designation	DC	DCON-MS	APMX	LU	LF
ESRE	71204030	4	6	6	30	70
	71204035	4	6	6	35	70
	71204040	4	6	6	40	80
	71204045	4	6	6	45	90
	71204050	4	6	6	50	100
	71204060	4	6	6	60	100
	71205016	5	6	8	16	60
	71205020	5	6	8	20	60
	71205026	5	6	8	26	65
	71205030	5	6	8	30	70
	71205035	5	6	8	35	75
	71205040	5	6	8	40	80
	71205050	5	6	8	50	90
	71205060	5	6	8	60	100
	71206015	6	6	9	15	60
	71206020	6	6	9	20	60
	71206030	6	6	9	30	70
	71206032	6	6	9	32	90
	71208025	8	8	12	25	70
	71208030	8	8	12	30	80
	71208042	8	8	12	42	100
	71210030	10	10	15	30	75
	71210035	10	10	15	35	80
	71210045	10	10	15	45	100
	71212035	12	12	20	35	80
	71212040	12	12	20	40	90
	71212050	12	12	20	50	110

ESRE714

4 flutes rib flat



CARBIDE	4	H-A 30°	Grade PC305H	h5 shank	DC	Tolerance
					Ø0.5 ~ Ø6 Ø8 ~ Ø12	0.000 ~ -0.012 0.000 ~ -0.015



	Designation	DC	DCON-MS	APMX	LU	LF
ESRE	71400501	0.5	4	0.5	1	40
	71400502	0.5	4	0.5	2	40
	71400503	0.5	4	0.5	3	45
	71400504	0.5	4	0.5	4	45
	71400505	0.5	4	0.5	5	45
	71400506	0.5	4	0.5	6	45
	71400508	0.5	4	0.5	8	45
	71400510	0.5	4	0.5	10	50
	71400601	0.6	4	0.6	1	45
	71400602	0.6	4	0.6	2	45
	71400603	0.6	4	0.6	3	45
	71400604	0.6	4	0.6	4	45
	71400605	0.6	4	0.6	5	45
	71400606	0.6	4	0.6	6	45
	71400608	0.6	4	0.6	8	45
	71400610	0.6	4	0.6	10	50
	71400612	0.6	4	0.6	12	50
	71400702	0.7	4	0.7	2	45
	71400704	0.7	4	0.7	4	45
	71400706	0.7	4	0.7	6	45
	71400708	0.7	4	0.7	8	45
	71400710	0.7	4	0.7	10	50
	71400801	0.8	4	0.8	1	40
	71400802	0.8	4	0.8	2	40
	71400803	0.8	4	0.8	3	40
	71400804	0.8	4	0.8	4	40
	71400805	0.8	4	0.8	5	40
	71400806	0.8	4	0.8	6	40
	71400808	0.8	4	0.8	8	40
	71400810	0.8	4	0.8	10	50
	71400812	0.8	4	0.8	12	50
	71400816	0.8	4	0.8	16	50
	71401002	1	4	1	2	45
	71401003	1	4	1	3	45
	71401004	1	4	1	4	45
	71401006	1	4	1	6	45
	71401008	1	4	1	8	45
	71401010	1	4	1	10	50
	71401012	1	4	1	12	50
	71401014	1	4	1	14	50

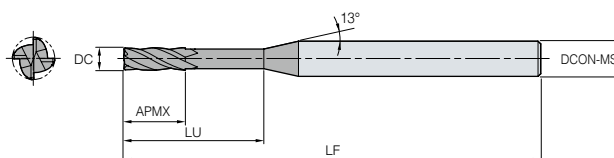
	Designation	DC	DCON-MS	APMX	LU	LF
ESRE	71401016	1	4	1	16	50
	71401018	1	4	1	18	60
	71401020	1	4	1	20	60
	71401204	1.2	4	1.2	4	45
	71401206	1.2	4	1.2	6	45
	71401208	1.2	4	1.2	8	45
	71401210	1.2	4	1.2	10	50
	71401212	1.2	4	1.2	12	50
	71401216	1.2	4	1.2	16	50
	71401218	1.2	4	1.2	18	60
	71401220	1.2	4	1.2	20	60
	71401406	1.4	4	1.4	6	45
	71401408	1.4	4	1.4	8	45
	71401410	1.4	4	1.4	10	50
	71401412	1.4	4	1.4	12	50
	71401414	1.4	4	1.4	14	50
	71401416	1.4	4	1.4	16	50
	71401504	1.5	4	1.5	4	45
	71401506	1.5	4	1.5	6	45
	71401508	1.5	4	1.5	8	45
	71401510	1.5	4	1.5	10	50
	71401512	1.5	4	1.5	12	50
	71401516	1.5	4	1.5	16	50
	71401518	1.5	4	1.5	18	60
	71401520	1.5	4	1.5	20	60
	71401525	1.5	4	1.5	25	60
	71401530	1.5	4	1.5	30	70
	71401606	1.6	4	1.6	6	45
	71401608	1.6	4	1.6	8	45
	71401610	1.6	4	1.6	10	50
	71401612	1.6	4	1.6	12	50
	71401614	1.6	4	1.6	14	50
	71401616	1.6	4	1.6	16	50
	71401618	1.6	4	1.6	18	60
	71401620	1.6	4	1.6	20	60
	71401625	1.6	4	1.6	25	70
	71401806	1.8	4	1.8	6	45
	71401808	1.8	4	1.8	8	45
	71401810	1.8	4	1.8	10	50
	71401812	1.8	4	1.8	12	50

ESRE714

4 flutes rib flat



DC	Tolerance
Ø0.5 ~ Ø6	0.000 ~ -0.012
Ø8 ~ Ø12	0.000 ~ -0.015



(mm)

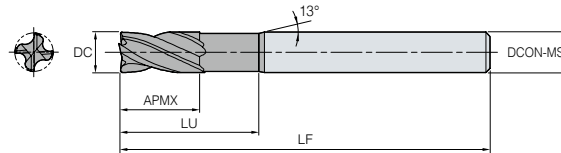
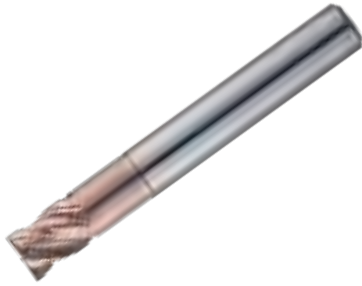
Designation	DC	DCON-MS	APMX	LU	LF
ESRE 71401816	1.8	4	1.8	16	50
71401820	1.8	4	1.8	20	60
71401825	1.8	4	1.8	25	70
71402004	2	4	2	4	45
71402006	2	4	2	6	45
71402008	2	4	2	8	45
71402010	2	4	2	10	50
71402012	2	4	2	12	50
71402014	2	4	2	14	50
71402016	2	4	2	16	50
71402018	2	4	2	18	50
71402020	2	4	2	20	50
71402022	2	4	2	22	60
71402025	2	4	2	25	60
71402030	2	4	2	30	70
71402510	2.5	4	2.5	10	50
71402512	2.5	4	2.5	12	50
71402516	2.5	4	2.5	16	50
71402520	2.5	4	2.5	20	50
71402525	2.5	4	2.5	25	60
71402530	2.5	4	2.5	30	70
71403006	3	6	3	6	45
71403008	3	6	3	8	45
71403010	3	6	3	10	50
71403012	3	6	3	12	50
71403016	3	6	3	16	55
71403020	3	6	3	20	60
71403025	3	6	3	25	65
71403030	3	6	3	30	70
71403035	3	6	3	35	75
71403040	3	6	3	40	80
71403045	3	6	3	45	90
71403050	3	6	3	50	100
71403060	3	6	3	60	110
71403512	3.5	6	3.5	12	50
71403516	3.5	6	3.5	16	55
71403520	3.5	6	3.5	20	60
71403525	3.5	6	3.5	25	65
71403530	3.5	6	3.5	30	70
71403535	3.5	6	3.5	35	75

Designation	DC	DCON-MS	APMX	LU	LF
ESRE 71403540	3.5	6	3.5	40	80
71404006	4	6	4	6	50
71404008	4	6	4	8	50
71404010	4	6	4	10	50
71404012	4	6	4	12	50
71404016	4	6	4	16	55
71404020	4	6	4	20	60
71404025	4	6	4	25	65
71404030	4	6	4	30	70
71404040	4	6	4	40	80
71404045	4	6	4	45	90
71404050	4	6	4	50	100
71404060	4	6	4	60	110
71404512	4.5	6	4.5	12	50
71404516	4.5	6	4.5	16	55
71404520	4.5	6	4.5	20	60
71404525	4.5	6	4.5	25	65
71404530	4.5	6	4.5	30	70
71404540	4.5	6	4.5	40	80
71405016	5	6	5	16	60
71405020	5	6	5	20	60
71405025	5	6	5	25	65
71405030	5	6	5	30	70
71405040	5	6	5	40	80
71405050	5	6	5	50	100
71405060	5	6	5	60	110
71406020	6	6	6	20	60
71406030	6	6	6	30	75
71406040	6	6	6	40	80
71406050	6	6	6	50	90
71406060	6	6	6	60	100
71408025	8	8	12	25	65
71408040	8	8	12	40	100
71408050	8	8	12	50	110
71410030	10	10	15	30	70
71410050	10	10	15	50	100
71410060	10	10	15	60	120
71412040	12	12	18	40	80
71412060	12	12	18	60	110
71412070	12	12	18	70	130

ESXE704

4 flutes neck type flat

CARBIDE	4	H-A Multi	Grade PC305H	h5 shank	DC	Tolerance
					ALL	0.000 ~ -0.020



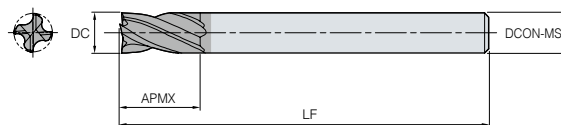
(mm)

Designation	DC	DCON-MS	APMX	LU	LF
ESXE 704010	1	4	1.5	4	45
704020	2	4	3	6	45
704030	3	6	4	7	45
704040	4	6	5	9	45
704060	6	6	7	14	50
704080	8	8	9	18	60
704100	10	10	12	25	75
704120	12	12	15	30	75

ESXE714

4 flutes flat

CARBIDE	4	H-A Multi	Grade PC305H	h5 shank	DC	Tolerance
					ALL	0.000 ~ -0.020

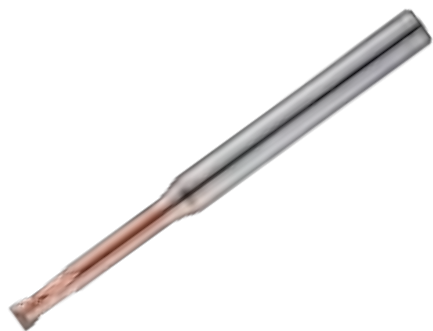


(mm)

Designation	DC	DCON-MS	APMX	LF
ESXE 714020	2	4	5	45
714030	3	6	8	45
714040	4	6	10	45
714040S4	4	4	10	45
714060	6	6	16	50
714080	8	8	20	60
714100	10	10	25	75
714120	12	12	35	85

ESLNS20

2 flutes long neck type flat



CARBIDE

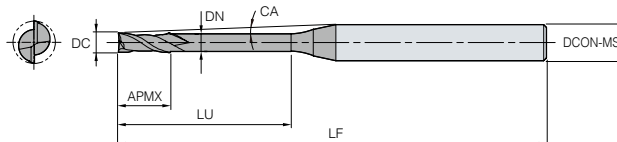
2

H-A
45°

Grade
PC305H

h5
shank

DC	Tolerance
ALL	0.000 ~ -0.012

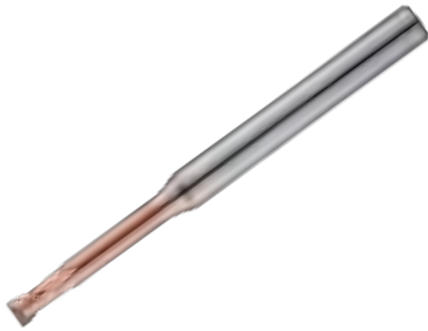


(mm)

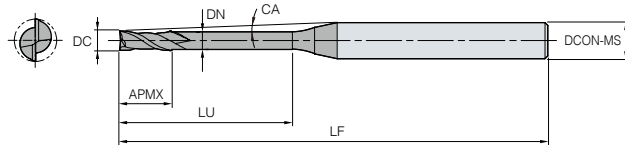
Designation	DC	DCON-MS	APMX	LU	LF	DN	CA	Effective length by inclination angle				
								0.5°	1°	1.5°	2°	3°
ESLNS 2001-0.3	0.1	4	0.15	0.3	45	0.08	11.6	0.4	0.4	0.5	0.5	0.5
2001-0.5	0.1	4	0.15	0.5	45	0.08	11.4	0.6	0.7	0.7	0.7	0.8
2001-1	0.1	4	0.15	1	45	0.08	10.9	1.2	1.2	1.2	1.3	1.4
2002-0.5	0.2	4	0.3	0.5	50	0.17	11.3	1.2	1.3	1.5	1.7	2
2002-1	0.2	4	0.3	1	50	0.17	10.8	1.7	1.9	2.2	2.4	2.7
2002-1.5	0.2	4	0.3	1.5	50	0.17	10.3	2.3	2.5	2.8	3	3.4
2003-1	0.3	4	0.45	1	50	0.27	10.8	1.7	1.9	2.2	2.4	2.7
2003-1.5	0.3	4	0.45	1.5	50	0.27	10.3	2.3	2.5	2.8	3	3.4
2003-2	0.3	4	0.45	2	50	0.27	9.8	2.8	3.1	3.4	3.6	4.1
2003-2.5	0.3	4	0.45	2.5	50	0.27	9.4	3.4	3.7	4	4.3	4.7
2003-3	0.3	4	0.45	3	50	0.27	9	3.9	4.3	4.6	4.9	5.4
2004-1	0.4	4	0.6	1	50	0.37	10.7	1.7	1.9	2.2	2.4	2.7
2004-1.5	0.4	4	0.6	1.5	50	0.37	10.2	2.3	2.5	2.8	3	3.4
2004-2	0.4	4	0.6	2	50	0.37	9.7	2.8	3.1	3.4	3.6	4.1
2004-2.5	0.4	4	0.6	2.5	50	0.37	9.3	3.4	3.7	4	4.3	4.7
2004-3	0.4	4	0.6	3	50	0.37	8.9	3.9	4.3	4.6	4.9	5.4
2004-3.5	0.4	4	0.6	3.5	50	0.37	8.6	4.5	4.9	5.2	5.5	6
2004-4	0.4	4	0.6	4	50	0.37	8.2	5	5.4	5.8	6.1	6.6
2004-5	0.4	4	0.6	5	50	0.37	7.6	6.1	6.6	6.9	7.3	7.8
2004-6	0.4	4	0.6	6	50	0.37	7.1	7.2	7.7	8.1	8.4	9
2005-1	0.5	4	0.75	1	50	0.47	10.7	1.7	1.9	2.2	2.4	2.7
2005-1.5	0.5	4	0.75	1.5	50	0.47	10.2	2.3	2.5	2.8	3	3.4
2005-2	0.5	4	0.75	2	50	0.47	9.7	2.8	3.1	3.4	3.6	4.1
2005-2.5	0.5	4	0.75	2.5	50	0.47	9.3	3.4	3.7	4	4.3	4.7
2005-3	0.5	4	0.75	3	50	0.47	8.9	3.9	4.3	4.6	4.9	5.4
2005-4	0.5	4	0.75	4	50	0.47	8.1	5	5.4	5.8	6.1	6.6
2005-5	0.5	4	0.75	5	50	0.47	7.5	6.1	6.6	6.9	7.3	7.8
2005-6	0.5	4	0.75	6	50	0.47	7	7.2	7.7	8.1	8.4	9
2005-8	0.5	4	0.75	8	50	0.47	6.2	9.3	9.9	10.3	10.7	11.4
2006-2	0.6	4	0.9	2	50	0.57	9.6	2.8	3.1	3.4	3.6	4.1
2006-4	0.6	4	0.9	4	50	0.57	6.9	7.2	7.7	8.1	8.4	9
2006-6	0.6	4	0.9	6	50	0.57	6.1	9.3	9.9	10.3	10.7	11.4
2006-8	0.6	4	0.9	8	50	0.57	5.4	11.5	12.1	12.6	13	13.7
2006-10	0.6	4	0.9	10	50	0.57	9.6	2.8	3.1	3.4	3.6	4.1
2007-2	0.7	4	1.05	2	50	0.67	8	5	5.4	5.8	6.1	6.6
2007-4	0.7	4	1.05	4	50	0.67	6.9	7.2	7.7	8.1	8.4	9
2007-6	0.7	4	1.05	6	50	0.67	6	9.3	9.9	10.3	10.7	11.4
2007-8	0.7	4	1.05	8	50	0.67	5.3	11.5	12.1	12.6	13	13.7
2007-10	0.7	4	1.2	10	50	0.77	7.9	5	5.4	5.8	6.1	6.6
2008-4	0.8	4	1.2	4	50	0.77	6.8	7.2	7.7	8.1	8.4	9

ESLNS20

2 flutes long neck type flat



CARBIDE	2	H-A 45°	Grade PC305H	h5 shank	DC	Tolerance
					ALL	0.000 ~ -0.012

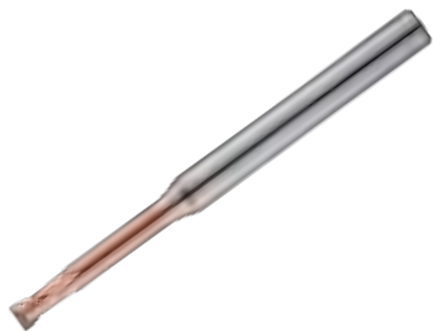


(mm)

Designation	DC	DCON-MS	APMX	LU	LF	DN	CA	Effective length by inclination angle				
								0.5°	1°	1.5°	2°	3°
ESLNS 2008-6	0.8	4	1.2	6	50	0.77	5.9	9.3	9.9	10.3	10.7	11.4
2008-8	0.8	4	1.2	8	50	0.77	5.2	11.5	12.1	12.6	13	13.7
2008-10	0.8	4	1.2	10	50	0.77	4.7	13.6	14.2	14.8	15.2	16
2008-12	0.8	4	1.2	12	55	0.77	6.7	7.2	7.7	8.1	8.4	9.1
2009-6	0.9	4	1.35	6	50	0.86	5.8	9.4	9.9	10.4	10.7	11.4
2009-8	0.9	4	1.35	8	50	0.86	5.1	11.5	12.1	12.6	13	13.7
2009-10	0.9	4	1.35	10	55	0.86	4.6	13.6	14.3	14.8	15.2	16
2009-12	0.9	4	1.35	12	55	0.86	9.4	2.9	3.2	3.4	3.7	4.1
2010-2	1	4	1.5	2	50	0.96	7.7	5.1	5.5	5.8	6.1	6.6
2010-4	1	4	1.5	4	50	0.96	6.6	7.2	7.7	8.1	8.4	9.1
2010-6	1	4	1.5	6	50	0.96	5.7	9.4	9.9	10.4	10.7	11.4
2010-8	1	4	1.5	8	50	0.96	5	11.5	12.1	12.6	13	13.7
2010-10	1	4	1.5	10	55	0.96	4.5	13.6	14.3	14.8	15.2	16
2010-12	1	4	1.5	12	55	0.96	4.1	15.7	16.4	17	17.4	18.7
2010-14	1	4	1.5	14	60	0.96	3.8	17.8	18.6	19.1	19.6	21.3
2010-16	1	4	1.5	16	60	0.96	3.2	22	22.8	23.5	24	26.6
2010-20	1	4	1.5	20	60	0.96	6.3	7.3	7.7	8.1	8.5	9.1
2012-6	1.2	4	1.8	6	50	1.15	5.5	9.4	9.9	10.4	10.8	11.4
2012-8	1.2	4	1.8	8	50	1.15	4.8	11.5	12.1	12.6	13	13.7
2012-10	1.2	4	1.8	10	50	1.15	11.5	12.1	12.6	13	13.7	16
2012-12	1.2	4	1.8	12	55	1.15	4.3	13.6	14.3	14.8	15.2	16
2012-16	1.2	4	1.8	16	55	1.15	3.6	17.8	18.6	19.2	19.7	21.3
2014-6	1.4	4	2.1	6	50	1.34	6.1	7.3	7.8	8.1	8.5	9.1
2014-8	1.4	4	2.1	8	50	1.34	5.3	9.4	10	10.4	10.8	11.5
2014-10	1.4	4	2.1	10	50	1.34	4.6	11.6	12.1	12.6	13	13.8
2014-12	1.4	4	2.1	12	55	1.34	4.1	13.7	14.3	14.8	15.3	16.1
2014-14	1.4	4	2.1	14	55	1.34	3.7	15.8	16.5	17	17.5	18.7
2014-16	1.4	4	2.1	16	55	1.34	3.4	17.9	18.6	19.2	19.7	21.4
2015-4	1.5	4	2.25	4	50	1.44	7.2	5.2	5.5	5.9	6.2	6.7
2015-6	1.5	4	2.25	6	50	1.44	6	7.3	7.8	8.1	8.5	9.1
2015-8	1.5	4	2.25	8	50	1.44	5.1	9.4	10	10.4	10.8	11.5
2015-10	1.5	4	2.25	10	50	1.44	4.5	11.6	12.1	12.6	13	13.8
2015-12	1.5	4	2.25	12	55	1.44	4	13.7	14.3	14.8	15.3	16.1
2015-14	1.5	4	2.25	14	55	1.44	3.6	15.8	16.5	17	17.5	18.7
2015-16	1.5	4	2.25	16	55	1.44	3.3	17.9	18.6	19.2	19.7	-
2015-18	1.5	4	2.25	18	60	1.44	3	20	20.7	21.3	21.9	-
2015-20	1.5	4	2.25	20	60	1.44	2.8	22	22.9	23.5	24.1	-
2015-25	1.5	4	2.25	25	65	1.44	2.4	27.3	28.1	28.8	30	-
2016-6	1.6	4	2.4	6	50	1.54	5.9	7.3	7.8	8.1	8.5	9.1
2016-8	1.6	4	2.4	8	50	1.54	5	9.4	10	10.4	10.8	11.5

ESLNS20

2 flutes long neck type flat



CARBIDE

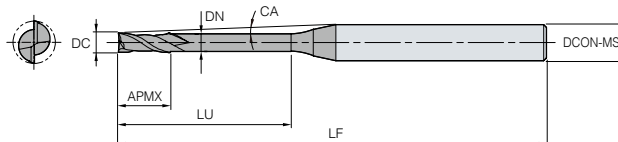
2

H-A 45°

Grade PC305H

h5 shank

DC	Tolerance
ALL	0.000 ~ -0.012

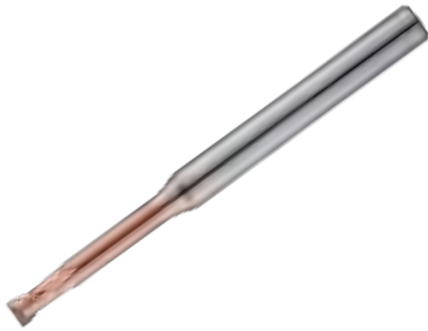


(mm)

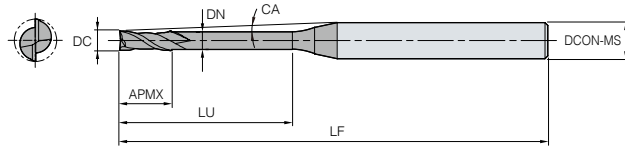
	Designation	DC	DCON-MS	APMX	LU	LF	DN	CA	Effective length by inclination angle				
									0.5°	1°	1.5°	2°	3°
ESLNS	2016-10	1.6	4	2.4	10	50	1.54	4.4	11.6	12.1	12.6	13	13.8
	2016-12	1.6	4	2.4	12	55	1.54	3.9	13.7	14.3	14.8	15.3	16.1
	2016-14	1.6	4	2.4	14	55	1.54	3.5	15.8	16.5	17	17.5	18.7
	2016-16	1.6	4	2.4	16	55	1.54	3.2	17.9	18.6	19.2	19.7	21.4
	2016-18	1.6	4	2.4	18	60	1.54	2.9	20	20.7	21.3	21.9	-
	2016-20	1.6	4	2.4	20	60	1.54	2.7	22	22.9	23.5	24.1	-
	2018-6	1.8	4	2.7	6	50	1.73	5.6	7.4	7.8	8.2	8.5	9.1
	2018-8	1.8	4	2.7	8	50	1.73	4.8	9.5	10	10.4	10.8	11.5
	2018-10	1.8	4	2.7	10	50	1.73	4.2	11.6	12.2	12.6	13	13.8
	2018-12	1.8	4	2.7	12	55	1.73	3.7	13.7	14.3	14.8	15.3	16.1
	2018-14	1.8	4	2.7	14	55	1.73	3.3	15.8	16.5	17	17.5	18.8
	2018-16	1.8	4	2.7	16	55	1.73	3	17.9	18.6	19.2	19.7	-
	2018-18	1.8	4	2.7	18	60	1.73	2.7	20	20.7	21.3	21.9	-
	2018-20	1.8	4	2.7	20	60	1.73	2.5	22.1	22.9	23.5	24.1	-
	2020-4	2	4	3	4	50	1.92	6.5	5.3	5.6	5.9	6.2	6.7
	2020-6	2	4	3	6	50	1.92	5.3	7.4	7.8	8.2	8.5	9.1
	2020-8	2	4	3	8	50	1.92	4.5	9.5	10	10.4	10.8	11.5
	2020-10	2	4	3	10	50	1.92	3.9	11.6	12.2	12.7	13.1	13.8
	2020-12	2	4	3	12	55	1.92	3.4	13.7	14.3	14.9	15.3	16.1
	2020-14	2	4	3	14	55	1.92	3.1	15.8	16.5	17	17.5	18.8
	2020-16	2	4	3	16	55	1.92	2.8	17.9	18.6	19.2	19.7	-
	2020-18	2	4	3	18	60	1.92	2.6	20	20.8	21.4	21.9	-
	2020-20	2	4	3	20	60	1.92	2.4	22.1	22.9	23.5	24.1	-
	2020-25	2	4	3	25	65	1.92	2	27.3	28.2	28.9	-	-
	2020-30	2	4	3	30	70	1.92	1.7	32.5	33.4	34.4	-	-
	2025-8	2.5	4	3.75	8	50	2.4	3.7	9.6	10.1	10.5	10.9	11.5
	2025-10	2.5	4	3.75	10	50	2.4	3.1	11.7	12.2	12.7	13.1	13.8
	2025-12	2.5	4	3.75	12	55	2.4	2.7	13.8	14.4	14.9	15.3	-
	2025-14	2.5	4	3.75	14	55	2.4	2.4	15.9	16.5	17.1	17.5	-
	2025-16	2.5	4	3.75	16	55	2.4	2.2	18	18.7	19.2	19.7	-
	2025-18	2.5	4	3.75	18	55	2.4	2	20.1	20.8	21.4	-	-
	2025-20	2.5	4	3.75	20	60	2.4	1.8	22.1	22.9	23.5	-	-
	2025-25	2.5	4	3.75	25	60	2.4	1.5	27.3	28.2	-	-	-
	2025-30	2.5	4	3.75	30	70	2.4	1.3	32.6	33.5	-	-	-
	2030-8	3	6	4.5	8	55	2.88	5.6	9.6	10.1	10.5	10.9	11.5
	2030-10	3	6	4.5	10	55	2.88	5	11.7	12.3	12.7	13.1	13.8
	2030-12	3	6	4.5	12	60	2.88	4.5	13.8	14.4	14.9	15.4	16.3
	2030-14	3	6	4.5	14	60	2.88	4.1	15.9	16.6	17.1	17.6	18.9
	2030-16	3	6	4.5	16	60	2.88	3.7	18	18.7	19.3	19.8	21.6
	2030-18	3	6	4.5	18	60	2.88	3.4	20.1	20.8	21.4	21.9	24.2

ESLNS20

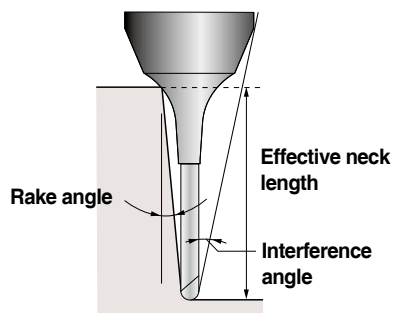
2 flutes long neck type flat



CARBIDE	2	H-A 45°	Grade PC305H	h5 shank	DC	Tolerance
					ALL	0.000 ~ -0.012



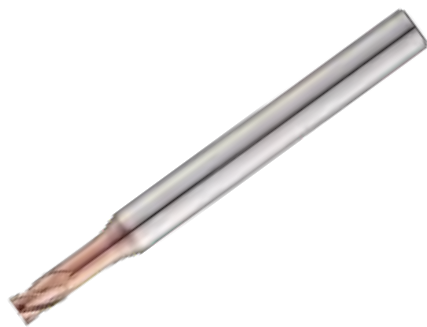
Designation		DC	DCON-MS	APMX	LU	LF	DN	CA	Effective length by inclination angle				
									0.5°	1°	1.5°	2°	3°
ESLNS	2030-20	3	6	4.5	20	65	2.88	3.2	22.2	23	23.6	24.2	26.9
	2030-25	3	6	4.5	25	70	2.88	2.7	27.4	28.2	28.9	30.2	-
	2030-30	3	6	4.5	30	75	2.88	2.4	32.6	33.5	34.5	36.2	-
	2030-35	3	6	4.5	35	80	2.88	2.1	37.7	38.7	40.2	42.2	-
	2030-40	3	6	4.5	40	90	2.88	1.9	42.9	43.9	45.9	-	-
	2040-12	4	6	6	12	60	3.85	3.4	13.9	14.5	15	15.4	16.3
	2040-16	4	6	6	16	60	3.85	2.8	18.1	18.8	19.3	19.8	-
	2040-20	4	6	6	20	70	3.85	2.3	22.3	23	23.6	24.3	-
	2040-25	4	6	6	25	70	3.85	2	27.4	28.3	28.9	-	-
	2040-30	4	6	6	30	80	3.85	1.7	32.6	33.5	34.6	-	-
	2040-35	4	6	6	35	80	3.85	1.5	37.8	38.8	-	-	-
	2040-40	4	6	6	40	90	3.85	1.3	42.9	44	-	-	-
	2040-45	4	6	6	45	90	3.85	1.2	48.1	49.4	-	-	-
	2040-50	4	6	6	50	100	3.85	1.1	53.2	54.8	-	-	-
	2050-16	5	6	7.5	16	60	4.85	1.5	18.1	18.8	-	-	-
	2050-20	5	6	7.5	20	60	4.85	1.3	22.3	23	-	-	-
	2050-25	5	6	7.5	25	70	4.85	1.1	27.4	28.3	-	-	-
	2050-30	5	6	7.5	30	70	4.85	0.9	32.6	-	-	-	-
	2050-35	5	6	7.5	35	80	4.85	0.8	37.8	-	-	-	-
	2050-40	5	6	7.5	40	90	4.85	0.7	42.9	-	-	-	-
	2050-50	5	6	7.5	50	100	4.85	0.6	53.2	-	-	-	-



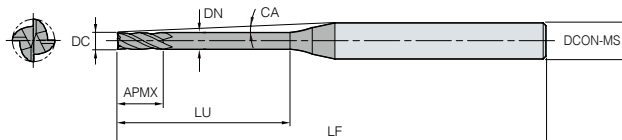
※ The marked effective neck length is the default value to prevent interference with the workpiece.
Proper control of the processing environment is required.

ESLNS40

4 flutes long neck type flat



DC	Tolerance
ALL	0.000 ~ -0.012

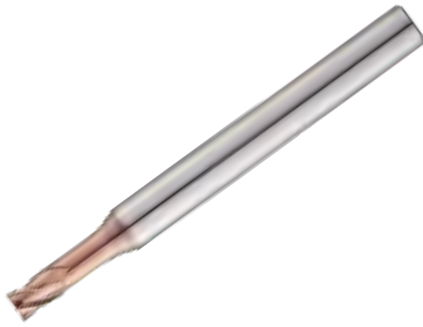


(mm)

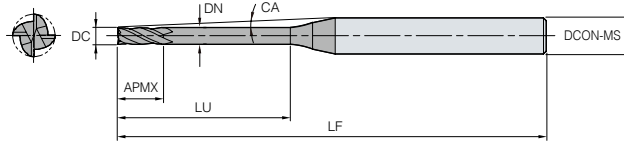
Designation		DC	DCON-MS	APMX	LU	LF	DN	CA	Effective length by inclination angle				
									0.5°	1°	1.5°	2°	3°
ESLNS	4010-4	1	4	1.5	4	50	0.96	7.7	5.1	5.5	5.8	6.1	6.6
	4010-6	1	4	1.5	6	50	0.96	6.6	7.2	7.7	8.1	8.4	9.1
	4010-8	1	4	1.5	8	50	0.96	5.7	9.4	9.9	10.4	10.7	11.4
	4010-10	1	4	1.5	10	50	0.96	5	11.5	12.1	12.6	13	13.7
	4015-4	1.5	4	2.25	4	50	1.44	7.2	5.2	5.5	5.9	6.2	6.7
	4015-6	1.5	4	2.25	6	50	1.44	6	7.3	7.8	8.1	8.5	9.1
	4015-8	1.5	4	2.25	8	50	1.44	5.1	9.4	10	10.4	10.8	11.5
	4015-10	1.5	4	2.25	10	50	1.44	4.5	11.6	12.1	12.6	13	13.8
	4015-12	1.5	4	2.25	12	55	1.44	4	13.7	14.3	14.8	15.3	16.1
	4015-14	1.5	4	2.25	14	55	1.44	3.6	15.8	16.5	17	17.5	18.7
	4015-16	1.5	4	2.25	16	55	1.44	3.3	17.9	18.6	19.2	19.7	-
	4015-18	1.5	4	2.25	18	60	1.44	3	20	20.7	21.3	21.9	-
	4015-20	1.5	4	2.25	20	60	1.44	2.8	22	22.9	23.5	24.1	-
	4015-25	1.5	4	2.25	25	65	1.44	2.4	27.3	28.1	28.8	30	-
	4020-4	2	4	3	4	50	1.92	6.5	5.3	5.6	5.9	6.2	6.7
	4020-6	2	4	3	6	50	1.92	5.3	7.4	7.8	8.2	8.5	9.1
	4020-8	2	4	3	8	50	1.92	4.5	9.5	10	10.4	10.8	11.5
	4020-10	2	4	3	10	50	1.92	3.9	11.6	12.2	12.7	13.1	13.8
	4020-12	2	4	3	12	55	1.92	3.4	13.7	14.3	14.9	15.3	16.1
	4020-14	2	4	3	14	55	1.92	3.1	15.8	16.5	17	17.5	18.8
	4020-16	2	4	3	16	55	1.92	2.8	17.9	18.6	19.2	19.7	-
	4020-18	2	4	3	18	60	1.92	2.6	20	20.8	21.4	21.9	-
	4020-20	2	4	3	20	60	1.92	2.4	22.1	22.9	23.5	24.1	-
	4020-25	2	4	3	25	65	1.92	2	27.3	28.2	28.9	-	-
	4020-30	2	4	3	30	70	1.92	1.7	32.5	33.4	34.4	-	-
	4025-8	2.5	4	3.75	8	50	2.4	3.7	9.6	10.1	10.5	10.9	11.5
	4025-10	2.5	4	3.75	10	50	2.4	3.1	11.7	12.2	12.7	13.1	13.8
	4025-12	2.5	4	3.75	12	55	2.4	2.7	13.8	14.4	14.9	15.3	-
	4025-14	2.5	4	3.75	14	55	2.4	2.4	15.9	16.5	17.1	17.5	-
	4025-16	2.5	4	3.75	16	55	2.4	2.2	18	18.7	19.2	19.7	-
	4025-18	2.5	4	3.75	18	60	2.4	2	20.1	20.8	21.4	-	-
	4025-20	2.5	4	3.75	20	60	2.4	1.8	22.1	22.9	23.5	-	-
	4025-25	2.5	4	3.75	25	65	2.4	1.5	27.3	28.2	-	-	-
	4025-30	2.5	4	3.75	30	70	2.4	1.3	32.6	33.5	-	-	-
	4030-8	3	6	4.5	8	55	2.88	5.6	9.6	10.1	10.5	10.9	11.5
	4030-10	3	6	4.5	10	55	2.88	5	11.7	12.3	12.7	13.1	13.8

ESLNS40

4 flutes long neck type flat



CARBIDE	4	H-A 30°	Grade PC305H	h5 shank	DC	Tolerance
					ALL	0.000 ~ -0.012

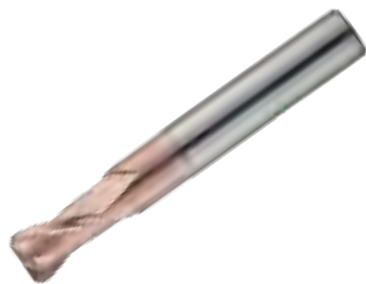


(mm)

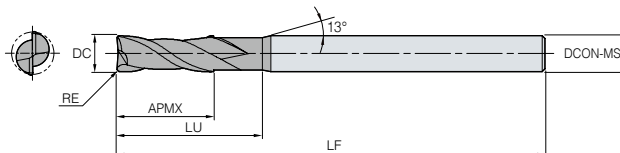
Designation		DC	DCON-MS	APMX	LU	LF	DN	CA	Effective length by inclination angle				
									0.5°	1°	1.5°	2°	3°
ESLNS	4030-12	3	6	4.5	12	60	2.88	4.5	13.8	14.4	14.9	15.4	16.3
	4030-14	3	6	4.5	14	60	2.88	4.1	15.9	16.6	17.1	17.6	18.9
	4030-16	3	6	4.5	16	60	2.88	3.7	18	18.7	19.3	19.8	21.6
	4030-18	3	6	4.5	18	60	2.88	3.4	20.1	20.8	21.4	21.9	24.2
	4030-20	3	6	4.5	20	65	2.88	3.2	22.2	23	23.6	24.2	26.9
	4030-25	3	6	4.5	25	70	2.88	2.7	27.4	28.2	28.9	30.2	-
	4030-30	3	6	4.5	30	75	2.88	2.4	32.6	33.5	34.5	36.2	-
	4030-35	3	6	4.5	35	80	2.88	2.1	37.7	38.7	40.2	42.2	-
	4030-40	3	6	4.5	40	90	2.88	1.9	42.9	43.9	45.9	-	-
	4040-12	4	6	6	12	60	3.85	3.4	13.9	14.5	15	15.4	16.3
	4040-16	4	6	6	16	60	3.85	2.8	18.1	18.8	19.3	19.8	-
	4040-20	4	6	6	20	70	3.85	2.3	22.3	23	23.6	24.3	-
	4040-25	4	6	6	25	70	3.85	2	27.4	28.3	28.9	-	-
	4040-30	4	6	6	30	80	3.85	1.7	32.6	33.5	34.6	-	-
	4040-35	4	6	6	35	80	3.85	1.5	37.8	38.8	-	-	-
	4040-40	4	6	6	40	90	3.85	1.3	42.9	44	-	-	-
	4040-45	4	6	6	45	90	3.85	1.2	48.1	49.4	-	-	-
	4040-50	4	6	6	50	100	3.85	1.1	53.2	54.8	-	-	-
	4050-16	5	6	7.5	16	60	4.85	1.5	18.1	18.8	-	-	-
	4050-20	5	6	7.5	20	60	4.85	1.3	22.3	23	-	-	-
	4050-25	5	6	7.5	25	70	4.85	1.1	27.4	28.3	-	-	-
	4050-30	5	6	7.5	30	70	4.85	0.9	32.6	-	-	-	-
	4050-35	5	6	7.5	35	80	4.85	0.8	37.8	-	-	-	-
	4050-40	5	6	7.5	40	90	4.85	0.7	42.9	-	-	-	-
	4050-50	5	6	7.5	50	100	4.85	0.6	53.2	-	-	-	-

ESR702

2 flutes neck type radius



CARBIDE	2	H-A 30°	Grade PC305H	h5 shank	RE ±0.01	RE ±0.015	DC	Tolerance
					DC6 or below	Above DC6	Ø1 ~ Ø6 Ø8 ~ Ø12	0.000 ~ -0.012 0.000 ~ -0.015



(mm)

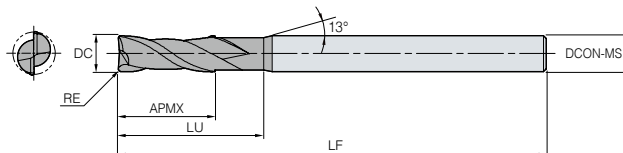
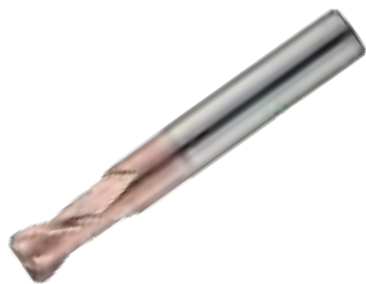
	Designation	DC	DCON-MS	APMX	LU	LF	RE
ESR	70201000503S4	1	4	1.5	3	50	0.05
	70201000504S4	1	4	1.5	4	50	0.05
	70201000506S4	1	4	1.5	6	50	0.05
	70201000508S4	1	4	1.5	8	50	0.05
	70201000510S4	1	4	1.5	10	50	0.05
	7020100103S4	1	4	1.5	3	50	0.1
	7020100104	1	6	1.5	4	50	0.1
	7020100104S4	1	4	1.5	4	50	0.1
	7020100106	1	6	1.5	6	50	0.1
	7020100106S4	1	4	1.5	6	50	0.1
	7020100108S4	1	4	1.5	8	50	0.1
	7020100110S4	1	4	1.5	10	50	0.1
	7020100203S4	1	4	1.5	3	50	0.2
	7020100204	1	6	1.5	4	50	0.2
	7020100204S4	1	4	1.5	4	50	0.2
	7020100206	1	6	1.5	6	50	0.2
	7020100206S4	1	4	1.5	6	50	0.2
	7020100208S4	1	4	1.5	8	50	0.2
	7020100210	1	6	1.5	10	50	0.2
	7020100210S4	1	4	1.5	10	50	0.2
	7020100212	1	6	1.5	12	50	0.2
	7020100303S4	1	4	1.5	3	50	0.3
	7020100304S4	1	4	1.5	4	50	0.3
	7020100306S4	1	4	1.5	6	50	0.3
	7020100308S4	1	4	1.5	8	50	0.3
	7020100310S4	1	4	1.5	10	50	0.3
	7020120208	1.2	6	2	8	50	0.2
	7020120212	1.2	6	2	12	50	0.2
	70201500504S4	1.5	4	2.5	4	50	0.05
	70201500506S4	1.5	4	2.5	6	50	0.05
	70201500508S4	1.5	4	2.5	8	50	0.05
	70201500510S4	1.5	4	2.5	10	50	0.05
	70201500512S4	1.5	4	2.5	12	50	0.05
	7020150104S4	1.5	4	2.5	4	50	0.1
	7020150106S4	1.5	4	2.5	6	50	0.1
	7020150108S4	1.5	4	2.5	8	50	0.1
	7020150110S4	1.5	4	2.5	10	50	0.1
	7020150112S4	1.5	4	2.5	12	50	0.1
	7020150204	1.5	6	2.5	4	50	0.2

	Designation	DC	DCON-MS	APMX	LU	LF	RE
ESR	7020150204S4	1.5	4	2.5	4	50	0.2
	7020150206	1.5	6	2.5	6	50	0.2
	7020150206S4	1.5	4	2.5	6	50	0.2
	7020150208	1.5	6	2.5	8	50	0.2
	7020150208S4	1.5	4	2.5	8	50	0.2
	7020150210	1.5	6	2.5	10	50	0.2
	7020150210S4	1.5	4	2.5	10	50	0.2
	7020150212S4	1.5	4	2.5	12	50	0.2
	7020150215	1.5	6	2.5	15	50	0.2
	7020150304S4	1.5	4	2.5	4	50	0.3
	7020150306S4	1.5	4	2.5	6	50	0.3
	7020150308S4	1.5	4	2.5	8	50	0.3
	7020150310S4	1.5	4	2.5	10	50	0.3
	7020150312S4	1.5	4	2.5	12	50	0.3
	7020150504S4	1.5	4	2.5	4	50	0.5
	7020150506S4	1.5	4	2.5	6	50	0.5
	7020150508S4	1.5	4	2.5	8	50	0.5
	7020150510S4	1.5	4	2.5	10	50	0.5
	7020150512S4	1.5	4	2.5	12	50	0.5
	7020200106S4	2	4	3	6	50	0.1
	7020200108	2	6	3	8	50	0.1
	7020200108S4	2	4	3	8	50	0.1
	7020200110S4	2	4	3	10	50	0.1
	7020200112	2	6	3	12	50	0.1
	7020200112S4	2	4	3	12	50	0.1
	7020200116S4	2	4	3	16	50	0.1
	7020200120S4	2	4	3	20	50	0.1
	7020200206	2	6	3	6	50	0.2
	7020200206S4	2	4	3	6	50	0.2
	7020200208S4	2	4	3	8	50	0.2
	7020200209	2	6	3	9	50	0.2
	7020200210S4	2	4	3	10	50	0.2
	7020200212S4	2	4	3	12	50	0.2
	7020200216	2	6	3	16	50	0.2
	7020200216S4	2	4	3	16	50	0.2
	7020200220S4	2	4	3	20	50	0.2
	7020200306	2	6	3	6	50	0.3
	7020200306S4	2	4	3	6	50	0.3
	7020200308S4	2	4	3	8	50	0.3

ESR702**2 flutes neck type radius****CARBIDE****2****H-A
30°****Grade
PC305H****h5
shank****RE
±0.01****RE
±0.015**

DC6 or below Above DC6

DC	Tolerance
Ø1 ~ Ø6	0.000 ~ -0.012
Ø8 ~ Ø12	0.000 ~ -0.015

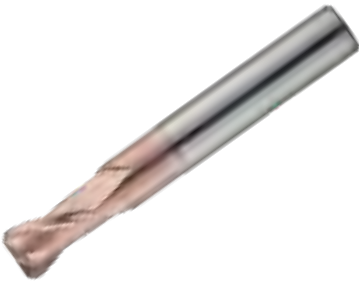


Designation	DC	DCON-MS	APMX	LU	LF	RE
ESR 7020200310S4	2	4	3	10	50	0.3
7020200312S4	2	4	3	12	50	0.3
7020200316S4	2	4	3	16	50	0.3
7020200320S4	2	4	3	20	50	0.3
7020200506	2	6	3	6	50	0.5
7020200506S4	2	4	3	6	50	0.5
7020200508S4	2	4	3	8	50	0.5
7020200509	2	6	3	9	50	0.5
7020200510S4	2	4	3	10	50	0.5
7020200512	2	6	3	12	50	0.5
7020200512S4	2	4	3	12	50	0.5
7020200516	2	6	3	16	50	0.5
7020200516S4	2	4	3	16	50	0.5
7020200520S4	2	4	3	20	50	0.5
7020250208S4	2.5	4	3.5	8	50	0.2
7020250210S4	2.5	4	3.5	10	50	0.2
7020250212S4	2.5	4	3.5	12	50	0.2
7020250216S4	2.5	4	3.5	16	50	0.2
7020250308S4	2.5	4	3.5	8	50	0.3
7020250310S4	2.5	4	3.5	10	50	0.3
7020250312S4	2.5	4	3.5	12	50	0.3
7020250316S4	2.5	4	3.5	16	50	0.3
7020250508S4	2.5	4	3.5	8	50	0.5
7020250510S4	2.5	4	3.5	10	50	0.5
7020250512S4	2.5	4	3.5	12	50	0.5
7020250516S4	2.5	4	3.5	16	50	0.5
7020300108	3	6	4.5	8	55	0.1
7020300110	3	6	4.5	10	55	0.1
7020300112	3	6	4.5	12	55	0.1
7020300116	3	6	4.5	16	55	0.1
7020300120	3	6	4.5	20	60	0.1
7020300208	3	6	4.5	8	55	0.2
7020300209	3	6	4.5	9	55	0.2
7020300210	3	6	4.5	10	55	0.2
7020300212	3	6	4.5	12	55	0.2
7020300216	3	6	4.5	16	55	0.2
7020300208	3	6	4.5	8	55	0.2
7020300209	3	6	4.5	9	55	0.2
7020300210	3	6	4.5	10	55	0.2

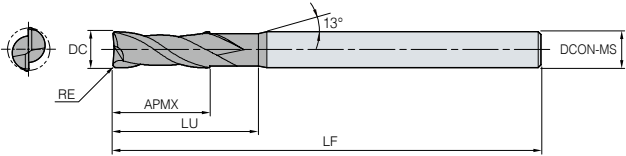
Designation	DC	DCON-MS	APMX	LU	LF	RE
ESR 7020300212	3	6	4.5	12	55	0.2
7020300216	3	6	4.5	16	55	0.2
7020300220	3	6	4.5	20	60	0.2
7020300308	3	6	4.5	8	55	0.3
7020300309	3	6	4.5	9	55	0.3
7020300310	3	6	4.5	10	55	0.3
7020300312	3	6	4.5	12	55	0.3
7020300314	3	6	4.5	14	55	0.3
7020300316	3	6	4.5	16	55	0.3
7020300320	3	6	4.5	20	60	0.3
7020300508	3	6	4.5	8	55	0.5
7020300509	3	6	4.5	9	55	0.5
7020300510	3	6	4.5	10	55	0.5
7020300512	3	6	4.5	12	55	0.5
7020300516	3	6	4.5	16	55	0.5
7020300520	3	6	4.5	20	60	0.5
7020301008	3	6	4.5	8	55	1
7020301010	3	6	4.5	10	55	1
7020301012	3	6	4.5	12	55	1
7020301016	3	6	4.5	16	55	1
7020301020	3	6	4.5	20	60	1
7020301025	3	6	4.5	25	60	1
7020400110	4	6	6	10	55	0.1
7020400112	4	6	6	12	55	0.1
7020400116	4	6	6	16	55	0.1
7020400120	4	6	6	20	60	0.1
7020400125	4	6	6	25	60	0.1
7020400210	4	6	6	10	55	0.2
7020400212	4	6	6	12	55	0.2
7020400216	4	6	6	16	55	0.2
7020400220	4	6	6	20	60	0.2
7020400225	4	6	6	25	60	0.2
7020400310	4	6	6	10	55	0.3
7020400312	4	6	6	12	55	0.3
7020400316	4	6	6	16	55	0.3
7020400320	4	6	6	20	60	0.3
7020400325	4	6	6	25	60	0.3
7020400510	4	6	6	10	55	0.5
7020400512	4	6	6	12	55	0.5

ESR702

2 flutes neck type radius



CARBIDE	2	H-A 30°	Grade PC305H	h5 shank	RE ±0.01	RE ±0.015	DC	Tolerance
					DC6 or below	Above DC6	Ø1 ~ Ø6 Ø8 ~ Ø12	0.000 ~ -0.012 0.000 ~ -0.015

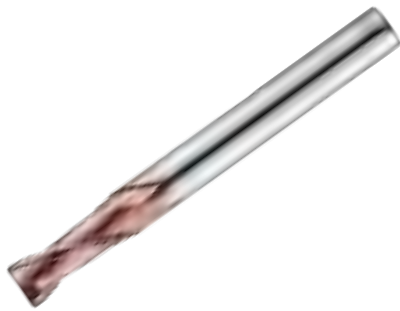


	Designation	DC	DCON-MS	APMX	LU	LF	RE
ESR	7020400516	4	6	6	16	55	0.5
	7020400520	4	6	6	20	60	0.5
	7020400525	4	6	6	25	60	0.5
	7020400530	4	6	6	30	70	0.5
	7020401010	4	6	6	10	55	1
	7020401012	4	6	6	12	55	1
	7020401016	4	6	6	16	55	1
	7020401020	4	6	6	20	60	1
	7020401025	4	6	6	25	60	1
	7020401030	4	6	6	30	70	1
	7020500318	5	6	8	18	60	0.3
	7020600220	6	6	9	20	60	0.2
	7020600320	6	6	9	20	60	0.3
	7020600520	6	6	9	20	60	0.5
	7020601020	6	6	9	20	60	1
	7020601520	6	6	9	20	60	1.5
	7020602020	6	6	9	20	60	2

	Designation	DC	DCON-MS	APMX	LU	LF	RE
ESR	7020800225	8	8	12	25	60	0.2
	7020800325	8	8	12	25	60	0.3
	7020800525	8	8	12	25	60	0.5
	7020801025	8	8	12	25	60	1
	7020801525	8	8	12	25	60	1.5
	7021000232	10	10	15	32	70	0.2
	7021000332	10	10	15	32	70	0.3
	7021000532	10	10	15	32	70	0.5
	7021001032	10	10	15	32	70	1
	7021001532	10	10	15	32	70	1.5
	7021002032	10	10	15	32	70	2
	7021200338	12	12	18	38	80	0.3
	7021200538	12	12	18	38	80	0.5
	7021201038	12	12	18	38	80	1
	7021201538	12	12	18	38	80	1.5
	7021202038	12	12	18	38	80	2

ESR732

2 flutes long shank radius



CARBIDE

2

H-A
30°

Grade
PC305H

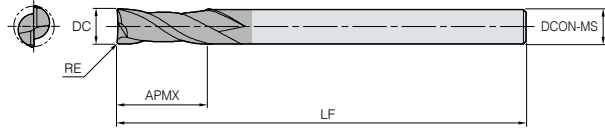
h5
shank

RE
±0.01

RE
±0.015

DC6 or below Above DC6

DC	Tolerance
Ø1 - Ø6	0.000 ~ -0.012
Ø8 - Ø12	0.000 ~ -0.015



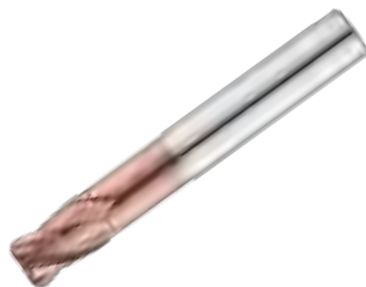
Designation	DC	DCON-MS	APMX	LF	RE
ESR 73201001	1	6	2	50	0.1
73201002	1	6	2	50	0.2
73201003	1	6	2	50	0.3
73201501	1.5	6	3	50	0.1
73201502	1.5	6	3	50	0.2
73201503	1.5	6	3	50	0.3
73201505	1.5	6	3	50	0.5
73202001	2	6	5	50	0.1
73202002	2	6	5	50	0.2
73202003	2	6	5	50	0.3
73202005	2	6	5	50	0.5
73202501	2.5	6	7	60	0.1
73202502	2.5	6	7	60	0.2
73202503	2.5	6	7	60	0.3
73202505	2.5	6	7	60	0.5
73203001	3	6	8	60	0.1
73203002	3	6	8	60	0.2
73203003	3	6	8	60	0.3
73203005	3	6	8	60	0.5
73204001	4	6	10	70	0.1
73204002	4	6	10	70	0.2
73204003	4	6	10	70	0.3
73204005	4	6	10	70	0.5
73204010	4	6	10	70	1
73205001	5	6	13	80	0.1

Designation	DC	DCON-MS	APMX	LF	RE
ESR 73205002	5	6	13	80	0.2
73205003	5	6	13	80	0.3
73205005	5	6	13	80	0.5
73205010	5	6	13	80	1
73206001	6	6	15	90	0.1
73206002	6	6	15	90	0.2
73206003	6	6	15	90	0.3
73206005	6	6	15	90	0.5
73206010	6	6	15	90	1
73208001	8	8	20	100	0.1
73208002	8	8	20	100	0.2
73208003	8	8	20	100	0.3
73208005	8	8	20	100	0.5
73208010	8	8	20	100	1
73208020	8	8	20	100	2
73210002	10	10	25	100	0.2
73210003	10	10	25	100	0.3
73210005	10	10	25	100	0.5
73210010	10	10	25	100	1
73210020	10	10	25	100	2
73212002	12	12	30	110	0.2
73212003	12	12	30	110	0.3
73212005	12	12	30	110	0.5
73212010	12	12	30	110	1
73212020	12	12	30	110	2

(mm)

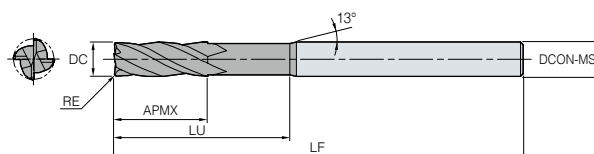
ESR704

4 flutes neck type radius



DC6 or below Above DC6

DC	Tolerance
Ø1 ~ Ø6	0.000 ~ -0.012
Ø8 ~ Ø12	0.000 ~ -0.015



(mm)

	Designation	DC	DCON-MS	APMX	LU	LF	RE
ESR	7040100103S4	1	4	2	3	50	0.1
	7040100104S4	1	4	2	4	50	0.1
	7040100106S4	1	4	2	6	50	0.1
	7040100203S4	1	4	2	3	50	0.2
	7040100204S4	1	4	2	4	50	0.2
	7040100206S4	1	4	2	6	50	0.2
	7040100303S4	1	4	2	3	50	0.3
	7040100304S4	1	4	2	4	50	0.3
	7040100306S4	1	4	2	6	50	0.3
	7040150104S4	1.5	4	2.5	4	50	0.1
	7040150106S4	1.5	4	2.5	6	50	0.1
	7040150204S4	1.5	4	2.5	4	50	0.2
	7040150206S4	1.5	4	2.5	6	50	0.2
	7040150304S4	1.5	4	2.5	4	50	0.3
	7040150306S4	1.5	4	2.5	6	50	0.3
	7040200106S4	2	4	3	6	50	0.1
	7040200108S4	2	4	3	8	50	0.1
	7040200206S4	2	4	3	6	50	0.2
	7040200208	2	6	3	8	50	0.2
	7040200208S4	2	4	3	8	50	0.2
	7040200210	2	6	3	10	50	0.2
	7040200212	2	6	3	12	50	0.2
	7040200306S4	2	4	3	6	50	0.3
	7040200308S4	2	4	3	8	50	0.3
	7040200506S4	2	4	3	6	50	0.5
	7040200508S4	2	4	3	8	50	0.5
	7040250106S4	2.5	4	3.5	6	50	0.1
	7040300108	3	6	4	8	55	0.1
	7040300110	3	6	4	10	55	0.1
	7040300112	3	6	4	12	55	0.1
	7040300116	3	6	4	16	55	0.1
	7040300120	3	6	4	20	60	0.1
	7040300208	3	6	4	8	55	0.2
	7040300210	3	6	4	10	55	0.2
	7040300212	3	6	4	12	55	0.2
	7040300216	3	6	4	16	55	0.2

	Designation	DC	DCON-MS	APMX	LU	LF	RE
ESR	7040300220	3	6	4	20	60	0.2
	7040300308	3	6	4	8	55	0.3
	7040300309	3	6	4	9	55	0.3
	7040300310	3	6	4	10	55	0.3
	7040300312	3	6	4	12	55	0.3
	7040300316	3	6	4	16	55	0.3
	7040300320	3	6	4	20	60	0.3
	7040300508	3	6	4	8	55	0.5
	7040300509	3	6	4	9	55	0.5
	7040300510	3	6	4	10	55	0.5
	7040300512	3	6	4	12	55	0.5
	7040300516	3	6	4	16	55	0.5
	7040300520	3	6	4	20	60	0.5
	7040301008	3	6	4	8	55	1
	7040301010	3	6	4	10	55	1
	7040301012	3	6	4	12	55	1
	7040301016	3	6	4	16	55	1
	7040301020	3	6	4	20	60	1
	7040400110	4	6	6	10	55	0.1
	7040400112	4	6	6	12	55	0.1
	7040400116	4	6	6	16	55	0.1
	7040400120	4	6	6	20	60	0.1
	7040400125	4	6	6	25	60	0.1
	7040400210	4	6	6	10	55	0.2
	7040400212	4	6	6	12	55	0.2
	7040400216	4	6	6	16	55	0.2
	7040400220	4	6	6	20	60	0.2
	7040400225	4	6	6	25	60	0.2
	7040400310	4	6	6	10	55	0.3
	7040400312	4	6	6	12	55	0.3
	7040400316	4	6	6	16	55	0.3
	7040400320	4	6	6	20	60	0.3
	7040400325	4	6	6	25	60	0.3
	7040400510	4	6	6	10	55	0.5
	7040400512	4	6	6	12	55	0.5
	7040400516	4	6	6	16	55	0.5

ESR704

4 flutes neck type radius

CARBIDE

4

H-A
30°

Grade
PC305H

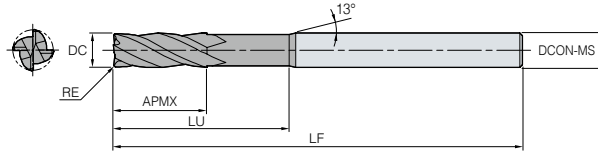
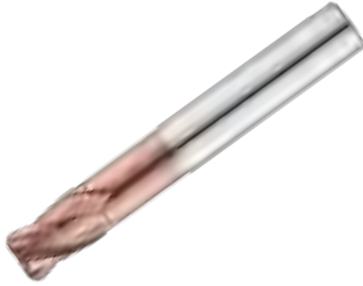
h5
shank

RE
±0.01

RE
±0.015

DC6 or below Above DC6

DC	Tolerance
Ø1 ~ Ø6	0.000 ~ -0.012
Ø8 ~ Ø12	0.000 ~ -0.015



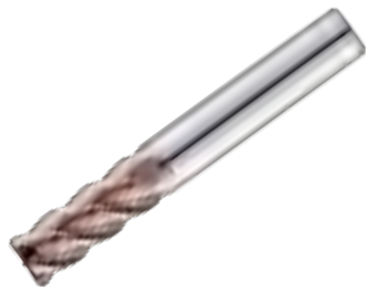
Designation	DC	DCON-MS	APMX	LU	LF	RE
ESR 7040400520	4	6	6	20	60	0.5
7040400525	4	6	6	25	60	0.5
7040401010	4	6	6	10	55	1
7040401012	4	6	6	12	55	1
7040401016	4	6	6	16	55	1
7040401020	4	6	6	20	60	1
7040401025	4	6	6	25	60	1
7040600220	6	6	9	20	60	0.2
7040600320	6	6	9	20	60	0.3
7040600520	6	6	9	20	60	0.5
7040601020	6	6	9	20	60	1
7040601520	6	6	9	20	60	1.5
7040602020	6	6	9	20	60	2
7040800225	8	8	12	25	60	0.2
7040800325	8	8	12	25	60	0.3

Designation	DC	DCON-MS	APMX	LU	LF	RE
ESR 7040800525	8	8	12	25	60	0.5
7040801025	8	8	12	25	60	1
7040801525	8	8	12	25	60	1.5
7040802025	8	8	12	25	60	2
7041000232	10	10	15	32	70	0.2
7041000332	10	10	15	32	70	0.3
7041000532	10	10	15	32	70	0.5
7041001032	10	10	15	32	70	1
7041001532	10	10	15	32	70	1.5
7041002032	10	10	15	32	70	2
7041200338	12	12	18	38	80	0.3
7041200538	12	12	18	38	80	0.5
7041201038	12	12	18	38	80	1
7041201538	12	12	18	38	80	1.5
7041202038	12	12	18	38	80	2

(mm)

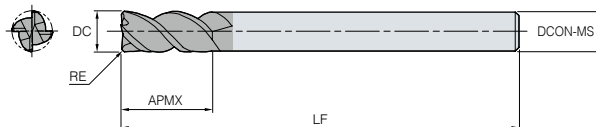
ESR714

4 flutes radius



DC	Tolerance
Ø3 ~ Ø6	0.000 ~ -0.012
Ø8 ~ Ø12	0.000 ~ -0.015

DC6 or below Above DC6



(mm)

Designation	DC	DCON-MS	APMX	LF	RE
ESR 7140303	3	6	8	50	0.3
7140305S4	3	4	8	50	0.5
7140305	3	6	8	50	0.5
7140403	4	6	11	50	0.3
7140405	4	6	11	50	0.5
7140405S4	4	4	11	50	0.5
7140410	4	6	11	50	1
7140603	6	6	15	60	0.3
7140605	6	6	15	60	0.5
7140610	6	6	15	60	1
7140803	8	8	20	60	0.3
7140805	8	8	20	60	0.5
7140810	8	8	20	60	1
7140815	8	8	20	60	1.5
7140820	8	8	20	60	2

Designation	DC	DCON-MS	APMX	LF	RE
ESR 7141003	10	10	25	70	0.3
7141005	10	10	25	70	0.5
7141010	10	10	25	70	1
7141015	10	10	25	70	1.5
7141020	10	10	25	70	2
7141025	10	10	25	70	2.5
7141030	10	10	25	70	3
7141203	12	12	30	80	0.3
7141205	12	12	30	80	0.5
7141210	12	12	30	80	1
7141215	12	12	30	80	1.5
7141220	12	12	30	80	2
7141225	12	12	30	80	2.5
7141230	12	12	30	80	3

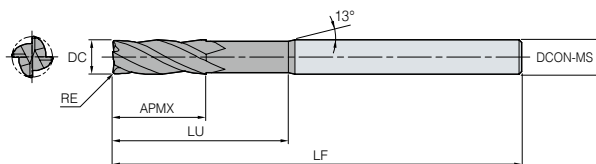
ESR724

4 flutes neck type radius



DC	Tolerance
Ø6	0.000 ~ -0.012
Ø8 ~ Ø12	0.000 ~ -0.015

DC6 or below Above DC6



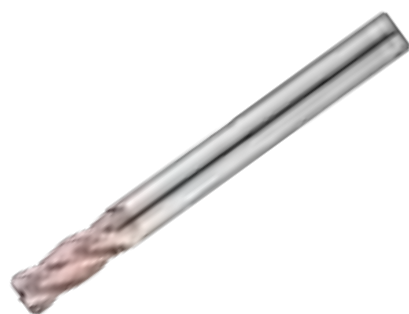
(mm)

Designation	DC	DCON-MS	APMX	LU	LF	RE
ESR 7240600520	6	6	9	20	90	0.5
7240601020	6	6	9	20	90	1
7240800525	8	8	12	25	100	0.5
7240801025	8	8	12	25	100	1
7241000532	10	10	15	32	100	0.5

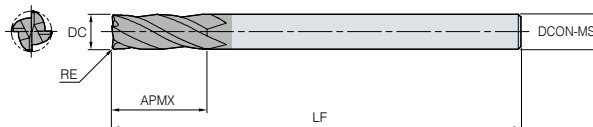
Designation	DC	DCON-MS	APMX	LU	LF	RE
ESR 7241001032	10	10	15	32	100	1
7241002032	10	10	15	32	100	2
7241200538	12	12	18	38	110	0.5
7241201038	12	12	18	38	110	1
7241202038	12	12	18	38	110	2

ESR734

4 flutes long shank radius



CARBIDE	4	H-A 30°	Grade PC305H	h5 shank	RE ±0.01	RE ±0.015	DC	Tolerance
					DC6 or below	Above DC6	Ø1 - Ø6 Ø8 - Ø12	0.000 ~ -0.012 0.000 ~ -0.015

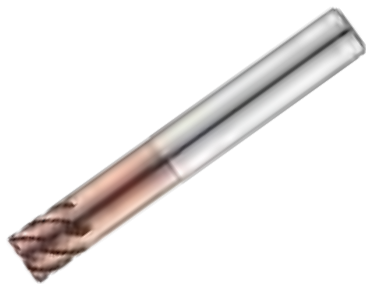


Designation	DC	DCON-MS	APMX	LF	RE
ESR 73401001	1	6	2	50	0.1
73401002	1	6	2	50	0.2
73401003	1	6	2	50	0.3
73401501	1.5	6	3	50	0.1
73401502	1.5	6	3	50	0.2
73401503	1.5	6	3	50	0.3
73401505	1.5	6	3	50	0.5
73402001	2	6	5	50	0.1
73402002	2	6	5	50	0.2
73402003	2	6	5	50	0.3
73402005	2	6	5	50	0.5
73402501	2.5	6	7	60	0.1
73402502	2.5	6	7	60	0.2
73402503	2.5	6	7	60	0.3
73402505	2.5	6	7	60	0.5
73403001	3	6	8	60	0.1
73403002	3	6	8	60	0.2
73403003	3	6	8	60	0.3
73403005	3	6	8	60	0.5
73404001	4	6	10	70	0.1
73404002	4	6	10	70	0.2
73404002S4	4	4	10	70	0.2
73404003	4	6	10	70	0.3
73404005	4	6	10	70	0.5
73404005S4	4	4	10	70	0.5
73404010	4	6	10	70	1
73405001	5	6	13	80	0.1

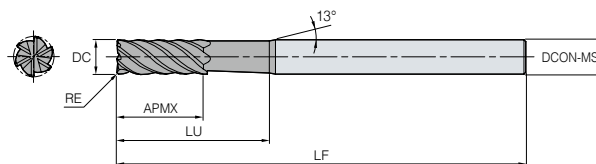
Designation	DC	DCON-MS	APMX	LF	RE
ESR 73405002	5	6	13	80	0.2
73405003	5	6	13	80	0.3
73405005	5	6	13	80	0.5
73405010	5	6	13	80	1
73406001	6	6	15	90	0.1
73406002	6	6	15	90	0.2
73406003	6	6	15	90	0.3
73406005	6	6	15	90	0.5
73406010	6	6	15	90	1
73408001	8	8	20	100	0.1
73408002	8	8	20	100	0.2
73408003	8	8	20	100	0.3
73408005	8	8	20	100	0.5
73408010	8	8	20	100	1
73408020	8	8	20	100	2
73410002	10	10	25	100	0.2
73410003	10	10	25	100	0.3
73410005	10	10	25	100	0.5
73410010	10	10	25	100	1
73410020	10	10	25	100	2
73412002	12	12	30	110	0.2
73412003	12	12	30	110	0.3
73412005	12	12	30	110	0.5
73412010	12	12	30	110	1
73412010L	12	12	30	150	1
73412020	12	12	30	110	2

ESR706

6 flutes neck type radius



CARBIDE	6	H-A 45°	Grade PC305H	h5 shank	RE ±0.01	RE ±0.015	DC Ø6 Ø8 ~ Ø12	Tolerance 0.000 ~ -0.012 0.000 ~ -0.015
					DC6 or below Above DC6			

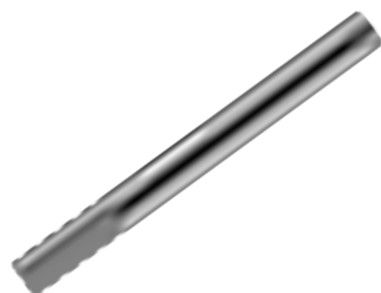


(mm)

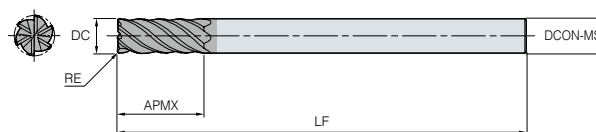
Designation		DC	DCON-MS	APMX	LU	LF	RE
ESR	7060600314	6	6	6	14	50	0.3
	7060600514	6	6	6	14	50	0.5
	7060800524	8	8	8	24	60	0.5
	7060801024	8	8	8	24	60	1
	7061000530	10	10	10	30	70	0.5
	7061001030	10	10	10	30	70	1
	7061200530	12	12	12	30	75	0.5
	7061201030	12	12	12	30	75	1

ESR736

6 flutes radius



CARBIDE	6	H-A 45°	Grade PC305H	h5 shank	RE ±0.01	RE ±0.015	DC Ø6 Ø8 ~ Ø12	Tolerance 0.000 ~ -0.012 0.000 ~ -0.015
					DC6 or below Above DC6			

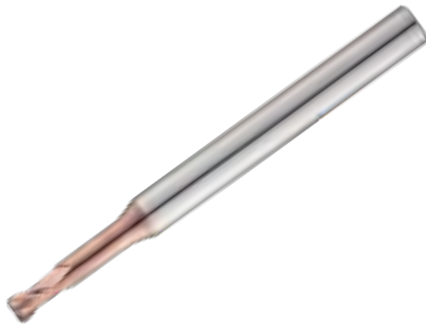


(mm)

Designation		DC	DCON-MS	APMX	LF	RE
ESR	73606005	6	6	15	90	0.5
	73606010	6	6	15	90	1
	73608005	8	8	20	100	0.5
	73608010	8	8	20	100	1
	73610005	10	10	25	100	0.5
	73610010	10	10	25	100	1
	73612005	12	12	30	110	0.5
	73612010	12	12	30	110	1

ESRR712

2 flutes rib ball



CARBIDE

2

H-A
30°

Grade
PC305H

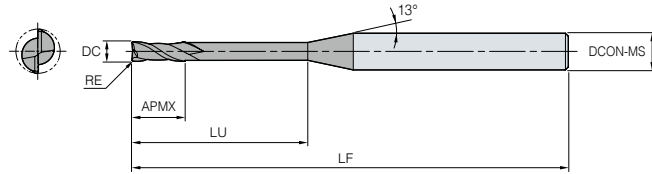
h5
shank

RE
±0.01

RE
±0.015

DC6 or below Above DC6

DC	Tolerance
Ø0.2 ~ Ø6	0.000 ~ -0.012
Ø8 ~ Ø16	0.000 ~ -0.015

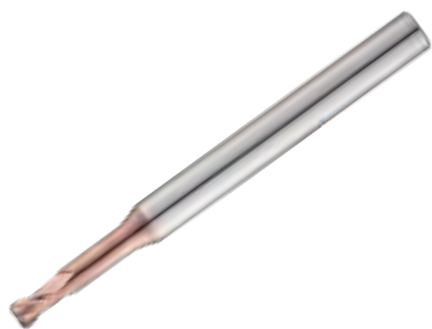


Designation	DC	DCON-MS	APMX	LU	LF	RE
ESRR 712002002005	0.2	4	0.2	0.5	40	0.02
71200200201	0.2	4	0.2	1	40	0.02
712002002015	0.2	4	0.2	1.5	40	0.02
712002005005	0.2	4	0.2	0.5	40	0.05
71200200501	0.2	4	0.2	1	40	0.05
712002005015	0.2	4	0.2	1.5	40	0.05
71200300201	0.3	4	0.3	1	40	0.02
71200300202	0.3	4	0.3	2	40	0.02
71200300203	0.3	4	0.3	3	40	0.02
71200300501	0.3	4	0.3	1	40	0.05
71200300502	0.3	4	0.3	2	40	0.05
71200300503	0.3	4	0.3	3	40	0.05
71200400201	0.4	4	0.4	1	40	0.02
71200400202	0.4	4	0.4	2	40	0.02
71200400203	0.4	4	0.4	3	40	0.02
71200400204	0.4	4	0.4	4	40	0.02
71200400501	0.4	4	0.4	1	40	0.05
71200400502	0.4	4	0.4	2	40	0.05
71200400503	0.4	4	0.4	3	40	0.05
71200400504	0.4	4	0.4	4	40	0.05
71200401001	0.4	4	0.4	1	40	0.1
712004010015	0.4	4	0.4	1.5	40	0.1
71200401002	0.4	4	0.4	2	40	0.1
71200401003	0.4	4	0.4	3	40	0.1
71200401004	0.4	4	0.4	4	40	0.1
71200500201	0.5	4	0.5	1	45	0.02
712005002015	0.5	4	0.5	1.5	45	0.02
71200500202	0.5	4	0.5	2	45	0.02
712005002025	0.5	4	0.5	2.5	45	0.02
71200500203	0.5	4	0.5	3	45	0.02
71200500204	0.5	4	0.5	4	45	0.02
71200500205	0.5	4	0.5	5	45	0.02
71200500206	0.5	4	0.5	6	45	0.02
71200500208	0.5	4	0.5	8	45	0.02
71200500210	0.5	4	0.5	10	45	0.02
71200500501	0.5	4	0.5	1	45	0.05
712005005015	0.5	4	0.5	1.5	45	0.05
71200500502	0.5	4	0.5	2	45	0.05
712005005025	0.5	4	0.5	2.5	45	0.05

Designation	DC	DCON-MS	APMX	LU	LF	RE
ESRR 71200500503	0.5	4	0.5	3	45	0.05
71200500504	0.5	4	0.5	4	45	0.05
71200500505	0.5	4	0.5	5	45	0.05
71200500506	0.5	4	0.5	6	45	0.05
71200500508	0.5	4	0.5	8	45	0.05
71200500510	0.5	4	0.5	10	45	0.05
71200501001	0.5	4	0.5	1	45	0.1
712005010015	0.5	4	0.5	1.5	45	0.1
71200501002	0.5	4	0.5	2	45	0.1
712005010025	0.5	4	0.5	2.5	45	0.1
71200501003	0.5	4	0.5	3	45	0.1
71200501004	0.5	4	0.5	4	45	0.1
71200501005	0.5	4	0.5	5	45	0.1
71200501006	0.5	4	0.5	6	45	0.1
71200501008	0.5	4	0.5	8	45	0.1
71200501010	0.5	4	0.5	10	45	0.1
71200600202	0.6	4	0.6	2	45	0.02
71200600203	0.6	4	0.6	3	45	0.02
71200600204	0.6	4	0.6	4	45	0.02
71200600206	0.6	4	0.6	6	45	0.02
71200600208	0.6	4	0.6	8	45	0.02
71200600210	0.6	4	0.6	10	45	0.02
71200600212	0.6	4	0.6	12	50	0.02
71200600502	0.6	4	0.6	2	45	0.05
71200600503	0.6	4	0.6	3	45	0.05
71200600504	0.6	4	0.6	4	45	0.05
71200600506	0.6	4	0.6	6	45	0.05
71200600508	0.6	4	0.6	8	45	0.05
71200600510	0.6	4	0.6	10	45	0.05
71200600512	0.6	4	0.6	12	50	0.05
71200601002	0.6	4	0.6	2	45	0.1
71200601003	0.6	4	0.6	3	45	0.1
71200601004	0.6	4	0.6	4	45	0.1
71200601006	0.6	4	0.6	6	45	0.1
71200601008	0.6	4	0.6	8	45	0.1
71200601010	0.6	4	0.6	10	45	0.1
71200601012	0.6	4	0.6	12	50	0.1
71200701002	0.7	4	0.7	2	45	0.1
71200701004	0.7	4	0.7	4	45	0.1

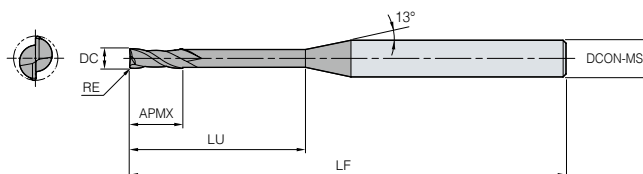
ESRR712

2 flutes rib ball



DC6 or below Above DC6

DC	Tolerance
Ø0.2 ~ Ø6	0.000 ~ -0.012
Ø8 ~ Ø16	0.000 ~ -0.015



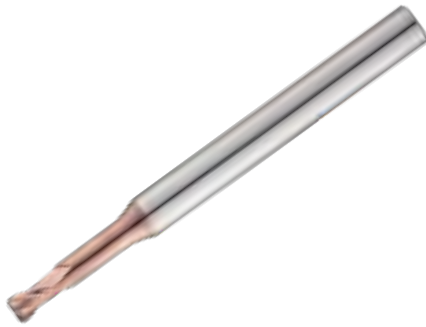
(mm)

	Designation	DC	DCON-MS	APMX	LU	LF	RE
ESRR	71200701006	0.7	4	0.7	6	45	0.1
	71200701008	0.7	4	0.7	8	45	0.1
	71200701010	0.7	4	0.7	10	45	0.1
	71200800202	0.8	4	0.8	2	45	0.02
	71200800204	0.8	4	0.8	4	45	0.02
	71200800206	0.8	4	0.8	6	45	0.02
	71200800208	0.8	4	0.8	8	45	0.02
	71200800210	0.8	4	0.8	10	45	0.02
	71200800212	0.8	4	0.8	12	50	0.02
	71200800502	0.8	4	0.8	2	45	0.05
	71200800504	0.8	4	0.8	4	45	0.05
	71200800506	0.8	4	0.8	6	45	0.05
	71200800508	0.8	4	0.8	8	45	0.05
	71200800510	0.8	4	0.8	10	45	0.05
	71200800512	0.8	4	0.8	12	50	0.05
	71200801002	0.8	4	0.8	2	45	0.1
	71200801004	0.8	4	0.8	4	45	0.1
	71200801006	0.8	4	0.8	6	45	0.1
	71200801008	0.8	4	0.8	8	45	0.1
	71200801010	0.8	4	0.8	10	45	0.1
	71200801012	0.8	4	0.8	12	50	0.1
	71200802002	0.8	4	0.8	2	45	0.2
	71200802004	0.8	4	0.8	4	45	0.2
	71200802006	0.8	4	0.8	6	45	0.2
	71200802008	0.8	4	0.8	8	45	0.2
	71200802010	0.8	4	0.8	10	45	0.2
	71200802012	0.8	4	0.8	12	50	0.2
	71201000204	1	4	1	4	45	0.02
	71201000206	1	4	1	6	45	0.02
	71201000208	1	4	1	8	45	0.02
	71201000210	1	4	1	10	50	0.02
	71201000212	1	4	1	12	50	0.02
	71201000214	1	4	1	14	50	0.02
	71201000216	1	4	1	16	50	0.02
	71201000220	1	4	1	20	50	0.02
	71201000504	1	4	1	4	45	0.05
	71201000506	1	4	1	6	45	0.05
	71201000508	1	4	1	8	45	0.05
	71201000510	1	4	1	10	50	0.05

	Designation	DC	DCON-MS	APMX	LU	LF	RE
ESRR	71201000512	1	4	1	12	50	0.05
	71201000514	1	4	1	14	50	0.05
	71201000516	1	4	1	16	50	0.05
	71201000520	1	4	1	20	50	0.05
	71201001004	1	4	1	4	45	0.1
	71201001006	1	4	1	6	45	0.1
	71201001008	1	4	1	8	45	0.1
	71201001010	1	4	1	10	50	0.1
	71201001012	1	4	1	12	50	0.1
	71201001014	1	4	1	14	50	0.1
	71201001016	1	4	1	16	50	0.1
	71201001020	1	4	1	20	50	0.1
	71201002004	1	4	1	4	45	0.2
	71201002006	1	4	1	6	45	0.2
	71201002008	1	4	1	8	45	0.2
	71201002010	1	4	1	10	50	0.2
	71201002012	1	4	1	12	50	0.2
	71201002014	1	4	1	14	50	0.2
	71201002016	1	4	1	16	50	0.2
	71201002020	1	4	1	20	50	0.2
	71201003004	1	4	1	4	45	0.3
	71201003006	1	4	1	6	45	0.3
	71201003008	1	4	1	8	45	0.3
	71201003010	1	4	1	10	50	0.3
	71201003012	1	4	1	12	50	0.3
	71201003014	1	4	1	14	50	0.3
	71201003016	1	4	1	16	50	0.3
	71201003020	1	4	1	20	50	0.3
	71201200204	1.2	4	1.2	4	45	0.02
	71201200206	1.2	4	1.2	6	45	0.02
	71201200208	1.2	4	1.2	8	45	0.02
	71201200210	1.2	4	1.2	10	50	0.02
	71201200212	1.2	4	1.2	12	50	0.02
	71201200214	1.2	4	1.2	14	50	0.02
	71201200216	1.2	4	1.2	16	50	0.02
	71201200220	1.2	4	1.2	20	50	0.02
	71201200504	1.2	4	1.2	4	45	0.05
	71201200506	1.2	4	1.2	6	45	0.05
	71201200508	1.2	4	1.2	8	45	0.05

ESRR712

2 flutes rib ball



CARBIDE

2

H-A
30°

Grade
PC305H

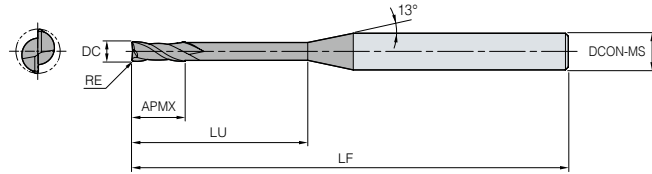
h5
shank

RE
±0.01

RE
±0.015

DC6 or below Above DC6

DC	Tolerance
Ø0.2 ~ Ø6	0.000 ~ -0.012
Ø8 ~ Ø16	0.000 ~ -0.015

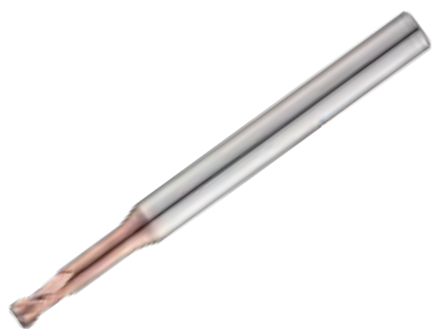


Designation	DC	DCON-MS	APMX	LU	LF	RE
ESRR 71201200510	1.2	4	1.2	10	50	0.05
71201200512	1.2	4	1.2	12	50	0.05
71201200514	1.2	4	1.2	14	50	0.05
71201200516	1.2	4	1.2	16	50	0.05
71201200520	1.2	4	1.2	20	50	0.05
71201201004	1.2	4	1.2	4	45	0.1
71201201006	1.2	4	1.2	6	45	0.1
71201201008	1.2	4	1.2	8	45	0.1
71201201010	1.2	4	1.2	10	50	0.1
71201201012	1.2	4	1.2	12	50	0.1
71201201014	1.2	4	1.2	14	50	0.1
71201201016	1.2	4	1.2	16	50	0.1
71201201020	1.2	4	1.2	20	50	0.1
71201202004	1.2	4	1.2	4	45	0.2
71201202006	1.2	4	1.2	6	45	0.2
71201202008	1.2	4	1.2	8	45	0.2
71201202010	1.2	4	1.2	10	50	0.2
71201202012	1.2	4	1.2	12	50	0.2
71201202014	1.2	4	1.2	14	50	0.2
71201202016	1.2	4	1.2	16	50	0.2
71201202020	1.2	4	1.2	20	50	0.2
71201203004	1.2	4	1.2	4	45	0.3
71201203006	1.2	4	1.2	6	45	0.3
71201203008	1.2	4	1.2	8	45	0.3
71201203010	1.2	4	1.2	10	50	0.3
71201203012	1.2	4	1.2	12	50	0.3
71201203014	1.2	4	1.2	14	50	0.3
71201203016	1.2	4	1.2	16	50	0.3
71201203020	1.2	4	1.2	20	50	0.3
71201500204	1.5	4	1.5	4	45	0.02
71201500206	1.5	4	1.5	6	45	0.02
71201500208	1.5	4	1.5	8	45	0.02
71201500210	1.5	4	1.5	10	50	0.02
71201500212	1.5	4	1.5	12	50	0.02
71201500214	1.5	4	1.5	14	50	0.02
71201500216	1.5	4	1.5	16	50	0.02
71201500220	1.5	4	1.5	20	50	0.02
71201500504	1.5	4	1.5	4	45	0.05
71201500506	1.5	4	1.5	6	45	0.05

Designation	DC	DCON-MS	APMX	LU	LF	RE
ESRR 71201500508	1.5	4	1.5	8	45	0.05
71201500510	1.5	4	1.5	10	50	0.05
71201500512	1.5	4	1.5	12	50	0.05
71201500514	1.5	4	1.5	14	50	0.05
71201500516	1.5	4	1.5	16	50	0.05
71201500520	1.5	4	1.5	20	50	0.05
71201501004	1.5	4	1.5	4	45	0.1
71201501006	1.5	4	1.5	6	45	0.1
71201501008	1.5	4	1.5	8	45	0.1
71201501010	1.5	4	1.5	10	50	0.1
71201501012	1.5	4	1.5	12	50	0.1
71201501014	1.5	4	1.5	14	50	0.1
71201501016	1.5	4	1.5	16	50	0.1
71201501020	1.5	4	1.5	20	50	0.1
71201502004	1.5	4	1.5	4	45	0.2
71201502006	1.5	4	1.5	6	45	0.2
71201502008	1.5	4	1.5	8	45	0.2
71201502010	1.5	4	1.5	10	50	0.2
71201502012	1.5	4	1.5	12	50	0.2
71201502014	1.5	4	1.5	14	50	0.2
71201502016	1.5	4	1.5	16	50	0.2
71201502020	1.5	4	1.5	20	50	0.2
71201503004	1.5	4	1.5	4	45	0.3
71201503006	1.5	4	1.5	6	45	0.3
71201503008	1.5	4	1.5	8	45	0.3
71201503010	1.5	4	1.5	10	50	0.3
71201503012	1.5	4	1.5	12	50	0.3
71201503014	1.5	4	1.5	14	50	0.3
71201503016	1.5	4	1.5	16	50	0.3
71201503020	1.5	4	1.5	20	50	0.3
71201505004	1.5	4	1.5	4	45	0.5
71201505006	1.5	4	1.5	6	45	0.5
71201505008	1.5	4	1.5	8	45	0.5
71201505010	1.5	4	1.5	10	50	0.5
71201505012	1.5	4	1.5	12	50	0.5
71201505014	1.5	4	1.5	14	50	0.5
71201505016	1.5	4	1.5	16	50	0.5
71201505020	1.5	4	1.5	20	50	0.5
71202000206	2	4	2	6	45	0.02

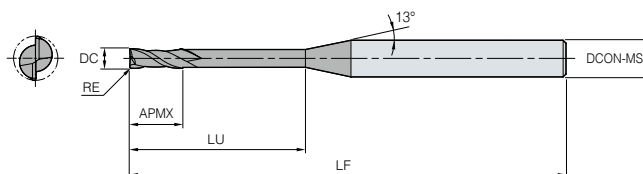
ESRR712

2 flutes rib ball



DC6 or below Above DC6

DC	Tolerance
Ø0.2 ~ Ø6	0.000 ~ -0.012
Ø8 ~ Ø16	0.000 ~ -0.015



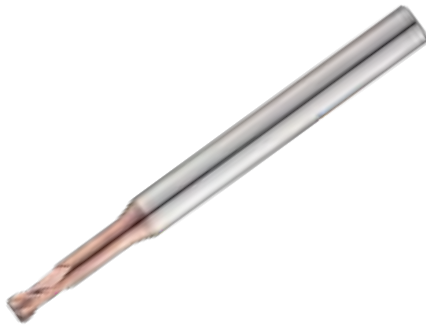
(mm)

	Designation	DC	DCON-MS	APMX	LU	LF	RE
ESRR	71202000208	2	4	2	8	45	0.02
	71202000210	2	4	2	10	50	0.02
	71202000212	2	4	2	12	50	0.02
	71202000214	2	4	2	14	50	0.02
	71202000216	2	4	2	16	50	0.02
	71202000220	2	4	2	20	50	0.02
	71202000225	2	4	2	25	60	0.02
	71202000506	2	4	2	6	45	0.05
	71202000508	2	4	2	8	45	0.05
	71202000510	2	4	2	10	50	0.05
	71202000512	2	4	2	12	50	0.05
	71202000514	2	4	2	14	50	0.05
	71202000516	2	4	2	16	50	0.05
	71202000520	2	4	2	20	50	0.05
	71202000525	2	4	2	25	60	0.05
	71202001006	2	4	2	6	45	0.1
	71202001008	2	4	2	8	45	0.1
	71202001010	2	4	2	10	50	0.1
	71202001012	2	4	2	12	50	0.1
	71202001014	2	4	2	14	50	0.1
	71202001016	2	4	2	16	50	0.1
	71202001020	2	4	2	20	50	0.1
	71202001025	2	4	2	25	60	0.1
	71202001030	2	4	2	30	70	0.1
	71202002006	2	4	2	6	45	0.2
	71202002008	2	4	2	8	45	0.2
	71202002010	2	4	2	10	50	0.2
	71202002012	2	4	2	12	50	0.2
	71202002014	2	4	2	14	50	0.2
	71202002016	2	4	2	16	50	0.2
	71202002020	2	4	2	20	50	0.2
	71202002025	2	4	2	25	60	0.2
	71202002030	2	4	2	30	70	0.2
	71202003006	2	4	2	6	45	0.3
	71202003008	2	4	2	8	45	0.3
	71202003010	2	4	2	10	50	0.3
	71202003012	2	4	2	12	50	0.3
	71202003014	2	4	2	14	50	0.3
	71202003016	2	4	2	16	50	0.3

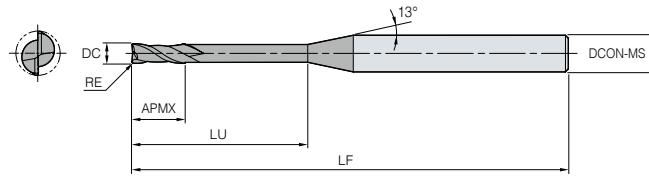
	Designation	DC	DCON-MS	APMX	LU	LF	RE
ESRR	71202003020	2	4	2	20	50	0.3
	71202003025	2	4	2	25	60	0.3
	71202003030	2	4	2	30	70	0.3
	71202005006	2	4	2	6	45	0.5
	71202005008	2	4	2	8	45	0.5
	71202005010	2	4	2	10	50	0.5
	71202005012	2	4	2	12	50	0.5
	71202005014	2	4	2	14	50	0.5
	71202005016	2	4	2	16	50	0.5
	71202005020	2	4	2	20	50	0.5
	71202005025	2	4	2	25	60	0.5
	71202005030	2	4	2	30	70	0.5
	71202501010	2.5	4	2.5	10	50	0.1
	71202501016	2.5	4	2.5	16	50	0.1
	71202501020	2.5	4	2.5	20	50	0.1
	71202501025	2.5	4	2.5	25	60	0.1
	71202501030	2.5	4	2.5	30	70	0.1
	71202502010	2.5	4	2.5	10	50	0.2
	71202502016	2.5	4	2.5	16	50	0.2
	71202502020	2.5	4	2.5	20	50	0.2
	71202503010	2.5	4	2.5	10	50	0.3
	71202503016	2.5	4	2.5	16	50	0.3
	71202503020	2.5	4	2.5	20	50	0.3
	71202505010	2.5	4	2.5	10	50	0.5
	71202505016	2.5	4	2.5	16	50	0.5
	71202505020	2.5	4	2.5	20	50	0.5
	71203001010	3	6	3	10	50	0.1
	71203001012	3	6	3	12	50	0.1
	71203001016	3	6	3	16	55	0.1
	71203001020	3	6	3	20	60	0.1
	71203001025	3	6	3	25	65	0.1
	71203001030	3	6	3	30	70	0.1
	71203001035	3	6	3	35	75	0.1
	71203001040	3	6	3	40	80	0.1
	71203002010	3	6	3	10	50	0.2
	71203002012	3	6	3	12	50	0.2
	71203002016	3	6	3	16	55	0.2
	71203002020	3	6	3	20	60	0.2
	71203002025	3	6	3	25	65	0.2

ESRR712

2 flutes rib ball



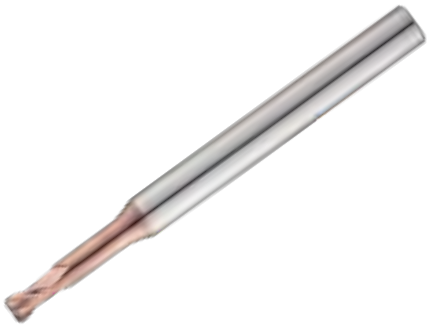
CARBIDE	2	H-A 30°	Grade PC305H	h5 shank	RE ±0.01	RE ±0.015	DC Ø0.2 ~ Ø6 Ø8 ~ Ø16	Tolerance 0.000 ~ -0.012 0.000 ~ -0.015
					DC6 or below Above DC6			



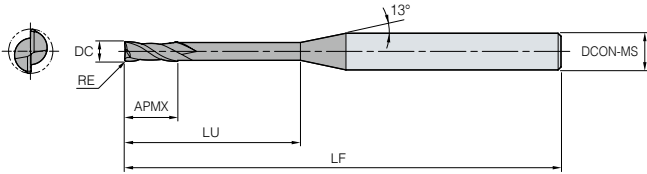
Designation		DC	DCON-MS	APMX	LU	LF	RE		
ESRR	71203002030	3	6	3	30	70	0.2		
	71203002035	3	6	3	35	75	0.2		
	71203002040	3	6	3	40	80	0.2		
	71203003010	3	6	3	10	50	0.3		
	71203003012	3	6	3	12	50	0.3		
	71203003016	3	6	3	16	55	0.3		
	71203003020	3	6	3	20	60	0.3		
	71203003025	3	6	3	25	65	0.3		
	71203003030	3	6	3	30	70	0.3		
	71203003035	3	6	3	35	75	0.3		
	71203003040	3	6	3	40	80	0.3		
	71203005010	3	6	3	10	50	0.5		
	71203005012	3	6	3	12	50	0.5		
	71203005016	3	6	3	16	55	0.5		
	71203005020	3	6	3	20	60	0.5		
	71203005025	3	6	3	25	65	0.5		
	71203005030	3	6	3	30	70	0.5		
	71203005035	3	6	3	35	75	0.5		
	71203005040	3	6	3	40	80	0.5		
	71203010010	3	6	3	10	50	1		
	71203010012	3	6	3	12	50	1		
	71203010016	3	6	3	16	55	1		
	71203010020	3	6	3	20	60	1		
	71203010025	3	6	3	25	65	1		
	71203010030	3	6	3	30	70	1		
	71203010035	3	6	3	35	75	1		
	71203010040	3	6	3	40	80	1		
	71204001012	4	6	4	12	50	0.1		
	71204001016	4	6	4	16	55	0.1		
	71204001020	4	6	4	20	60	0.1		
	71204001025	4	6	4	25	65	0.1		
	71204001030	4	6	4	30	70	0.1		
	71204001035	4	6	4	35	75	0.1		
	71204001040	4	6	4	40	80	0.1		
	71204002012	4	6	4	12	50	0.2		
	71204002016	4	6	4	16	55	0.2		
	71204002020	4	6	4	20	60	0.2		
	71204002025	4	6	4	25	65	0.2		
	71204002030	4	6	4	30	70	0.2		
ESRR	71204002035	4	6	4	35	75	0.2		
	71204002040	4	6	4	40	80	0.2		
	71204003012	4	6	4	12	50	0.3		
	71204003016	4	6	4	16	55	0.3		
	71204003020	4	6	4	20	60	0.3		
	71204003025	4	6	4	25	65	0.3		
	71204003030	4	6	4	30	70	0.3		
	71204003035	4	6	4	35	75	0.3		
	71204003040	4	6	4	40	80	0.3		
	71204005012	4	6	4	12	50	0.5		
	71204005016	4	6	4	16	55	0.5		
	71204005020	4	6	4	20	60	0.5		
	71204005025	4	6	4	25	65	0.5		
	71204005030	4	6	4	30	70	0.5		
	71204005035	4	6	4	35	75	0.5		
	71204005040	4	6	4	40	80	0.5		
	71204010012	4	6	4	12	50	1		
	71204010016	4	6	4	16	55	1		
	71204010020	4	6	4	20	60	1		
	71204010025	4	6	4	25	65	1		
	71204010030	4	6	4	30	70	1		
	71204010035	4	6	4	35	75	1		
	71204010040	4	6	4	40	80	1		
	71205002015	5	6	6	15	60	0.2		
	71205002025	5	6	6	25	70	0.2		
	71205002030	5	6	6	30	70	0.2		
	71205002040	5	6	6	40	80	0.2		
	71205005015	5	6	6	15	60	0.5		
	71205005025	5	6	6	25	70	0.5		
	71205005030	5	6	6	30	70	0.5		
	71205005040	5	6	6	40	80	0.5		
	71205010015	5	6	6	15	60	1		
	71205010025	5	6	6	25	70	1		
	71205010030	5	6	6	30	70	1		
	71205010040	5	6	6	40	80	1		
	71206001020	6	6	7	20	60	0.1		
	71206001040	6	6	7	40	80	0.1		
	71206002020	6	6	7	20	60	0.2		
	71206002040	6	6	7	40	80	0.2		

ESRR712

2 flutes rib ball



CARBIDE	2	H-A 30°	Grade PC305H	h5 shank	RE ±0.01	RE ±0.015	DC	Tolerance
							Ø0.2 ~ Ø6	0.000 ~ -0.012
							Ø8 ~ Ø16	0.000 ~ -0.015
							DC6 or below Above DC6	



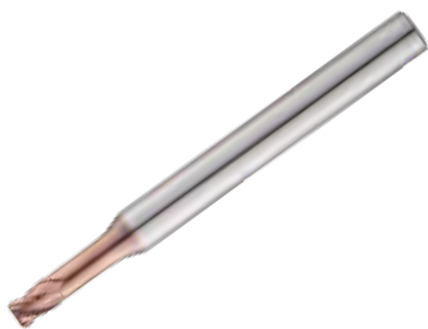
Designation		DC	DCON-MS	APMX	LU	LF	RE
ESRR	71206003020	6	6	7	20	60	0.3
	71206003040	6	6	7	40	80	0.3
	71206005020	6	6	7	20	60	0.5
	71206005040	6	6	7	40	80	0.5
	71206010020	6	6	7	20	60	1
	71206010040	6	6	7	40	80	1
	71206015020	6	6	7	20	60	1.5
	71206015040	6	6	7	40	80	1.5
	71208002022	8	8	9	22	65	0.2
	71208003022	8	8	9	22	65	0.3
	71208005022	8	8	9	22	65	0.5
	71208010022	8	8	9	22	65	1
	71208015022	8	8	9	22	65	1.5
	71210002024	10	10	11	24	70	0.2

(mm)

Designation		DC	DCON-MS	APMX	LU	LF	RE
ESRR	71210003024	10	10	11	24	70	0.3
	71210005024	10	10	11	24	70	0.5
	71210010024	10	10	11	24	70	1
	71210015024	10	10	11	24	70	1.5
	71210020024	10	10	11	24	70	2
	71212002026	12	12	13	26	80	0.2
	71212003026	12	12	13	26	80	0.3
	71212005026	12	12	13	26	80	0.5
	71212010026	12	12	13	26	80	1
	71212015026	12	12	13	26	80	1.5
	71212020026	12	12	13	26	80	2
	71212030026	12	12	13	26	80	3
	71216005035	16	16	20	35	110	0.5
	71216010035	16	16	20	35	110	1

ESRR714

4 flutes rib ball



CARBIDE

4

H-A
30°

Grade
PC305H

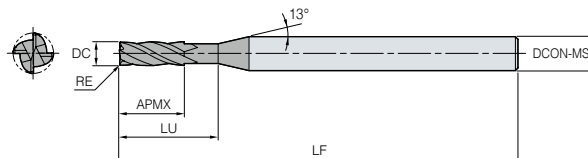
h5
shank

RE
±0.01

RE
±0.015

DC6 or below Above DC6

DC	Tolerance
Ø0.5 ~ Ø6	0.000 ~ -0.012
Ø8 ~ Ø20	0.000 ~ -0.015

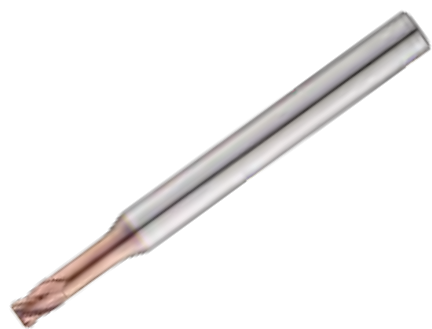


Designation	DC	DCON-MS	APMX	LU	LF	RE
ESRR 71400500502	0.5	4	0.5	2	45	0.05
71400500504	0.5	4	0.5	4	45	0.05
71400500506	0.5	4	0.5	6	45	0.05
71400500508	0.5	4	0.5	8	45	0.05
71400501002	0.5	4	0.5	2	45	0.1
71400501004	0.5	4	0.5	4	45	0.1
71400501006	0.5	4	0.5	6	45	0.1
71400501008	0.5	4	0.5	8	45	0.1
71400600502	0.6	4	0.6	2	45	0.05
71400600504	0.6	4	0.6	4	45	0.05
71400600506	0.6	4	0.6	6	45	0.05
71400600508	0.6	4	0.6	8	45	0.05
71400601002	0.6	4	0.6	2	45	0.1
71400601004	0.6	4	0.6	4	45	0.1
71400601006	0.6	4	0.6	6	45	0.1
71400601008	0.6	4	0.6	8	45	0.1
71400700502	0.7	4	0.7	2	45	0.05
71400700504	0.7	4	0.7	4	45	0.05
71400700506	0.7	4	0.7	6	45	0.05
71400700508	0.7	4	0.7	8	45	0.05
71400701002	0.7	4	0.7	2	45	0.1
71400701004	0.7	4	0.7	4	45	0.1
71400701006	0.7	4	0.7	6	45	0.1
71400701008	0.7	4	0.7	8	45	0.1
71400800202	0.8	4	0.8	2	45	0.02
71400800204	0.8	4	0.8	4	45	0.02
71400800206	0.8	4	0.8	6	45	0.02
71400800208	0.8	4	0.8	8	45	0.02
71400800210	0.8	4	0.8	10	45	0.02
71400800212	0.8	4	0.8	12	50	0.02
71400800502	0.8	4	0.8	2	45	0.05
71400800504	0.8	4	0.8	4	45	0.05
71400800506	0.8	4	0.8	6	45	0.05
71400800508	0.8	4	0.8	8	45	0.05
71400800510	0.8	4	0.8	10	45	0.05
71400800512	0.8	4	0.8	12	50	0.05
71400801002	0.8	4	0.8	2	45	0.1
71400801004	0.8	4	0.8	4	45	0.1
71400801006	0.8	4	0.8	6	45	0.1

Designation	DC	DCON-MS	APMX	LU	LF	RE
ESRR 71400801008	0.8	4	0.8	8	45	0.1
71400801010	0.8	4	0.8	10	45	0.1
71400801012	0.8	4	0.8	12	50	0.1
71401000204	1	4	1	4	45	0.02
71401000206	1	4	1	6	45	0.02
71401000208	1	4	1	8	45	0.02
71401000210	1	4	1	10	50	0.02
71401000212	1	4	1	12	50	0.02
71401000214	1	4	1	14	50	0.02
71401000216	1	4	1	16	50	0.02
71401000220	1	4	1	20	50	0.02
71401000503	1	4	1	3	45	0.05
71401000504	1	4	1	4	45	0.05
71401000506	1	4	1	6	45	0.05
71401000508	1	4	1	8	45	0.05
71401000510	1	4	1	10	50	0.05
71401000512	1	4	1	12	50	0.05
71401000514	1	4	1	14	50	0.05
71401000516	1	4	1	16	50	0.05
71401000520	1	4	1	20	50	0.05
71401001003	1	4	1	3	45	0.1
71401001004	1	4	1	4	45	0.1
71401001006	1	4	1	6	45	0.1
71401001008	1	4	1	8	45	0.1
71401001010	1	4	1	10	50	0.1
71401001012	1	4	1	12	50	0.1
71401001014	1	4	1	14	50	0.1
71401001016	1	4	1	16	50	0.1
71401001020	1	4	1	20	50	0.1
71401002003	1	4	1	3	45	0.2
71401002004	1	4	1	4	45	0.2
71401002006	1	4	1	6	45	0.2
71401002008	1	4	1	8	45	0.2
71401002010	1	4	1	10	50	0.2
71401002012	1	4	1	12	50	0.2
71401002014	1	4	1	14	50	0.2
71401002016	1	4	1	16	50	0.2
71401002020	1	4	1	20	50	0.2
71401003003	1	4	1	3	45	0.3

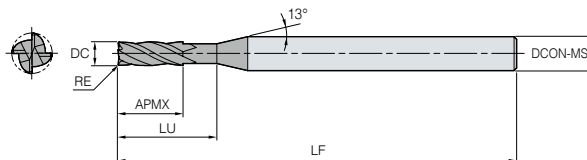
ESRR714

4 flutes rib ball



DC6 or below Above DC6

DC	Tolerance
Ø0.5 ~ Ø6	0.000 ~ -0.012
Ø8 ~ Ø20	0.000 ~ -0.015



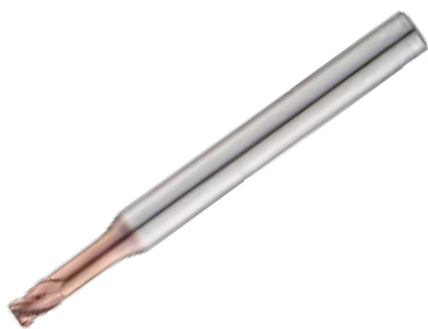
(mm)

	Designation	DC	DCON-MS	APMX	LU	LF	RE
ESRR	71401003004	1	4	1	4	45	0.3
	71401003006	1	4	1	6	45	0.3
	71401003008	1	4	1	8	45	0.3
	71401003010	1	4	1	10	50	0.3
	71401003012	1	4	1	12	50	0.3
	71401003014	1	4	1	14	50	0.3
	71401003016	1	4	1	16	50	0.3
	71401003020	1	4	1	20	50	0.3
	71401200204	1.2	4	1.2	4	45	0.02
	71401200206	1.2	4	1.2	6	45	0.02
	71401200208	1.2	4	1.2	8	45	0.02
	71401200210	1.2	4	1.2	10	50	0.02
	71401200212	1.2	4	1.2	12	50	0.02
	71401200214	1.2	4	1.2	14	50	0.02
	71401200216	1.2	4	1.2	16	50	0.02
	71401200220	1.2	4	1.2	20	50	0.02
	71401200503	1.2	4	1.2	3	45	0.05
	71401200504	1.2	4	1.2	4	45	0.05
	71401200506	1.2	4	1.2	6	45	0.05
	71401200508	1.2	4	1.2	8	45	0.05
	71401200510	1.2	4	1.2	10	50	0.05
	71401200512	1.2	4	1.2	12	50	0.05
	71401200514	1.2	4	1.2	14	50	0.05
	71401200516	1.2	4	1.2	16	50	0.05
	71401200520	1.2	4	1.2	20	50	0.05
	71401201003	1.2	4	1.2	3	45	0.1
	71401201004	1.2	4	1.2	4	45	0.1
	71401201006	1.2	4	1.2	6	45	0.1
	71401201008	1.2	4	1.2	8	45	0.1
	71401201010	1.2	4	1.2	10	50	0.1
	71401201012	1.2	4	1.2	12	50	0.1
	71401201014	1.2	4	1.2	14	50	0.1
	71401201016	1.2	4	1.2	16	50	0.1
	71401201020	1.2	4	1.2	20	50	0.1
	71401202003	1.2	4	1.2	3	45	0.2
	71401202004	1.2	4	1.2	4	45	0.2
	71401202006	1.2	4	1.2	6	45	0.2
	71401202008	1.2	4	1.2	8	45	0.2
	71401202010	1.2	4	1.2	10	50	0.2

	Designation	DC	DCON-MS	APMX	LU	RE	LF
ESRR	71401202012	1.2	4	1.2	12	0.2	50
	71401202014	1.2	4	1.2	14	0.2	50
	71401202016	1.2	4	1.2	16	0.2	50
	71401202020	1.2	4	1.2	20	0.2	50
	71401203003	1.2	4	1.2	3	0.3	45
	71401203004	1.2	4	1.2	4	0.3	45
	71401203006	1.2	4	1.2	6	0.3	45
	71401203008	1.2	4	1.2	8	0.3	45
	71401203010	1.2	4	1.2	10	0.3	50
	71401203012	1.2	4	1.2	12	0.3	50
	71401203016	1.2	4	1.2	16	0.3	50
	71401203020	1.2	4	1.2	20	0.3	50
	71401500206	1.5	4	1.5	6	0.02	45
	71401500208	1.5	4	1.5	8	0.02	45
	71401500210	1.5	4	1.5	10	0.02	50
	71401500212	1.5	4	1.5	12	0.02	50
	71401500214	1.5	4	1.5	14	0.02	50
	71401500216	1.5	4	1.5	16	0.02	50
	71401500220	1.5	4	1.5	20	0.02	50
	71401500222	1.5	4	1.5	22	0.02	60
	71401500504	1.5	4	1.5	4	0.05	45
	71401500506	1.5	4	1.5	6	0.05	45
	71401500508	1.5	4	1.5	8	0.05	45
	71401500510	1.5	4	1.5	10	0.05	50
	71401500512	1.5	4	1.5	12	0.05	50
	71401500514	1.5	4	1.5	14	0.05	50
	71401500516	1.5	4	1.5	16	0.05	50
	71401500520	1.5	4	1.5	20	0.05	50
	71401500522	1.5	4	1.5	22	0.05	60
	71401500526	1.5	4	1.5	26	0.05	60
	71401501004	1.5	4	1.5	4	0.1	45
	71401501006	1.5	4	1.5	6	0.1	45
	71401501008	1.5	4	1.5	8	0.1	45
	71401501010	1.5	4	1.5	10	0.1	50
	71401501012	1.5	4	1.5	12	0.1	50
	71401501014	1.5	4	1.5	14	0.1	50
	71401501016	1.5	4	1.5	16	0.1	50
	71401501020	1.5	4	1.5	20	0.1	50
	71401501022	1.5	4	1.5	22	0.1	60

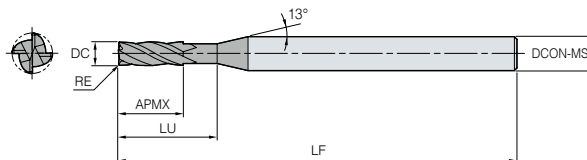
ESRR714

4 flutes rib ball



DC	Tolerance
Ø0.5 ~ Ø6	0.000 ~ -0.012
Ø8 ~ Ø20	0.000 ~ -0.015

DC6 or below Above DC6

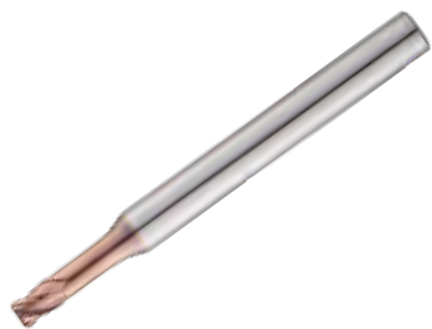


Designation	DC	DCON-MS	APMX	LU	LF	RE
ESRR 71401501026	1.5	4	1.5	26	60	0.1
71401502004	1.5	4	1.5	4	45	0.2
71401502006	1.5	4	1.5	6	45	0.2
71401502008	1.5	4	1.5	8	45	0.2
71401502010	1.5	4	1.5	10	50	0.2
71401502012	1.5	4	1.5	12	50	0.2
71401502014	1.5	4	1.5	14	50	0.2
71401502016	1.5	4	1.5	16	50	0.2
71401502020	1.5	4	1.5	20	50	0.2
71401502022	1.5	4	1.5	22	60	0.2
71401502025	1.5	4	1.5	25	60	0.2
71401503004	1.5	4	1.5	4	45	0.3
71401503006	1.5	4	1.5	6	45	0.3
71401503008	1.5	4	1.5	8	45	0.3
71401503010	1.5	4	1.5	10	50	0.3
71401503012	1.5	4	1.5	12	50	0.3
71401503014	1.5	4	1.5	14	50	0.3
71401503016	1.5	4	1.5	16	50	0.3
71401503020	1.5	4	1.5	20	50	0.3
71401503022	1.5	4	1.5	22	60	0.3
71401503025	1.5	4	1.5	25	60	0.3
71401505004	1.5	4	1.5	4	45	0.5
71401505006	1.5	4	1.5	6	45	0.5
71401505008	1.5	4	1.5	8	45	0.5
71401505010	1.5	4	1.5	10	50	0.5
71401505012	1.5	4	1.5	12	50	0.5
71401505014	1.5	4	1.5	14	50	0.5
71401505016	1.5	4	1.5	16	50	0.5
71401505020	1.5	4	1.5	20	50	0.5
71401505022	1.5	4	1.5	22	60	0.5
71401505025	1.5	4	1.5	25	60	0.5
71402000206	2	4	2	6	45	0.02
71402000208	2	4	2	8	45	0.02
71402000210	2	4	2	10	50	0.02
71402000212	2	4	2	12	50	0.02
71402000214	2	4	2	14	50	0.02
71402000216	2	4	2	16	50	0.02
71402000220	2	4	2	20	50	0.02
71402000225	2	4	2	25	60	0.02

Designation	DC	DCON-MS	APMX	LU	LF	RE
ESRR 71402000230	2	4	2	30	70	0.02
71402000506	2	4	2	6	45	0.05
71402000508	2	4	2	8	45	0.05
71402000510	2	4	2	10	50	0.05
71402000512	2	4	2	12	50	0.05
71402000514	2	4	2	14	50	0.05
71402000516	2	4	2	16	50	0.05
71402000520	2	4	2	20	50	0.05
71402000525	2	4	2	25	60	0.05
71402000530	2	4	2	30	70	0.05
71402001006	2	4	2	6	45	0.1
71402001008	2	4	2	8	45	0.1
71402001010	2	4	2	10	50	0.1
71402001012	2	4	2	12	50	0.1
71402001014	2	4	2	14	50	0.1
71402001016	2	4	2	16	50	0.1
71402001020	2	4	2	20	50	0.1
71402001022	2	4	2	22	60	0.1
71402001025	2	4	2	25	60	0.1
71402001030	2	4	2	30	70	0.1
71402002006	2	4	2	6	45	0.2
71402002008	2	4	2	8	45	0.2
71402002010	2	4	2	10	50	0.2
71402002012	2	4	2	12	50	0.2
71402002014	2	4	2	14	50	0.2
71402002016	2	4	2	16	50	0.2
71402002020	2	4	2	20	50	0.2
71402002022	2	4	2	22	60	0.2
71402002025	2	4	2	25	60	0.2
71402002030	2	4	2	30	70	0.2
71402003006	2	4	2	6	45	0.3
71402003008	2	4	2	8	45	0.3
71402003010	2	4	2	10	50	0.3
71402003012	2	4	2	12	50	0.3
71402003014	2	4	2	14	50	0.3
71402003016	2	4	2	16	50	0.3
71402003020	2	4	2	20	50	0.3
71402003022	2	4	2	22	60	0.3
71402003025	2	4	2	25	60	0.3

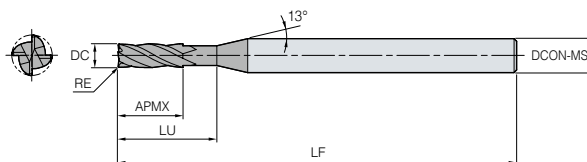
ESRR714

4 flutes rib ball



DC	Tolerance
Ø0.5 ~ Ø6	0.000 ~ -0.012
Ø8 ~ Ø20	0.000 ~ -0.015

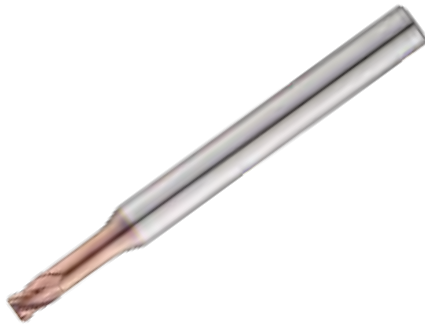
DC6 or below Above DC6



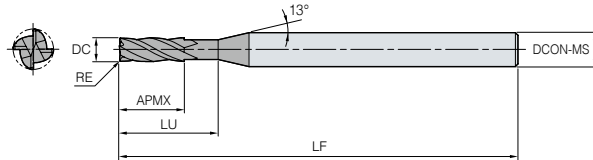
(mm)

	Designation	DC	DCON-MS	APMX	LU	LF	RE
ESRR	71402003030	2	4	2	30	70	0.3
	71402005006	2	4	3	6	45	0.5
	71402005008	2	4	2	8	45	0.5
	71402005010	2	4	2	10	50	0.5
	71402005012	2	4	2	12	50	0.5
	71402005014	2	4	2	14	50	0.5
	71402005016	2	4	2	16	50	0.5
	71402005020	2	4	2	20	50	0.5
	71402005022	2	4	2	22	60	0.5
	71402005025	2	4	2	25	60	0.5
	71402005030	2	4	2	30	70	0.5
	71402501008	2.5	4	2.5	8	45	0.1
	71402501010	2.5	4	2.5	10	50	0.1
	71402501012	2.5	4	2.5	12	50	0.1
	71402501014	2.5	4	2.5	14	50	0.1
	71402501016	2.5	4	2.5	16	50	0.1
	71402501020	2.5	4	2.5	20	50	0.1
	71402501025	2.5	4	2.5	25	60	0.1
	71402501030	2.5	4	2.5	30	70	0.1
	71402502008	2.5	4	2.5	8	45	0.2
	71402502010	2.5	4	2.5	10	50	0.2
	71402502012	2.5	4	2.5	12	50	0.2
	71402502014	2.5	4	2.5	14	50	0.2
	71402502016	2.5	4	2.5	16	50	0.2
	71402502020	2.5	4	2.5	20	50	0.2
	71402502025	2.5	4	2.5	25	60	0.2
	71402502030	2.5	4	2.5	30	70	0.2
	71402503008	2.5	4	2.5	8	45	0.3
	71402503010	2.5	4	2.5	10	50	0.3
	71402503012	2.5	4	2.5	12	50	0.3
	71402503014	2.5	4	2.5	14	50	0.3
	71402503016	2.5	4	2.5	16	50	0.3
	71402503020	2.5	4	2.5	20	50	0.3
	71402503025	2.5	4	2.5	25	60	0.3
	71402503030	2.5	4	2.5	30	70	0.3
	71402505008	2.5	4	2.5	8	45	0.5
	71402505010	2.5	4	2.5	10	50	0.5
	71402505012	2.5	4	2.5	12	50	0.5
	71402505014	2.5	4	2.5	14	50	0.5

	Designation	DC	DCON-MS	APMX	LU	LF	RE
ESRR	71402505016	2.5	4	2.5	16	50	0.5
	71402505020	2.5	4	2.5	20	50	0.5
	71402505025	2.5	4	2.5	25	60	0.5
	71402505030	2.5	4	2.5	30	70	0.5
	71403001008	3	6	3	8	45	0.1
	71403001010	3	6	3	10	50	0.1
	71403001012	3	6	3	12	50	0.1
	71403001014	3	6	3	14	50	0.1
	71403001016	3	6	3	16	55	0.1
	71403001020	3	6	3	20	60	0.1
	71403001025	3	6	3	25	65	0.1
	71403001030	3	6	3	30	70	0.1
	71403001035	3	6	3	35	75	0.1
	71403001040	3	6	3	40	80	0.1
	71403001045	3	6	3	45	90	0.1
	71403002008	3	6	3	8	45	0.2
	71403002010	3	6	3	10	50	0.2
	71403002012	3	6	3	12	50	0.2
	71403002014	3	6	3	14	50	0.2
	71403002016	3	6	3	16	55	0.2
	71403002020	3	6	3	20	60	0.2
	71403002025	3	6	3	25	65	0.2
	71403002030	3	6	3	30	70	0.2
	71403002035	3	6	3	35	75	0.2
	71403002040	3	6	3	40	80	0.2
	71403002045	3	6	3	45	90	0.2
	71403003008	3	6	3	8	45	0.3
	71403003010	3	6	3	10	50	0.3
	71403003012	3	6	3	12	50	0.3
	71403003014	3	6	3	14	50	0.3
	71403003016	3	6	3	16	55	0.3
	71403003020	3	6	3	20	60	0.3
	71403003025	3	6	3	25	65	0.3
	71403003030	3	6	3	30	70	0.3
	71403003035	3	6	3	35	75	0.3
	71403003040	3	6	3	40	80	0.3
	71403003045	3	6	3	45	90	0.3
	71403005008	3	6	3	8	45	0.5
	71403005010	3	6	3	10	50	0.5

ESRR714**4 flutes rib ball**

CARBIDE	4	H-A 30°	Grade PC305H	h5 shank	RE ±0.01	RE ±0.015	DC Ø0.5 ~ Ø6 Ø8 ~ Ø20	Tolerance 0.000 ~ -0.012 0.000 ~ -0.015
					DC6 or below Above DC6			

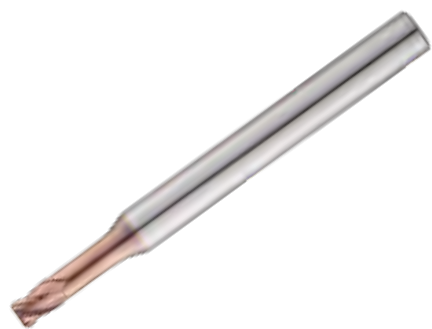


Designation		DC	DCON-MS	APMX	LU	LF	RE
ESRR	71403005012	3	6	3	12	50	0.5
	71403005014	3	6	3	14	50	0.5
	71403005016	3	6	3	16	55	0.5
	71403005020	3	6	3	20	60	0.5
	71403005025	3	6	3	25	65	0.5
	71403005030	3	6	3	30	70	0.5
	71403005035	3	6	3	35	75	0.5
	71403005040	3	6	3	40	80	0.5
	71403005045	3	6	3	45	90	0.5
	71403005050	3	6	3	50	100	0.5
	71403010008	3	6	3	8	45	1
	71403010010	3	6	3	10	50	1
	71403010012	3	6	3	12	50	1
	71403010014	3	6	3	14	50	1
	71403010016	3	6	3	16	55	1
	71403010020	3	6	3	20	60	1
	71403010025	3	6	3	25	65	1
	71403010030	3	6	3	30	70	1
	71403010035	3	6	3	35	75	1
	71403010040	3	6	3	40	80	1
	71403010045	3	6	3	45	90	1
	71403010050	3	6	3	50	100	1
	71404001010	4	6	4	10	50	0.1
	71404001012	4	6	4	12	50	0.1
	71404001013	4	6	4	13	55	0.1
	71404001016	4	6	4	16	55	0.1
	71404001020	4	6	4	20	60	0.1
	71404001025	4	6	4	25	65	0.1
	71404001030	4	6	4	30	70	0.1
	71404001035	4	6	4	35	75	0.1
	71404001040	4	6	4	40	80	0.1
	71404001045	4	6	4	45	90	0.1
	71404001050	4	6	4	50	100	0.1
	71404002010	4	6	4	10	50	0.2
	71404002012	4	6	4	12	50	0.2
	71404002013	4	6	4	13	55	0.2
	71404002016	4	6	4	16	55	0.2
	71404002020	4	6	4	20	60	0.2
	71404002025	4	6	4	25	65	0.2

Designation		DC	DCON-MS	APMX	LU	LF	RE
ESRR	71404002030	4	6	4	30	70	0.2
	71404002035	4	6	4	35	75	0.2
	71404002040	4	6	4	40	80	0.2
	71404002045	4	6	4	45	90	0.2
	71404002050	4	6	4	50	100	0.2
	71404003010	4	6	4	10	50	0.3
	71404003012	4	6	4	12	50	0.3
	71404003013	4	6	4	13	55	0.3
	71404003016	4	6	4	16	55	0.3
	71404003020	4	6	4	20	60	0.3
	71404003025	4	6	4	25	65	0.3
	71404003030	4	6	4	30	70	0.3
	71404003035	4	6	4	35	75	0.3
	71404003040	4	6	4	40	80	0.3
	71404003045	4	6	4	45	90	0.3
	71404003050	4	6	4	50	100	0.3
	71404005010	4	6	4	10	50	0.5
	71404005012	4	6	4	12	50	0.5
	71404005013	4	6	4	13	55	0.5
	71404005016	4	6	4	16	55	0.5
	71404005020	4	6	4	20	60	0.5
	71404005025	4	6	4	25	65	0.5
	71404005030	4	6	4	30	70	0.5
	71404005035	4	6	4	35	75	0.5
	71404005040	4	6	4	40	80	0.5
	71404005045	4	6	4	45	90	0.5
	71404005050	4	6	4	50	100	0.5
	71404005055	4	6	4	55	100	0.5
	71404010010	4	6	4	10	50	1
	71404010012	4	6	4	12	50	1
	71404010013	4	6	4	13	55	1
	71404010016	4	6	4	16	55	1
	71404010020	4	6	4	20	60	1
	71404010025	4	6	4	25	65	1
	71404010030	4	6	4	30	70	1
	71404010035	4	6	4	35	75	1
	71404010040	4	6	4	40	80	1
	71404010045	4	6	4	45	90	1
	71404010050	4	6	4	50	100	1

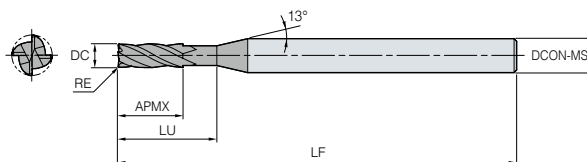
ESRR714

4 flutes rib ball



DC	Tolerance
Ø0.5 ~ Ø6	0.000 ~ -0.012
Ø8 ~ Ø20	0.000 ~ -0.015

DC6 or below Above DC6



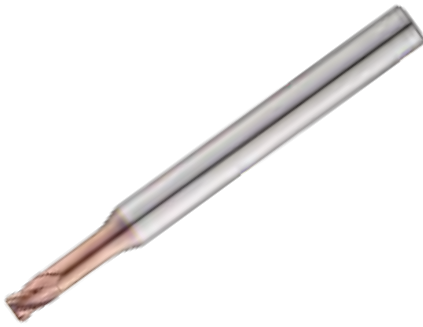
(mm)

	Designation	DC	DCON-MS	APMX	LU	LF	RE
ESRR	71404010055	4	6	4	55	100	1
	71405001016	5	6	5	16	60	0.1
	71405001030	5	6	5	30	70	0.1
	71405001040	5	6	5	40	80	0.1
	71405002016	5	6	5	16	60	0.2
	71405002030	5	6	5	30	70	0.2
	71405002040	5	6	5	40	80	0.2
	71405003016	5	6	5	16	60	0.3
	71405003030	5	6	5	30	70	0.3
	71405003040	5	6	5	40	80	0.3
	71405005016	5	6	5	16	60	0.5
	71405005030	5	6	5	30	70	0.5
	71405005040	5	6	5	40	80	0.5
	71405005050	5	6	5	50	100	0.5
	71405005060	5	6	5	60	110	0.5
	71405010016	5	6	5	16	60	1
	71405010030	5	6	5	30	70	1
	71405010040	5	6	5	40	80	1
	71405010050	5	6	5	50	100	1
	71405010060	5	6	5	60	110	1
	71405015015	5	6	5	15	60	1.5
	71405020015	5	6	5	15	60	2
	71406001020	6	6	7	20	60	0.1
	71406001040	6	6	7	40	80	0.1
	71406001050	6	6	7	50	100	0.1
	71406002020	6	6	7	20	60	0.2
	71406002040	6	6	7	40	80	0.2
	71406002050	6	6	7	50	100	0.2
	71406003020	6	6	7	20	60	0.3
	71406003030	6	6	7	30	70	0.3
	71406003040	6	6	7	40	80	0.3
	71406003050	6	6	7	50	100	0.3
	71406005020	6	6	7	20	60	0.5
	71406005030	6	6	7	30	70	0.5
	71406005040	6	6	7	40	80	0.5
	71406005050	6	6	7	50	100	0.5
	71406005060	6	6	7	60	110	0.5
	71406010020	6	6	7	20	60	1
	71406010030	6	6	7	30	70	1

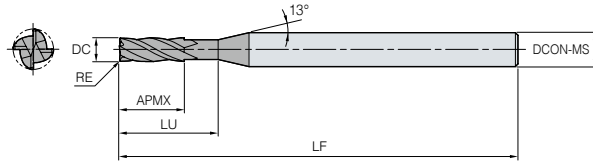
	Designation	DC	DCON-MS	APMX	LU	LF	RE
ESRR	71406010040	6	6	7	40	80	1
	71406010050	6	6	7	50	100	1
	71406010060	6	6	7	60	110	1
	71406015020	6	6	7	20	60	1.5
	71406015040	6	6	7	40	80	1.5
	71406015050	6	6	7	50	100	1.5
	71406020020	6	6	7	20	60	2
	71406020030	6	6	7	30	70	2
	71406020040	6	6	7	40	80	2
	71406020050	6	6	7	50	100	2
	71408001025	8	8	9	25	70	0.1
	71408002022	8	8	9	22	65	0.2
	71408002040	8	8	9	40	100	0.2
	71408003022	8	8	9	22	65	0.3
	71408003040	8	8	9	40	100	0.3
	71408005022	8	8	9	22	65	0.5
	71408005035	8	8	9	35	100	0.5
	71408005040	8	8	9	40	100	0.5
	71408005050	8	8	9	50	120	0.5
	71408005060	8	8	9	60	120	0.5
	71408010022	8	8	9	22	65	1
	71408010035	8	8	9	35	100	1
	71408010040	8	8	9	40	100	1
	71408010050	8	8	9	50	120	1
	71408010060	8	8	9	60	120	1
	71408015022	8	8	9	22	65	1.5
	71408015040	8	8	9	40	100	1.5
	71408020022	8	8	9	22	65	2
	71408020040	8	8	9	40	100	2
	71408020050	8	8	9	50	120	2
	71410001030	10	10	11	30	75	0.1
	71410002024	10	10	11	24	70	0.2
	71410002040	10	10	11	40	100	0.2
	71410003024	10	10	11	24	70	0.3
	71410003040	10	10	11	40	100	0.3
	71410005024	10	10	11	24	70	0.5
	71410005040	10	10	11	40	100	0.5
	71410005050	10	10	11	50	120	0.5
	71410005060	10	10	11	60	120	0.5

ESRR714

4 flutes rib ball



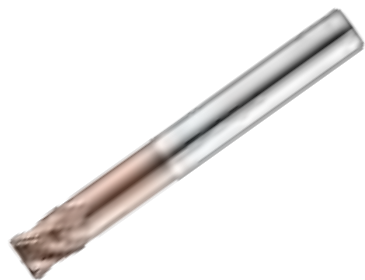
CARBIDE	4	H-A 30°	Grade PC305H	h5 shank	RE ±0.01	RE ±0.015	DC Ø0.5 ~ Ø6 Ø8 ~ Ø20	Tolerance 0.000 ~ -0.012 0.000 ~ -0.015
					DC6 or below Above DC6			



								(mm)
Designation		DC	DCON-MS	APMX	LU	LF	RE	
ESRR	71410010024	10	10	11	24	70	1	
	71410010040	10	10	11	40	100	1	
	71410010050	10	10	11	50	120	1	
	71410010060	10	10	11	60	120	1	
	71410015024	10	10	11	24	70	1.5	
	71410015040	10	10	11	40	100	1.5	
	71410020024	10	10	11	24	70	2	
	71410020040	10	10	11	40	100	2	
	71410020050	10	10	11	50	120	2	
	71410025024	10	10	11	24	70	2.5	
	71412002032	12	12	13	32	80	0.2	
	71412003026	12	12	13	26	80	0.3	
	71412003045	12	12	13	45	110	0.3	
	71412005026	12	12	13	26	80	0.5	
	71412005040	12	12	13	40	110	0.5	
	71412005060	12	12	13	60	130	0.5	
Designation		DC	DCON-MS	APMX	LU	LF	RE	
ESRR	71412010026	12	12	13	26	80	1	
	71412010040	12	12	13	40	110	1	
	71412010060	12	12	13	60	130	1	
	71412015026	12	12	13	26	80	1.5	
	71412020026	12	12	13	26	80	2	
	71412020040	12	12	13	40	110	2	
	71412030026	12	12	13	26	80	3	
	71416005035	16	16	20	35	100	0.5	
	71416005050	16	20	35	50	150	0.5	
	71416010035	16	16	20	35	100	1	
	71416010050	16	20	35	50	150	1	
	71420005040	20	20	25	40	100	0.5	
	71420005055	20	20	40	55	150	0.5	
	71420010040	20	20	25	40	100	1	
	71420010055	20	20	40	55	150	1	

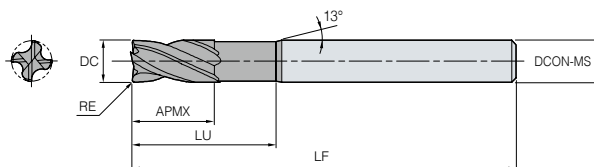
ESXR704

4 flutes neck type radius



DC6 or below Above DC6

DC	Tolerance
ALL	0.000 ~ -0.020



(mm)

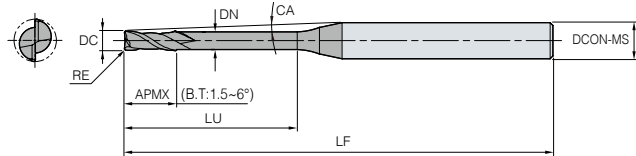
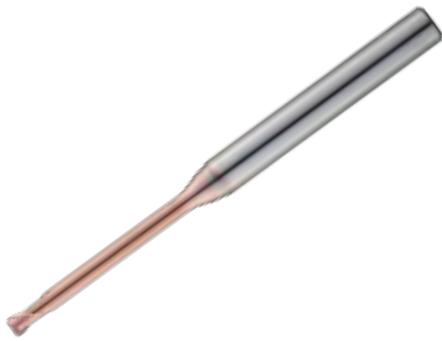
	Designation	DC	DCON-MS	APMX	LU	LF	RE
ESXR	70401000504	1	4	1.5	4	45	0.05
	70402000506	2	4	3	6	45	0.05
	70402000507	2	4	2.5	7	50	0.05
	7040200107	2	4	2.5	7	50	0.1
	7040300107	3	6	4	7	45	0.1
	7040300109	3	6	4	9	55	0.1
	7040300209	3	6	4	9	55	0.2
	7040300309	3	6	4	9	55	0.3
	7040300312	3	6	4	12	55	0.3
	7040300316	3	6	4	16	55	0.3
	7040400109	4	6	5	9	45	0.1
	7040400212	4	6	5	12	55	0.2
	7040400212S4	4	4	5	12	55	0.2
	7040400312	4	6	5	12	55	0.3
	7040400316	4	6	5	16	55	0.3
	7040400320	4	6	5	20	55	0.3
	7040400512	4	6	5	12	55	0.5
	7040400516	4	6	5	16	55	0.5
	7040400516S4	4	4	5	16	55	0.5
	7040400520	4	6	5	20	55	0.5
	7040401012	4	6	5	12	55	1
	7040500116	5	6	6	16	60	0.1
	7040500216	5	6	6	16	60	0.2
	7040500316	5	6	6	16	60	0.3
	7040500516	5	6	6	16	60	0.5
	7040501016	5	6	6	16	60	1
	7040600120	6	6	7	20	60	0.1
	7040600214	6	6	7	14	50	0.2

	Designation	DC	DCON-MS	APMX	LU	LF	RE
ESXR	7040600220	6	6	7	20	60	0.2
	7040600320	6	6	7	20	60	0.3
	7040600520	6	6	7	20	60	0.5
	7040601020	6	6	7	20	60	1
	7040601520	6	6	7	20	60	1.5
	7040800125	8	8	9	25	60	0.1
	7040800218	8	8	9	18	60	0.2
	7040800225	8	8	9	25	60	0.2
	7040800325	8	8	9	25	60	0.3
	7040800525	8	8	9	25	60	0.5
	7040801025	8	8	9	25	60	1
	7040801525	8	8	9	25	60	1.5
	7040802025	8	8	9	25	60	2
	7041000225	10	10	12	25	75	0.2
	7041000232	10	10	11	32	75	0.2
	7041000332	10	10	11	32	75	0.3
	7041000532	10	10	11	32	75	0.5
	7041001032	10	10	11	32	75	1
	7041001532	10	10	11	32	75	1.5
	7041002032	10	10	11	32	75	2
	7041200238	12	12	12	38	75	0.2
	7041200330	12	12	15	30	75	0.3
	7041200338	12	12	12	38	75	0.3
	7041200538	12	12	12	38	75	0.5
	7041201038	12	12	12	38	75	1
	7041201538	12	12	12	38	75	1.5
	7041202038	12	12	12	38	75	2

ESLNR20

2 flutes long neck type radius

CARBIDE	2	H-A 30°	Grade PC305H	h5 shank	RE ±0.01	DC	Tolerance
						ALL	0.000 ~ -0.012

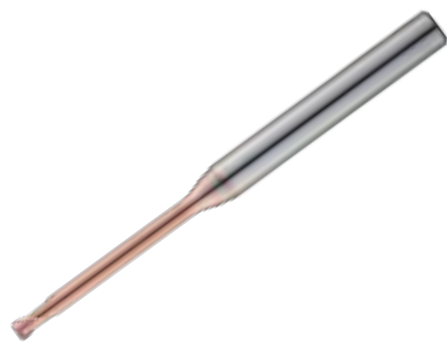


(mm)

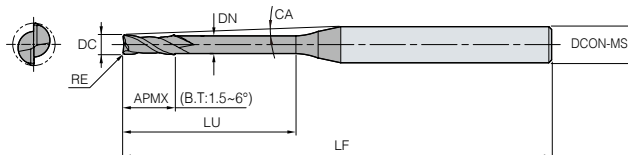
Designation		DC	DCON-MS	APMX	LU	LF	DN	CA	RE	Effective length by inclination angle				
										0.5°	1°	1.5°	2°	3°
ESLNR	2002-0.5-005	0.2	4	0.15	0.5	50	0.17	11.4	0.05	0.9	1	1	1.1	12
	2002-1-005	0.2	4	0.15	1	50	0.17	10.9	0.05	1.6	1.7	1.9	2	2.3
	2002-1.5-005	0.2	4	0.15	1.5	50	0.17	10.3	0.05	2.1	2.3	2.5	2.7	3
	2002-2-005	0.2	4	0.15	2	50	0.17	9.9	0.05	2.8	3.1	3.4	3.6	4.1
	2003-1-005	0.3	4	0.25	1	50	0.27	10.8	0.05	1.4	1.5	1.6	1.7	1.9
	2003-1.5-005	0.3	4	0.25	1.5	50	0.27	10.3	0.05	2.1	2.3	2.5	2.7	3
	2003-2.5-005	0.3	4	0.25	2.5	50	0.27	9.8	0.05	2.7	2.9	3.1	3.3	3.6
	2003-2-005	0.3	4	0.25	2	50	0.27	9.4	0.05	3.2	3.5	3.7	3.9	4.3
	2003-3-005	0.3	4	0.25	3	50	0.27	9	0.05	3.9	4.3	4.6	4.9	5.4
	2004-1-005	0.4	4	0.3	1	50	0.37	10.8	0.05	1.4	1.5	1.6	1.7	1.9
	2004-1.5-005	0.4	4	0.3	1.5	50	0.37	10.3	0.05	2	2.1	2.2	2.3	2.5
	2004-2-005	0.4	4	0.3	2	50	0.37	9.8	0.05	2.7	2.9	3.1	3.3	3.6
	2004-2.5-005	0.4	4	0.3	2.5	50	0.37	9.4	0.05	3.2	3.5	3.7	3.9	4.3
	2004-3-005	0.4	4	0.3	3	50	0.37	9	0.05	3.8	4	4.3	4.5	4.9
	2004-3.5-005	0.4	4	0.3	3.5	50	0.37	8.6	0.05	4.3	4.6	4.9	5.1	5.5
	2004-4-005	0.4	4	0.3	4	50	0.37	8.3	0.05	5	5.4	5.8	6.1	6.6
	2004-2-01	0.4	4	0.3	2	50	0.37	9.8	0.1	2.7	2.9	3.1	3.3	3.6
	2004-3-01	0.4	4	0.3	3	50	0.37	9	0.1	3.8	4	4.3	4.5	4.9
	2004-4-01	0.4	4	0.3	4	50	0.37	8.3	0.1	5	5.4	5.8	6.1	6.6
	2005-1-005	0.5	4	0.35	1	50	0.47	10.8	0.05	1.4	1.5	1.6	1.7	1.9
	2005-2-005	0.5	4	0.35	2	50	0.47	9.7	0.05	2.5	2.6	2.8	2.9	3.1
	2005-3-005	0.5	4	0.35	3	50	0.47	8.9	0.05	3.8	4	4.3	4.5	4.9
	2005-4-005	0.5	4	0.35	4	50	0.47	8.2	0.05	4.8	5.2	5.4	5.7	6.1
	2005-5-005	0.5	4	0.35	5	50	0.47	7.6	0.05	6.1	6.6	6.9	7.3	7.8
	2005-6-005	0.5	4	0.35	6	50	0.47	7	0.05	7.2	7.7	8.1	8.4	9
	2005-1-01	0.5	4	0.35	1	50	0.47	10.8	0.1	1.4	1.5	1.6	1.7	1.9
	2005-2-01	0.5	4	0.35	2	50	0.47	9.8	0.1	2.5	2.6	2.8	2.9	3.1
	2005-3-01	0.5	4	0.35	3	50	0.47	8.9	0.1	3.8	4	4.3	4.5	4.9
	2005-4-01	0.5	4	0.35	4	50	0.47	8.2	0.1	4.8	5.2	5.4	5.7	6.1
	2005-5-01	0.5	4	0.35	5	50	0.47	7.6	0.1	6.1	6.5	6.9	7.2	7.8
	2005-6-01	0.5	4	0.35	6	50	0.47	7.1	0.1	7.2	7.7	8.1	8.4	9
	2006-2-01	0.6	4	0.4	2	50	0.57	9.7	0.1	2.5	2.6	2.8	2.9	3.1
	2006-4-01	0.6	4	0.4	4	50	0.57	8.1	0.1	4.8	5.2	5.4	5.7	6.1
	2006-6-01	0.6	4	0.4	6	50	0.57	7	0.1	7.2	7.7	8.1	8.4	9

ESLNR20

2 flutes long neck type radius



CARBIDE	2	H-A 30°	Grade PC305H	h5 shank	RE ±0.01	DC ALL	Tolerance 0.000 ~ -0.012
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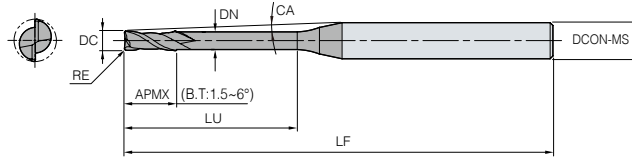
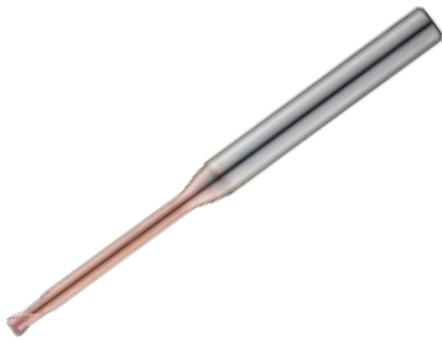
(mm)

Designation		DC	DCON-MS	APMX	LU	LF	DN	CA	RE	Effective length by inclination angle				
										0.5°	1°	1.5°	2°	3°
ESLNR	2006-8-01	0.6	4	0.4	8	50	0.57	6.1	0.1	9.3	9.9	10.3	10.7	11
	2006-10-01	0.6	4	0.4	10	50	0.57	5.5	0.1	11.5	12.1	12.5	13	13.7
	2008-4-01	0.8	4	0.5	4	50	4	8	0.1	4.8	5.2	5.4	5.7	6.1
	2008-6-01	0.8	4	0.5	6	50	6	6.8	0.1	7	7.4	7.7	7.9	8.4
	2008-8-01	0.8	4	0.5	8	50	8	5.9	0.1	9.3	9.9	10.3	10.7	11.4
	2008-12-01	0.8	4	0.5	12	50	12	4.7	0.1	13.6	14.2	14.7	15.2	16
	2008-4-02	0.8	4	0.5	4	50	4	8	0.2	4.8	5.1	5.4	5.6	6.1
	2008-6-02	0.8	4	0.5	6	50	6	6.9	0.2	7	7.3	7.7	7.9	8.4
	2010-4-01	1	4	0.8	4	50	0.94	7.7	0.1	4.7	4.9	5.1	5.2	5.5
	2010-6-01	1	4	0.8	6	50	0.94	6.6	0.1	7.1	7.4	7.7	8	8.5
	2010-8-01	1	4	0.8	8	50	0.94	5.7	0.1	9.2	9.6	9.9	10.2	10.8
	2010-10-01	1	4	0.8	10	50	0.94	5.1	0.1	11.6	12.1	12.6	13	13.7
	2010-12-01	1	4	0.8	12	55	0.94	4.5	0.1	13.7	14.3	14.8	15.3	16
	2010-16-01	1	4	0.8	16	60	0.94	3.8	0.1	17.9	18.6	19.2	19.7	21.3
	2010-20-01	1	4	0.8	20	60	0.94	3.2	0.1	22	22.8	23.5	24	26.7
	2010-4-02	1	4	0.8	4	50	0.94	7.8	0.2	4.7	4.9	5.1	5.2	5.5
	2010-6-02	1	4	0.8	6	50	0.94	6.6	0.2	7.1	7.4	7.7	8	8.5
	2010-8-02	1	4	0.8	8	50	0.94	5.8	0.2	9.2	9.6	9.9	10.2	10.8
	2010-10-02	1	4	0.8	10	50	0.94	5.1	0.2	11.6	12.1	12.6	13	13.7
	2010-12-02	1	4	0.8	12	55	0.94	4.6	0.2	13.7	14.3	14.8	15.2	16
	2010-16-02	1	4	0.8	16	60	0.94	3.8	0.2	17.9	18.6	19.2	19.7	21.3
	2010-20-02	1	4	0.8	20	60	0.94	3.2	0.2	22	22.8	23.5	24	26.6
	2010-6-03	1	4	0.8	6	50	0.94	6.7	0.3	7.1	7.4	7.7	8	8.4
	2010-10-03	1	4	0.8	10	50	0.94	5.1	0.3	11.5	12.1	12.6	13	13.7
	2010-16-03	1	4	0.8	16	60	0.94	3.8	0.3	17.9	18.6	19.1	19.6	21.3
	2010-20-03	1	4	0.8	20	60	0.94	3.2	0.3	22	22.8	23.5	24	26.6
	2015-4-01	1.5	4	1.35	4	50	1.42	7.2	0.1	4.8	4.9	5.1	5.3	5.5
	2015-8-01	1.5	4	1.35	8	50	1.42	5.2	0.1	9.2	9.6	10	10.3	10.8
	2015-12-01	1.5	4	1.35	12	55	1.42	4	0.1	13.4	13.9	14.3	14.7	16.1
	2015-15-01	1.5	4	1.35	15	55	1.42	3.5	0.1	16.9	17.6	18.1	18.6	20.1
	2015-20-01	1.5	4	1.35	20	60	1.42	2.8	0.1	22.1	22.9	23.5	24.1	-
	2015-4-02	1.5	4	1.35	4	50	1.42	7.3	0.2	4.7	4.9	5.1	5.3	5.5
	2015-8-02	1.5	4	1.35	8	50	1.42	5.2	0.2	9.2	9.6	10	10.3	10.8
	2015-12-02	1.5	4	1.35	12	55	1.42	4.1	0.2	13.4	13.9	14.3	14.7	16.1

ESLNR20

2 flutes long neck type radius

CARBIDE	2	H-A 30°	Grade PC305H	h5 shank	RE ±0.01	DC	Tolerance
						ALL	0.000 ~ -0.012

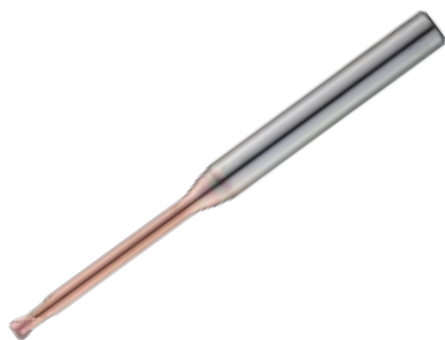


(mm)

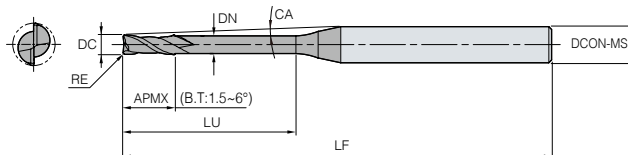
Designation		DC	DCON-MS	APMX	LU	LF	DN	CA	RE	Effective length by inclination angle				
										0.5°	1°	1.5°	2°	3°
ESLNR	2015-15-02	1.5	4	1.35	15	55	1.42	3.5	0.2	16.9	17.5	18.1	18.6	20
	2015-20-02	1.5	4	1.35	20	60	1.42	2.8	0.2	22.1	22.9	23.5	24.1	-
	2015-8-03	1.5	4	1.35	8	50	1.42	5.2	0.3	9.2	9.6	10	10.3	10.8
	2015-15-03	1.5	4	1.35	15	55	1.42	3.5	0.3	16.9	17.5	18.1	18.6	20
	2015-20-03	1.5	4	1.35	20	60	1.42	2.8	0.3	22.1	22.9	23.5	24	-
	2020-6-02	2	4	1.7	6	50	1.92	5.4	0.2	6.8	7.1	7.3	7.5	8.1
	2020-8-02	2	4	1.7	8	50	1.92	4.6	0.2	8.9	9.2	9.4	9.7	10.8
	2020-12-02	2	4	1.7	12	55	1.92	3.5	0.2	13.4	13.9	14.3	14.7	16.1
	2020-16-02	2	4	1.7	16	55	1.92	2.8	0.2	17.6	18.1	18.6	19.3	-
	2020-20-02	2	4	1.7	20	60	1.92	2.4	0.2	22.1	22.9	23.5	24.1	-
	2020-25-02	2	4	1.7	25	65	1.92	2	0.2	27.3	28.2	28.8	-	-
	2020-30-02	2	4	1.7	30	70	1.92	1.7	0.2	32.5	33.4	34.4	-	-
	2020-8-03	2	4	1.7	8	50	1.92	4.6	0.3	8.9	9.2	9.4	9.7	10.7
	2020-16-03	2	4	1.7	16	55	1.92	2.8	0.3	17.6	18.1	18.6	19.3	-
	2020-20-03	2	4	1.7	20	60	1.92	2.4	0.3	22.1	22.9	23.5	24	-
	2020-6-05	2	4	1.7	6	50	1.92	5.5	0.5	6.8	7.1	7.3	7.4	8
	2020-8-05	2	4	1.7	8	50	1.92	4.7	0.5	8.9	9.2	9.4	9.6	10.7
	2020-12-05	2	4	1.7	12	55	1.92	3.5	0.5	13.4	13.9	14.3	14.6	16
	2020-16-05	2	4	1.7	16	55	1.92	2.9	0.5	17.6	18.1	18.6	19.2	-
	2020-20-05	2	4	1.7	20	60	1.92	2.4	0.5	22.1	22.9	23.5	24	-
	2020-25-05	2	4	1.7	25	65	1.92	2	0.5	27.3	28.1	28.8	-	-
	2020-30-05	2	4	1.7	30	70	1.92	1.7	0.5	32.5	33.4	34.3	-	-
	2020-8-08	2	4	1.7	8	50	1.92	4.8	0.8	8.9	9.2	9.4	9.6	10.6
	2020-16-08	2	4	1.7	16	55	1.92	2.9	0.8	17.6	18.1	18.6	19.2	-
	2020-20-08	2	4	1.7	20	60	1.92	2.4	0.8	22.1	22.8	23.5	24	-
	2030-8-02	3	6	2.5	8	55	2.86	5.7	0.2	9	9.3	9.5	9.9	10.9
	2030-12-02	3	6	2.5	12	60	2.86	4.5	0.2	13.1	13.5	14	14.7	16.2
	2030-16-02	3	6	2.5	16	60	2.86	3.8	0.2	17.7	18.2	18.7	19.5	21.6
	2030-20-02	3	6	2.5	20	65	2.86	3.2	0.2	21.8	22.4	23.1	24.2	26.9
	2030-30-02	3	6	2.5	30	75	2.86	2.4	0.2	32.6	33.5	34.5	36.2	-
	2030-35-02	3	6	2.5	35	80	2.86	2.1	0.2	37.7	38.7	40.2	42.2	-
	2030-8-03	3	6	2.5	8	55	2.86	5.7	0.3	9	9.3	9.5	9.9	10.9
	2030-16-03	3	6	2.5	16	60	2.86	3.8	0.3	17.7	18.2	18.7	19.4	21.5
	2030-20-03	3	6	2.5	20	65	2.86	3.2	0.3	21.8	22.4	23.1	24.2	26.8

ESLNR20

2 flutes long neck type radius

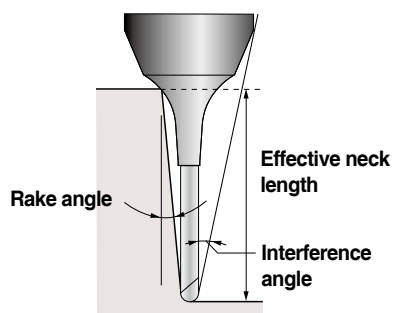


CARBIDE	2	H-A 30°	Grade PC305H	h5 shank	RE ±0.01	DC	Tolerance
						ALL	0.000 ~ -0.012



(mm)

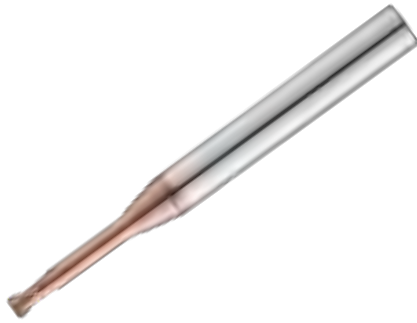
Designation		DC	DCON-MS	APMX	LU	LF	DN	CA	RE	Effective length by inclination angle				
										0.5°	1°	1.5°	2°	3°
ESLNR	2030-30-03	3	6	2.5	30	75	2.86	2.4	0.3	32.6	33.5	34.5	36.2	-
	2030-8-05	3	6	2.5	8	55	2.86	5.8	0.5	9	9.3	9.5	9.8	10.8
	2030-12-05	3	6	2.5	12	60	2.86	4.6	0.5	13.1	13.5	13.9	14.6	16.2
	2030-16-05	3	6	2.5	16	60	2.86	3.8	0.5	17.7	18.2	18.7	19.4	21.5
	2030-20-05	3	6	2.5	20	65	2.86	3.2	0.5	21.8	22.4	23.1	24.2	26.8
	2030-30-05	3	6	2.5	30	75	2.86	2.4	0.5	32.6	33.5	34.5	36.1	-
	2030-35-05	3	6	2.5	35	80	2.86	2.1	0.5	37.7	38.7	40.2	42.1	-



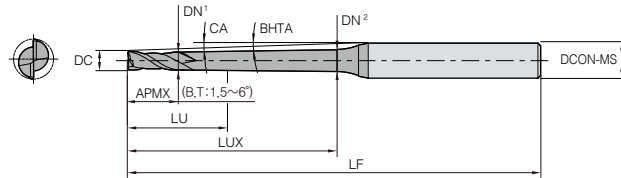
※ The marked effective neck length is the default value to prevent interference with the workpiece.
Proper control of the processing environment is required.

ESTNR20

2 flutes tapered neck type radius



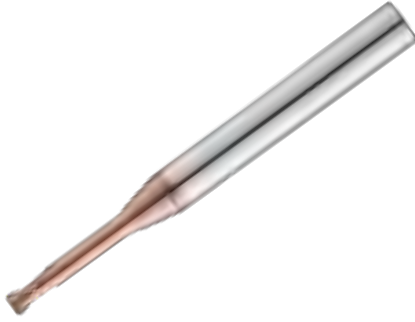
CARBIDE	2	H-A 30°	Grade PC305H	h5 shank	RE ±0.01	DC	Tolerance
						ALL	0.000 ~ -0.012



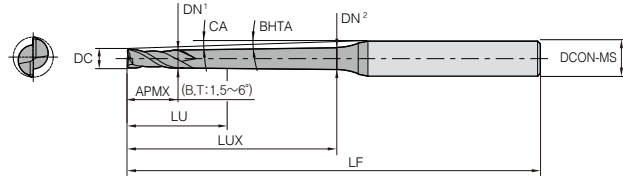
													(mm)				
ESTNR	Designation	DC	DCON-MS	APMX	LU	LUX	CA	BHTA(°)	DN1	DN2	LF	RE	Effective length by inclination angle				
													0.5°	1°	1.5°	2°	3°
	2002-2-09005	0.2	4	0.15	1.1	2	0.9	10	0.17	0.23	50	0.05	-	2.8	3.1	3.4	3.9
	2004-4-09005	0.4	4	0.3	1.25	4	0.9	8.4	0.37	0.49	50	0.05	-	4.9	5.3	5.7	6.3
	2004-5-09005	0.4	4	0.3	1.25	5	0.9	7.8	0.37	0.52	50	0.05	-	5.9	6.4	6.8	7.5
	2004-4-0901	0.4	4	0.3	1.25	4	0.9	8.5	0.37	0.49	50	0.1	-	4.9	5.3	5.7	6.3
	2004-5-0901	0.4	4	0.3	1.25	5	0.9	7.9	0.37	0.52	50	0.1	-	5.9	6.4	6.8	7.5
	2005-5-0901	0.5	4	0.35	1.3	5	0.9	7.8	0.47	0.62	50	0.1	-	5.9	6.4	6.8	7.5
	2005-8-0901	0.5	4	0.35	1.3	8	0.9	6.4	0.47	0.71	50	0.1	-	9	9.7	10.2	11
	2005-10-0901	0.5	4	0.35	1.3	10	0.9	5.8	0.47	0.77	55	0.1	-	11	11.8	12.4	13.2
	2006-12-0901	0.6	4	0.4	1.35	12	0.9	5.1	0.57	0.93	55	0.1	-	13	13.9	14.5	15.5
	2006-15-0901	0.6	4	0.4	1.35	15	0.9	4.5	0.57	1.3	55	0.1	-	16.1	17.1	17.8	18.8
	2008-6-0402	0.8	4	0.5	2.64	6	0.4	7	0.77	0.85	50	0.2	6.6	7.1	7.5	7.8	8.3
	2008-12-0902	0.8	4	0.5	1.45	12	0.9	5	0.77	1.13	55	0.2	-	13	13.9	14.5	15.5
	2010-8-0402	1	6	0.8	5.09	8	0.4	7.4	0.94	1.4	55	0.2	8.8	9.3	9.7	10.1	10.6
	2010-10-0902	1	6	0.8	5.09	10	0.9	6.8	0.94	1.23	55	0.2	-	11.2	11.9	12.4	13.3
	2010-15-0902	1	6	0.8	2.7	15	0.9	5.6	0.94	1.39	60	0.2	-	16.3	17.2	17.8	18.8
	2010-20-0902	1	6	0.8	2.7	20	0.9	4.8	0.94	1.54	65	0.2	-	21.3	22.4	23.2	24.7
	2010-25-0902	1	6	0.8	2.7	25	0.9	4.1	0.94	1.7	70	0.2	-	26.4	27.6	28.5	30.9
	2010-30-0902	1	6	0.8	2.7	30	0.9	3.7	0.94	1.86	75	0.2	-	31.5	32.8	33.7	37
	2010-35-0902	1	6	0.8	2.7	35	0.9	3.3	0.94	2.2	80	0.2	-	36.5	38	39	43.2
	2010-8-0403	1	6	0.8	2.7	8	0.4	7.4	0.94	1.4	55	0.3	8.8	9.3	9.7	10	10.6
	2010-15-0903	1	6	0.8	2.7	15	0.9	5.6	0.94	1.39	60	0.3	-	16.3	17.2	17.8	18.8
	2010-25-0903	1	6	0.8	2.7	25	0.9	4.2	0.94	1.7	70	0.3	-	26.4	27.6	28.5	30.8
	2010-30-0903	1	6	0.8	2.7	30	0.9	3.7	0.94	1.86	75	0.3	-	31.5	32.8	33.7	37
	2015-10-0402	1.5	6	1.35	7.07	10	0.4	6.4	1.42	1.54	55	0.2	11	11.5	11.9	12.3	13
	2015-15-0902	1.5	6	1.35	7.07	15	0.9	5.3	1.42	1.85	60	0.2	-	16.4	17.3	17.9	18.9
	2015-20-0902	1.5	6	1.35	3.89	20	0.9	4.5	1.42	2.1	65	0.2	-	21.5	22.5	23.2	24.9
	2015-25-0902	1.5	6	1.35	3.89	25	0.9	3.9	1.42	2.16	70	0.2	-	26.6	27.7	28.5	31
	2015-30-0902	1.5	6	1.35	3.89	30	0.9	3.4	1.42	2.32	75	0.2	-	31.6	32.9	33.8	37.1
	2015-10-0403	1.5	6	1.35	3.89	10	0.4	6.4	1.42	1.54	55	0.3	11	11.5	11.9	12.3	13
	2015-20-0903	1.5	6	1.35	3.89	20	0.9	4.5	1.42	2.1	65	0.3	-	21.5	22.5	23.2	24.8
	2015-25-0903	1.5	6	1.35	3.89	25	0.9	3.9	1.42	2.16	70	0.3	-	26.5	27.7	28.5	31
	2015-30-0903	1.5	6	1.35	3.89	30	0.9	3.4	1.42	2.32	75	0.3	-	31.6	32.9	33.8	37.1
	2020-30-0902	2	6	1.7	7.42	30	0.9	3.1	1.92	2.81	70	0.2	-	31.6	32.9	33.8	37.2
	2020-40-0902	2	6	1.7	7.42	40	0.9	2.5	1.92	3.12	80	0.2	-	41.8	43.3	44.6	-
	2020-50-0902	2	6	1.7	7.42	50	0.9	2.1	1.92	3.44	90	0.2	-	51.9	53.6	55.7	-
	2020-12-0403	2	6	1.7	7.42	12	0.4	5.5	1.92	2.06	55	0.3	13	13.6	14.1	14.5	15.6
	2020-20-0903	2	6	1.7	4.24	20	0.9	4.1	1.92	2.5	65	0.3	-	21.5	22.5	23.2	24.9
	2020-30-0903	2	6	1.7	4.24	30	0.9	3.1	1.92	2.81	70	0.3	-	31.6	32.9	33.8	37.1

ESTNR20

2 flutes tapered neck type radius

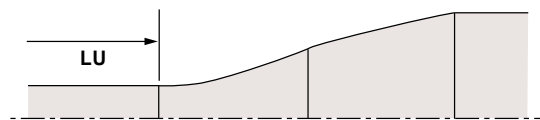
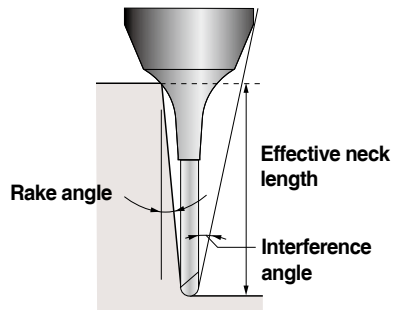


CARBIDE	2	H-A 30°	Grade PC305H	h5 shank	RE ±0.01	DC	Tolerance
						ALL	0.000 ~ -0.012



(mm)

Designation	DC	DCON-MS	APMX	LU	LUX	CA	BHTA(°)	DN¹	DN²	LF	RE	Effective length by inclination angle				
												0.5°	1°	1.5°	2°	3°
ESTNR 2020-40-0903	2	6	1.7	4.24	40	0.9	2.5	1.92	3.12	80	0.3	-	41.7	43.3	44.6	-
2020-50-0903	2	6	1.7	4.24	50	0.9	2.1	1.92	3.44	90	0.3	-	51.8	53.6	55.7	-
2020-8-0405	2	6	1.7	4.24	8	0.4	6.8	1.92	2.01	50	0.5	8.7	9	9.3	9.5	10.4
2020-12-0405	2	6	1.7	4.24	12	0.4	5.6	1.92	2.06	55	0.5	13	13.6	14.1	14.4	15.5
2020-16-0405	2	6	1.7	4.24	16	0.4	4.7	1.92	2.12	60	0.5	17	17.8	18.3	18.7	20.7
2020-20-0905	2	6	1.7	4.24	20	0.9	4.2	1.92	2.5	65	0.5	-	21.5	22.5	23.2	24.8
2020-25-0905	2	6	1.7	4.24	25	0.9	3.6	1.92	2.65	65	0.5	-	26.6	27.7	28.5	30.9
2020-30-0905	2	6	1.7	4.24	30	0.9	3.1	1.92	2.81	70	0.5	-	31.6	32.9	33.8	37.1
2020-40-0905	2	6	1.7	4.24	40	0.9	2.5	1.92	3.12	80	0.5	-	41.7	43.2	44.6	-
2020-50-0905	2	6	1.7	4.24	50	0.9	2.1	1.92	3.44	90	0.5	-	51.8	53.6	55.6	-
2030-40-0902	3	6	2.5	6.95	40	0.9	2	2.86	4.04	80	0.2	-	42	43.4	-	-
2030-50-0902	3	6	2.5	6.95	50	0.9	1.6	2.86	4.35	90	0.2	-	52.1	53.7	-	-
2030-60-0902	3	6	2.5	6.95	60	0.9	1.4	2.86	4.67	100	0.2	-	62.2	-	-	-
2030-40-0903	3	6	2.5	6.95	40	0.9	2	2.86	4.04	80	0.3	-	42	43.4	-	-
2030-50-0903	3	6	2.5	6.95	50	0.9	1.7	2.86	4.35	90	0.3	-	52.1	53.7	-	-
2030-60-0903	3	6	2.5	6.95	60	0.9	1.4	2.86	4.67	100	0.3	-	62.2	-	-	-
2030-40-0905	3	6	2.5	6.95	40	0.9	2	2.86	4.04	80	0.5	-	42	43.4	-	-
2030-50-0905	3	6	2.5	6.95	50	0.9	1.7	2.86	4.35	90	0.5	-	52.1	53.7	-	-
2030-60-0905	3	6	2.5	6.95	60	0.9	1.4	2.86	4.67	100	0.5	-	62.1	-	-	-

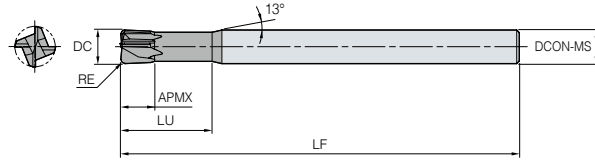
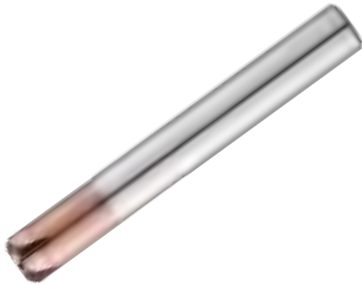


※ The marked effective neck length is the default value to prevent interference with the workpiece.
Proper control of the processing environment is required.

ESPM4

4 flutes neck type radius

CARBIDE	4	H-A 0°	Grade PC305H	h5 shank	RE ±0.01	RE ±0.015	DC	Tolerance
					DC6 or below	Above DC6	Ø3 - Ø6 Ø8 - Ø12	0.000 ~ -0.012 0.000 ~ -0.015

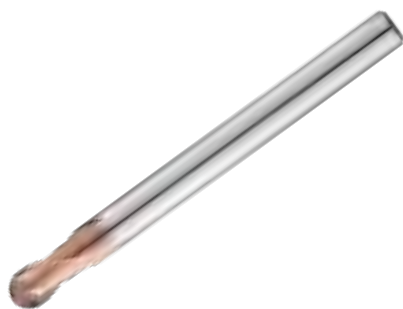


(mm)

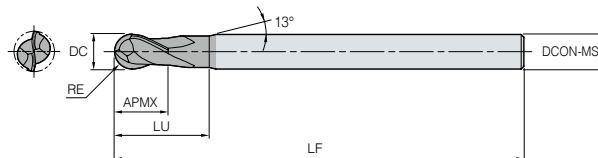
	Designation	DC	DCON-MS	APMX	LU	LF	RE
ESPM	4030-05	3	6	1.2	8	50	0.5
	4040-05	4	6	1.5	10	50	0.5
	4060-05	6	6	2.5	12	60	0.5
	4060-10	6	6	2.5	12	60	1
	4060-15	6	6	2.5	12	60	1.5
	4060-15L	6	6	2.5	12	90	1.5
	4080-10	8	8	3.5	16	60	1
	4080-20	8	8	3.5	16	60	2
	4080-20L	8	8	3.5	16	100	2
	4100-10	10	10	4	20	70	1
	4100-20	10	10	4	20	70	2
	4100-20L	10	10	4	20	100	2
	4120-20	12	12	5	25	80	2
	4120-30	12	12	5	25	80	3
	4120-30L	12	12	5	25	110	3

ESB702

2 flutes neck type ball



CARBIDE	2	H-A 30°	Grade PC305H	h5 shank	RE ±0.005	RE ±0.008	DC	Tolerance
					DC6 or below	Above DC6	Ø0.1 ~ Ø6 Ø8 ~ Ø12	0.000 ~ -0.012 0.000 ~ -0.015



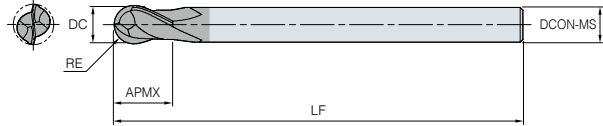
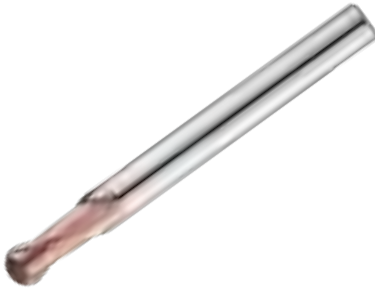
(mm)

	Designation	DC	DCON-MS	APMX	LU	LF	RE
ESB	702001	0.1	4	0.15	-	40	0.05
	702002	0.2	4	0.3	-	40	0.1
	702003	0.3	4	0.5	-	40	0.15
	702004	0.4	4	0.6	-	40	0.2
	702005	0.5	4	0.7	-	40	0.25
	702006	0.6	4	0.9	-	40	0.3
	702007	0.7	4	1.1	-	40	0.35
	702008	0.8	4	1.2	-	40	0.4
	702009	0.9	4	1.4	-	40	0.45
	702010	1	6	1.5	3	50	0.5
	702010S4	1	4	1.5	-	45	0.5
	702015	1.5	6	2.0	4	50	0.75
	702015S4	1.5	4	2.0	-	45	0.75
	702020	2	6	2.5	5	50	1
	702020S4	2	4	2.5	-	45	1
	702025	2.5	6	3.0	7	50	1.25
	702030	3	6	4.0	10	60	1.5
	702030S	3	6	4.0	10	50	1.5
	702030S4	3	4	4.0	-	45	1.5
	702031	3	6	4.0	10	70	1.5
	702040	4	6	5.0	10	60	2
	702040S	4	6	5.0	10	50	2
	702040S4	4	4	5.0	-	45	2
	702041	4	6	5.0	10	70	2
	702050	5	6	6.0	12	60	2.5
	702060	6	6	7.0	12	60	3
	702061	6	6	7.0	12	90	3
	702080	8	8	9.0	15	70	4
	702081	8	8	9.0	15	100	4
	702100	10	10	11.0	25	75	5
	702101	10	10	11.0	25	100	5
	702120	12	12	12	25	80	6
	702121	12	12	12	25	110	6

ESB712

2 flutes ball

CARBIDE	2	H-A 30°	Grade PC305H	h5 shank	RE ±0.005	RE ±0.008	DC	Tolerance
					DC6 or below	Above DC6	Ø1 ~ Ø6 Ø8 ~ Ø12	0.000 ~ -0.012 0.000 ~ -0.015

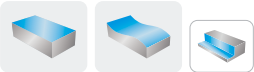
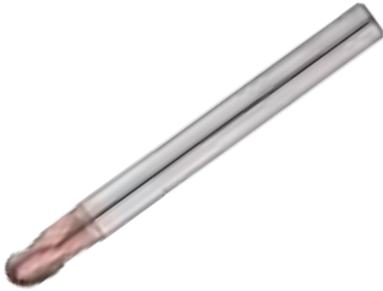


(mm)

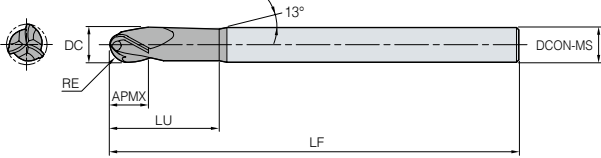
Designation	DC	DCON-MS	APMX	LF	RE
ESB 712010	1	6	2.5	50	0.5
712010S	1	6	1.5	40	0.5
712010S4	1	4	2.5	50	0.5
712012	1.2	6	3	50	0.6
712015	1.5	6	4	50	0.75
712015S	1.5	6	2.5	40	0.75
712015S4	1.5	4	4	50	0.75
712020	2	6	5	50	1
712020S	2	6	3	40	1
712020S4	2	4	5	50	1
712025	2.5	6	7	60	1.25
712030	3	6	8	60	1.5
712030S	3	6	4.5	50	1.5
712030S4	3	4	8	60	1.5
712040	4	6	8	70	2
712040S	4	6	6	50	2
712050	5	6	10	80	2.5
712050S	5	6	7.5	50	2.5
712060	6	6	12	90	3
712060S	6	6	9	50	3
712080S	8	8	12	50	4
712081	8	8	14	100	4
712100	10	10	18	100	5
712100S	10	10	15	60	5
712120	12	12	22	110	6
712120S	12	12	18	60	6

ESB703

3 flutes neck type ball



CARBIDE	3	H-A 30°	Grade PC305H	h5 shank	RE ±0.005	RE ±0.008	DC	Tolerance
					DC6 or below	Above DC6	Ø2 ~ Ø6 Ø8 ~ Ø12	0.000 ~ -0.012 0.000 ~ -0.015



(mm)							
Designation		DC	DCON-MS	APMX	LU	LF	RE
ESB	703020	2	6	2.5	5	50	1
	703025	2.5	6	3	7	50	1.25
	703030	3	6	4	10	60	1.5
	703030S	3	6	4	10	50	1.5
	703031	3	6	4	10	70	1.5
	703040	4	6	5	10	60	2
	703040S	4	6	5	10	50	2
	703041	4	6	5	10	70	2
	703050	5	6	6	12	60	2.5
	703060	6	6	7	12	60	3
	703061	6	6	7	12	90	3
	703080	8	8	9	15	70	4
	703081	8	8	9	15	100	4
	703100	10	10	11	25	75	5
	703101	10	10	11	25	100	5
	703120	12	12	12	25	80	6
	703121	12	12	12	25	110	6

ESB734

4 flutes 15° helix ball

CARBIDE

4

H-A
15°

Grade
PC305H

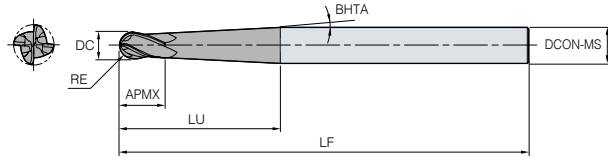
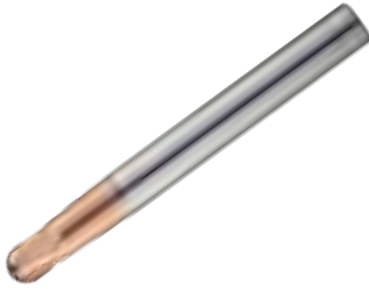
h5
shank

RE
±0.005

RE
±0.008

DC6 or below Above DC6

DC	Tolerance
Ø2 - Ø6	0.000 ~ -0.012
Ø8 - Ø10	0.000 ~ -0.015

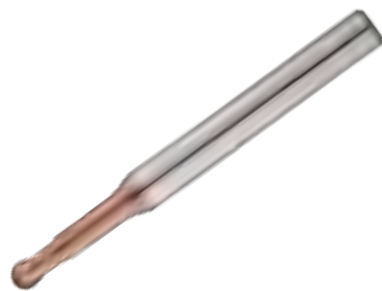


(mm)

Designation	DC	DCON-MS	APMX	LU	LF	RE	BHTA(°)
ESB 734020-2.5	2	4	2	25	60	1	2.5
734020-3.5	2	4	2	18	60	1	3.5
734025-2.5	2.5	4	3	20	60	1.25	2.5
734025-3.0	2.5	4	3	17	60	1.25	3
734030-2.0	3	6	3	46	70	1.5	2
734030-2.5	3	6	3	37	70	1.5	2.5
734040-2.0	4	6	4	33	70	2	2
734040-2.5	4	6	4	27	70	2	2.5
734050-2.5	5	6	5	16	70	2.5	2.5
734060-1.5	6	8	6	44	100	3	1.5
734060-2.5	6	8	6	29	100	3	2.5
734080-1.5	8	10	8	46	100	4	1.5
734080-2.5	8	10	8	31	100	4	2.5
734100-1.5	10	12	10	48	110	5	1.5
734100-2.5	10	12	10	33	110	5	2.5

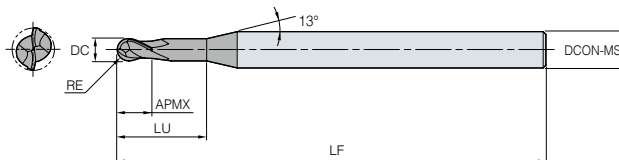
ESRB712

2 flutes rib ball



DC6 or below Above DC6

DC	Tolerance
Ø0.1 ~ Ø6	0.000 ~ -0.012
Ø8 ~ Ø12	0.000 ~ -0.015



(mm)

Designation	DC	DCON-MS	APMX	LU	LF	RE
ESRB 712001002	0.1	4	0.1	0.2	40	0.05
712001003	0.1	4	0.1	0.3	40	0.05
712001005	0.1	4	0.1	0.5	40	0.05
71200101	0.1	4	0.1	1	40	0.05
712002005	0.2	4	0.2	0.5	40	0.1
712002015	0.2	4	0.2	1.5	40	0.1
71200201	0.2	4	0.2	1	40	0.1
71200202	0.2	4	0.2	2	40	0.1
71200203	0.2	4	0.2	3	40	0.1
712003015	0.3	4	0.3	1.5	40	0.15
71200301	0.3	4	0.3	1	40	0.15
712003025	0.3	4	0.3	2.5	40	0.15
71200302	0.3	4	0.3	2	40	0.15
71200303	0.3	4	0.3	3	40	0.15
71200304	0.3	4	0.3	4	40	0.15
71200305	0.3	4	0.3	5	40	0.15
712004015	0.4	4	0.4	1.5	40	0.2
71200401	0.4	4	0.4	1	40	0.2
712004025	0.4	4	0.4	2.5	40	0.2
71200402	0.4	4	0.4	2	40	0.2
71200403	0.4	4	0.4	3	40	0.2
71200404	0.4	4	0.4	4	40	0.2
71200405	0.4	4	0.4	5	40	0.2
71200406	0.4	4	0.4	6	40	0.2
71200408	0.4	4	0.4	8	40	0.2
71200410	0.4	4	0.4	10	40	0.2
712005015	0.5	4	0.5	1.5	45	0.25
71200501	0.5	4	0.5	1	45	0.25
71200501S6	0.5	6	0.5	1	45	0.25
712005025	0.5	4	0.5	2.5	45	0.25
71200502	0.5	4	0.5	2	45	0.25
71200502S6	0.5	6	0.5	2	45	0.25
71200503	0.5	4	0.5	3	45	0.25
71200504	0.5	4	0.5	4	45	0.25
71200504S6	0.5	6	0.5	4	45	0.25
71200505	0.5	4	0.5	5	45	0.25
71200506	0.5	4	0.5	6	45	0.25
71200508	0.5	4	0.5	8	45	0.25
71200510	0.5	4	0.5	10	45	0.25

Designation	DC	DCON-MS	APMX	LU	LF	RE
ESRB 71200512	0.5	4	0.5	12	45	0.25
71200514	0.5	4	0.5	14	45	0.25
71200516	0.5	4	0.5	16	45	0.25
71200601	0.6	4	0.6	1	45	0.3
71200601S6	0.6	6	0.6	1	45	0.3
71200602	0.6	4	0.6	2	45	0.3
71200602S6	0.6	6	0.6	2	45	0.3
71200603	0.6	4	0.6	3	45	0.3
71200603S6	0.6	6	0.6	3	45	0.3
71200604	0.6	4	0.6	4	45	0.3
71200604S6	0.6	6	0.6	4	45	0.3
71200605	0.6	4	0.6	5	45	0.3
71200605S6	0.6	6	0.6	5	45	0.3
71200606	0.6	4	0.6	6	45	0.3
71200606S6	0.6	6	0.6	6	45	0.3
71200608	0.6	4	0.6	8	45	0.3
71200608S6	0.6	6	0.6	8	45	0.3
71200610	0.6	4	0.6	10	45	0.3
71200610S6	0.6	6	0.6	10	45	0.3
71200612	0.6	4	0.6	12	45	0.3
71200612S6	0.6	6	0.6	12	45	0.3
71200614	0.6	4	0.6	14	45	0.3
71200614S6	0.6	6	0.6	14	45	0.3
71200616	0.6	4	0.6	16	45	0.3
71200616S6	0.6	6	0.6	16	50	0.3
71200702	0.7	4	0.7	2	45	0.35
71200704	0.7	4	0.7	4	45	0.35
71200706	0.7	4	0.7	6	45	0.35
71200708	0.7	4	0.7	8	45	0.35
71200710	0.7	4	0.7	10	45	0.35
71200712	0.7	4	0.7	12	45	0.35
71200801	0.8	4	0.8	1	45	0.4
71200801S6	0.8	6	0.8	1	45	0.4
71200802	0.8	4	0.8	2	45	0.4
71200802S6	0.8	6	0.8	2	45	0.4
71200803	0.8	4	0.8	3	45	0.4
71200803S6	0.8	6	0.8	3	45	0.4
71200804	0.8	4	0.8	4	45	0.4
71200804S6	0.8	6	0.8	4	45	0.4

ESRB712

2 flutes rib ball

CARBIDE

2

H-A
30°

Grade
PC305H

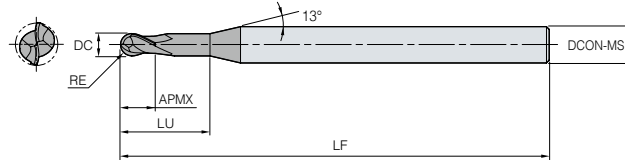
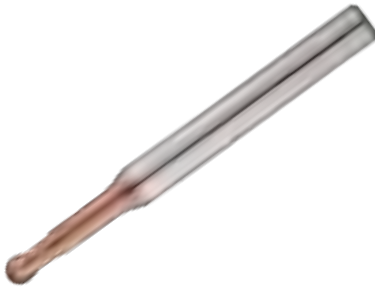
h5
shank

RE
±0.005

RE
±0.008

DC6 or below Above DC6

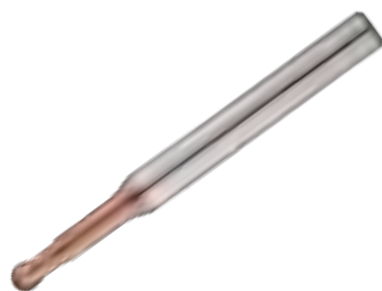
DC	Tolerance
Ø0.1 ~ Ø6	0.000 ~ -0.012
Ø8 ~ Ø12	0.000 ~ -0.015



																(mm)
Designation		DC	DCON-MS	APMX	LU	LF	RE	Designation		DC	DCON-MS	APMX	LU	LF	RE	
ESRB	71200805	0.8	4	0.8	5	45	0.4	ESRB	71201012S6	1	6	1	12	50	0.5	
	71200805S6	0.8	6	0.8	5	45	0.4		71201014	1	4	1	14	50	0.5	
	71200806	0.8	4	0.8	6	45	0.4		71201014S6	1	6	1	14	50	0.5	
	71200806S6	0.8	6	0.8	6	45	0.4		71201016	1	4	1	16	50	0.5	
	71200808	0.8	4	0.8	8	45	0.4		71201016S6	1	6	1	16	50	0.5	
	71200808S6	0.8	6	0.8	8	45	0.4		71201018	1	4	1	18	50	0.5	
	71200810	0.8	4	0.8	10	45	0.4		71201018S6	1	6	1	18	50	0.5	
	71200810S6	0.8	6	0.8	10	45	0.4		71201020	1	4	1	20	55	0.5	
	71200812	0.8	4	0.8	12	45	0.4		71201020S6	1	6	1	20	55	0.5	
	71200812S6	0.8	6	0.8	12	45	0.4		71201022	1	4	1	22	60	0.5	
	71200814	0.8	4	0.8	14	45	0.4		71201022S6	1	6	1	22	60	0.5	
	71200814S6	0.8	6	0.8	14	45	0.4		71201026	1	4	1	26	60	0.5	
	71200816	0.8	4	0.8	16	45	0.4		71201026S6	1	6	1	26	60	0.5	
	71200816S6	0.8	6	0.8	16	50	0.4		71201030	1	4	1	30	70	0.5	
	71200820	0.8	4	0.8	20	50	0.4		71201030S6	1	6	1	30	70	0.5	
	71200820S6	0.8	6	0.8	20	55	0.4		71201040	1	4	1	40	80	0.5	
	71200904	0.9	4	0.9	4	45	0.45		71201050	1	4	1	50	100	0.5	
	71200906	0.9	4	0.9	6	45	0.45		71201204	1.2	4	1.2	4	50	0.6	
	71200908	0.9	4	0.9	8	45	0.45		71201206	1.2	4	1.2	6	50	0.6	
	71200910	0.9	4	0.9	10	45	0.45		71201208	1.2	4	1.2	8	50	0.6	
	71201002	1	4	1	2	50	0.5		71201210	1.2	4	1.2	10	50	0.6	
	71201002S6	1	6	1	2	50	0.5		71201212	1.2	4	1.2	12	50	0.6	
	71201003	1	4	1	3	50	0.5		71201216	1.2	4	1.2	16	50	0.6	
	71201003S6	1	6	1	3	50	0.5		71201220	1.2	4	1.2	20	50	0.6	
	71201004	1	4	1	4	50	0.5		71201226	1.2	4	1.2	26	60	0.6	
	71201004S6	1	6	1	4	50	0.5		71201406	1.4	4	1.4	6	50	0.7	
	71201005	1	4	1	5	50	0.5		71201408	1.4	4	1.4	8	50	0.7	
	71201005S6	1	6	1	5	50	0.5		71201410	1.4	4	1.4	10	50	0.7	
	71201006	1	4	1	6	50	0.5		71201412	1.4	4	1.4	12	50	0.7	
	71201006S6	1	6	1	6	50	0.5		71201416	1.4	4	1.4	16	50	0.7	
	71201007	1	4	1	7	50	0.5		71201503	1.5	4	1.5	3	50	0.75	
	71201007S6	1	6	1	7	50	0.5		71201503S6	1.5	6	1.5	3	50	0.75	
	71201008	1	4	1	8	50	0.5		71201504	1.5	4	1.5	4	50	0.75	
	71201008S6	1	6	1	8	50	0.5		71201504S6	1.5	6	1.5	4	50	0.75	
	71201009	1	4	1	9	50	0.5		71201505	1.5	4	1.5	5	50	0.75	
	71201009S6	1	6	1	9	50	0.5		71201506	1.5	4	1.5	6	50	0.75	
	71201010	1	4	1	10	50	0.5		71201506S6	1.5	6	1.5	6	50	0.75	
	71201010S6	1	6	1	10	50	0.5		71201507	1.5	4	1.5	7	50	0.75	
	71201012	1	4	1	12	50	0.5		71201508	1.5	4	1.5	8	50	0.75	

ESRB712

2 flutes rib ball



CARBIDE

2

H-A
30°

Grade
PC305H

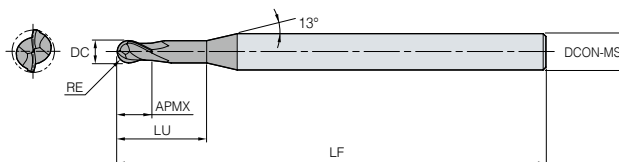
h5 shank

RE
±0.005

RE
±0.008

DC6 or below Above DC6

DC	Tolerance
Ø0.1 ~ Ø6	0.000 ~ -0.012
Ø8 ~ Ø12	0.000 ~ -0.015



(mm)

	Designation	DC	DCON-MS	APMX	LU	LF	RE
ESRB	71201508S6	1.5	6	1.5	8	50	0.75
	71201510	1.5	4	1.5	10	50	0.75
	71201510S6	1.5	6	1.5	10	50	0.75
	71201512	1.5	4	1.5	12	50	0.75
	71201512S6	1.5	6	1.5	12	50	0.75
	71201514	1.5	4	1.5	14	50	0.75
	71201514S6	1.5	6	1.5	14	50	0.75
	71201516	1.5	4	1.5	16	50	0.75
	71201516S6	1.5	6	1.5	16	50	0.75
	71201518	1.5	4	1.5	18	50	0.75
	71201518S6	1.5	6	1.5	18	50	0.75
	71201520	1.5	4	1.5	20	55	0.75
	71201520S6	1.5	6	1.5	20	55	0.75
	71201522	1.5	4	1.5	22	60	0.75
	71201522S6	1.5	6	1.5	22	60	0.75
	71201526	1.5	4	1.5	26	60	0.75
	71201526S6	1.5	6	1.5	26	60	0.75
	71201530	1.5	4	1.5	30	70	0.75
	71201530S6	1.5	6	1.5	30	70	0.75
	71201535	1.5	4	1.5	35	70	0.75
	71201535S6	1.5	6	1.5	35	70	0.75
	71201540	1.5	4	1.5	40	80	0.75
	71201540S6	1.5	6	1.5	40	80	0.75
	71201604	1.6	4	1.6	4	50	0.8
	71201606	1.6	4	1.6	6	50	0.8
	71201608	1.6	4	1.6	8	50	0.8
	71201610	1.6	4	1.6	10	50	0.8
	71201612	1.6	4	1.6	12	50	0.8
	71201616	1.6	4	1.6	16	50	0.8
	71201620	1.6	4	1.6	20	50	0.8
	71201804	1.8	4	1.8	4	50	0.9
	71201806	1.8	4	1.8	6	50	0.9
	71201808	1.8	4	1.8	8	50	0.9
	71201810	1.8	4	1.8	10	50	0.9
	71201812	1.8	4	1.8	12	50	0.9
	71201816	1.8	4	1.8	16	50	0.9
	71201820	1.8	4	1.8	20	50	0.9
	71202004	2	4	2	4	50	1
	71202004S6	2	6	2	4	50	1

	Designation	DC	DCON-MS	APMX	LU	LF	RE
ESRB	71202006	2	4	2	6	50	1
	71202006S6	2	6	2	6	50	1
	71202008	2	4	2	8	50	1
	71202008S6	2	6	2	8	50	1
	71202010	2	4	2	10	50	1
	71202010S6	2	6	2	10	50	1
	71202012	2	4	2	12	50	1
	71202012S6	2	6	2	12	50	1
	71202014	2	4	2	14	50	1
	71202014S6	2	6	2	14	50	1
	71202016	2	4	2	16	50	1
	71202016S6	2	6	2	16	50	1
	71202018	2	4	2	18	55	1
	71202018S6	2	6	2	18	55	1
	71202020	2	4	2	20	55	1
	71202020S6	2	6	2	20	55	1
	71202022	2	4	2	22	60	1
	71202022S6	2	6	2	22	60	1
	71202026	2	4	2	26	60	1
	71202026S6	2	6	2	26	60	1
	71202030	2	4	2	30	70	1
	71202030S6	2	6	2	30	70	1
	71202035	2	4	2	35	70	1
	71202035S6	2	6	2	35	70	1
	71202040	2	4	2	40	80	1
	71202040S6	2	6	2	40	80	1
	71202045	2	4	2	45	90	1
	71202045S6	2	6	2	45	90	1
	71202050	2	4	2	50	100	1
	71202050S6	2	6	2	50	100	1
	71202060	2	4	2	60	110	1
	71202508	2.5	4	2.5	8	50	1.25
	71202510	2.5	4	2.5	10	50	1.25
	71202512	2.5	4	2.5	12	50	1.25
	71202516	2.5	4	2.5	16	50	1.25
	71202520	2.5	4	2.5	20	50	1.25
	71202522	2.5	4	2.5	22	60	1.25
	71202526	2.5	4	2.5	26	60	1.25
	71202530	2.5	4	2.5	30	70	1.25

ESRB712

2 flutes rib ball

CARBIDE

2

H-A
30°

Grade
PC305H

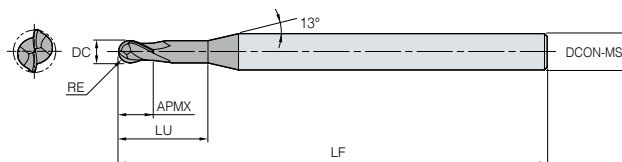
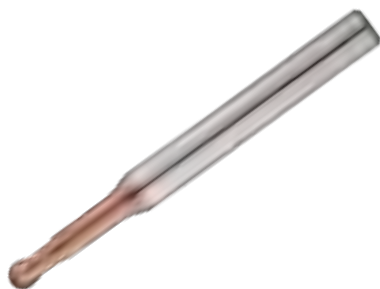
h5
shank

RE
±0.005

RE
±0.008

DC6 or below Above DC6

DC	Tolerance
Ø0.1 ~ Ø6	0.000 ~ -0.012
Ø8 ~ Ø12	0.000 ~ -0.015

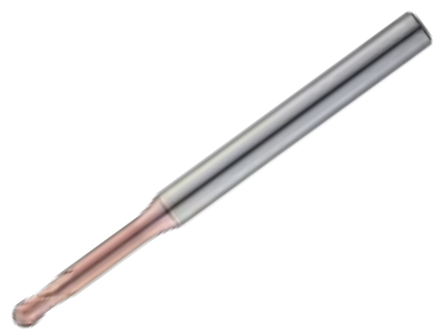


Designation	DC	DCON-MS	APMX	LU	LF	RE
ESRB 71202535	2.5	4	2.5	35	70	1.25
71202540	2.5	4	2.5	40	80	1.25
71202545	2.5	4	2.5	45	90	1.25
71202550	2.5	4	2.5	50	100	1.25
71203006	3	6	3	6	50	1.5
71203008	3	6	3	8	50	1.5
71203010	3	6	3	10	50	1.5
71203012	3	6	3	12	50	1.5
71203014	3	6	3	14	60	1.5
71203016	3	6	3	16	60	1.5
71203018	3	6	3	18	60	1.5
71203020	3	6	3	20	60	1.5
71203022	3	6	3	22	65	1.5
71203026	3	6	3	26	65	1.5
71203030	3	6	3	30	70	1.5
71203035	3	6	3	35	70	1.5
71203040	3	6	3	40	80	1.5
71203045	3	6	3	45	90	1.5
71203050	3	6	3	50	100	1.5
71203060	3	6	3	60	100	1.5
71203510	3.5	6	3	10	50	1.75
71203516	3.5	6	3	16	60	1.75
71203520	3.5	6	3	20	60	1.75
71203526	3.5	6	3	26	65	1.75
71203530	3.5	6	3	30	70	1.75
71204008	4	6	4	8	50	2
71204010	4	6	4	10	50	2
71204012	4	6	4	12	50	2
71204014	4	6	4	14	60	2
71204016	4	6	4	16	60	2
71204018	4	6	4	18	60	2
71204020	4	6	4	20	60	2
71204022	4	6	4	22	65	2
71204026	4	6	4	26	65	2

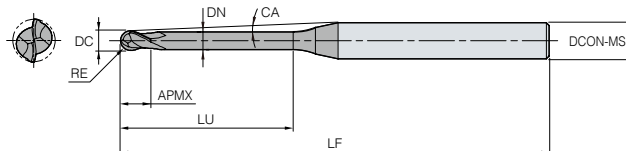
Designation	DC	DCON-MS	APMX	LU	LF	RE
ESRB 71204030	4	6	4	30	70	2
71204035	4	6	4	35	70	2
71204040	4	6	4	40	80	2
71204045	4	6	4	45	90	2
71204050	4	6	4	50	100	2
71204055	4	6	4	55	100	2
71204060	4	6	4	60	100	2
71205015	5	6	6	15	60	2.5
71205020	5	6	6	20	60	2.5
71205026	5	6	6	26	65	2.5
71205030	5	6	6	30	70	2.5
71205035	5	6	6	35	70	2.5
71205040	5	6	6	40	80	2.5
71205045	5	6	6	45	90	2.5
71205050	5	6	6	50	100	2.5
71205055	5	6	6	55	100	2.5
71205060	5	6	6	60	100	2.5
7120602090	6	6	12	20	90	3
71206020	6	6	8	20	60	3
7120603090	6	6	12	30	90	3
71206030	6	6	8	30	60	3
71208025100	8	8	14	25	100	4
71208025	8	8	10	25	70	4
71208035100	8	8	14	35	100	4
71208035	8	8	10	35	70	4
71210030100	10	10	18	30	100	5
71210030	10	10	12	30	75	5
71210040100	10	10	18	40	100	5
71210040	10	10	12	40	75	5
71212032110	12	12	22	32	110	6
71212032	12	12	14	32	80	6
71212045110	12	12	22	45	110	6
71212045	12	12	14	45	80	6

ESLNB20

2 flutes long neck type ball



CARBIDE	2	H-A 30°	Grade PC305H	h5 shank	RE ±0.005	RE ±0.008	DC	Tolerance
					DC6 or below	Above DC6	ALL	0.000 ~ -0.012

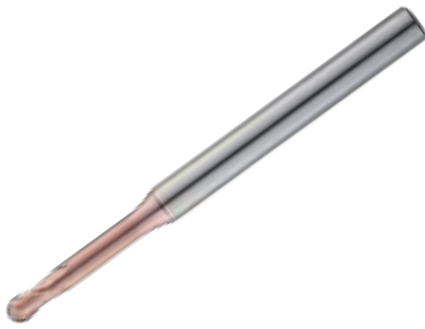


(mm)

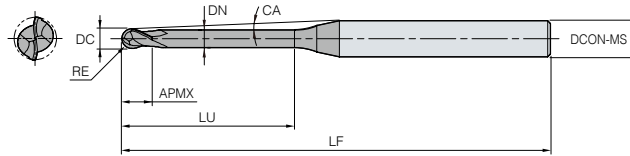
Designation	DC	DCON-MS	APMX	LU	LF	DN	CA	RE	Effective length by inclination angle				
									0.5°	1°	1.5°	2°	3°
ESLNB 2001-0.2	0.1	4	0.08	0.2	45	0.08	11.8	0.05	0.3	0.3	0.3	0.4	0.4
2001-0.3	0.1	4	0.08	0.3	45	0.08	11.7	0.05	0.4	0.4	0.5	0.5	0.5
2001-0.5	0.1	4	0.08	0.5	45	0.08	11.4	0.05	0.6	0.7	0.7	0.7	0.8
2002-0.5	0.2	4	0.15	0.5	50	0.17	11.5	0.1	1.2	1.3	1.5	1.6	2
2002-1	0.2	4	0.15	1	50	0.17	10.9	0.1	1.7	1.9	2.1	2.3	2.7
2002-1.5	0.2	4	0.15	1.5	50	0.17	10.4	0.1	2.3	2.5	2.8	3	3.4
2002-2	0.2	4	0.15	2	50	0.17	9.9	0.1	2.8	3.1	3.4	3.6	4.1
2002-2.5	0.2	4	0.15	2.5	50	0.17	9.5	0.1	3.4	3.7	4	4.2	4.7
2002-3.0	0.2	4	0.15	3	50	0.17	9.1	0.1	3.9	4.3	4.6	4.9	5.4
2003-1	0.3	4	0.25	1	50	0.27	10.9	0.15	1.7	1.9	2.1	2.3	2.7
2003-1.5	0.3	4	0.25	1.5	50	0.27	10.4	0.15	2.3	2.5	2.7	3	3.4
2003-2	0.3	4	0.25	2	50	0.27	9.9	0.15	2.8	3.1	3.4	3.6	4
2003-2.5	0.3	4	0.25	2.5	50	0.27	9.5	0.15	3.4	3.7	4	4.2	4.7
2003-3	0.3	4	0.25	3	50	0.27	9.1	0.15	3.9	4.3	4.6	4.8	5.3
2004-1	0.4	4	0.3	1	50	0.37	11	0.2	1.7	1.9	2.1	2.3	2.7
2004-1.5	0.4	4	0.3	1.5	50	0.37	10.4	0.2	2.3	2.5	2.7	2.9	3.4
2004-2	0.4	4	0.3	2	50	0.37	9.9	0.2	2.8	3.1	3.4	3.6	4
2004-2.5	0.4	4	0.3	2.5	50	0.37	9.5	0.2	3.4	3.7	4	4.2	4.7
2004-3	0.4	4	0.3	3	50	0.37	9.1	0.2	3.9	4.3	4.6	4.8	5.3
2004-3.5	0.4	4	0.3	3.5	50	0.37	8.7	0.2	4.5	4.8	5.2	5.4	6
2004-4	0.4	4	0.3	4	50	0.37	8.3	0.2	5	5.4	5.7	6	6.6
2004-4.5	0.4	4	0.3	4.5	50	0.37	8	0.2	5.6	6	6.3	6.6	7.2
2005-1	0.5	4	0.35	1	50	0.47	11	0.25	1.7	1.9	2.1	2.3	2.6
2005-2	0.5	4	0.35	2	50	0.47	9.9	0.25	2.8	3.1	3.3	3.6	4
2005-3	0.5	4	0.35	3	50	0.47	9	0.25	3.9	4.3	4.6	4.8	5.3
2005-4	0.5	4	0.35	4	50	0.47	8.3	0.25	5	5.4	5.7	6	6.6
2005-5	0.5	4	0.35	5	50	0.47	7.7	0.25	6.1	6.5	6.9	7.2	7.8
2005-6	0.5	4	0.35	6	50	0.47	7.1	0.25	7.2	7.6	8	8.4	9
2005-8	0.5	4	0.35	8	50	0.47	6.3	0.25	9.3	9.9	10.3	10.7	11.4
2006-1	0.6	4	0.4	1	50	0.57	11	0.3	1.7	1.9	2.1	2.3	2.6
2006-2	0.6	4	0.4	2	50	0.57	9.9	0.3	2.8	3.1	3.3	3.6	4
2006-3	0.6	4	0.4	3	50	0.57	9	0.3	3.9	4.3	4.5	4.8	5.3
2006-4	0.6	4	0.4	4	50	0.57	8.3	0.3	5	5.4	5.7	6	6.6
2006-5	0.6	4	0.4	5	50	0.57	7.6	0.3	6.1	6.5	6.9	7.2	7.8
2006-6	0.6	4	0.4	6	50	0.57	7.1	0.3	7.2	7.6	8	8.4	9
2006-7	0.6	4	0.4	7	50	0.57	6.6	0.3	8.3	8.8	9.2	9.5	10.2

ESLNB20

2 flutes long neck type ball



CARBIDE	2	H-A 30°	Grade PC305H	h5 shank	RE ±0.005	RE ±0.008	DC	Tolerance
					DC6 or below	Above DC6	ALL	0.000 ~ -0.012

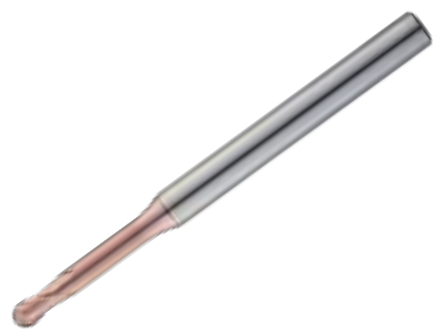


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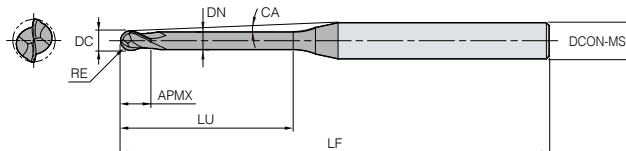
	Designation	DC	DCON-MS	APMX	LU	LF	DN	CA	RE	Effective length by inclination angle				
										0.5°	1°	1.5°	2°	3°
ESLNB	2006-8	0.6	4	0.4	8	50	0.57	6.2	0.3	9.3	9.9	10.3	10.7	11.4
	2006-9	0.6	4	0.4	9	50	0.57	5.8	0.3	10.4	10.9	11.4	11.8	12.5
	2006-10	0.6	4	0.4	10	50	0.57	5.5	0.3	11.4	12	12.5	12.9	13.7
	2006-12	0.6	4	0.4	12	50	0.57	5	0.3	13.6	14.2	14.7	15.2	16
	2008-2	0.8	4	0.5	2	50	0.77	9.9	0.4	2.8	3.1	3.3	3.5	4
	2008-4	0.8	4	0.5	4	50	0.77	8.2	0.4	5	5.4	5.7	6	6.5
	2008-5	0.8	4	0.5	5	50	0.77	7.5	0.4	6.1	6.5	6.9	7.2	7.8
	2008-6	0.8	4	0.5	6	50	0.77	7	0.4	7.2	7.6	8	8.4	9
	2008-8	0.8	4	0.5	8	50	0.77	6.1	0.4	9.3	9.8	10.3	10.7	11.3
	2008-10	0.8	4	0.5	10	50	0.77	5.4	0.4	11.4	12	12.5	12.9	13.7
	2010-2	1	4	0.8	2	50	0.96	9.9	0.5	2.9	3.1	3.3	3.5	4
	2010-3	1	4	0.8	3	50	0.96	8.9	0.5	4	4.3	4.5	4.8	5.3
	2010-4	1	4	0.8	4	50	0.96	8.1	0.5	5	5.4	5.7	6	6.5
	2010-5	1	4	0.8	5	50	0.96	7.4	0.5	6.1	6.5	6.9	7.2	7.8
	2010-6	1	4	0.8	6	50	0.96	6.8	0.5	7.2	7.7	8	8.4	9
	2010-7	1	4	0.8	7	50	0.96	6.3	0.5	8.3	8.8	9.2	9.5	10.2
	2010-8	1	4	0.8	8	50	0.96	5.9	0.5	9.3	9.9	10.3	10.7	11.3
	2010-9	1	4	0.8	9	50	0.96	5.5	0.5	10.4	11	11.4	11.8	12.5
	2010-10	1	4	0.8	10	50	0.96	5.2	0.5	11.5	12	12.5	12.9	13.7
	2010-12	1	4	0.8	12	55	0.96	4.6	0.5	13.6	14.2	14.7	15.2	15.9
	2010-14	1	4	0.8	14	55	0.96	4.2	0.5	15.7	16.4	16.9	17.4	18.5
	2010-16	1	4	0.8	16	55	0.96	3.8	0.5	17.8	18.5	19.1	19.6	21.2
	2010-18	1	4	0.8	18	60	0.96	3.5	0.5	19.9	20.7	21.3	21.8	23.8
	2010-20	1	4	0.8	20	60	0.96	3.3	0.5	22	22.8	23.4	24	26.5
	2012-4	1.2	4	1.1	4	50	1.15	7.9	0.6	5.1	5.4	5.7	6	6.5
	2012-6	1.2	4	1.1	6	50	1.15	6.6	0.6	7.2	7.7	8	8.4	9
	2012-8	1.2	4	1.1	8	50	1.15	5.7	0.6	9.4	9.9	10.3	10.7	11.3
	2012-10	1.2	4	1.1	10	50	1.15	5	0.6	11.5	12.1	12.5	12.9	13.7
	2012-12	1.2	4	1.1	12	55	1.15	4.5	0.6	13.6	14.2	14.7	15.2	15.9
	2014-8	1.4	4	1.3	8	50	1.34	5.5	0.7	9.4	9.9	10.3	10.7	11.3
	2014-12	1.4	4	1.3	12	55	1.34	4.3	0.7	13.6	14.2	14.7	15.2	15.9
	2014-16	1.4	4	1.3	16	55	1.34	3.5	0.7	17.8	18.5	19.1	19.6	21.2
	2015-4	1.5	4	1.35	4	50	1.44	7.7	0.75	5.1	5.4	5.7	6	6.5
	2015-6	1.5	4	1.35	6	50	1.44	6.4	0.75	7.3	7.7	8	8.4	9
	2015-8	1.5	4	1.35	8	50	1.44	5.4	0.75	9.4	9.9	10.3	10.7	11.3
	2015-10	1.5	4	1.35	10	50	1.44	4.7	0.75	11.5	12.1	12.5	12.9	13.7

ESLNB20

2 flutes long neck type ball



CARBIDE	2	H-A 30°	Grade PC305H	h5 shank	RE ±0.005	RE ±0.008	DC	Tolerance
					DC6 or below	Above DC6	ALL	0.000 ~ -0.012

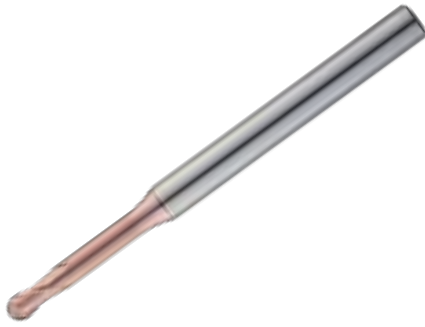


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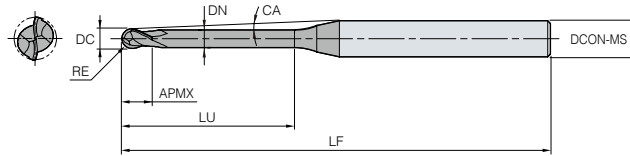
	Designation	DC	DCON-MS	APMX	LU	LF	DN	CA	RE	Effective length by inclination angle				
										0.5°	1°	1.5°	2°	3°
ESLNB	2015-12	1.5	4	1.35	12	55	1.44	4.2	0.75	13.6	14.2	14.7	15.2	15.9
	2015-14	1.5	4	1.35	14	55	1.44	3.8	0.75	15.7	16.4	16.9	17.4	18.5
	2015-16	1.5	4	1.35	16	55	1.44	3.4	0.75	17.8	18.5	19.1	19.6	21.1
	2015-20	1.5	4	1.35	20	60	1.44	2.9	0.75	22	22.8	23.4	24	-
	2016-8	1.6	4	1.4	8	50	1.54	5.3	0.8	9.4	9.9	10.3	10.7	11.3
	2016-10	1.6	4	1.4	10	55	1.54	4.6	0.8	11.5	12.1	12.5	12.9	13.7
	2016-12	1.6	4	1.4	12	55	1.54	4.1	0.8	13.6	14.2	14.7	15.2	15.9
	2016-16	1.6	4	1.4	16	55	1.54	3.3	0.8	17.8	18.5	19.1	19.6	21.1
	2016-20	1.6	4	1.4	20	60	1.54	2.8	0.8	22	22.8	23.4	24	-
	2018-8	1.8	4	1.6	8	50	1.73	5.1	0.9	9.4	9.9	10.3	10.7	11.3
	2018-12	1.8	4	1.6	12	55	1.73	3.9	0.9	13.7	14.3	14.7	15.2	15.9
	2018-16	1.8	4	1.6	16	55	1.73	3.1	0.9	17.9	18.6	19.1	19.6	21.1
	2018-20	1.8	4	1.6	20	60	1.73	2.6	0.9	22	22.8	23.4	24	-
	2020-3	2	4	1.7	3	50	1.92	8.3	1	4.1	4.4	4.6	4.8	5.2
	2020-4	2	4	3	4	50	1.92	7.3	1	5.2	5.5	5.8	6	6.5
	2020-6	2	4	3	6	50	1.92	5.8	1	7.3	7.7	8.1	8.4	9
	2020-8	2	4	3	8	50	1.92	4.9	1	9.5	9.9	10.3	10.7	11.3
	2020-10	2	4	3	10	50	1.92	4.2	1	11.6	12.1	12.6	12.9	13.6
	2020-12	2	4	3	12	55	1.92	3.7	1	13.7	14.3	14.8	15.2	15.9
	2020-14	2	4	3	14	55	1.92	3.2	1	15.8	16.4	16.9	17.4	18.5
	2020-16	2	4	3	16	55	1.92	2.9	1	17.9	18.6	19.1	19.6	-
	2020-18	2	4	3	18	60	1.92	2.7	1	20	20.7	21.3	21.8	-
	2020-20	2	4	3	20	60	1.92	2.4	1	22.1	22.8	23.4	24	-
	2020-22	2	4	3	22	60	1.92	2.3	1	24.1	24.9	25.6	26.3	-
	2020-25	2	4	3	25	65	1.92	2	1	27.3	28.1	28.8	-	-
	2020-30	2	4	3	30	70	1.92	1.7	1	32.4	33.4	34.2	-	-
	2020-35	2	4	3	35	75	1.92	1.5	1	37.6	38.6	-	-	-
	2020-40	2	4	3	40	80	1.92	1.4	1	42.8	43.8	-	-	-
	2025-10	2.5	4	4	10	50	2.4	3.4	1.25	11.6	12.1	12.6	13	13.6
	2025-16	2.5	4	4	16	55	2.4	2.3	1.25	17.9	18.6	19.1	19.6	-
	2025-20	2.5	4	4	20	60	2.4	1.9	1.25	22.1	22.8	23.5	-	-
	2030-8	3	6	4	8	55	2.88	6.2	1.5	9.6	10	10.4	10.7	11.3
	2030-10	3	6	4	10	55	2.88	5.5	1.5	11.7	12.2	12.6	13	13.6
	2030-13	3	6	4	13	60	2.88	4.6	1.5	14.8	15.4	15.9	16.3	17.1
	2030-16	3	6	4	16	60	2.88	4	1.5	18	18.6	19.1	19.6	21.1
	2030-18	3	6	4	18	60	2.88	3.6	1.5	20	20.7	21.3	21.8	23.7

ESLNB20

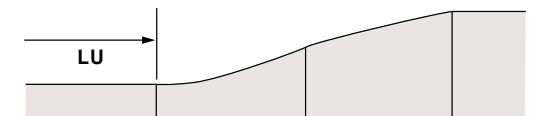
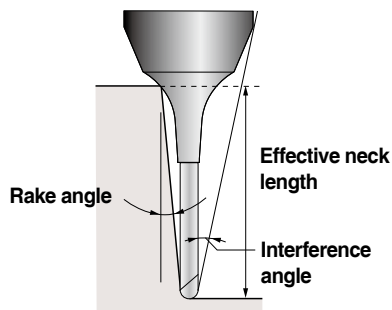
2 flutes long neck type ball



CARBIDE	2	H-A 30°	Grade PC305H	h5 shank	RE ±0.005	RE ±0.008	DC	Tolerance
					DC6 or below	Above DC6	ALL	0.000 ~ -0.012



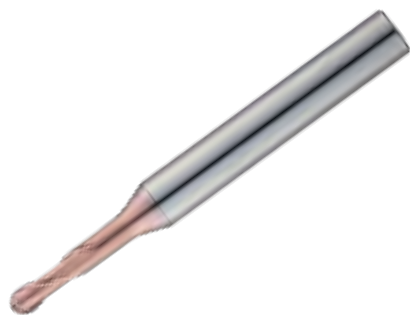
										(mm)				
Designation		DC	DCON-MS	APMX	LU	LF	DN	CA	RE	Effective length by inclination angle				
										0.5°	1°	1.5°	2°	3°
ESLNB	2030-20	3	6	4	20	65	2.88	3.4	1.5	22.1	22.9	23.5	24	26.4
	2030-25	3	6	4	25	70	2.88	2.8	1.5	27.3	28.2	28.8	29.9	-
	2030-30	3	6	4	30	75	2.88	2.5	1.5	32.5	33.4	34.3	35.9	-
	2030-35	3	6	4	35	80	2.88	2.2	1.5	37.7	38.7	40	41.9	-
	2040-10	4	6	5	10	55	3.9	4.5	2	11.6	12.1	12.5	12.9	13.5
	2040-13	4	6	5	13	60	3.9	3.6	2	14.7	15.3	15.8	16.2	17
	2040-16	4	6	5	16	60	3.9	3.1	2	17.9	18.5	19.1	19.5	20.9
	2040-20	4	6	5	20	65	3.9	2.5	2	22.1	22.8	23.4	23.9	-
	2040-25	4	6	5	25	70	3.9	2.1	2	27.3	28.1	28.8	29.8	-
	2040-30	4	6	5	30	75	3.9	1.8	2	32.4	33.4	34.2	-	-
	2040-35	4	6	5	35	80	3.9	1.6	2	37.6	38.6	39.9	-	-
	2040-40	4	6	5	40	80	3.9	1.4	2	42.8	43.8	-	-	-
	2040-45	4	6	5	45	90	3.9	1.2	2	47.9	49.1	-	-	-
	2040-50	4	6	5	50	100	3.9	1.1	2	53.1	54.5	-	-	-
	2050-20	5	6	6	20	65	4.9	1.4	2.5	22	22.8	-	-	-
	2050-25	5	6	6	25	70	4.9	1.2	2.5	27.2	28.1	-	-	-
	2050-30	5	6	6	30	75	4.9	1	2.5	32.4	-	-	-	-
	2050-35	5	6	6	35	80	4.9	0.8	2.5	42.8	-	-	-	-
	2050-40	5	6	6	40	90	4.9	0.7	2.5	42.8	-	-	-	-



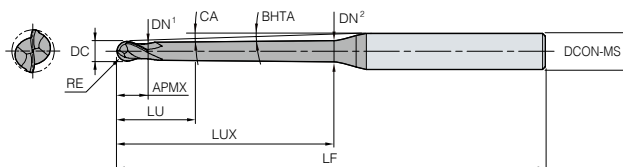
※ The marked effective neck length is the default value to prevent interference with the workpiece.
Proper control of the processing environment is required.

ESTNB20

2 flutes tapered neck type ball



CARBIDE	2	H-A 30°	Grade PC305H	h5 shank	RE ±0.01	RE ±0.015	DC	Tolerance
					DC6 or below	Above DC6	Ø0.2 ~ Ø6 Ø8 ~ Ø10	0.000 ~ -0.012 0.000 ~ -0.015

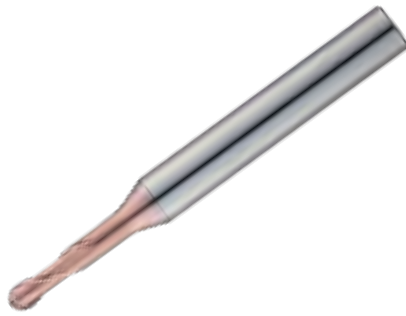


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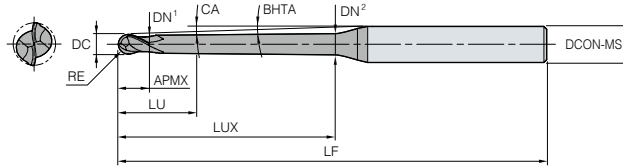
Designation	DC	DCON-MS	APMX	LU	LUX	LF	DN1	DN2	RE	BHTA(°)	CA	Effective length by inclination angle				
												0.5°	1°	1.5°	2°	3°
ESTNB 2002-1-04	0.2	4	0.15	1.35	1	50	0.17	0.18	0.1	0.4	10.9	1.5	1.7	1.8	2	2.3
2002-1.5-04	0.2	4	0.15	1.77	1.5	50	0.17	0.19	0.1	0.4	10.4	2	2.2	2.4	2.6	2.9
2002-2-09	0.2	4	0.15	1.1	2	50	0.17	0.23	0.1	0.9	10.1	-	2.8	3.1	3.4	3.9
2002-2.5-09	0.2	4	0.15	1.1	2.5	50	0.17	0.24	0.1	0.9	9.6	-	3.3	3.7	4	4.5
2003-2-04	0.3	4	0.25	2.19	2	50	0.28	0.29	0.15	0.4	10	2.5	2.8	3	3.2	3.5
2003-3-09	0.3	4	0.25	1.2	3	50	0.28	0.36	0.15	0.9	9.3	-	3.8	4.2	4.5	5.1
2003-4-09	0.3	4	0.25	1.2	4	50	0.28	0.39	0.15	0.9	8.6	-	4.8	5.3	5.7	6.3
2004-2-04	0.4	4	0.3	2.2	2	50	0.37	0.39	0.2	0.4	10	2.5	2.8	3	3.2	3.5
2004-3-04	0.4	4	0.3	2.44	3	50	0.37	0.41	0.2	0.4	9.1	3.6	3.9	4.1	4.4	4.8
2004-4-04	0.4	4	0.3	2.44	4	50	0.37	0.42	0.2	0.4	8.4	4.7	5.2	5.6	5.9	6.5
2004-4-09	0.4	4	0.3	1.25	4	50	0.37	0.49	0.2	0.9	8.5	-	4.8	5.3	5.7	6.3
2004-5-04	0.4	4	0.3	2.44	5	50	0.37	0.44	0.2	0.4	7.8	5.7	6.3	6.7	7.1	7.7
2004-5-09	0.4	4	0.3	1.25	5	50	0.37	0.52	0.2	0.9	7.9	-	5.9	6.4	6.8	7.5
2005-4-04	0.5	4	0.35	2.49	4	50	0.47	0.52	0.25	0.4	8.4	4.6	5	5.3	5.5	5.9
2005-8-09	0.5	4	0.35	1.3	8	50	0.47	0.71	0.25	0.9	6.5	-	8.9	9.6	10.1	10.9
2005-12-09	0.5	4	0.35	1.3	12	50	0.47	0.84	0.25	0.9	5.3	-	13	13.9	14.5	15.4
20054-2-04	0.54	4	0.37	1.8	2	50	0.52	0.54	0.27	0.4	10	2.3	2.5	2.7	2.8	3
20054-4-04	0.54	4	0.37	1.8	4	50	0.52	0.57	0.27	0.4	8.4	4.5	4.9	5.2	5.5	5.9
20054-5-04	0.54	4	0.37	1.8	5	50	0.52	0.59	0.27	0.4	7.8	5.5	6	6.3	6.6	7.1
20054-6-04	0.54	4	0.37	1.8	6	50	0.52	0.6	0.27	0.4	7.2	6.7	7.3	7.8	8.2	8.8
20054-6.5-04	0.54	4	0.37	1.8	6.5	50	0.52	0.61	0.27	0.4	7	7.2	7.9	8.3	8.7	9.4
20054-7-04	0.54	4	0.37	1.8	7	50	0.52	0.61	0.27	0.4	6.8	7.7	8.4	8.9	9.3	10
2006-2-04	0.6	4	0.4	2.17	2	50	0.57	0.59	0.3	0.4	10	2.4	2.5	2.7	2.8	3
2006-4-04	0.6	4	0.4	2.54	4	50	0.57	0.62	0.3	0.4	8.4	4.6	5	5.2	5.5	5.9
2006-6-04	0.6	4	0.4	2.54	6	50	0.57	0.65	0.3	0.4	7.2	6.8	7.4	7.8	8.2	8.8
2006-6-09	0.6	4	0.4	1.35	6	50	0.57	0.75	0.3	0.9	7.3	-	6.9	7.5	7.9	8.6
2006-8-09	0.6	4	0.4	1.35	8	50	0.57	0.81	0.3	0.9	6.4	-	8.9	9.6	10.1	10.9
2006-10-04	0.6	4	0.4	2.54	10	50	0.57	0.7	0.3	0.4	5.6	10.8	11.7	12.2	12.7	13.5
2006-10-09	0.6	4	0.4	1.35	10	50	0.57	0.87	0.3	0.9	5.7	-	11	11.8	12.3	13.2
2006-12-09	0.6	4	0.4	1.35	12	55	0.57	0.93	0.3	0.9	5.2	-	13	13.9	14.5	15.4
2006-15-04	0.6	4	0.4	2.54	15	55	0.57	0.77	0.3	0.4	4.4	15.9	17	17.6	18.2	19.2
2006-15-09	0.6	4	0.4	1.35	15	55	0.57	1.03	0.3	0.9	4.5	-	16.1	17.1	17.7	18.8
2008-4-04	0.8	4	0.5	2.64	4	50	0.77	0.82	0.4	0.4	8.3	4.6	4.9	5.2	5.5	5.9
2008-6-04	0.8	4	0.5	2.64	6	50	0.77	0.85	0.4	0.4	7.1	6.6	7.1	7.5	7.7	8.3
2008-8-09	0.8	4	0.5	1.45	8	50	0.77	1.01	0.4	0.9	6.3	-	8.9	9.6	10.1	10.9
2008-12-09	0.8	4	0.5	1.45	12	55	0.77	1.13	0.4	0.9	5	-	13	13.9	14.5	15.4

ESTNB20

2 flutes tapered neck type ball



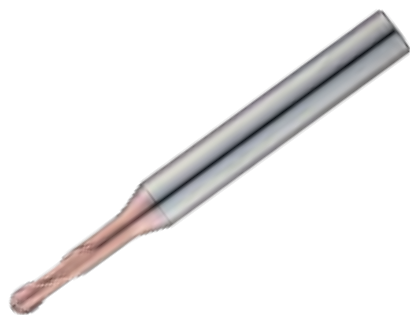
CARBIDE	2	H-A 30°	Grade PC305H	h5 shank	RE ±0.01	RE ±0.015	DC Ø0.2 ~ Ø6 Ø8 ~ Ø10	Tolerance 0.000 ~ -0.012 0.000 ~ -0.015
					DC6 or below Above DC6			



													(mm)				
ESTNB	Designation	DC	DCON-MS	APMX	LU	LUX	LF	DN1	DN2	RE	BHTA(°)	CA	Effective length by inclination angle				
													0.5°	1°	1.5°	2°	3°
	2008-16-09	0.8	4	0.5	1.45	16	55	0.77	1.26	0.4	0.9	4.2	-	17.1	18.1	18.8	19.9
	2009-4-04	0.9	4	0.6	3.46	4	50	0.86	0.91	0.45	0.4	8.2	4.5	4.7	4.9	5.1	5.4
	2009-8-04	0.9	4	0.6	3.46	8	55	0.86	0.96	0.45	0.4	6.1	8.7	9.3	9.7	10	10.6
	2009-12-04	0.9	4	0.6	3.46	12	55	0.86	1.02	0.45	0.4	4.8	12.9	13.8	14.4	14.9	15.7
	2009-16-04	0.9	4	0.6	3.46	16	60	0.86	1.08	0.45	0.4	4	17	18	18.7	19.3	20.5
	2009-18-04	0.9	4	0.6	3.46	18	65	0.86	1.1	0.45	0.4	3.7	19.1	20.1	20.9	21.5	23.1
	2009-20-04	0.9	4	0.6	3.46	20	65	0.86	1.13	0.45	0.4	3.4	21.1	22.2	23	23.6	25.6
	2009-22-04	0.9	4	0.6	3.46	22	65	0.86	1.16	0.45	0.4	3.2	23.1	24.3	25.1	25.8	28.2
	2009-24-04	0.9	4	0.6	3.46	24	70	0.86	1.19	0.45	0.4	3	25.2	26.4	27.2	27.9	-
	2010-6-04	1	6	0.8	5.09	6	50	0.94	1.01	0.5	0.4	8.3	6.8	7.2	7.5	7.8	8.3
	2010-8-04	1	6	0.8	5.09	8	55	0.94	1.04	0.5	0.4	7.5	8.8	9.3	9.7	10	10.6
	2010-10-04	1	6	0.8	5.09	10	55	0.94	1.07	0.5	0.4	6.8	11	11.7	12.3	12.7	13.5
	2010-10-09	1	6	0.8	2.7	10	55	0.94	1.23	0.5	0.9	6.9	-	11.2	11.9	12.4	13.2
	2010-15-09	1	6	0.8	2.7	15	60	0.94	1.39	0.5	0.9	5.7	-	16.2	17.1	17.8	18.8
	2010-20-04	1	6	0.8	5.09	20	65	0.94	1.21	0.5	0.4	4.7	21.2	22.3	23	23.6	25.7
	2010-20-09	1	6	0.8	2.7	20	65	0.94	1.54	0.5	0.4	4.8	-	21.3	22.4	23.1	24.6
	2010-25-09	1	6	0.8	2.7	25	70	0.94	1.7	0.5	0.4	4.2	-	26.4	27.6	28.4	30.8
	2010-30-04	1	6	0.8	5.09	30	75	0.94	1.35	0.5	0.4	3.6	31.3	32.7	33.6	34.8	38.5
	2010-30-09	1	6	0.8	2.7	30	75	0.94	1.86	0.5	0.9	3.7	-	31.4	32.8	33.7	36.9
	2010-35-09	1	6	0.8	2.7	35	80	0.94	2.02	0.5	0.9	3.3	-	36.5	38	39	43.1
	2010-40-09	1	6	0.8	2.7	40	85	0.94	2.17	0.5	0.9	3	-	41.6	43.2	44.4	-
	2010-50-09	1	6	0.8	2.7	50	95	0.94	2.49	0.5	0.9	2.5	-	51.7	53.5	55.5	-
	2010-60-09	1	6	0.8	2.7	60	105	0.94	2.8	0.5	0.9	2.2	-	61.8	63.8	66.6	-
	2010-70-09	1	6	0.8	2.7	70	115	0.94	3.11	0.5	0.9	1.9	-	71.9	74	-	-
	2015-8-04	1.5	6	1.35	7.07	8	55	1.42	1.51	0.75	0.4	7.3	8.9	9.4	9.7	10	10.6
	2015-10-04	1.5	6	1.35	7.07	10	55	1.42	1.54	0.75	0.4	6.6	10.9	11.5	11.9	12.2	12.9
	2015-12-04	1.5	6	1.35	7.07	12	55	1.42	1.57	0.75	0.4	6	13	13.6	14	14.4	15.4
	2015-15-09	1.5	6	1.35	3.89	15	60	1.42	1.85	0.75	0.9	5.4	-	16.4	17.2	17.8	18.8
	2015-20-09	1.5	6	1.35	3.89	20	65	1.42	2.01	0.75	0.9	4.5	-	21.4	22.4	23.2	24.7
	2015-30-09	1.5	6	1.35	3.89	30	75	1.42	2.32	0.75	0.9	3.4	-	31.5	32.9	33.7	37
	2018-4-04	1.8	6	1.6	4.38	4	50	1.73	1.76	0.9	0.4	9.2	4.6	4.8	4.9	5.1	5.4
	2018-8-04	1.8	6	1.6	6.61	8	50	1.73	1.82	0.9	0.4	7.1	8.6	9	9.2	9.4	10.2
	2018-12-04	1.8	6	1.6	6.61	12	55	1.73	1.88	0.9	0.4	5.8	12.9	13.5	14	14.4	15.4
	2018-16-04	1.8	6	1.6	6.61	16	60	1.73	1.93	0.9	0.4	4.9	17	17.7	18.3	18.7	20.5
	2018-20-04	1.8	6	1.6	6.61	20	65	1.73	1.99	0.9	0.4	4.3	21.2	22.3	23	23.6	25.6
	2018-24-04	1.8	6	1.6	6.61	24	65	1.73	2.04	0.9	0.4	3.8	25.3	26.5	27.3	27.9	30.8

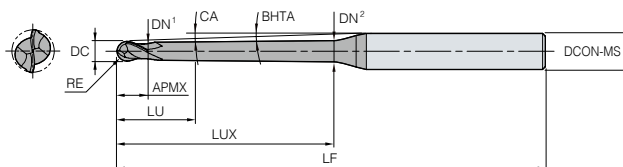
ESTNB20

2 flutes tapered neck type ball



DC6 or below Above DC6

DC	Tolerance
Ø0.2 ~ Ø6	0.000 ~ -0.012
Ø8 ~ Ø10	0.000 ~ -0.015

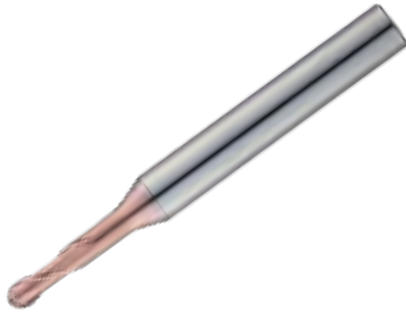


(mm)

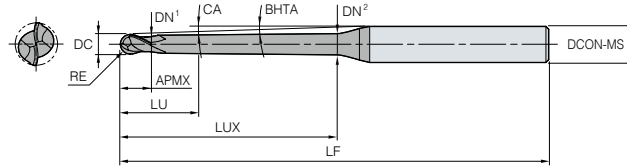
	Designation	DC	DCON-MS	APMX	LU	LUX	LF	DN ¹	DN ²	RE	BHTA(°)	CA	Effective length by inclination angle				
													0.5°	1°	1.5°	2°	3°
ESTNB	2018-28-04	1.8	6	1.6	6.61	28	70	1.73	2.1	0.9	0.4	3.4	29.4	30.6	31.5	32.4	35.9
	2018-32-04	1.8	6	1.6	6.61	32	70	1.73	2.15	0.9	0.4	3	33.4	34.8	35.7	37.1	-
	2018-36-04	1.8	6	1.6	6.61	36	75	1.73	2.21	0.9	0.4	2.8	37.5	38.9	39.9	41.7	-
	2018-38-04	1.8	6	1.6	6.61	38	80	1.73	2.24	0.9	0.4	2.7	39.5	41	42	44	-
	2018-40-04	1.8	6	1.6	6.61	40	80	1.73	2.27	0.9	0.4	2.6	41.5	43.1	44.2	46.3	-
	2020-8-04	2	6	1.7	7.42	8	50	1.92	2.01	1	0.4	7	8.7	9	9.2	9.5	10.2
	2020-12-04	2	6	1.7	7.42	12	55	1.92	2.06	1	0.4	5.7	13	13.6	14	14.4	15.4
	2020-16-04	2	6	1.7	7.42	16	60	1.92	2.12	1	0.4	4.8	17	17.7	18.3	18.7	20.5
	2020-20-04	2	6	1.7	7.42	20	65	1.92	2.18	1	0.4	4.1	21.3	22.3	23	23.6	25.6
	2020-20-09	2	6	1.7	4.24	20	65	1.92	2.5	1	0.9	4.2	-	21.4	22.4	23.2	24.6
	2020-25-09	2	6	1.7	4.24	25	65	1.92	2.65	1	0.9	3.6	-	26.5	27.7	28.5	30.8
	2020-30-04	2	6	1.7	7.42	30	70	1.92	2.32	1	0.4	3.1	31.4	32.7	33.6	34.8	38.5
	2020-30-09	2	6	1.7	4.24	30	70	1.92	2.81	1	0.9	3.2	-	31.6	32.9	33.7	36.9
	2020-35-09	2	6	1.7	4.24	35	75	1.92	2.97	1	0.9	2.8	-	36.6	38	39	-
	2020-40-04	2	6	1.7	7.42	40	80	1.92	2.46	1	0.4	2.5	41.5	43.1	44.2	46.3	-
	2020-40-09	2	6	1.7	4.24	40	80	1.92	3.12	1	0.9	2.6	-	41.7	43.2	44.5	-
	2020-50-09	2	6	1.7	4.24	50	90	1.92	3.44	1	0.9	2.1	-	51.5	53.5	55.5	-
	2020-60-09	2	6	1.7	4.24	60	100	1.92	3.75	1	0.9	1.8	-	61.9	63.8	-	-
	2020-70-09	2	6	1.7	4.24	70	110	1.92	4.07	1	0.9	1.8	-	72	74.1	-	-
	2030-8-04	3	6	2.5	8.5	8	50	2.86	2.94	1.5	0.4	6.3	8.8	9.1	9.3	9.5	10.3
	2030-16-04	3	6	2.5	12.52	16	55	2.86	3.05	1.5	0.4	4.1	17.2	17.8	18.3	18.7	20.6
	2030-20-04	3	6	2.5	12.52	20	60	2.86	3.1	1.5	0.4	3.4	21.2	22	22.6	23.3	25.7
	2030-30-04	3	6	2.5	12.52	30	70	2.86	3.24	1.5	0.4	2.5	31.6	32.8	33.7	34.9	-
	2030-30-09	3	6	2.5	6.95	30	70	2.86	3.72	1.5	0.9	2.6	-	31.8	33	33.8	-
	2030-40-04	3	6	2.5	12.52	40	80	2.86	3.38	1.5	0.4	2	41.7	43.2	44.3	-	-
	2030-40-09	3	6	2.5	6.95	40	80	2.86	4.04	1.5	0.9	2	-	41.9	43.3	-	-
	2030-50-09	3	6	2.5	6.95	50	90	2.86	4.35	1.5	0.9	1.7	-	52	53.6	-	-
	2030-60-09	3	6	2.5	6.95	60	100	2.86	4.67	1.5	0.9	1.4	-	62.1	-	-	-
	2030-70-09	3	6	2.5	6.95	70	110	2.86	4.98	1.5	0.9	1.2	-	72.1	-	-	-
	2040-20-10	4	8	8	12.01	20	70	3.86	4.28	2	1	5	20.5	21.6	22.3	22.8	23.5
	2040-30-10	4	8	8	12.01	30	80	3.86	4.63	2	1	3.51	22	31.6	32.5	33.2	34.16
	2040-40-10	4	8	8	12.01	40	90	3.86	4.98	2	1	2.7	22	42	43.4	44.3	-
	2040-50-10	4	8	8	12.01	50	100	3.86	5.33	2	1	2.2	22	52	53.6	54.7	-
	2040-60-10	4	8	8	12.01	60	110	3.86	5.68	2	1	1.9	22	62	63.8	-	-
	2050-30-10	5	8	10	14.01	30	80	4.86	5.56	2.5	1	2.8	25.5	31.7	32.6	33.2	-
	2050-40-10	5	8	10	14.01	40	90	4.86	5.91	2.5	1	2.1	25.5	41.7	42.8	43.5	-

ESTNB20

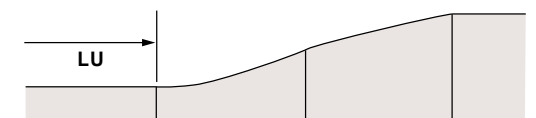
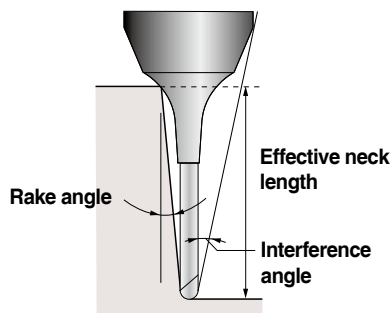
2 flutes tapered neck type ball



CARBIDE	2	H-A 30°	Grade PC305H	h5 shank	RE ±0.01	RE ±0.015	DC	Tolerance
					DC6 or below	Above DC6	Ø0.2 ~ Ø6 Ø8 ~ Ø10	0.000 ~ -0.012 0.000 ~ -0.015



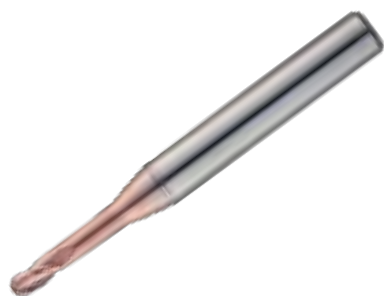
													(mm)				
ESTNB	Designation	DC	DCON-MS	APMX	LU	LUX	LF	DN1	DN2	RE	BHTA(°)	CA	Effective length by inclination angle				
													0.5°	1°	1.5°	2°	3°
	2050-60-10	5	8	10	14.01	60	110	4.86	6.61	2.5	1	1.5	25.5	62.1	-	-	-
	2060-30-10	6	8	12	16.01	30	80	5.86	6.49	3	1	1.9	29	31.8	32.6	-	-
	2060-40-10	6	8	12	16.01	40	90	5.86	6.84	3	1	1.5	29	41.8	-	-	-
	2060-50-10	6	8	12	16.01	50	100	5.86	7.19	3	1	1.2	29	51.8	-	-	-
	2060-60-10	6	10	12	16.01	60	110	5.86	7.54	3	1	1.9	29	62.2	63.9	-	-
	2060-70-10	6	10	12	16.01	70	120	5.86	7.89	3	1	1.7	29	72.2	74.1	-	-
	2060-80-10	6	10	12	16.01	80	130	5.86	8.23	3	1	1.5	29	82.2	-	-	-
	2080-50-10	8	10	14	18.01	50	110	7.86	9.12	4	1	1.2	32	51.9	-	-	-
	2080-60-10	8	10	14	18.01	60	120	7.86	9.47	4	1	1	32	-	-	-	-
	2080-70-10	8	10	14	18.01	70	130	7.86	9.82	4	1	0.9	32	-	-	-	-
	2080-80-10	8	12	14	18.01	80	140	7.86	10.16	4	1	1.5	32	82.3	-	-	-
	2100-60-10	10	12	18	22.01	60	130	9.86	11.33	5	1	1.1	39	62.1	-	-	-
	2100-75-10	10	12	18	22.01	75	140	9.86	11.85	5	1	0.9	39	-	-	-	-



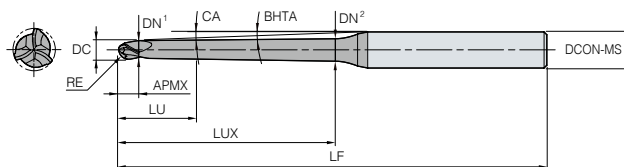
※ The marked effective neck length is the default value to prevent interference with the workpiece.
Proper control of the processing environment is required.

ESTNB30

3 flutes tapered neck type ball



CARBIDE	3	H-A 30°	Grade PC305H	h5 shank	RE ±0.005	DC ALL	Tolerance 0.000 ~ -0.015
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(mm)

	Designation	DC	DCON-MS	APMX	LU	LUX	LF	DN ¹	DN ²	RE	BHTA(°)	CA	Effective length by inclination angle				
													0.5°	1°	1.5°	2°	3°
ESTNB	3020-8-04	2	6	1.7	7.42	8	50	1.92	2.01	1	0.4	7	8.7	9	9.2	9.5	10.2
	3020-12-04	2	6	1.7	7.42	12	55	1.92	2.06	1	0.4	5.7	13	13.6	14	14.4	15.4
	3020-16-04	2	6	1.7	7.42	16	60	1.92	2.12	1	0.4	4.8	17	17.7	18.3	18.7	20.5
	3020-20-04	2	6	1.7	7.42	20	65	1.92	2.18	1	0.4	4.1	21.3	22.3	23	23.6	25.6
	3020-20-09	2	6	1.7	4.24	20	65	1.92	2.5	1	0.9	4.2	-	21.4	22.4	23.2	24.6
	3020-25-09	2	6	1.7	4.24	25	65	1.92	2.65	1	0.9	3.6	-	26.5	27.7	28.5	30.8
	3020-30-04	2	6	1.7	7.42	30	70	1.92	2.32	1	0.4	3.1	31.4	32.7	33.6	34.8	38.5
	3020-30-09	2	6	1.7	4.24	30	70	1.92	2.81	1	0.9	3.2	-	31.6	32.9	33.7	36.9
	3020-35-09	2	6	1.7	4.24	35	75	1.92	2.97	1	0.9	2.8	-	36.6	38	39	-
	3020-40-04	2	6	1.7	7.42	40	80	1.92	2.46	1	0.4	2.5	41.5	43.1	44.2	46.3	-
	3020-40-09	2	6	1.7	4.24	40	80	1.92	3.12	1	0.9	2.6	-	41.7	43.2	44.5	-
	3020-50-09	2	6	1.7	4.24	50	90	1.92	3.44	1	0.9	2.1	-	51.8	53.5	55.5	-
	3020-60-09	2	6	1.7	4.24	60	100	1.92	3.75	1	0.9	1.8	-	61.9	63.8	-	-
	3020-70-09	2	6	1.7	4.24	70	110	1.92	4.07	1	0.9	1.6	-	72	74.1	-	-
	3030-8-04	3	6	2.5	8.5	8	50	2.86	2.94	1.5	0.4	6.3	8.8	9.1	9.3	9.5	10.3
	3030-16-04	3	6	2.5	12.52	16	55	2.86	3.05	1.5	0.4	4.1	17.2	17.8	18.3	18.7	20.6
	3030-20-04	3	6	2.5	12.52	20	60	2.86	3.1	1.5	0.4	3.4	21.2	22	22.6	23.3	25.7
	3030-30-04	3	6	2.5	12.52	30	70	2.86	3.24	1.5	0.4	2.5	31.6	32.8	33.7	34.9	-
	3030-30-09	3	6	2.5	6.95	30	70	2.86	3.72	1.5	0.9	2.6	-	31.8	33	33.8	-
	3030-40-04	3	6	2.5	12.52	40	80	2.86	3.38	1.5	0.4	2	41.7	43.2	44.3	-	-
	3030-40-09	3	6	2.5	6.95	40	80	2.86	4.04	1.5	0.9	2	-	41.9	43.3	-	-
	3030-50-09	3	6	2.5	6.95	50	90	2.86	4.35	1.5	0.9	1.7	-	52	53.6	-	-
	3030-60-09	3	6	2.5	6.95	60	100	2.86	4.67	1.5	0.9	1.4	-	62.1	-	-	-
	3030-70-09	3	6	2.5	6.95	70	110	2.86	4.98	1.5	0.9	1.2	-	72.1	-	-	-
	3040-20-10	4	8	8	12.01	20	70	3.86	4.28	2	1	5	20.5	21.6	22.3	22.8	23.5
	3040-30-10	4	8	8	12.01	30	80	3.86	4.63	2	1	3.6	22	31.6	32.5	33.2	34.1
	3040-40-10	4	8	8	12.01	40	90	3.86	4.98	2	1	2.7	22	42	43.4	44.3	-
	3040-50-10	4	8	8	12.01	50	100	3.86	5.33	2	1	2.2	22	52	53.6	54.7	-
	3040-60-10	4	8	8	12.01	60	110	3.86	5.68	2	1	1.9	22	62	63.8	-	-
	3050-30-10	5	8	10	14.01	30	80	4.86	5.56	2.5	1	2.8	25.5	31.7	32.6	33.2	-
	3050-40-10	5	8	10	14.01	40	90	4.86	5.91	2.5	1	2.1	25.5	41.7	42.8	43.5	-
	3050-60-10	5	8	10	12.52	60	110	4.86	6.61	2.5	1	1.5	25.5	62.1	-	-	-

For medium hardness steel(HRC30~50)

U-Star Endmill

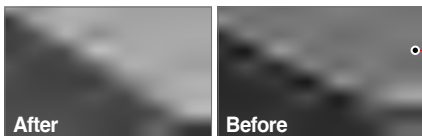
- Enhanced cutting edge strength of ball Endmill applying ultra-fine substrate (PC303W)
- Higher chipping resistance of flat Endmill applying high toughness substrate (PC315W)
- Various shaped line-ups for complicated mold machining
- Suitable for precision cutting with high precision Range of h5 shank, flute and radius

Code system

U	R	51	2	010	10	50	100	S3
U-Star Endmill		Length, Shank type		Tool diameter	Corner R		Overall length	
		50: Straight 51: Neck 52: Long shank 53: Lollipop 54: Taper neck		10: Ø1mm 60: Ø6mm 250: Ø25mm	5: 0.05mm 10: 0.1mm 20: 0.2mm		50: 50mm 100: 100mm 150: 150mm	
Type			No. of flute			Effective length		Shank diameter
B: Ball SB: Staright ball E: Flat R: Radius SPM: High feed XE: Heavy cut flat XR: Multi helix radius TE: Taper flat TR: Taper radius TB: Taper ball DR: Double radius F: Roughing LE: Flat (lathe)			2: 2 flutes 3: 3 flutes 4: 4 flutes 5: 5 flutes 6: 6 flutes			5: 0.5mm 10: 10mm 50: 50mm		S3: Ø3mm S6: Ø6mm S20: Ø20mm

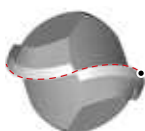
Features

• Edge treatment



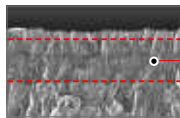
- Enhanced chipping resistance in the beginning of cutting
- Guiding stable cutting for managing the properties of mold machining

• Applying S-curved gash shape



- Increased cutting performance and wear resistance due to dispersing cutting force

• AlCrN base new coating



- Increased wear and oxidation resistance due to multi layer
- Enhanced lubrication with Cr containing
- Stable cutting under frictional heat

• Grade PC315W

• Applying substrate for medium hardness steel cutting



- Separating the substrate (PC303W and PC315W) maximizes the features of tool and ensures general use

• Grade PC303W

B Technical Information for U-Star Endmill

Line-up

Designation	Picture	Product name	Range	Irregular flute spacing	Page
UE502		2 flutes flat Endmill	Ø0.1 ~ 25		B94 ~ B95
UE512		2 flutes long neck flat Endmill	Ø0.1 ~ 12.		B96 ~ B98
UE522		2 flutes flat Endmill	Ø1 ~ 25		B99 ~ B100
UXE502		2 flutes flat for heavy cuts	Ø0.1 ~ 20		B101 ~ 102
UE504H		4 flutes 45° helix flat Endmill	Ø1 ~ 20		B103
UE514		4 flutes long neck flat Endmill	Ø1 ~ 12		B104 ~ B105
UE524		4 flutes flat Endmill	Ø1 ~ 25		B106 ~ B107
ULE504		4 flutes automatic cnc flat Endmill	Ø3 ~ 16		B108
UE504		4 flutes flat Endmill	Ø0.8 ~ 25	●	B109
UXE504		4 flutes flat for heavy cuts	Ø1 ~ 20	●	B110
UE506		6 flutes flat Endmill	Ø6 ~ 25		B111
UTE502		2 flutes tapered flat Endmill	Ø0.3 ~ 10		B112 ~ B113
UTE504		4 flutes tapered flat Endmill	Ø0.8 ~ 10		B114 ~ B116
UR502		2 flutes radius Endmill	Ø0.2 ~ 20		B117 ~ B119
UR512		2 flutes neck type radius Endmill	Ø0.2 ~ 20		B120 ~ B125
UR542		2 flutes tapered neck radius Endmill	Ø0.2 ~ 4		B126 ~ B131
UR504		4 flutes radius Endmill	Ø3 ~ 20	●	B132
UR544		4 flutes tapered neck radius Endmill	Ø1 ~ 4		B133 ~ B136
UXR504		4 flutes multi helix radius Endmill	Ø1 ~ 20	●	B137 ~ B139
UXR514		4 flutes multi helix neck radius Endmill	Ø1 ~ 20	●	B140 ~ B144
UR506		6 flutes radius Endmill	Ø6 ~ 20		B145
UDR503		3 flutes double radius Endmill	Ø6 ~ 20		B146
USPM4		4 flutes radius for high speed machining	Ø1 ~ 20		B147
UTR504		4 flutes tapered radius Endmill	Ø0.8 ~ 2.5		B148 ~ B151
UB502		2 flutes ball Endmill	Ø0.1 ~ 25		B152 ~ B153
UB502P		2 flutes high precision ball Endmill	Ø0.1 ~ 12		B154
UB512		2 flutes long neck ball Endmill	Ø0.1 ~ 12		B155 ~ B158
UB512S6		2 flutes long neck ball Endmill (shank 6)	Ø0.5 ~ 2		B159
UB532		2 flutes lollipop style ball Endmill	Ø3 ~ 12		B160
UB542		2 flutes tapered neck ball Endmill	Ø0.1 ~ 12		B161 ~ B167
USB502		2 flutes straight ball Endmill	Ø3 ~ 20		B168
UB503		3 flutes ball Endmill	Ø1 ~ 12		B169
UB504		4 flutes ball Endmill	Ø1 ~ 12		B169
UTB502		2 flutes tapered ball Endmill	Ø0.3 ~ 2		B170 ~ B171
UF50		3~5 flutes chamfer pitch roughing Endmill	Ø3 ~ 25		B172
UF51		3~5 flutes fine pitch roughing Endmill	Ø3 ~ 25		B173
UF51H		3~5 flutes 45° helix fine pitch roughing Endmill	Ø3 ~ 25		B174

UE502

2 flutes flat



CARBIDE

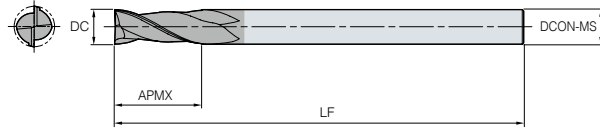
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H-A
30°

Grade
PC315W

h5
shank

DC	Tolerance
Ø0.1 ~ Ø6	0.000 ~ -0.012
Ø6.5 ~ Ø25	0.000 ~ -0.015



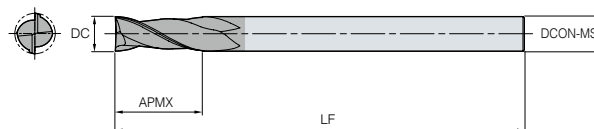
											(mm)
	Designation	DC	DCON-MS	APMX	LF		Designation	DC	DCON-MS	APMX	LF
UE	502001S3	0.1	3	0.2	40	UE	502015S3	1.5	3	4	50
	502001	0.1	4	0.2	40		502015S4	1.5	4	4	50
	5020015	0.15	4	0.3	40		502015	1.5	6	4	50
	502002S3	0.2	3	0.4	40		502016S4	1.6	4	4	50
	502002	0.2	4	0.4	40		502017S4	1.7	4	4	50
	5020025	0.25	4	0.5	40		502018S4	1.8	4	5	50
	502003S3	0.3	3	0.6	40		502019S4	1.9	4	5	50
	502003	0.3	4	0.6	40		502020S3	2	3	6	50
	5020035	0.35	4	0.7	40		502020S4	2	4	6	50
	502004S3	0.4	3	0.8	40		502020	2	6	6	50
	502004	0.4	4	0.8	40		502021S4	2.1	4	6	50
	5020045	0.45	4	0.9	40		502022S4	2.2	4	6	50
	502005S3	0.5	3	1	40		502023S4	2.3	4	6	50
	502005	0.5	4	1	40		502024S4	2.4	4	6	50
	5020055	0.55	4	1.1	40		502025S3	2.5	3	7	50
	502006S3	0.6	3	1.2	40		502025	2.5	6	7	50
	502006	0.6	4	1.2	40		502025S4	2.5	4	8	50
	5020065	0.65	4	1.3	40		502026S4	2.6	4	8	50
	502007S3	0.7	3	1.4	40		502027S4	2.7	4	8	50
	502007	0.7	4	1.4	40		502028S4	2.8	4	8	50
	5020075	0.75	4	1.5	40		502029S4	2.9	4	8	50
	502008S3	0.8	3	1.6	40		502030S3	3	3	8	50
	502008	0.8	4	1.6	40		502030S4	3	4	8	50
	5020085	0.85	4	1.7	40		502030	3	6	8	50
	502009S3	0.9	3	1.8	40		502035S4	3.5	4	10	50
	502009	0.9	4	1.8	40		502035	3.5	6	10	50
	5020095	0.95	4	2	40		502040080S4	4	4	10	80
	502010S3	1	3	2.5	50		502040S4	4	4	10	50
	502010S4	1	4	2.5	50		502040	4	6	10	50
	502010	1	6	2.5	50		502045	4.5	6	14	50
	502011S4	1.1	4	3	50		502050	5	6	15	60
	502012S3	1.2	3	3	50		502055	5.5	6	15	60
	502012S4	1.2	4	3	50		502060	6	6	15	60
	502012	1.2	6	3	50		502065	6.5	8	18	60
	502013S4	1.3	4	3	50		502070	7	8	20	60
	502014S4	1.4	4	4	50		502075	7.5	8	20	60

UE502

2 flutes flat



CARBIDE	2	H-A 30°	Grade PC315W	h5 shank	DC	Tolerance
					Ø0.1 ~ Ø6	0.000 ~ -0.012
					Ø6.5 ~ Ø25	0.000 ~ -0.015



(mm)

	Designation	DC	DCON-MS	APMX	LF
UE	502080	8	8	20	70
	502085	8.5	10	22	70
	502090	9	10	22	70
	502095	9.5	10	24	70
	502100	10	10	25	75
	502105	10.5	12	26	75
	502110	11	12	30	75
	502115	11.5	12	30	80
	502120	12	12	30	80
	502130	13	12	35	100
	502140S16	14	16	35	100
	502140	14	14	35	100
	502140S12	14	12	35	100

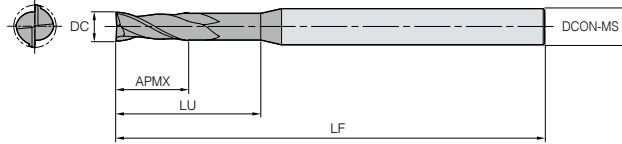
	Designation	DC	DCON-MS	APMX	LF
UE	502150	15	16	38	100
	502160	16	16	40	100
	502170	17	16	42	100
	502180	18	18	45	100
	502180S16	18	16	45	100
	502190	19	20	45	100
	502200	20	20	45	100
	502210	21	20	45	100
	502220	22	20	45	100
	502230	23	25	50	120
	502240	24	25	50	120
	502250	25	25	50	120

UE512

2 flutes long neck flat



CARBIDE	2	H-A 30°	Grade PC315W	h5 shank	DC Ø0.1 ~ Ø6 Ø8 ~ Ø12	Tolerance 0.000 ~ -0.012 0.000 ~ -0.015
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		(mm)											
Designation		DC	DCON-MS	APMX	LU	LF	Designation		DC	DCON-MS	APMX	LU	LF
UE	512001003	0.1	4	0.15	0.3	40	UE	51200602	0.6	4	0.9	2	45
	512001005	0.1	4	0.15	0.5	40		51200603	0.6	4	0.9	3	45
	51200101	0.1	4	0.15	1	40		51200604	0.6	4	0.9	4	45
	512002005	0.2	4	0.3	0.5	40		51200605	0.6	4	0.9	5	45
	51200201	0.2	4	0.3	1	40		51200606	0.6	4	0.9	6	45
	512002015	0.2	4	0.3	1.5	40		51200608	0.6	4	0.9	8	45
	51200202	0.2	4	0.3	2	40		51200610	0.6	4	0.9	10	45
	51200301	0.3	4	0.5	1	40		51200612	0.6	4	0.9	12	45
	512003015	0.3	4	0.5	1.5	40		51200614	0.6	4	0.9	14	45
	51200302	0.3	4	0.5	2	40		51200616	0.6	4	0.9	16	45
	512003025	0.3	4	0.5	2.5	40		51200702	0.7	4	1.2	2	45
	51200303	0.3	4	0.5	3	40		51200704	0.7	4	1.2	4	45
	51200304	0.3	4	0.5	4	40		51200706	0.7	4	1.2	6	45
	51200305	0.3	4	0.5	5	40		51200708	0.7	4	1.2	8	45
	51200401	0.4	4	0.6	1	40		51200710	0.7	4	1.2	10	45
	512004015	0.4	4	0.6	1.5	40		51200712	0.7	4	1.2	12	45
	51200402	0.4	4	0.6	2	40		51200802	0.8	4	1.2	2	45
	512004025	0.4	4	0.6	2.5	40		51200803	0.8	4	1.2	3	45
	51200403	0.4	4	0.6	3	40		51200804	0.8	4	1.2	4	45
	51200404	0.4	4	0.6	4	40		51200805	0.8	4	1.2	5	45
	51200405	0.4	4	0.6	5	40		51200806	0.8	4	1.2	6	45
	51200406	0.4	4	0.6	6	40		51200808	0.8	4	1.2	8	45
	51200408	0.4	4	0.6	8	40		51200810	0.8	4	1.2	10	45
	51200410	0.4	4	0.6	10	40		51200812	0.8	4	1.2	12	45
	51200501	0.5	4	0.7	1	45		51200814	0.8	4	1.2	14	45
	512005015	0.5	4	0.7	1.5	45		51200816	0.8	4	1.2	16	45
	51200502	0.5	4	0.7	2	45		51200820	0.8	4	1.2	20	45
	512005025	0.5	4	0.7	2.5	45		51200906	0.9	4	1.3	6	45
	51200503	0.5	4	0.7	3	45		51200908	0.9	4	1.3	8	45
	51200504	0.5	4	0.7	4	45		51200910	0.9	4	1.3	10	45
	51200505	0.5	4	0.7	5	45		51201002	1	4	1.5	2	50
	51200506	0.5	4	0.7	6	45		51201003	1	4	1.5	3	50
	51200508	0.5	4	0.7	8	45		51201004	1	4	1.5	4	50
	51200510	0.5	4	0.7	10	45		51201005	1	4	1.5	5	50
	51200512	0.5	4	0.7	12	45		51201006	1	4	1.5	6	50
	51200514	0.5	4	0.7	14	45		51201007	1	4	1.5	7	50
	51200516	0.5	4	0.7	16	45		51201008	1	4	1.5	8	50

UE512

2 flutes long neck flat



CARBIDE

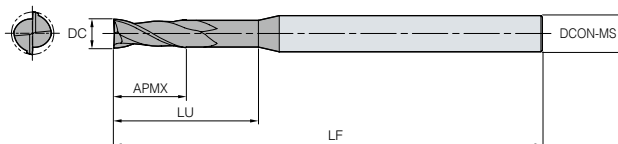
2

H-A
30°

Grade
PC315W

h5 shank

DC	Tolerance
Ø0.1 ~ Ø6	0.000 ~ -0.012
Ø8 ~ Ø12	0.000 ~ -0.015



(mm)

	Designation	DC	DCON-MS	APMX	LU	LF
UE	51201010	1	4	1.5	10	50
	51201012	1	4	1.5	12	50
	51201014	1	4	1.5	14	50
	51201016	1	4	1.5	16	50
	51201018	1	4	1.5	18	50
	51201020	1	4	1.5	20	50
	51201022	1	4	1.5	22	60
	51201026	1	4	1.5	26	60
	51201030	1	4	1.5	30	70
	51201040	1	4	1.5	40	80
	51201050	1	4	1.5	50	100
	51201204	1.2	4	1.8	4	50
	51201206	1.2	4	1.8	6	50
	51201208	1.2	4	1.8	8	50
	51201210	1.2	4	1.8	10	50
	51201212	1.2	4	1.8	12	50
	51201214	1.2	4	1.8	14	50
	51201216	1.2	4	1.8	16	50
	51201220	1.2	4	1.8	20	50
	51201226	1.2	4	1.8	26	60
	51201230	1.2	4	1.8	30	70
	51201406	1.4	4	2.1	6	50
	51201408	1.4	4	2.1	8	50
	51201410	1.4	4	2.1	10	50
	51201414	1.4	4	2.1	14	50
	51201416	1.4	4	2.1	16	50
	51201420	1.4	4	2.1	20	50
	51201504	1.5	4	2.3	4	50
	51201505	1.5	4	2.3	5	50
	51201506	1.5	4	2.3	6	50
	51201507	1.5	4	2.3	7	50
	51201508	1.5	4	2.3	8	50
	51201510	1.5	4	2.3	10	50
	51201512	1.5	4	2.3	12	50
	51201514	1.5	4	2.3	14	50
	51201516	1.5	4	2.3	16	50
	51201518	1.5	4	2.3	18	50

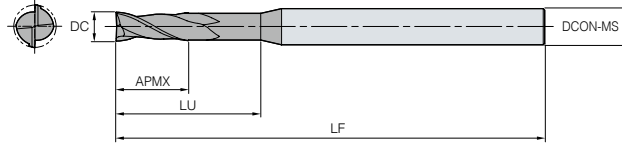
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UE	51201520	1.5	4	2.3	20	50
	51201522	1.5	4	2.3	22	60
	51201526	1.5	4	2.3	26	60
	51201530	1.5	4	2.3	30	70
	51201608	1.6	4	2.3	8	50
	51201610	1.6	4	2.3	10	50
	51201612	1.6	4	2.3	12	50
	51201616	1.6	4	2.3	16	50
	51201620	1.6	4	2.3	20	50
	51201808	1.8	4	2.7	8	50
	51201810	1.8	4	2.7	10	50
	51201812	1.8	4	2.7	12	50
	51201816	1.8	4	2.7	16	50
	51201820	1.8	4	2.7	20	50
	51202006	2	4	3	6	50
	51202008	2	4	3	8	50
	51202010	2	4	3	10	50
	51202012	2	4	3	12	50
	51202014	2	4	3	14	50
	51202016	2	4	3	16	50
	51202018	2	4	3	18	50
	51202020	2	4	3	20	50
	51202022	2	4	3	22	60
	51202026	2	4	3	26	60
	51202030	2	4	3	30	70
	51202035	2	4	3	35	70
	51202040	2	4	3	40	80
	51202045	2	4	3	45	90
	51202050	2	4	3	50	100
	51202060	2	4	3	60	110
	51202508	2.5	4	4	8	50
	51202510	2.5	4	4	10	50
	51202512	2.5	4	4	12	50
	51202514	2.5	4	4	14	50
	51202516	2.5	4	4	16	50
	51202518	2.5	4	4	18	50
	51202520	2.5	4	4	20	50

UE512

2 flutes long neck flat



CARBIDE	2	H-A 30°	Grade PC315W	h5 shank	DC Ø0.1 ~ Ø6 Ø8 ~ Ø12	Tolerance 0.000 ~ -0.012 0.000 ~ -0.015
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							(mm)						
Designation		DC	DCON-MS	APMX	LU	LF	Designation		DC	DCON-MS	APMX	LU	LF
UE	51202522	2.5	4	4	22	60	UE	51204022	4	6	6	22	65
	51202526	2.5	4	4	26	60		51204026	4	6	6	26	65
	51202530	2.5	4	4	30	70		51204030	4	6	6	30	70
	51202535	2.5	4	4	35	70		51204035	4	6	6	35	70
	51202540	2.5	4	4	40	80		51204040	4	6	6	40	80
	51202545	2.5	4	4	45	90		51204045	4	6	6	45	90
	51202550	2.5	4	4	50	100		51204050	4	6	6	50	100
	51203006	3	6	4.5	6	50		51204060	4	6	6	60	100
	51203008	3	6	4.5	8	50		51205016	5	6	8	16	60
	51203010	3	6	4.5	10	50		51205020	5	6	8	20	60
	51203012	3	6	4.5	12	50		51205026	5	6	8	26	65
	51203014	3	6	4.5	14	60		51205030	5	6	8	30	70
	51203016	3	6	4.5	16	60		51205035	5	6	8	35	75
	51203018	3	6	4.5	18	60		51205040	5	6	8	40	80
	51203020	3	6	4.5	20	60		51205050	5	6	8	50	90
	51203022	3	6	4.5	22	65		51205060	5	6	8	60	100
	51203026	3	6	4.5	26	65		51206015	6	6	9	15	60
	51203030	3	6	4.5	30	70		51206020	6	6	9	20	60
	51203035	3	6	4.5	35	70		51206030	6	6	9	30	70
	51203040	3	6	4.5	40	80		51206032	6	6	9	32	90
	51203045	3	6	4.5	45	90		51208025	8	8	12	25	70
	51203050	3	6	4.5	50	100		51208030	8	8	12	30	80
	51203060	3	6	4.5	60	100		51208042	8	8	12	42	100
	51204008	4	6	6	8	50		51210030	10	10	15	30	75
	51204010	4	6	6	10	50		51210035	10	10	15	35	80
	51204012	4	6	6	12	50		51210045	10	10	15	45	100
	51204014	4	6	6	14	60		51212035	12	12	20	35	80
	51204016	4	6	6	16	60		51212040	12	12	20	40	90
	51204018	4	6	6	18	60		51212050	12	12	20	50	110
	51204020	4	6	6	20	60							

UE522

2 flutes flat



CARBIDE

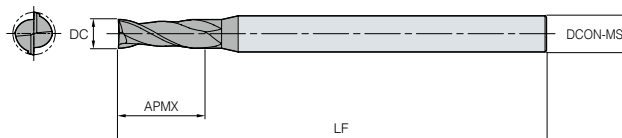
2

H-A
30°

Grade
PC315W

h5
shank

DC	Tolerance
Ø1 ~ Ø6	0.000 ~ -0.012
Ø6.5 ~ Ø25	0.000 ~ -0.015



(mm)

	Designation	DC	DCON-MS	APMX	LF
UE	52201003	1	6	3	60
	52201004	1	6	4	60
	52201005	1	6	5	60
	52201006	1	6	6	60
	52201007	1	6	7	60
	52201008	1	6	8	60
	52201010	1	6	10	60
	52201012	1	6	12	60
	52201204	1.2	6	4	60
	52201206	1.2	6	6	60
	52201208	1.2	6	8	60
	52201210	1.2	6	10	60
	52201212	1.2	6	12	60
	52201506	1.5	6	6	60
	52201508	1.5	6	8	60
	52201510	1.5	6	10	60
	52201512	1.5	6	12	60
	52201514	1.5	6	14	60
	52201516	1.5	6	16	60
	52202008	2	6	8	60
	52202010	2	6	10	60
	52202012	2	6	12	60
	52202014	2	6	14	60
	52202016	2	6	16	60
	52202510	2.5	6	10	60
	52202512	2.5	6	12	60
	52202516	2.5	6	16	60
	52202520	2.5	6	20	60
	52202526	2.5	6	26	60
	52203010	3	6	10	70
	52203012	3	6	12	70
	52203014	3	6	14	70
	52203016	3	6	16	70
	52203016S3	3	3	16	100
	52203020	3	6	20	70
	52203026	3	6	26	70
	52203030	3	6	30	70

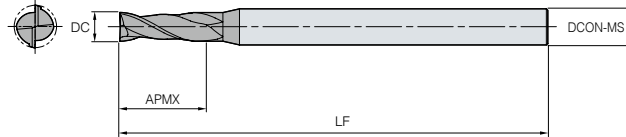
	Designation	DC	DCON-MS	APMX	LF
UE	52204012	4	6	12	70
	52204016	4	6	16	70
	52204020	4	6	20	70
	52204020S4	4	4	20	100
	52204026	4	6	26	70
	52204030	4	6	30	70
	52205015	5	6	15	70
	52205020	5	6	20	70
	52205025100	5	6	25	100
	52205025	5	6	25	70
	52205030	5	6	30	80
	52205035	5	6	35	90
	52205040	5	6	40	100
	52206015080	6	6	15	80
	52206015	6	6	15	60
	52206020090	6	6	20	90
	52206020	6	6	20	70
	52206025	6	6	25	75
	52206030150	6	6	30	150
	52206030100	6	6	30	100
	52206030	6	6	30	80
	52206035	6	6	35	90
	52206040120	6	6	40	120
	52206040	6	6	40	90
	52206045	6	6	45	150
	52207035	7	8	35	85
	52208020	8	8	20	100
	52208025	8	8	25	80
	52208030100	8	8	30	100
	52208030	8	8	30	80
	52208035	8	8	35	90
	52208040150	8	8	40	150
	52208040120	8	8	40	120
	52208040	8	8	40	90
	52208045	8	8	45	100
	52208050150	8	8	50	150
	52208050	8	8	50	100

UE522

2 flutes flat



CARBIDE	2	H-A 30°	Grade PC315W	h5 shank	DC Ø1 ~ Ø6 Ø6.5 ~ Ø25	Tolerance 0.000 ~ -0.012 0.000 ~ -0.015
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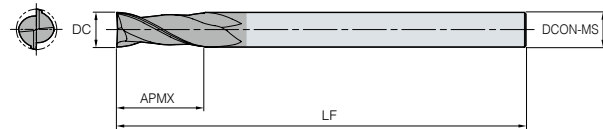
						(mm)					
Designation		DC	DCON-MS	APMX	LF	Designation		DC	DCON-MS	APMX	LF
UE	52209045	9	10	45	100	UE	52214060	14	16	60	120
	52210025	10	10	25	100		52216040150	16	16	40	150
	52210030100	10	10	30	100		52216040	16	16	40	120
	52210030	10	10	30	80		52216050150	16	16	50	150
	52210035	10	10	35	90		52216050	16	16	50	110
	52210040120	10	10	40	120		52216060	16	16	60	120
	52210040	10	10	40	90		52216070200	16	16	70	200
	52210045	10	10	45	100		52216070150	16	16	70	150
	52210050200	10	10	50	200		52216070	16	16	70	130
	52210050150	10	10	50	150		52216080	16	16	80	150
	52210050	10	10	50	100		52216090	16	16	90	150
	52210055	10	10	55	150		522160110	16	16	110	200
	52210060200	10	10	60	200		522160120	16	16	120	250
	52210060155	10	10	60	155		52218050	18	20	50	120
	52210060	10	10	60	110		52218060	18	18	60	120
	52211050	11	12	50	110		52218070	18	20	70	130
	52212035	12	12	35	90		522180100	18	20	100	200
	52212040120	12	12	40	120		52220050150	20	20	50	150
	52212040	12	12	40	100		52220050	20	20	50	110
	52212045	12	12	45	130		52220060	20	20	60	130
	52212050150	12	12	50	150		52220070	20	20	70	130
	52212050	12	12	50	100		52220080	20	20	80	150
	52212055	12	12	55	110		52220090200	20	20	90	200
	52212060200	12	12	60	200		52220090	20	20	90	150
	52212060150	12	12	60	150		522200110	20	20	110	200
	52212060	12	12	60	110		522200120	20	20	120	250
	52212065	12	12	65	150		52222075	22	20	75	150
	52212070200	12	12	70	200		522220110	22	20	110	200
	52212070	12	12	70	120		52225070	25	25	70	150
	52214040	14	16	40	110		52225090	25	25	90	150
	52214050	14	16	50	110		522250110	25	25	110	200
	52214060150	14	16	60	150		522250120	25	25	120	250

UXE502

2 flutes flat for heavy cuts



CARBIDE	2	H-A 30°	Grade PC315W	h5 shank	DC	Tolerance
					Ø0.1 ~ Ø7 Ø8 ~ Ø20	0.000 ~ -0.012 0.000 ~ -0.015



(mm)

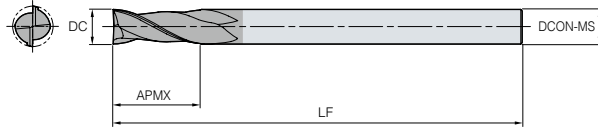
	Designation	DC	DCON-MS	APMX	LF
UXE	502001001	0.1	4	0.1	40
	502001	0.1	4	0.2	40
	502001003	0.1	4	0.3	40
	502002002	0.2	4	0.2	40
	502002	0.2	4	0.4	40
	502002006	0.2	4	0.6	40
	502003003	0.3	4	0.3	40
	502003	0.3	4	0.6	40
	502003009	0.3	4	0.9	40
	502004004	0.4	4	0.4	40
	502004	0.4	4	0.8	40
	502004012	0.4	4	1.2	40
	502005005	0.5	4	0.5	40
	502005	0.5	4	1	40
	502005015	0.5	4	1.5	40
	502006006	0.6	4	0.6	40
	502006	0.6	4	1.2	40
	502006018	0.6	4	1.8	40
	502007007	0.7	4	0.7	40
	502007	0.7	4	1.4	40
	502007021	0.7	4	2.1	40
	502008008	0.8	4	0.8	40
	502008	0.8	4	1.6	40
	502008024	0.8	4	2.4	40
	502009009	0.9	4	0.9	40
	502009	0.9	4	1.8	40
	502009027	0.9	4	2.7	40
	50201001	1	6	1	40
	50201002	1	6	2	40
	502010	1	6	2.5	50
	50201003	1	6	3	50
	50201004	1	6	4	50
	50201006	1	6	6	50
	50201202	1.2	6	2	40
	502012	1.2	6	3	50
	50201204	1.2	6	4	50
	50201206	1.2	6	6	50

	Designation	DC	DCON-MS	APMX	LF
UXE	502015015	1.5	6	1.5	40
	50201503	1.5	6	3	40
	502015	1.5	6	4	50
	50201506	1.5	6	6	50
	50201508	1.5	6	8	50
	50201510	1.5	6	10	50
	50202002	2	6	2	40
	50202004	2	6	4	40
	502020	2	6	6	50
	50202008	2	6	8	50
	50202010	2	6	10	50
	50202012	2	6	12	50
	502025025	2.5	6	2.5	40
	50202505	2.5	6	5	40
	502025	2.5	6	7	50
	50202510	2.5	6	10	50
	50202512	2.5	6	12	50
	50203003	3	6	3	40
	50203006	3	6	6	40
	502030	3	6	8	50
	50203010	3	6	10	50
	50203012	3	6	12	50
	50203014	3	6	14	50
	50203510	3.5	6	10	50
	50204004	4	6	4	40
	50204008	4	6	8	40
	502040	4	6	10	50
	50204012	4	6	12	50
	50204014	4	6	14	50
	50204016	4	6	16	50
	50204511	4.5	6	11	50
	50205005	5	6	5	50
	50205010	5	6	10	50
	502050	5	6	15	60
	50205020	5	6	20	60
	50205025	5	6	25	60
	50205513	5.5	6	13	50

UXE502

2 flutes flat for heavy cuts

CARBIDE	2	H-A 30°	Grade PC315W	h5 shank	DC Ø0.1 ~ Ø7 Ø8 ~ Ø20	Tolerance 0.000 ~ -0.012 0.000 ~ -0.015
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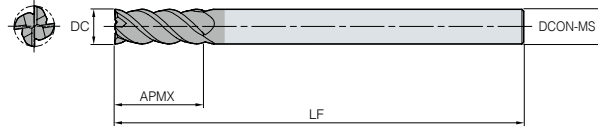
						(mm)					
Designation		DC	DCON-MS	APMX	LF	Designation		DC	DCON-MS	APMX	LF
UXE	50206006	6	6	6	50	UXE	502100	10	10	25	75
	50206012	6	6	12	50		50210030	10	10	30	75
	502060	6	6	15	60		50210035	10	10	35	75
	50206020	6	6	20	60		50212026	12	12	26	70
	50206025	6	6	25	60		502120	12	12	30	80
	50206513	6.5	8	13	60		50212035	12	12	35	80
	50207018	7	8	18	60		50212040	12	12	40	80
	50208016	8	8	16	60		502140	14	16	35	100
	502080	8	8	20	70		502160	16	16	32	100
	50208025	8	8	25	70		50216040	16	16	40	100
	50208030	8	8	30	70		502180	18	20	45	100
	50210022	10	10	22	65		502200	20	20	45	100

UE504H

4 flutes 45° helix flat



CARBIDE	4	H-A 45°	Grade PC315W	h5 shank	DC	Tolerance
					Ø1 ~ Ø20	0.000 ~ -0.030



(mm)

	Designation	DC	DCON-MS	APMX	LF
UE	504H010	1	6	2.5	50
	504H01004	1	6	4	60
	504H01006	1	6	6	60
	504H015	1.5	6	4	50
	504H01506	1.5	6	6	60
	504H01508	1.5	6	8	60
	504H020	2	6	6	50
	504H02008	2	6	8	60
	504H02010	2	6	10	60
	504H030	3	6	8	50
	504H03010	3	6	10	70
	504H03012	3	6	12	70
	504H03016	3	6	16	70
	504H040	4	6	10	50
	504H04012	4	6	12	70
	504H04016	4	6	16	70
	504H04020	4	6	20	70
	504H050	5	6	15	50
	504H05030	5	6	30	80
	504H060	6	6	15	60
	504H06020	6	6	20	70

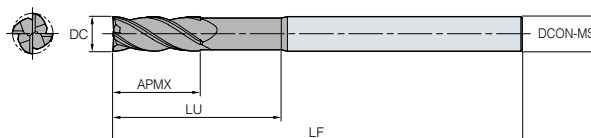
	Designation	DC	DCON-MS	APMX	LF
UE	504H06030	6	6	30	80
	504H080	8	8	20	70
	504H08030	8	8	30	80
	504H08035	8	8	35	90
	504H08040	8	8	40	90
	504H100	10	10	25	75
	504H10030	10	10	30	80
	504H10040	10	10	40	90
	504H10050	10	10	50	100
	504H120	12	12	30	80
	504H12040	12	12	40	90
	504H12050	12	12	50	100
	504H12060	12	12	60	110
	504H160	16	16	40	100
	504H16050	16	16	50	110
	504H16060	16	16	60	120
	504H160110	16	16	110	200
	504H200	20	20	45	100
	504H20060	20	20	60	120
	504H20070	20	20	70	130
	504H200110	20	20	110	200

UE514

4 flutes long neck flat



CARBIDE	4	H-A 35°	Grade PC315W	h5 shank	DC Ø1 ~ Ø12	Tolerance 0.000 ~ -0.030
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	Designation	DC	DCON-MS	APMX	LU	LF
UE	51401002	1	4	1.5	2	50
	51401003	1	4	1.5	3	50
	51401004	1	4	1.5	4	50
	51401005	1	4	1.5	5	50
	51401006	1	4	1.5	6	50
	51401007	1	4	1.5	7	50
	51401008	1	4	1.5	8	50
	51401010	1	4	1.5	10	50
	51401012	1	4	1.5	12	50
	51401014	1	4	1.5	14	50
	51401016	1	4	1.5	16	50
	51401018	1	4	1.5	18	50
	51401020	1	4	1.5	20	50
	51401022	1	4	1.5	22	60
	51401026	1	4	1.5	26	60
	51401030	1	4	1.5	30	70
	51401040	1	4	1.5	40	80
	51401050	1	4	1.5	50	100
	51401204	1.2	4	1.8	4	50
	51401206	1.2	4	1.8	6	50
	51401208	1.2	4	1.8	8	50
	51401210	1.2	4	1.8	10	50
	51401212	1.2	4	1.8	12	50
	51401214	1.2	4	1.8	14	50
	51401216	1.2	4	1.8	16	50
	51401220	1.2	4	1.8	20	50
	51401226	1.2	4	1.8	26	60
	51401230	1.2	4	1.8	30	70
	51401504	1.5	4	2.3	4	50
	51401505	1.5	4	2.3	5	50
	51401506	1.5	4	2.3	6	50
	51401507	1.5	4	2.3	7	50
	51401508	1.5	4	2.3	8	50
	51401510	1.5	4	2.3	10	50
	51401512	1.5	4	2.3	12	50
	51401514	1.5	4	2.3	14	50
	51401516	1.5	4	2.3	16	50

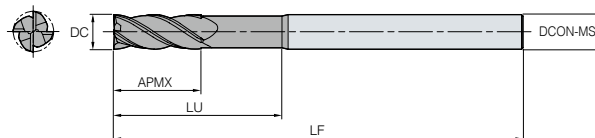
	Designation	DC	DCON-MS	APMX	LU	LF
UE	51401518	1.5	4	2.3	18	50
	51401520	1.5	4	2.3	20	50
	51401522	1.5	4	2.3	22	60
	51401526	1.5	4	2.3	26	60
	51401530	1.5	4	2.3	30	70
	51402006	2	4	3	6	50
	51402008	2	4	3	8	50
	51402010	2	4	3	10	50
	51402012	2	4	3	12	50
	51402014	2	4	3	14	50
	51402016	2	4	3	16	50
	51402018	2	4	3	18	50
	51402020	2	4	3	20	50
	51402022	2	4	3	22	60
	51402026	2	4	3	26	60
	51402030	2	4	3	30	70
	51402035	2	4	3	35	70
	51402040	2	4	3	40	80
	51402045	2	4	3	45	90
	51402050	2	4	3	50	100
	51402060	2	4	3	60	110
	51402508	2.5	4	4	8	50
	51402510	2.5	4	4	10	50
	51402512	2.5	4	4	12	50
	51402514	2.5	4	4	14	50
	51402516	2.5	4	4	16	50
	51402518	2.5	4	4	18	50
	51402520	2.5	4	4	20	50
	51402522	2.5	4	4	22	60
	51402526	2.5	4	4	26	60
	51402530	2.5	4	4	30	70
	51402535	2.5	4	4	35	70
	51402540	2.5	4	4	40	80
	51402545	2.5	4	4	45	90
	51402550	2.5	4	4	50	100
	51403006	3	6	4.5	6	50
	51403008	3	6	4.5	8	50

UE514

4 flutes long neck flat



DC	Tolerance
Ø1 ~ Ø12	0.000 ~ -0.030



(mm)

	Designation	DC	DCON-MS	APMX	LU	LF
UE	51403010	3	6	4.5	10	50
	51403012	3	6	4.5	12	50
	51403014	3	6	4.5	14	60
	51403016	3	6	4.5	16	60
	51403018	3	6	4.5	18	60
	51403020	3	6	4.5	20	60
	51403022	3	6	4.5	22	65
	51403026	3	6	4.5	26	65
	51403030	3	6	4.5	30	70
	51403035	3	6	4.5	35	70
	51403040	3	6	4.5	40	80
	51403045	3	6	4.5	45	90
	51403050	3	6	4.5	50	100
	51403060	3	6	4.5	60	100
	51404008	4	6	4.5	8	50
	51404010	4	6	4.5	10	50
	51404012	4	6	4.5	12	50
	51404014	4	6	4.5	14	60
	51404016	4	6	4.5	16	60
	51404018	4	6	4.5	18	60
	51404020	4	6	4.5	20	60
	51404022	4	6	4.5	22	65
	51404026	4	6	4.5	26	65
	51404030	4	6	4.5	30	70
	51404035	4	6	4.5	35	70

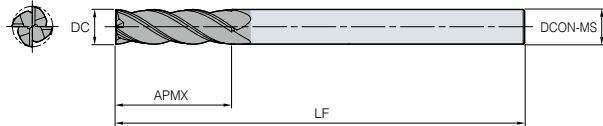
	Designation	DC	DCON-MS	APMX	LU	LF
UE	51404040	4	6	4.5	40	80
	51404045	4	6	4.5	45	90
	51404050	4	6	4.5	50	100
	51404060	4	6	4.5	60	100
	51405016	5	6	8	16	60
	51405020	5	6	8	20	60
	51405026	5	6	8	26	65
	51405030	5	6	8	30	70
	51405035	5	6	8	35	75
	51405040	5	6	8	40	80
	51405050	5	6	8	50	90
	51405060	5	6	8	60	100
	51406015	6	6	9	15	60
	51406020	6	6	9	20	60
	51406030	6	6	9	30	70
	51406032	6	6	9	32	90
	51408025	8	8	12	25	70
	51408030	8	8	12	30	80
	51408042	8	8	12	42	100
	51410030	10	10	15	30	75
	51410035	10	10	15	35	80
	51410045	10	10	15	45	100
	51412035	12	12	20	35	80
	51412040	12	12	20	40	90
	51412050	12	12	20	50	110

UE524

4 flutes flat



CARBIDE	4	H-A 35°	Grade PC315W	h5 shank	DC Ø1 ~ Ø25	Tolerance 0.000 ~ -0.030
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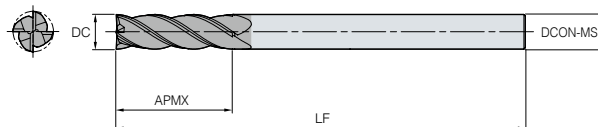
											(mm)
Designation		DC	DCON-MS	APMX	LF	Designation		DC	DCON-MS	APMX	LF
UE	52401003	1	6	3	60	UE	52404012	4	6	12	70
	52401004	1	6	4	60		52404016	4	6	16	70
	52401005	1	6	5	60		52404020	4	6	20	70
	52401006	1	6	6	60		52404020S4	4	4	20	100
	52401007	1	6	7	60		52404026	4	6	26	70
	52401008	1	6	8	60		52404030	4	6	30	70
	52401010	1	6	10	60		52405015	5	6	15	70
	52401012	1	6	12	60		52405020	5	6	20	70
	52401204	1.2	6	4	60		52405025100	5	6	25	100
	52401206	1.2	6	6	60		52405025	5	6	25	70
	52401208	1.2	6	8	60		52405030	5	6	30	80
	52401210	1.2	6	10	60		52405035	5	6	35	90
	52401212	1.2	6	12	60		52405040	5	6	40	100
	52401506	1.5	6	6	60		52406015080	6	6	15	80
	52401508	1.5	6	8	60		52406015	6	6	15	60
	52401510	1.5	6	10	60		52406020090	6	6	20	90
	52401512	1.5	6	12	60		52406020	6	6	20	70
	52401514	1.5	6	14	60		52406025	6	6	25	75
	52401516	1.5	6	16	60		52406030150	6	6	30	150
	52401520	1.5	6	20	60		52406030100	6	6	30	100
	52401526	1.5	6	26	60		52406030	6	6	30	80
	52402008	2	6	8	60		52406030100	6	6	30	100
	52402008S4	2	4	8	40		52406030	6	6	30	80
	52402010	2	6	10	60		52406035	6	6	35	90
	52402012	2	6	12	60		52406040120	6	6	40	120
	52402014	2	6	14	60		52406040	6	6	40	90
	52402016	2	6	16	60		52406045	6	6	45	150
	52402510	2.5	6	10	60		52407035	7	8	35	85
	52402512	2.5	6	12	60		52408020	8	8	20	100
	52402516	2.5	6	16	60		52408025	8	8	25	80
	52402520	2.5	6	20	60		52408030100	8	8	30	100
	52402526	2.5	6	26	60		52408030	8	8	30	80
	52403010	3	6	10	70		52408035	8	8	35	90
	52403012	3	6	12	70		52408040150	8	8	40	150
	52403014	3	6	14	70		52408040120	8	8	40	120
	52403016	3	6	16	70		52408040	8	8	40	90
	52403016S3	3	3	16	100		52408045	8	8	45	100
	52403020	3	6	20	70		52408050150	8	8	50	150
	52403026	3	6	26	70		52408050	8	8	50	100
	52403030	3	6	30	70		52408060	8	8	60	155
	52403035	3	6	35	90		52408080	8	8	80	200

UE524

4 flutes flat



DC	Tolerance
Ø1 ~ Ø25	0.000 ~ -0.030



(mm)

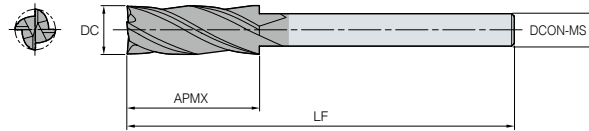
	Designation	DC	DCON-MS	APMX	LF
UE	52409045	9	10	45	100
	52410025	10	10	25	100
	524100100	10	10	30	100
	52410030	10	10	30	80
	52410035	10	10	35	90
	52410040120	10	10	40	120
	52410040	10	10	40	90
	52410045	10	10	45	100
	52410050200	10	10	50	200
	52410050150	10	10	50	150
	52410050	10	10	50	100
	52410055	10	10	55	150
	52410060200	10	10	60	200
	52410060155	10	10	60	155
	52410060	10	10	60	110
	52410080200	10	10	80	200
	52411050	11	12	50	110
	52412030	12	12	30	110
	52412035	12	12	35	90
	52412040120	12	12	40	120
	52412040	12	12	40	100
	52412045	12	12	45	130
	52412050150	12	12	50	150
	52412050	12	12	50	100
	52412055	12	12	55	110
	52412060200	12	12	60	200
	52412060150	12	12	60	150
	52412060	12	12	60	110
	52412065	12	12	65	150
	52412070200	12	12	70	200
	52412070	12	12	70	120
	52412080	12	12	80	200
	52414040	14	16	40	110
	52414050	14	16	50	110
	52414060	14	16	60	150

	Designation	DC	DCON-MS	APMX	LF
UE	52416040	16	16	40	150
	52416050150	16	16	50	150
	52416050	16	16	50	110
	52416060	16	16	60	120
	52416070200	16	16	70	200
	52416070150	16	16	70	150
	52416070	16	16	70	130
	52416080	16	16	80	150
	52416090	16	16	90	150
	524160100	16	16	100	200
	524160110	16	16	110	200
	524160120	16	16	120	250
	52418050	18	20	50	120
	52418070	18	20	70	130
	524180100	18	20	100	200
	52420050150	20	20	50	150
	52420050	20	20	50	110
	52420060	20	20	60	130
	52420070	20	20	70	130
	52420080	20	20	80	150
	52420090200	20	20	90	200
	52420090	20	20	90	150
	524200100	20	20	100	200
	524200110	20	20	110	200
	524200120	20	20	120	250
	524200130	20	20	130	250
	52422075	22	20	75	150
	524220110	22	20	110	200
	52425070	25	25	70	150
	52425090	25	25	90	150
	524250100	25	25	100	200
	524250110	25	25	110	200
	524250120	25	25	120	250
	524250150	25	25	150	250

ULE504

4 flutes automatic CNC flat

CARBIDE	4	H-A 30°	Grade PC315W	h5 shank	DC	Tolerance
					Ø3 ~ Ø16	0.000 ~ -0.020



(mm)

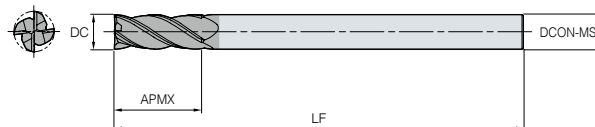
Designation	DC	DCON-MS	APMX	LF
ULE 504030S04	3	4	3	35
504040S04	4	4	4	35
504060S06	6	6	6	35
504070S07	7	7	20	45
504080S08	8	8	8	45
504080S06	8	6	8	35
504080S07	8	7	15	45
504090S07	9	7	15	45
504100S06	10	6	10	35
504100S10	10	10	15	45
504100S07	10	7	15	45
504110S07	11	7	15	45
504120S10	12	10	12	45
504120S06	12	6	12	35
504120S07	12	7	15	45
504120-25S10	12	10	25	55
504130S10	13	10	25	55
504140S10	14	10	25	55
504140S07	14	7	25	55
504160S10	16	10	25	60

UE504

4 flutes flat



DC	Tolerance
Ø0.8 ~ Ø25	0.000 ~ -0.030



(mm)

	Designation	DC	DCON-MS	APMX	LF
UE	504008	0.8	4	1.6	40
	504009	0.9	4	1.8	40
	504010	1	6	2.5	50
	504012	1.2	6	3	50
	504015	1.5	6	4	50
	504020	2	6	6	50
	504025	2.5	6	7	50
	504030	3	6	8	50
	504035	3.5	6	10	50
	504040	4	6	10	50
	504045	4.5	6	14	50
	504050	5	6	15	60
	504055	5.5	6	15	60
	504060	6	6	15	60
	504065	6.5	8	18	60
	504070	7	8	20	60
	504075	7.5	8	20	60
	504080	8	8	20	70
	504085	8.5	10	22	70
	504090	9	10	22	70
	504095	9.5	10	24	70
	504100	10	10	25	75

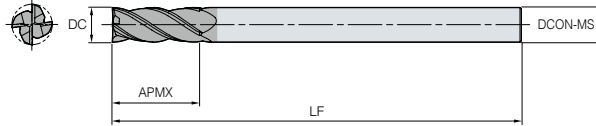
	Designation	DC	DCON-MS	APMX	LF
UE	504105	10.5	12	26	75
	504110	11	12	30	75
	504115	11.5	12	30	80
	504120	12	12	30	80
	504125	12.5	12	30	80
	504130	13	12	35	100
	504140	14	16	35	100
	504140S14	14	14	35	100
	504140S12	14	12	35	100
	504150	15	16	38	100
	504160	16	16	40	100
	504170	17	16	42	100
	504180	18	18	45	100
	504180S16	18	16	45	100
	504190	19	20	45	100
	504200	20	20	45	100
	504210	21	20	45	100
	504220	22	20	45	100
	504230	23	25	50	120
	504240	24	25	50	120
	504250	25	25	50	120

UXE504

4 flutes flat Endmill for heavy cuts



CARBIDE	4	H-A 34°/36°	Grade PC315W	h5 shank	DC Ø1 ~ Ø20	Tolerance 0.000 ~ -0.030
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											(mm)
Designation		DC	DCON-MS	APMX	LF	Designation		DC	DCON-MS	APMX	LF
UXE	50401001	1	6	1	40	UXE	50404004	4	6	4	40
	50401002	1	6	2	40		50404008	4	6	8	40
	504010	1	6	2.5	50		504040	4	6	10	50
	50401003	1	6	3	50		50404012	4	6	12	50
	50401004	1	6	4	50		50404014	4	6	14	50
	50401006	1	6	6	50		50404016	4	6	16	50
	50401202	1.2	6	2	40		50405005	5	6	5	50
	504012	1.2	6	3	50		50405010	5	6	10	50
	50401204	1.2	6	4	50		504050	5	6	15	60
	50401206	1.2	6	6	50		50405020	5	6	20	60
	504015015	1.5	6	1.5	40		50405025	5	6	25	60
	50401503	1.5	6	3	40		50406006	6	6	6	50
	504015	1.5	6	4	50		50406012	6	6	12	50
	50401506	1.5	6	6	50		504060	6	6	15	60
	50401508	1.5	6	8	50		50406020	6	6	20	60
	50401510	1.5	6	10	50		50406025	6	6	25	60
	50402002	2	6	2	40		50408016	8	8	16	60
	50402004	2	6	4	40		504080	8	8	20	70
	504020	2	6	6	50		50408025	8	8	25	70
	50402008	2	6	8	50		50408030	8	8	30	70
	50402010	2	6	10	50		50410022	10	10	22	65
	50402012	2	6	12	50		504100	10	10	25	75
	504025025	2.5	6	2.5	40		50410030	10	10	30	75
	50402505	2.5	6	5	40		50410035	10	10	35	75
	504025	2.5	6	7	50		50412026	12	12	26	70
	50402510	2.5	6	10	50		504120	12	12	30	80
	50402512	2.5	6	12	50		50412035	12	12	35	80
	50403003	3	6	3	40		50412040	12	12	40	80
	50403006	3	6	6	40		504140	14	16	35	100
	504030	3	6	8	50		50416032	16	16	32	100
	50403010	3	6	10	50		504160	16	16	40	100
	50403012	3	6	12	50		504180	18	20	45	100
	50403014	3	6	14	50		504200	20	20	45	100

UE506

6 flutes flat



CARBIDE

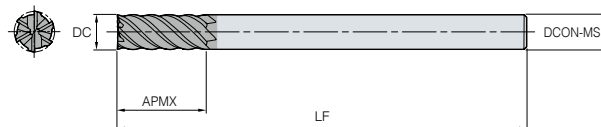
6

H-A
45°

Grade
PC315W

h5
shank

DC	Tolerance
Ø6 ~ Ø25	0.000 ~ -0.030



(mm)

	Designation	DC	DCON-MS	APMX	LF
UE	506060	6	6	15	60
	50606020	6	6	20	70
	50606030110	6	6	30	110
	50606030	6	6	30	80
	506070	7	8	18	60
	506080	8	8	20	70
	50608030	8	8	30	80
	50608035	8	8	35	90
	50608040130	8	8	40	130
	50608040	8	8	40	90
	506090	9	10	22	70
	506100	10	10	25	75
	50610030	10	10	30	80
	50610040	10	10	40	90
	50610050150	10	10	50	150
	50610050	10	10	50	100
	506110	11	12	26	75
	506120	12	12	30	80
	50612040	12	12	40	90
	50612050	12	12	50	100

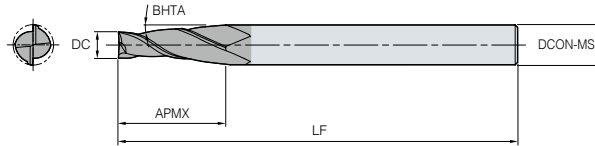
	Designation	DC	DCON-MS	APMX	LF
UE	50612060150	12	12	60	150
	50612060	12	12	60	110
	506130	13	14	32	85
	506140	14	14	32	85
	506160	16	16	40	100
	50616050	16	16	50	110
	50616060	16	16	60	120
	50616090	16	16	90	150
	506160110250	16	16	110	250
	506160110	16	16	110	200
	506180	18	18	44	100
	506200	20	20	45	100
	50620060	20	20	60	120
	50620070	20	20	70	130
	506200110300	20	20	110	300
	506200110250	20	20	110	250
	506200110	20	20	110	200
	506250	25	25	50	120
	506251	25	25	92	180

UTE502

2 flutes tapered flat



CARBIDE	2	H-A 30°	Grade PC315W	h5 shank	DC	Tolerance
					Ø0.3 ~ Ø10	0.000 ~ -0.030



							(mm)						
	Designation	DC	DCON-MS	APMX	BHTA(°)	LF		Designation	DC	DCON-MS	APMX	BHTA(°)	LF
UTE	502003005	0.3	4	1.2	0.5	40	UTE	50200703	0.7	4	2.8	3	45
	50200301	0.3	4	1.2	1	40		50200705	0.7	4	3.5	5	45
	502003015	0.3	4	1.2	1.5	40		50200707	0.7	4	3.5	7	45
	50200302	0.3	4	1.2	2	40		50200710	0.7	4	3.5	10	45
	50200303	0.3	4	1.5	3	40		502008005	0.8	4	3.2	0.5	45
	50200305	0.3	4	1.5	5	40		50200801	0.8	4	3.2	1	45
	50200307	0.3	4	1.5	7	40		502008015	0.8	4	3.2	1.5	45
	50200310	0.3	4	1.5	10	40		50200802	0.8	4	3.2	2	45
	502004005	0.4	4	1.6	0.5	40		50200803	0.8	4	3.2	3	45
	50200401	0.4	4	1.6	1	40		50200805	0.8	4	4	5	45
	502004015	0.4	4	1.6	1.5	40		50200807	0.8	4	4	7	45
	50200402	0.4	4	1.6	2	40		50200810	0.8	4	4	10	45
	50200403	0.4	4	1.6	3	40		502010005	1	4	4	0.5	50
	50200405	0.4	4	2	5	40		50201001	1	4	4	1	50
	50200407	0.4	4	2	7	40		502010015	1	4	4	1.5	50
	50200410	0.4	4	2	10	40		50201002	1	4	6	2	50
	502005005	0.5	4	2	0.5	45		50201003	1	4	6	3	50
	50200501	0.5	4	2	1	45		50201005	1	4	8	5	50
	502005015	0.5	4	2	1.5	45		50201007	1	4	8	7	50
	50200502	0.5	4	2	2	45		50201010	1	4	8	10	50
	50200503	0.5	4	2	3	45		502015005	1.5	4	6	0.5	50
	50200505	0.5	4	2.5	5	45		50201501	1.5	4	6	1	50
	50200507	0.5	4	2.5	7	45		502015015	1.5	4	6	1.5	50
	50200510	0.5	4	2.5	10	45		50201502	1.5	4	8	2	50
	502006005	0.6	4	2.4	0.5	45		50201503	1.5	4	8	3	50
	50200601	0.6	4	2.4	1	45		50201505	1.5	4	10	5	50
	502006015	0.6	4	2.4	1.5	45		50201507	1.5	4	10	7	50
	50200602	0.6	4	2.4	2	45		50201510	1.5	6	10	10	50
	50200603	0.6	4	2.4	3	45		502020005	2	4	8	0.5	50
	50200605	0.6	4	3	5	45		50202001	2	4	8	1	50
	50200607	0.6	4	3	7	45		502020015	2	4	8	1.5	50
	50200610	0.6	4	3	10	45		50202002	2	4	10	2	50
	502007005	0.7	4	2.8	0.5	45		50202003	2	4	10	3	50
	50200701	0.7	4	2.8	1	45		50202005	2	6	12	5	50
	502007015	0.7	4	2.8	1.5	45		50202007	2	6	12	7	50
	50200702	0.7	4	2.8	2	45		50202010	2	8	12	10	50

UTE502

2 flutes tapered flat



CARBIDE

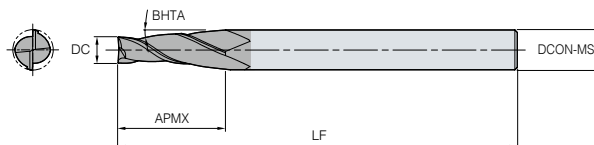
2

H-A
30°

Grade
PC315W

h5
shank

DC	Tolerance
Ø0.3 ~ Ø10	0.000 ~ -0.030



(mm)

	Designation	DC	DCON-MS	APMX	BHTA(°)	LF
UTE	502025005	2.5	6	10	0.5	50
	50202501	2.5	6	10	1	50
	502025015	2.5	6	10	1.5	50
	50202502	2.5	6	12	2	50
	50202503	2.5	6	12	3	50
	50202505	2.5	6	14	5	50
	50202507	2.5	6	14	7	50
	50202510	2.5	8	14	10	50
	502030005	3	6	12	0.5	50
	50203001	3	6	12	1	50
	502030015	3	6	12	1.5	50
	50203002	3	6	14	2	50
	50203003	3	6	14	3	50
	50203005	3	6	16	5	50
	50203007	3	8	16	7	50
	50203010	3	10	16	10	50
	502040005	4	6	16	0.5	60
	50204001	4	6	16	1	60
	502040015	4	6	16	1.5	60
	50204002	4	6	16	2	60
	50204003	4	6	19	3	60
	50204005	4	8	22	5	65
	50204007	4	8	16	7	65
	50204010	4	10	17	10	65
	502060005	6	8	20	0.5	65
	50206001	6	8	20	1	65
	502060015	6	8	20	1.5	65
	50206002	6	8	20	2	65

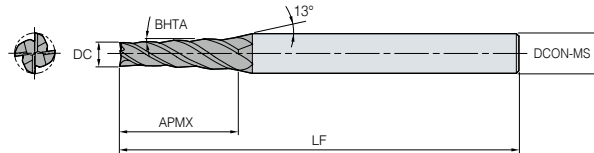
	Designation	DC	DCON-MS	APMX	BHTA(°)	LF
UTE	50206003	6	8	19	3	65
	50206005	6	10	22	5	75
	50206007	6	12	24	7	75
	50206010	6	12	17	10	75
	502070005	7	8	28	0.5	70
	50207001	7	8	28	1	70
	502070015	7	10	28	1.5	70
	50207002	7	10	28	2	80
	50207003	7	10	28	3	80
	50207005	7	12	28	5	80
	502080005	8	10	35	0.5	90
	50208001	8	10	35	1	90
	502080015	8	10	35	1.5	90
	50208002	8	10	28	2	90
	50208003	8	12	38	3	90
	50208005	8	16	45	5	100
	50208007	8	16	32	7	90
	50208010	8	20	34	10	100
	50208010S25	8	25	48	10	150
	502100005	10	12	40	0.5	90
	50210001	10	12	40	1	90
	502100015	10	12	38	1.5	90
	50210002	10	16	40	2	75
	50210003	10	16	40	3	100
	50210005	10	16	34	5	100
	50210007	10	20	40	7	90
	50210010	10	25	42	10	100

UTE504

4 flutes tapered flat



CARBIDE	4	H-A 30°	Grade PC315W	h5 shank	DC	Tolerance
					Ø0.8 ~ Ø10	0.000 ~ -0.030



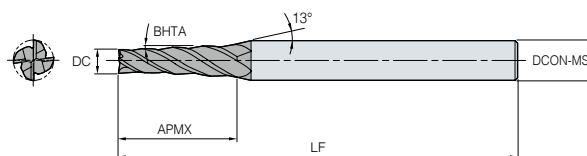
														(mm)
	Designation	DC	DCON-MS	APMX	BHTA(°)	LF		Designation	DC	DCON-MS	APMX	BHTA(°)	LF	
UTE	50400800504	0.8	4	4	0.5	45	UTE	50401001516	1	4	16	1.5	50	
	50400800506	0.8	4	6	0.5	45		50401002004	1	4	4	2	50	
	50400800508	0.8	4	8	0.5	45		50401002006	1	4	6	2	50	
	50400800510	0.8	4	10	0.5	45		50401002008	1	4	8	2	50	
	50400800512	0.8	4	12	0.5	45		50401002010	1	4	10	2	50	
	50400801004	0.8	4	4	1	45		50401002012	1	4	12	2	50	
	50400801006	0.8	4	6	1	45		50401002016	1	4	16	2	50	
	50400801008	0.8	4	8	1	45		50401003004	1	4	4	3	50	
	50400801010	0.8	4	10	1	45		50401003006	1	4	6	3	50	
	50400801012	0.8	4	12	1	45		50401003008	1	4	8	3	50	
	50400801504	0.8	4	4	1.5	45		50401003010	1	4	10	3	50	
	50400801506	0.8	4	6	1.5	45		50401003012	1	4	12	3	50	
	50400801508	0.8	4	8	1.5	45		50401003016	1	4	16	3	50	
	50400801510	0.8	4	10	1.5	45		50401200506	1.2	4	6	0.5	50	
	50400801512	0.8	4	12	1.5	45		50401200508	1.2	4	8	0.5	50	
	50400802004	0.8	4	4	2	45		50401200510	1.2	4	10	0.5	50	
	50400802006	0.8	4	6	2	45		50401200512	1.2	4	12	0.5	50	
	50400802008	0.8	4	8	2	45		50401200516	1.2	4	16	0.5	50	
	50400802010	0.8	4	10	2	45		50401201006	1.2	4	6	1	50	
	50400802012	0.8	4	12	2	45		50401201008	1.2	4	8	1	50	
UTE	50401000504	1	4	4	0.5	50	UTE	50401201010	1.2	4	10	1	50	
	50401000506	1	4	6	0.5	50		50401201012	1.2	4	12	1	50	
	50401000508	1	4	8	0.5	50		50401201016	1.2	4	16	1	50	
	50401000510	1	4	10	0.5	50		50401201506	1.2	4	6	1.5	50	
	50401000512	1	4	12	0.5	50		50401201508	1.2	4	8	1.5	50	
	50401000516	1	4	16	0.5	50		50401201510	1.2	4	10	1.5	50	
	50401001004	1	4	4	1	50		50401201512	1.2	4	12	1.5	50	
	50401001006	1	4	6	1	50		50401201516	1.2	4	16	1.5	50	
	50401001008	1	4	8	1	50		50401202006	1.2	4	6	2	50	
	50401001010	1	4	10	1	50		50401202008	1.2	4	8	2	50	
	50401001012	1	4	12	1	50		50401202010	1.2	4	10	2	50	
	50401001016	1	4	16	1	50		50401202012	1.2	4	12	2	50	
	50401001504	1	4	4	1.5	50		50401202016	1.2	4	16	2	50	
	50401001506	1	4	6	1.5	50		50401203006	1.2	4	6	3	50	
	50401001508	1	4	8	1.5	50		50401203008	1.2	4	8	3	50	
	50401001510	1	4	10	1.5	50		50401203010	1.2	4	10	3	50	
	50401001512	1	4	12	1.5	50		50401203012	1.2	4	12	3	50	

UTE504

4 flutes tapered flat



CARBIDE	4	H-A 30°	Grade PC315W	h5 shank	DC	Tolerance
					Ø0.8 ~ Ø10	0.000 ~ -0.030



(mm)

	Designation	DC	DCON-MS	APMX	BHTA(°)	LF
UTE	50401203016	1.2	4	16	3	50
	50401500506	1.5	4	6	0.5	50
	50401500508	1.5	4	8	0.5	50
	50401500510	1.5	4	10	0.5	50
	50401500512	1.5	4	12	0.5	50
	50401500516	1.5	4	16	0.5	50
	50401500520	1.5	4	20	0.5	60
	50401501006	1.5	4	6	1	50
	50401501008	1.5	4	8	1	50
	50401501010	1.5	4	10	1	50
	50401501012	1.5	4	12	1	50
	50401501016	1.5	4	16	1	50
	50401501020	1.5	4	20	1	60
	50401501506	1.5	4	6	1.5	50
	50401501508	1.5	4	8	1.5	50
	50401501510	1.5	4	10	1.5	50
	50401501512	1.5	4	12	1.5	50
	50401501516	1.5	4	16	1.5	50
	50401501520	1.5	4	20	1.5	60
	50401502006	1.5	4	6	2	50
	50401502008	1.5	4	8	2	50
	50401502010	1.5	4	10	2	50
	50401502012	1.5	4	12	2	50
	50401502016	1.5	4	16	2	50
	50401502020	1.5	4	20	2	60
	50401503006	1.5	4	6	3	50
	50401503008	1.5	4	8	3	50
	50401503010	1.5	4	10	3	50
	50401503012	1.5	4	12	3	50
	50401503016	1.5	4	16	3	50
	50401503020	1.5	4	20	3	60
	50402000508	2	4	8	0.5	50
	50402000510	2	4	10	0.5	50
	50402000512	2	4	12	0.5	50
	50402000516	2	4	16	0.5	50
	50402000520	2	4	20	0.5	60
	50402000525	2	4	25	0.5	60

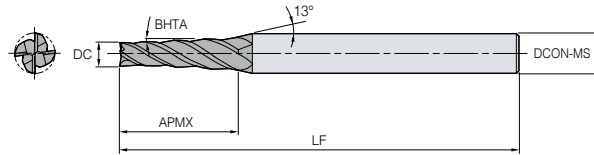
	Designation	DC	DCON-MS	APMX	BHTA(°)	LF
UTE	50402001008	2	4	8	1	50
	50402001010	2	4	10	1	50
	50402001012	2	4	12	1	50
	50402001016	2	4	16	1	50
	50402001020	2	4	20	1	60
	50402001025	2	4	25	1	60
	50402001508	2	4	8	1.5	50
	50402001510	2	4	10	1.5	50
	50402001512	2	4	12	1.5	50
	50402001516	2	4	16	1.5	50
	50402001520	2	4	20	1.5	60
	50402001525	2	4	25	1.5	60
	50402002008	2	4	8	2	50
	50402002010	2	4	10	2	50
	50402002012	2	4	12	2	50
	50402002016	2	4	16	2	50
	50402002020	2	4	20	2	60
	50402002025	2	4	25	2	60
	50402003008	2	4	8	3	50
	50402003010	2	4	10	3	50
	50402003012	2	4	12	3	50
	50402003016	2	4	16	3	50
	50402003020	2	6	20	3	60
	50402003025	2	6	25	3	60
	50402500510	2.5	4	10	0.5	50
	50402500512	2.5	4	12	0.5	50
	50402500516	2.5	4	16	0.5	50
	50402500520	2.5	4	20	0.5	60
	50402500525	2.5	4	25	0.5	60
	50402500530	2.5	4	30	0.5	60
	50402501010	2.5	4	10	1	50
	50402501012	2.5	4	12	1	50
	50402501016	2.5	4	16	1	50
	50402501020	2.5	4	20	1	60
	50402501025	2.5	4	25	1	60
	50402501030	2.5	4	30	1	60
	50402501510	2.5	4	10	1.5	50

UTE504

4 flutes tapered flat



CARBIDE	4	H-A 30°	Grade PC315W	h5 shank	DC Ø0.8 ~ Ø10	Tolerance 0.000 ~ -0.030
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														(mm)
Designation		DC	DCON-MS	APMX	BHTA(°)	LF	Designation		DC	DCON-MS	APMX	BHTA(°)	LF	
UTE	50402501512	2.5	4	12	1.5	50	UTE	50404010	4	10	17	10	65	
	50402501516	2.5	4	16	1.5	50		504060005	6	8	20	0.5	65	
	50402501520	2.5	4	20	1.5	60		50406001	6	8	20	1	65	
	50402501525	2.5	4	25	1.5	60		504060015	6	8	20	1.5	65	
	50402501530	2.5	6	30	1.5	60		50406002	6	8	20	2	65	
	50402502010	2.5	4	10	2	50		50406003	6	8	19	3	65	
	50402502012	2.5	4	12	2	50		50406005	6	10	22	5	75	
	50402502016	2.5	4	16	2	50		50406007	6	12	24	7	75	
	50402502020	2.5	4	20	2	60		50406010	6	12	17	10	75	
	50402502025	2.5	6	25	2	60		504070005	7	8	28	0.5	70	
	50402502030	2.5	6	30	2	60		50407001	7	8	28	1	70	
	50402503010	2.5	4	10	3	50		504070015	7	10	28	1.5	70	
	50402503012	2.5	4	12	3	50		50407002	7	10	28	2	80	
	50402503016	2.5	6	16	3	50		50407003	7	10	28	3	80	
	50402503020	2.5	6	20	3	60		50407005	7	12	28	5	80	
	50402503025	2.5	6	25	3	60		504080005	8	10	35	0.5	90	
	50402503030	2.5	6	30	3	60		50408001	8	10	35	1	90	
	504030005	3	6	12	0.5	50		504080015	8	10	35	1.5	90	
	50403001	3	6	12	1	50		50408002	8	10	28	2	90	
	504030015	3	6	12	1.5	50		50408003	8	12	38	3	90	
	50403002	3	6	14	2	50		50408005	8	16	45	5	100	
	50403003	3	6	14	3	50		50408007	8	16	32	7	90	
	50403005	3	6	16	5	50		50408010	8	20	34	10	100	
	50403007	3	8	16	7	50		504100005	10	12	40	0.5	90	
	50403010	3	10	16	10	50		50410001	10	12	40	1	90	
	504040005	4	6	16	0.5	60		504100015	10	12	38	1.5	90	
	50404001	4	6	16	1	60		50410002	10	16	40	2	90	
	504040015	4	6	16	1.5	60		50410003	10	16	40	3	100	
	50404002	4	6	16	2	60		50410005	10	16	34	5	100	
	50404003	4	6	19	3	60		50410007	10	20	40	7	90	
	50404005	4	8	22	5	65		50410010	10	25	42	10	100	
	50404007	4	8	16	7	65								

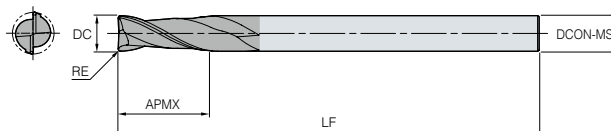
UR502

2 flutes radius



DC6 or below Above DC6

DC	Tolerance
Ø0.2 ~ Ø6	0.000 ~ -0.012
Ø7 ~ Ø20	0.000 ~ -0.015



(mm)

	Designation	DC	DCON-MS	APMX	LF	RE
UR	502002002	0.2	4	0.4	40	0.02
	502002005	0.2	4	0.4	40	0.05
	502003002	0.3	4	0.6	40	0.02
	502003005	0.3	4	0.6	40	0.05
	502004005	0.4	4	0.8	40	0.05
	50200401	0.4	4	0.8	40	0.1
	502005005	0.5	4	1	40	0.05
	50200501	0.5	4	1	40	0.1
	502006005	0.6	4	1.2	40	0.05
	50200601	0.6	4	1.2	40	0.1
	50200602	0.6	4	1.2	40	0.2
	502007005	0.7	4	1.4	40	0.05
	50200701	0.7	4	1.4	40	0.1
	50200702	0.7	4	1.4	40	0.2
	502008005	0.8	4	1.6	40	0.05
	50200801	0.8	4	1.6	40	0.1
	50200802	0.8	4	1.6	40	0.2
	502009005	0.9	4	1.8	40	0.05
	50200901	0.9	4	1.8	40	0.1
	502010005	1	6	2.5	50	0.05
	50201001	1	6	2.5	50	0.1
	50201002	1	6	2.5	50	0.2
	50201003	1	6	2.5	50	0.3
	502012005	1.2	6	3	50	0.05
	50201201	1.2	6	3	50	0.1
	50201202	1.2	6	3	50	0.2
	50201203	1.2	6	3	50	0.3
	502015005	1.5	6	4	50	0.05
	50201501	1.5	6	4	50	0.1
	50201502	1.5	6	4	50	0.2
	50201503	1.5	6	4	50	0.3
	50201505	1.5	6	4	50	0.5
	50202001	2	6	6	50	0.1
	50202002	2	6	6	50	0.2
	50202003	2	6	6	50	0.3
	50202005	2	6	6	50	0.5
	50202501	2.5	6	7	60	0.1
	50202502	2.5	6	7	60	0.2
	50202503	2.5	6	7	60	0.3

	Designation	DC	DCON-MS	APMX	LF	RE
UR	50202505	2.5	6	7	60	0.5
	50203001	3	6	8	60	0.1
	50203002	3	6	8	60	0.2
	50203003	3	6	8	60	0.3
	50203005	3	6	8	60	0.5
	50203010	3	6	8	60	1
	50203501	3.5	6	10	70	0.1
	50203502	3.5	6	10	70	0.2
	50203503	3.5	6	10	70	0.3
	50203505	3.5	6	10	70	0.5
	50204001	4	6	10	70	0.1
	50204001100S4	4	4	10	100	0.1
	50204001S4	4	4	10	70	0.1
	50204002	4	6	10	70	0.2
	50204002100S4	4	4	10	100	0.2
	50204002S4	4	4	10	70	0.2
	50204003	4	6	10	70	0.3
	50204003100S4	4	4	10	100	0.3
	50204003S4	4	4	10	70	0.3
	50204005	4	6	10	70	0.5
	50204005100S4	4	4	10	100	0.5
	50204005S4	4	4	10	70	0.5
	50204010	4	6	10	70	1
	50204010100S4	4	4	10	100	1
	50204010S4	4	4	10	70	1
	50204501	4.5	6	11	80	0.1
	50204502	4.5	6	11	80	0.2
	50204503	4.5	6	11	80	0.3
	50204505	4.5	6	11	80	0.5
	50205001	5	6	13	90	0.1
	50205002	5	6	13	90	0.2
	50205003	5	6	13	90	0.3
	50205005	5	6	13	90	0.5
	50205010	5	6	13	90	1
	50205501	5.5	6	13	90	0.1
	50205502	5.5	6	13	90	0.2
	50205503	5.5	6	13	90	0.3
	50205505	5.5	6	13	90	0.5
	50205510	5.5	6	13	90	1

UR502

2 flutes radius



CARBIDE

2

H-A
30°

Grade
PC315W

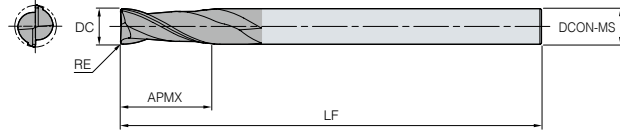
h5
shank

RE
±0.01

RE
±0.015

DC6 or below Above DC6

DC	Tolerance
Ø0.2 ~ Ø6	0.000 ~ -0.012
Ø7 ~ Ø20	0.000 ~ -0.015



Designation	DC	DCON-MS	APMX	LF	RE
UR 50206001	6	6	15	90	0.1
50206002	6	6	15	90	0.2
50206003	6	6	15	90	0.3
5020600360	6	6	15	60	0.3
50206005	6	6	15	90	0.5
50206005110	6	6	15	110	0.5
50206005130	6	6	15	130	0.5
5020600560	6	6	15	60	0.5
50206010	6	6	15	90	1
50206010110	6	6	15	110	1
50206010130	6	6	15	130	1
5020601060	6	6	15	60	1
50206015	6	6	15	90	1.5
50206020	6	6	15	90	2
50207001	7	8	16	90	0.1
50207002	7	8	16	90	0.2
50207003	7	8	16	90	0.3
50207005	7	8	16	90	0.5
50207010	7	8	16	90	1
50207020	7	8	16	90	2
50208001	8	8	20	100	0.1
50208002	8	8	20	100	0.2
50208003	8	8	20	100	0.3
5020800370	8	8	20	70	0.3
50208005	8	8	20	100	0.5
50208005120	8	8	20	120	0.5
50208005150	8	8	20	150	0.5
5020800570	8	8	20	70	0.5
50208010	8	8	20	100	1
50208010120	8	8	20	120	1
50208010150	8	8	20	150	1
5020801070	8	8	20	70	1
50208015	8	8	20	100	1.5
50208020	8	8	20	100	2
50208025	8	8	20	100	2.5
50208030	8	8	20	100	3
50210001	10	10	25	100	0.1
50210002	10	10	25	100	0.2
50210003	10	10	25	100	0.3

Designation	DC	DCON-MS	APMX	LF	RE
UR 5021000375	10	10	25	75	0.3
50210005	10	10	25	100	0.5
50210005130	10	10	25	130	0.5
50210005150	10	10	25	150	0.5
5021000575	10	10	25	75	0.5
50210010	10	10	25	100	1
50210010130	10	10	25	130	1
50210010150	10	10	25	150	1
5021001075	10	10	25	75	1
50210015	10	10	25	100	1.5
50210020	10	10	25	100	2
50210025	10	10	25	100	2.5
50210030	10	10	25	100	3
50210040	10	10	25	100	4
50211002	11	12	25	110	0.2
50211003	11	12	25	110	0.3
50211005	11	12	25	110	0.5
50211010	11	12	25	110	1
50211020	11	12	25	110	2
50212001	12	12	30	110	0.1
50212002	12	12	30	110	0.2
50212003	12	12	30	110	0.3
5021200380	12	12	30	80	0.3
50212005	12	12	30	110	0.5
50212005130	12	12	30	130	0.5
50212005150	12	12	30	150	0.5
5021200580	12	12	30	80	0.5
50212010	12	12	30	110	1
50212010130	12	12	30	130	1
50212010150	12	12	30	150	1
5021201080	12	12	30	80	1
50212015	12	12	30	110	1.5
50212020	12	12	30	110	2
50212025	12	12	30	110	2.5
50212030	12	12	30	110	3
50212040	12	12	30	110	4
50212050	12	12	30	110	5
50214005	14	16	30	150	0.5
50214010	14	16	30	150	1

UR502

2 flutes radius



CARBIDE

2

H-A
30°

Grade
PC315W

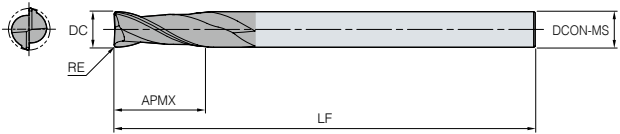
h5
shank

RE
±0.01

RE
±0.015

DC6 or below Above DC6

DC	Tolerance
Ø0.2 ~ Ø6	0.000 ~ -0.012
Ø7 ~ Ø20	0.000 ~ -0.015



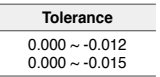
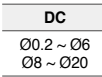
Designation		DC	DCON-MS	APMX	LF	RE
UR	50214020	14	16	30	150	2
	50216005	16	16	32	150	0.5
	50216010	16	16	32	150	1
	50216015	16	16	32	150	1.5
	50216020	16	2	32	150	16
	50210030	16	3	32	150	16

(mm)

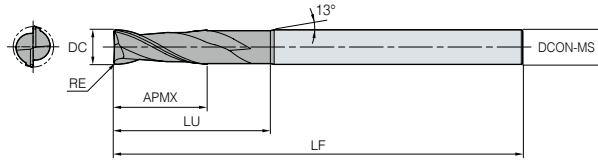
Designation		DC	DCON-MS	APMX	LF	RE
UR	50220005	20	0.5	38	150	20
	50220010	20	1	38	150	20
	50220015	20	1.5	38	150	20
	50220020	20	2	38	150	20
	50220030	20	3	38	150	20

UR512

2 flutes neck type radius



DC6 or below Above DC6



Designation	DC	DCON-MS	APMX	LU	LF	RE
UR 512002002005	0.2	4	0.3	0.5	40	0.02
51200200201	0.2	4	0.3	1	40	0.02
512002002015	0.2	4	0.3	1.5	40	0.02
51200200202	0.2	4	0.3	2	40	0.02
512002005005	0.2	4	0.3	0.5	40	0.05
51200200501	0.2	4	0.3	1	40	0.05
512002005015	0.2	4	0.3	1.5	40	0.05
51200200502	0.2	4	0.3	2	40	0.05
51200300201	0.3	4	0.5	1	40	0.02
51200300202	0.3	4	0.5	2	40	0.02
51200300203	0.3	4	0.5	3	40	0.02
51200300501	0.3	4	0.5	1	40	0.05
51200300502	0.3	4	0.5	2	40	0.05
51200300503	0.3	4	0.5	3	40	0.05
51200400501	0.4	4	0.6	1	40	0.05
512004005015	0.4	4	0.6	1.5	40	0.05
51200400502	0.4	4	0.6	2	40	0.05
512004005025	0.4	4	0.6	2.5	40	0.05
51200400503	0.4	4	0.6	3	40	0.05
51200400504	0.4	4	0.6	4	40	0.05
5120040101	0.4	4	0.6	1	40	0.1
51200401015	0.4	4	0.6	1.5	40	0.1
5120040102	0.4	4	0.6	2	40	0.1
51200401025	0.4	4	0.6	2.5	40	0.1
5120040103	0.4	4	0.6	3	40	0.1
5120040104	0.4	4	0.6	4	40	0.1
51200500501	0.5	4	0.7	1	45	0.05
512005005015	0.5	4	0.7	1.5	45	0.05
51200500502	0.5	4	0.7	2	45	0.05
512005005025	0.5	4	0.7	2.5	45	0.05
51200500503	0.5	4	0.7	3	45	0.05
51200500504	0.5	4	0.7	4	45	0.05
51200500505	0.5	4	0.7	5	45	0.05
51200500506	0.5	4	0.7	6	45	0.05
5120050101	0.5	4	0.7	1	45	0.1
51200501015	0.5	4	0.7	1.5	45	0.1
5120050102	0.5	4	0.7	2	45	0.1
51200501025	0.5	4	0.7	2.5	45	0.1

Designation	DC	DCON-MS	APMX	LU	LF	RE
UR 5120050103	0.5	4	0.7	3	45	0.1
5120050104	0.5	4	0.7	4	45	0.1
5120050105	0.5	4	0.7	5	45	0.1
5120050106	0.5	4	0.7	6	45	0.1
51200600502	0.6	4	0.9	2	45	0.05
51200600503	0.6	4	0.9	3	45	0.05
51200600504	0.6	4	0.9	4	45	0.05
51200600506	0.6	4	0.9	6	45	0.05
51200600508	0.6	4	0.9	8	45	0.05
51200600510	0.6	4	0.9	10	45	0.05
5120060102	0.6	4	0.9	2	45	0.1
5120060103	0.6	4	0.9	3	45	0.1
5120060104	0.6	4	0.9	4	45	0.1
5120060106	0.6	4	0.9	6	45	0.1
5120060108	0.6	4	0.9	8	45	0.1
5120060110	0.6	4	0.9	10	45	0.1
5120060202	0.6	4	0.9	2	45	0.2
5120060203	0.6	4	0.9	3	45	0.2
5120060204	0.6	4	0.9	4	45	0.2
5120060206	0.6	4	0.9	6	45	0.2
5120060208	0.6	4	0.9	8	45	0.2
5120060210	0.6	4	0.9	10	45	0.2
51200700502	0.7	4	1.2	2	45	0.05
51200700504	0.7	4	1.2	4	45	0.05
51200700506	0.7	4	1.2	6	45	0.05
51200700508	0.7	4	1.2	8	45	0.05
51200700510	0.7	4	1.2	10	45	0.05
5120070102	0.7	4	1.2	2	45	0.1
5120070104	0.7	4	1.2	4	45	0.1
5120070106	0.7	4	1.2	6	45	0.1
5120070108	0.7	4	1.2	8	45	0.1
5120070110	0.7	4	1.2	10	45	0.1
5120070202	0.7	4	1.2	2	45	0.2
5120070204	0.7	4	1.2	4	45	0.2
5120070206	0.7	4	1.2	6	45	0.2
5120070208	0.7	4	1.2	8	45	0.2
5120070210	0.7	4	1.2	10	45	0.2
51200800502	0.8	4	1.2	2	45	0.05

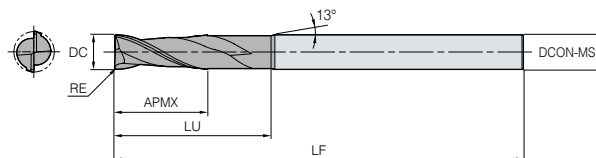
UR512

2 flutes neck type radius



DC6 or below Above DC6

DC	Tolerance
Ø0.2 ~ Ø6	0.000 ~ -0.012
Ø8 ~ Ø20	0.000 ~ -0.015



(mm)

	Designation	DC	DCON-MS	APMX	LU	LF	RE
UR	51200800503	0.8	4	1.2	3	45	0.05
	51200800504	0.8	4	1.2	4	45	0.05
	51200800506	0.8	4	1.2	6	45	0.05
	51200800508	0.8	4	1.2	8	45	0.05
	51200800510	0.8	4	1.2	10	45	0.05
	5120080102	0.8	4	1.2	2	45	0.1
	5120080103	0.8	4	1.2	3	45	0.1
	5120080104	0.8	4	1.2	4	45	0.1
	5120080106	0.8	4	1.2	6	45	0.1
	5120080108	0.8	4	1.2	8	45	0.1
	5120080110	0.8	4	1.2	10	45	0.1
	5120080202	0.8	4	1.2	2	45	0.2
	5120080203	0.8	4	1.2	3	45	0.2
	5120080204	0.8	4	1.2	4	45	0.2
	5120080206	0.8	4	1.2	6	45	0.2
	5120080208	0.8	4	1.2	8	45	0.2
	5120080210	0.8	4	1.2	10	45	0.2
	51201000503	1	4	1.5	3	50	0.05
	51201000504	1	4	1.5	4	50	0.05
	51201000506	1	4	1.5	6	50	0.05
	51201000508	1	4	1.5	8	50	0.05
	51201000510	1	4	1.5	10	50	0.05
	51201000512	1	4	1.5	12	50	0.05
	51201000514	1	4	1.5	14	50	0.05
	51201000516	1	4	1.5	16	50	0.05
	51201000520	1	4	1.5	20	50	0.05
	5120100103	1	4	1.5	3	50	0.1
	5120100104	1	4	1.5	4	50	0.1
	5120100106	1	4	1.5	6	50	0.1
	5120100108	1	4	1.5	8	50	0.1
	5120100110	1	4	1.5	10	50	0.1
	5120100112	1	4	1.5	12	50	0.1
	5120100114	1	4	1.5	14	50	0.1
	5120100116	1	4	1.5	16	50	0.1
	5120100120	1	4	1.5	20	50	0.1
	5120100203	1	4	1.5	3	50	0.2
	5120100204	1	4	1.5	4	50	0.2
	5120100206	1	4	1.5	6	50	0.2

	Designation	DC	DCON-MS	APMX	LU	LF	RE
UR	5120100208	1	4	1.5	8	50	0.2
	5120100210	1	4	1.5	10	50	0.2
	5120100212	1	4	1.5	12	50	0.2
	5120100214	1	4	1.5	14	50	0.2
	5120100216	1	4	1.5	16	50	0.2
	5120100220	1	4	1.5	20	50	0.2
	5120100303	1	4	1.5	3	50	0.3
	5120100304	1	4	1.5	4	50	0.3
	5120100306	1	4	1.5	6	50	0.3
	5120100308	1	4	1.5	8	50	0.3
	5120100310	1	4	1.5	10	50	0.3
	5120100312	1	4	1.5	12	50	0.3
	5120100314	1	4	1.5	14	50	0.3
	5120100316	1	4	1.5	16	50	0.3
	5120100320	1	4	1.5	20	50	0.3
	51201200503	1.2	4	1.8	3	50	0.05
	51201200504	1.2	4	1.8	4	50	0.05
	51201200506	1.2	4	1.8	6	50	0.05
	51201200508	1.2	4	1.8	8	50	0.05
	51201200510	1.2	4	1.8	10	50	0.05
	51201200512	1.2	4	1.8	12	50	0.05
	51201200516	1.2	4	1.8	16	50	0.05
	51201200520	1.2	4	1.8	20	50	0.05
	5120120103	1.2	4	1.8	3	50	0.1
	5120120104	1.2	4	1.8	4	50	0.1
	5120120106	1.2	4	1.8	6	50	0.1
	5120120108	1.2	4	1.8	8	50	0.1
	5120120110	1.2	4	1.8	10	50	0.1
	5120120112	1.2	4	1.8	12	50	0.1
	5120120116	1.2	4	1.8	16	50	0.1
	5120120120	1.2	4	1.8	20	50	0.1
	5120120203	1.2	4	1.8	3	50	0.2
	5120120204	1.2	4	1.8	4	50	0.2
	5120120206	1.2	4	1.8	6	50	0.2
	5120120208	1.2	4	1.8	8	50	0.2
	5120120210	1.2	4	1.8	10	50	0.2
	5120120212	1.2	4	1.8	12	50	0.2
	5120120216	1.2	4	1.8	16	50	0.2

UR512

2 flutes neck type radius



CARBIDE

2

H-A
30°

Grade
PC315W

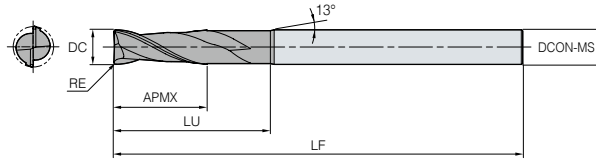
h5
shank

RE
±0.01

RE
±0.015

DC6 or below Above DC6

DC	Tolerance
Ø0.2 ~ Ø6	0.000 ~ -0.012
Ø8 ~ Ø20	0.000 ~ -0.015



Designation	DC	DCON-MS	APMX	LU	LF	RE
UR 5120120220	1.2	4	1.8	20	50	0.2
5120120303	1.2	4	1.8	3	50	0.3
5120120304	1.2	4	1.8	4	50	0.3
5120120306	1.2	4	1.8	6	50	0.3
5120120308	1.2	4	1.8	8	50	0.3
5120120310	1.2	4	1.8	10	50	0.3
5120120312	1.2	4	1.8	12	50	0.3
5120120316	1.2	4	1.8	16	50	0.3
5120120320	1.2	4	1.8	20	50	0.3
51201500504	1.5	4	2.3	4	50	0.05
51201500506	1.5	4	2.3	6	50	0.05
51201500508	1.5	4	2.3	8	50	0.05
51201500510	1.5	4	2.3	10	50	0.05
51201500512	1.5	4	2.3	12	50	0.05
51201500514	1.5	4	2.3	14	50	0.05
51201500516	1.5	4	2.3	16	50	0.05
51201500520	1.5	4	2.3	20	50	0.05
51201500522	1.5	4	2.3	22	60	0.05
51201500526	1.5	4	2.3	26	60	0.05
5120150104	1.5	4	2.3	4	50	0.1
5120150106	1.5	4	2.3	6	50	0.1
5120150108	1.5	4	2.3	8	50	0.1
5120150110	1.5	4	2.3	10	50	0.1
5120150112	1.5	4	2.3	12	50	0.1
5120150114	1.5	4	2.3	14	50	0.1
5120150116	1.5	4	2.3	16	50	0.1
5120150120	1.5	4	2.3	20	50	0.1
5120150122	1.5	4	2.3	22	60	0.1
5120150126	1.5	4	2.3	26	60	0.1
5120150204	1.5	4	2.3	4	50	0.2
5120150206	1.5	4	2.3	6	50	0.2
5120150208	1.5	4	2.3	8	50	0.2
5120150210	1.5	4	2.3	10	50	0.2
5120150212	1.5	4	2.3	12	50	0.2
5120150214	1.5	4	2.3	14	50	0.2
5120150216	1.5	4	2.3	16	50	0.2
5120150220	1.5	4	2.3	20	50	0.2
5120150222	1.5	4	2.3	22	60	0.2

Designation	DC	DCON-MS	APMX	LU	LF	RE
UR 5120150226	1.5	4	2.3	26	60	0.2
5120150304	1.5	4	2.3	4	50	0.3
5120150306	1.5	4	2.3	6	50	0.3
5120150308	1.5	4	2.3	8	50	0.3
5120150310	1.5	4	2.3	10	50	0.3
5120150312	1.5	4	2.3	12	50	0.3
5120150314	1.5	4	2.3	14	50	0.3
5120150316	1.5	4	2.3	16	50	0.3
5120150320	1.5	4	2.3	20	50	0.3
5120150322	1.5	4	2.3	22	60	0.3
5120150326	1.5	4	2.3	26	60	0.3
5120150504	1.5	4	2.3	4	50	0.5
5120150506	1.5	4	2.3	6	50	0.5
5120150508	1.5	4	2.3	8	50	0.5
5120150510	1.5	4	2.3	10	50	0.5
5120150512	1.5	4	2.3	12	50	0.5
5120150514	1.5	4	2.3	14	50	0.5
5120150516	1.5	4	2.3	16	50	0.5
5120150520	1.5	4	2.3	20	50	0.5
5120150522	1.5	4	2.3	22	60	0.5
5120150526	1.5	4	2.3	26	60	0.5
5120200106	2	4	3	6	50	0.1
5120200108	2	4	3	8	50	0.1
5120200110	2	4	3	10	50	0.1
5120200112	2	4	3	12	50	0.1
5120200114	2	4	3	14	50	0.1
5120200116	2	4	3	16	50	0.1
5120200120	2	4	3	20	50	0.1
5120200122	2	4	3	22	60	0.1
5120200126	2	4	3	26	60	0.1
5120200130	2	4	3	30	70	0.1
5120200206	2	4	3	6	50	0.2
5120200208	2	4	3	8	50	0.2
5120200210	2	4	3	10	50	0.2
5120200212	2	4	3	12	50	0.2
5120200214	2	4	3	14	50	0.2
5120200216	2	4	3	16	50	0.2
5120200220	2	4	3	20	50	0.2

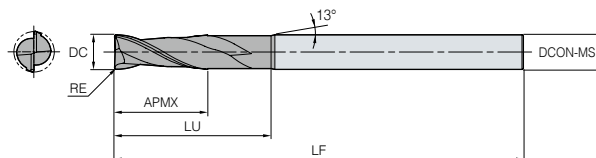
UR512

2 flutes neck type radius



DC6 or below Above DC6

DC	Tolerance
Ø0.2 ~ Ø6	0.000 ~ -0.012
Ø8 ~ Ø20	0.000 ~ -0.015



(mm)

	Designation	DC	DCON-MS	APMX	LU	LF	RE
UR	5120200222	2	4	3	22	60	0.2
	5120200226	2	4	3	26	60	0.2
	5120200230	2	4	3	30	70	0.2
	5120200306	2	4	3	6	50	0.3
	5120200308	2	4	3	8	50	0.3
	5120200310	2	4	3	10	50	0.3
	5120200312	2	4	3	12	50	0.3
	5120200314	2	4	3	14	50	0.3
	5120200316	2	4	3	16	50	0.3
	5120200320	2	4	3	20	50	0.3
	5120200322	2	4	3	22	60	0.3
	5120200326	2	4	3	26	60	0.3
	5120200330	2	4	3	30	70	0.3
	5120200506	2	4	3	6	50	0.5
	5120200508	2	4	3	8	50	0.5
	5120200510	2	4	3	10	50	0.5
	5120200512	2	4	3	12	50	0.5
	5120200514	2	4	3	14	50	0.5
	5120200516	2	4	3	16	50	0.5
	5120200520	2	4	3	20	50	0.5
	5120200522	2	4	3	22	60	0.5
	5120200526	2	4	3	26	60	0.5
	5120200530	2	4	3	30	70	0.5
	5120250108	2.5	4	4	8	50	0.1
	5120250110	2.5	4	4	10	50	0.1
	5120250112	2.5	4	4	12	50	0.1
	5120250114	2.5	4	4	14	50	0.1
	5120250116	2.5	4	4	16	50	0.1
	5120250120	2.5	4	4	20	50	0.1
	5120250126	2.5	4	4	26	60	0.1
	5120250130	2.5	4	4	30	70	0.1
	5120250208	2.5	4	4	8	50	0.2
	5120250210	2.5	4	4	10	50	0.2
	5120250212	2.5	4	4	12	50	0.2
	5120250214	2.5	4	4	14	50	0.2
	5120250216	2.5	4	4	16	50	0.2
	5120250220	2.5	4	4	20	50	0.2
	5120250226	2.5	4	4	26	60	0.2

	Designation	DC	DCON-MS	APMX	LU	LF	RE
UR	5120250230	2.5	4	4	30	70	0.2
	5120250308	2.5	4	4	8	50	0.3
	5120250310	2.5	4	4	10	50	0.3
	5120250312	2.5	4	4	12	50	0.3
	5120250314	2.5	4	4	14	50	0.3
	5120250316	2.5	4	4	16	50	0.3
	5120250320	2.5	4	4	20	50	0.3
	5120250326	2.5	4	4	26	60	0.3
	5120250330	2.5	4	4	30	70	0.3
	5120250508	2.5	4	4	8	50	0.5
	5120250510	2.5	4	4	10	50	0.5
	5120250512	2.5	4	4	12	50	0.5
	5120250514	2.5	4	4	14	50	0.5
	5120250516	2.5	4	4	16	50	0.5
	5120250520	2.5	4	4	20	50	0.5
	5120250526	2.5	4	4	26	60	0.5
	5120250530	2.5	4	4	30	70	0.5
	5120300108	3	6	4.5	8	50	0.1
	5120300110	3	6	4.5	10	50	0.1
	5120300112	3	6	4.5	12	50	0.1
	5120300114	3	6	4.5	14	60	0.1
	5120300116	3	6	4.5	16	60	0.1
	5120300120	3	6	4.5	20	60	0.1
	5120300126	3	6	4.5	26	65	0.1
	5120300130	3	6	4.5	30	70	0.1
	5120300135	3	6	4.5	35	70	0.1
	5120300140	3	6	4.5	40	80	0.1
	5120300208	3	6	4.5	8	50	0.2
	5120300210	3	6	4.5	10	50	0.2
	5120300212	3	6	4.5	12	50	0.2
	5120300214	3	6	4.5	14	60	0.2
	5120300216	3	6	4.5	16	60	0.2
	5120300220	3	6	4.5	20	60	0.2
	5120300226	3	6	4.5	26	65	0.2
	5120300230	3	6	4.5	30	70	0.2
	5120300235	3	6	4.5	35	70	0.2
	5120300240	3	6	4.5	40	80	0.2
	5120300308	3	6	4.5	8	50	0.3

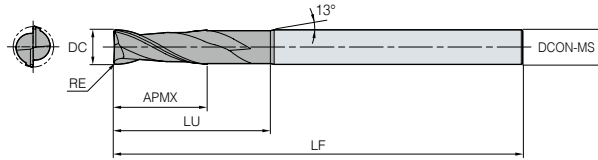
UR512

2 flutes neck type radius



DC	Tolerance
Ø0.2 ~ Ø6	0.000 ~ -0.012
Ø8 ~ Ø20	0.000 ~ -0.015

DC6 or below Above DC6



Designation	DC	DCON-MS	APMX	LU	LF	RE
UR 5120300310	3	6	4.5	10	50	0.3
5120300312	3	6	4.5	12	50	0.3
5120300314	3	6	4.5	14	60	0.3
5120300316	3	6	4.5	16	60	0.3
5120300320	3	6	4.5	20	60	0.3
5120300326	3	6	4.5	26	65	0.3
5120300330	3	6	4.5	30	70	0.3
5120300335	3	6	4.5	35	70	0.3
5120300340	3	6	4.5	40	80	0.3
5120300508	3	6	4.5	8	50	0.5
5120300510	3	6	4.5	10	50	0.5
5120300512	3	6	4.5	12	50	0.5
5120300514	3	6	4.5	14	60	0.5
5120300516	3	6	4.5	16	60	0.5
5120300520	3	6	4.5	20	60	0.5
5120300526	3	6	4.5	26	65	0.5
5120300530	3	6	4.5	30	70	0.5
5120300535	3	6	4.5	35	70	0.5
5120300540	3	6	4.5	40	80	0.5
5120301008	3	6	4.5	8	50	1
5120301010	3	6	4.5	10	50	1
5120301012	3	6	4.5	12	50	1
5120301014	3	6	4.5	14	60	1
5120301016	3	6	4.5	16	60	1
5120301020	3	6	4.5	20	60	1
5120301026	3	6	4.5	26	65	1
5120301030	3	6	4.5	30	70	1
5120301035	3	6	4.5	35	70	1
5120301040	3	6	4.5	40	80	1
5120400110	4	6	6	10	50	0.1
5120400112	4	6	6	12	50	0.1
5120400114	4	6	6	14	60	0.1
5120400116	4	6	6	16	60	0.1
5120400120	4	6	6	20	60	0.1
5120400126	4	6	6	26	65	0.1
5120400130	4	6	6	30	65	0.1
5120400135	4	6	6	35	70	0.1
5120400140	4	6	6	40	80	0.1

Designation	DC	DCON-MS	APMX	LU	LF	RE
UR 5120400145	4	6	6	45	90	0.1
5120400150	4	6	6	50	100	0.1
5120400210	4	6	6	10	50	0.2
5120400212	4	6	6	12	50	0.2
5120400214	4	6	6	14	60	0.2
5120400216	4	6	6	16	60	0.2
5120400220	4	6	6	20	60	0.2
5120400226	4	6	6	26	65	0.2
5120400230	4	6	6	30	65	0.2
5120400235	4	6	6	35	70	0.2
5120400240	4	6	6	40	80	0.2
5120400245	4	6	6	45	90	0.2
5120400250	4	6	6	50	100	0.2
5120400310	4	6	6	10	50	0.3
5120400312	4	6	6	12	50	0.3
5120400314	4	6	6	14	60	0.3
5120400316	4	6	6	16	60	0.3
5120400320	4	6	6	20	60	0.3
5120400326	4	6	6	26	65	0.3
5120400330	4	6	6	30	65	0.3
5120400335	4	6	6	35	70	0.3
5120400340	4	6	6	40	80	0.3
5120400345	4	6	6	45	90	0.3
5120400350	4	6	6	50	100	0.3
5120400510	4	6	6	10	50	0.5
5120400512	4	6	6	12	50	0.5
5120400514	4	6	6	14	60	0.5
5120400516	4	6	6	16	60	0.5
5120400520	4	6	6	20	60	0.5
5120400526	4	6	6	26	65	0.5
5120400530	4	6	6	30	65	0.5
5120400535	4	6	6	35	70	0.5
5120400540	4	6	6	40	80	0.5
5120400545	4	6	6	45	90	0.5
5120400550	4	6	6	50	100	0.5
5120401010	4	6	6	10	50	1
5120401012	4	6	6	12	50	1
5120401014	4	6	6	14	60	1

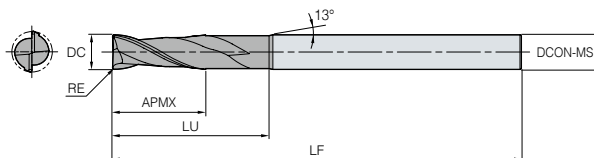
UR512

2 flutes neck type radius



DC6 or below Above DC6

DC	Tolerance
Ø0.2 ~ Ø6	0.000 ~ -0.012
Ø8 ~ Ø20	0.000 ~ -0.015



(mm)

	Designation	DC	DCON-MS	APMX	LU	LF	RE
UR	5120401016	4	6	6	16	60	1
	5120401020	4	6	6	20	60	1
	5120401026	4	6	6	26	65	1
	5120401030	4	6	6	30	65	1
	5120401035	4	6	6	35	70	1
	5120401040	4	6	6	40	80	1
	5120401045	4	6	6	45	90	1
	5120401050	4	6	6	50	100	1
	51205001	5	6	8	15	60	0.1
	51205002	5	6	8	15	60	0.2
	51205003	5	6	8	15	60	0.3
	51205005	5	6	8	15	60	0.5
	51205010	5	6	8	15	60	1
	51205015	5	6	8	15	60	1.5
	51205020	5	6	8	15	60	2
	51206001	6	6	9	20	60	0.1
	51206002	6	6	9	20	60	0.2
	51206003	6	6	9	20	60	0.3
	5120600390	6	6	15	30	90	0.3
	51206005	6	6	9	20	60	0.5
	5120600590	6	6	15	30	90	0.5
	51206010	6	6	9	20	60	1
	5120601090	6	6	15	30	90	1
	51206015	6	6	9	20	60	1.5
	51206020	6	6	9	20	60	2
	51208001	8	8	12	25	70	0.1
	51208002	8	8	12	25	70	0.2
	51208003	8	8	12	25	70	0.3
	51208003100	8	8	20	35	100	0.3
	51208005	8	8	12	25	70	0.5
	51208005100	8	8	20	35	100	0.5

	Designation	DC	DCON-MS	APMX	LU	LF	RE
UR	51208010	8	8	12	25	70	1
	51208010100	8	8	20	35	100	1
	51208015	8	8	12	25	70	1.5
	51208020	8	8	12	25	70	2
	51210001	10	10	15	30	75	0.1
	51210002	10	10	15	30	75	0.2
	51210003	10	10	15	30	75	0.3
	51210003100	10	10	25	40	100	0.3
	51210005	10	10	15	30	75	0.5
	51210005100	10	10	25	40	100	0.5
	51210010	10	10	15	30	75	1
	51210010100	10	10	25	40	100	1
	51210015	10	10	15	30	75	1.5
	51210020	10	10	15	30	75	2
	51212002	12	12	18	32	80	0.2
	51212003	12	12	18	32	80	0.3
	51212003110	12	12	30	45	110	0.3
	51212005	12	12	18	32	80	0.5
	51212005110	12	12	30	45	110	0.5
	51212010	12	12	18	32	80	1
	51212010110	12	12	30	45	110	1
	51212015	12	12	18	32	80	1.5
	51212020	12	12	18	32	80	2
	51216005	16	16	20	35	100	0.5
	51216005150	16	16	35	50	150	0.5
	51216010	16	16	20	35	100	1
	51216010150	16	16	35	50	150	1
	51220005	20	20	25	40	100	0.5
	51220005150	20	20	40	55	150	0.5
	51220010	20	20	25	40	100	1
	51220010150	20	20	40	55	150	1

UR542

2 flutes tapered neck



CARBIDE

2

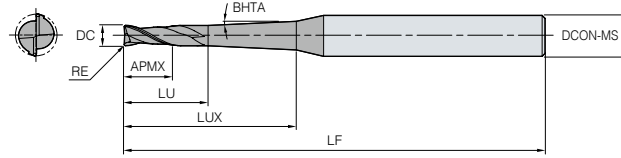
H-A
30°

Grade
PC315W

h5
shank

RE
±0.01

DC	Tolerance
Ø0.2 ~ Ø4	0.000 ~ -0.012



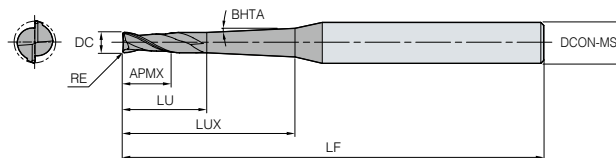
Designation	DC	D CON-MS	APMX	LU	LUX	LF	BHTA(°)	RE
UR 5420020050101	0.2	4	0.3	0.4	1	40	1	0.05
5420020050102	0.2	4	0.3	0.4	2	40	1	0.05
5420020050103	0.2	4	0.3	0.4	3	40	1	0.05
5420020050201	0.2	4	0.3	0.4	1	40	2	0.05
5420020050202	0.2	4	0.3	0.4	2	40	2	0.05
5420020050203	0.2	4	0.3	0.4	3	40	2	0.05
5420030050102	0.3	4	0.5	0.6	2	40	1	0.05
5420030050103	0.3	4	0.5	0.6	3	40	1	0.05
5420030050104	0.3	4	0.5	0.6	4	40	1	0.05
5420030050105	0.3	4	0.5	0.6	5	40	1	0.05
5420030050202	0.3	4	0.5	0.6	2	40	2	0.05
5420030050203	0.3	4	0.5	0.6	3	40	2	0.05
5420030050204	0.3	4	0.5	0.6	4	40	2	0.05
5420030050205	0.3	4	0.5	0.6	5	40	2	0.05
5420040050102	0.4	4	0.6	0.8	2	50	1	0.05
5420040050103	0.4	4	0.6	0.8	3	50	1	0.05
5420040050104	0.4	4	0.6	0.8	4	50	1	0.05
5420040050105	0.4	4	0.6	0.8	5	50	1	0.05
5420040050106	0.4	4	0.6	0.8	6	50	1	0.05
5420040050202	0.4	4	0.6	0.8	2	50	2	0.05
5420040050203	0.4	4	0.6	0.8	3	50	2	0.05
5420040050204	0.4	4	0.6	0.8	4	50	2	0.05
5420040050205	0.4	4	0.6	0.8	5	50	2	0.05
5420040050206	0.4	4	0.6	0.8	6	50	2	0.05
542004010102	0.4	4	0.6	0.8	2	50	1	0.1
542004010103	0.4	4	0.6	0.8	3	50	1	0.1
542004010104	0.4	4	0.6	0.8	4	50	1	0.1
542004010105	0.4	4	0.6	0.8	5	50	1	0.1
542004010106	0.4	4	0.6	0.8	6	50	1	0.1
542004010202	0.4	4	0.6	0.8	2	50	2	0.1
542004010203	0.4	4	0.6	0.8	3	50	2	0.1
542004010204	0.4	4	0.6	0.8	4	50	2	0.1
542004010205	0.4	4	0.6	0.8	5	50	2	0.1

Designation	DC	D CON-MS	APMX	LU	LUX	LF	BHTA(°)	RE
UR 542004010206	0.4	4	0.6	0.8	6	50	2	0.1
5420050050104	0.5	4	0.7	1	4	50	1	0.05
5420050050106	0.5	4	0.7	1	6	50	1	0.05
5420050050108	0.5	4	0.7	1	8	50	1	0.05
5420050050110	0.5	4	0.7	1	10	50	1	0.05
5420050050204	0.5	4	0.7	1	4	50	2	0.05
5420050050206	0.5	4	0.7	1	6	50	2	0.05
5420050050208	0.5	4	0.7	1	8	50	2	0.05
5420050050210	0.5	4	0.7	1	10	50	2	0.05
542005010104	0.5	4	0.7	1	4	50	1	0.1
542005010106	0.5	4	0.7	1	6	50	1	0.1
542005010108	0.5	4	0.7	1	8	50	1	0.1
542005010110	0.5	4	0.7	1	10	50	1	0.1
542005010204	0.5	4	0.7	1	4	50	2	0.1
542005010206	0.5	4	0.7	1	6	50	2	0.1
542005010208	0.5	4	0.7	1	8	50	2	0.1
542005010210	0.5	4	0.7	1	10	50	2	0.1
542006010104	0.6	4	0.9	1.2	4	50	1	0.1
542006010106	0.6	4	0.9	1.2	6	50	1	0.1
542006010108	0.6	4	0.9	1.2	8	50	1	0.1
542006010110	0.6	4	0.9	1.2	10	50	1	0.1
542006010112	0.6	4	0.9	1.2	12	50	1	0.1
542006010204	0.6	4	0.9	1.2	4	50	2	0.1
542006010206	0.6	4	0.9	1.2	6	50	2	0.1
542006010208	0.6	4	0.9	1.2	8	50	2	0.1
542006010210	0.6	4	0.9	1.2	10	50	2	0.1
542006010212	0.6	4	0.9	1.2	12	50	2	0.1
542006020104	0.6	4	0.9	1.2	4	50	1	0.2
542006020106	0.6	4	0.9	1.2	6	50	1	0.2
542006020108	0.6	4	0.9	1.2	8	50	1	0.2
542006020110	0.6	4	0.9	1.2	10	50	1	0.2
542006020112	0.6	4	0.9	1.2	12	50	1	0.2
542006020204	0.6	4	0.9	1.2	4	50	2	0.2

(mm)

UR542

2 flutes tapered neck



(mm)

	Designation	DC	DCON-MS	APMX	LU	LUX	LF	BHTA(°)	RE
UR	542006020206	0.6	4	0.9	1.2	6	50	2	0.2
	542006020208	0.6	4	0.9	1.2	8	50	2	0.2
	542006020210	0.6	4	0.9	1.2	10	50	2	0.2
	542006020212	0.6	4	0.9	1.2	12	50	2	0.2
	542008010104	0.8	4	1.2	1.6	4	50	1	0.1
	542008010106	0.8	4	1.2	1.6	6	50	1	0.1
	542008010108	0.8	4	1.2	1.6	8	50	1	0.1
	542008010110	0.8	4	1.2	1.6	10	50	1	0.1
	542008010112	0.8	4	1.2	1.6	12	50	1	0.1
	542008010116	0.8	4	1.2	1.6	16	50	1	0.1
	542008010204	0.8	4	1.2	1.6	4	50	2	0.1
	542008010206	0.8	4	1.2	1.6	6	50	2	0.1
	542008010208	0.8	4	1.2	1.6	8	50	2	0.1
	542008010210	0.8	4	1.2	1.6	10	50	2	0.1
	542008010212	0.8	4	1.2	1.6	12	50	2	0.1
	542008010216	0.8	4	1.2	1.6	16	50	2	0.1
	542008020104	0.8	4	1.2	1.6	4	50	1	0.2
	542008020106	0.8	4	1.2	1.6	6	50	1	0.2
	542008020108	0.8	4	1.2	1.6	8	50	1	0.2
	542008020110	0.8	4	1.2	1.6	10	50	1	0.2
	542008020112	0.8	4	1.2	1.6	12	50	1	0.2
	542008020116	0.8	4	1.2	1.6	16	50	1	0.2
	542008020204	0.8	4	1.2	1.6	4	50	2	0.2
	542008020206	0.8	4	1.2	1.6	6	50	2	0.2
	542008020208	0.8	4	1.2	1.6	8	50	2	0.2
	542008020210	0.8	4	1.2	1.6	10	50	2	0.2
	542008020212	0.8	4	1.2	1.6	12	50	2	0.2
	542008020216	0.8	4	1.2	1.6	16	50	2	0.2
	542010010106	1	4	1.5	2.5	6	50	1	0.1
	542010010108	1	4	1.5	2.5	8	50	1	0.1
	542010010110	1	4	1.5	2.5	10	50	1	0.1
	542010010112	1	4	1.5	2.5	12	50	1	0.1
	542010010116	1	4	1.5	2.5	16	50	1	0.1

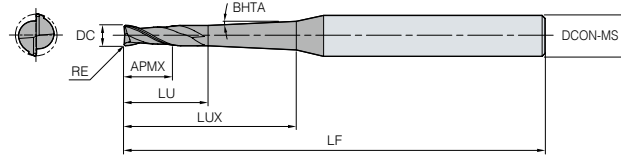
	Designation	DC	DCON-MS	APMX	LU	LUX	LF	BHTA(°)	RE
UR	542010010120	1	4	1.5	2.5	20	50	1	0.1
	542010010125	1	4	1.5	2.5	25	60	1	0.1
	542010010130	1	4	1.5	2.5	30	70	1	0.1
	542010010140	1	4	1.5	2.5	40	80	1	0.1
	542010010150	1	6	1.5	2.5	50	90	1	0.1
	542010010206	1	4	1.5	2.5	6	50	2	0.1
	542010010208	1	4	1.5	2.5	8	50	2	0.1
	542010010210	1	4	1.5	2.5	10	50	2	0.1
	542010010212	1	4	1.5	2.5	12	50	2	0.1
	542010010216	1	4	1.5	2.5	16	50	2	0.1
	542010010220	1	4	1.5	2.5	20	50	2	0.1
	542010010225	1	4	1.5	2.5	25	60	2	0.1
	542010010230	1	4	1.5	2.5	30	70	2	0.1
	542010010240	1	4	1.5	2.5	40	80	2	0.1
	542010010250	1	6	1.5	2.5	50	90	2	0.1
	542010020106	1	4	1.5	2.5	6	50	1	0.2
	542010020108	1	4	1.5	2.5	8	50	1	0.2
	542010020110	1	4	1.5	2.5	10	50	1	0.2
	542010020112	1	4	1.5	2.5	12	50	1	0.2
	542010020116	1	4	1.5	2.5	16	50	1	0.2
	542010020120	1	4	1.5	2.5	20	50	1	0.2
	542010020125	1	4	1.5	2.5	25	60	1	0.2
	542010020130	1	4	1.5	2.5	30	70	1	0.2
	542010020140	1	4	1.5	2.5	40	80	1	0.2
	542010020150	1	6	1.5	2.5	50	90	1	0.2
	542010020206	1	4	1.5	2.5	6	50	2	0.2
	542010020208	1	4	1.5	2.5	8	50	2	0.2
	542010020210	1	4	1.5	2.5	10	50	2	0.2
	542010020212	1	4	1.5	2.5	12	50	2	0.2
	542010020216	1	4	1.5	2.5	16	50	2	0.2
	542010020220	1	4	1.5	2.5	20	50	2	0.2
	542010020225	1	4	1.5	2.5	25	60	2	0.2
	542010020230	1	4	1.5	2.5	30	70	2	0.2

UR542

2 flutes tapered neck



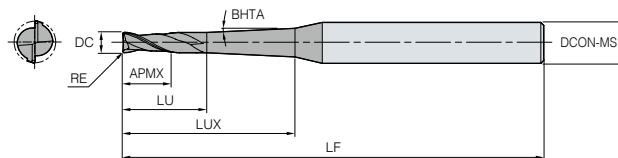
CARBIDE	2	H-A 30°	Grade PC315W	h5 shank	RE ±0.01	DC Ø0.2 ~ Ø4	Tolerance 0.000 ~ -0.012
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										(mm)									
Designation		DC	DCON-MS	APMX	LU	LUX	LF	BHTA(°)	RE	Designation		DC	DCON-MS	APMX	LU	LUX	LF	BHTA(°)	RE
UR	542010020240	1	4	1.5	2.5	40	80	2	0.2	UR	542015010140	1.5	4	2.3	3	40	80	1	0.1
	542010020250	1	6	1.5	2.5	50	90	2	0.2		542015010150	1.5	4	2.3	3	50	90	1	0.1
	542012010108	1.2	4	1.8	3	8	50	1	0.1		542015010208	1.5	4	2.3	3	8	50	2	0.1
	542012010112	1.2	4	1.8	3	12	50	1	0.1		542015010210	1.5	4	2.3	3	10	50	2	0.1
	542012010116	1.2	4	1.8	3	16	50	1	0.1		542015010212	1.5	4	2.3	3	12	50	2	0.1
	542012010120	1.2	4	1.8	3	20	50	1	0.1		542015010216	1.5	4	2.3	3	16	50	2	0.1
	542012010125	1.2	4	1.8	3	25	60	1	0.1		542015010220	1.5	4	2.3	3	20	50	2	0.1
	542012010130	1.2	4	1.8	3	30	70	1	0.1		542015010225	1.5	4	2.3	3	25	60	2	0.1
	542012010208	1.2	4	1.8	3	8	50	2	0.1		542015010230	1.5	4	2.3	3	30	70	2	0.1
	542012010212	1.2	4	1.8	3	12	50	2	0.1		542015010240	1.5	6	2.3	3	40	80	2	0.1
	542012010216	1.2	4	1.8	3	16	50	2	0.1		542015010250	1.5	6	2.3	3	50	90	2	0.1
	542012010220	1.2	4	1.8	3	20	50	2	0.1		542015020108	1.5	4	2.3	3	8	50	1	0.2
	542012010225	1.2	4	1.8	3	25	60	2	0.1		542015020110	1.5	4	2.3	3	10	50	1	0.2
	542012010230	1.2	4	1.8	3	30	70	2	0.1		542015020112	1.5	4	2.3	3	12	50	1	0.2
	542012020108	1.2	4	1.8	3	8	50	1	0.2		542015020116	1.5	4	2.3	3	16	50	1	0.2
	542012020112	1.2	4	1.8	3	12	50	1	0.2		542015020120	1.5	4	2.3	3	20	50	1	0.2
	542012020116	1.2	4	1.8	3	16	50	1	0.2		542015020125	1.5	4	2.3	3	25	60	1	0.2
	542012020120	1.2	4	1.8	3	20	50	1	0.2		542015020130	1.5	4	2.3	3	30	70	1	0.2
	542012020125	1.2	4	1.8	3	25	60	1	0.2		542015020140	1.5	4	2.3	3	40	80	1	0.2
	542012020130	1.2	4	1.8	3	30	70	1	0.2		542015020150	1.5	4	2.3	3	50	90	1	0.2
	542012020208	1.2	4	1.8	3	8	50	2	0.2		542015020208	1.5	4	2.3	3	8	50	2	0.2
	542012020212	1.2	4	1.8	3	12	50	2	0.2		542015020210	1.5	4	2.3	3	10	50	2	0.2
	542012020216	1.2	4	1.8	3	16	50	2	0.2		542015020212	1.5	4	2.3	3	12	50	2	0.2
	542012020220	1.2	4	1.8	3	20	50	2	0.2		542015020216	1.5	4	2.3	3	16	50	2	0.2
	542012020225	1.2	4	1.8	3	25	60	2	0.2		542015020220	1.5	4	2.3	3	20	50	2	0.2
	542012020230	1.2	4	1.8	3	30	70	2	0.2		542015020225	1.5	4	2.3	3	25	60	2	0.2
	542015010108	1.5	4	2.3	3	8	50	1	0.1		542015020230	1.5	4	2.3	3	30	70	2	0.2
	542015010110	1.5	4	2.3	3	10	50	1	0.1		542015020240	1.5	6	2.3	3	40	80	2	0.2
	542015010112	1.5	4	2.3	3	12	50	1	0.1		542015020250	1.5	6	2.3	3	50	90	2	0.2
	542015010116	1.5	4	2.3	3	16	50	1	0.1		542015030108	1.5	4	2.3	3	8	50	1	0.3
	542015010120	1.5	4	2.3	3	20	50	1	0.1		542015030110	1.5	4	2.3	3	10	50	1	0.3
	542015010125	1.5	4	2.3	3	25	60	1	0.1		542015030112	1.5	4	2.3	3	12	50	1	0.3
	542015010130	1.5	4	2.3	3	30	70	1	0.1		542015030116	1.5	4	2.3	3	16	50	1	0.3

UR542

2 flutes tapered neck



(mm)

	Designation	DC	DCON-MS	APMX	LU	LUX	LF	BHTA(°)	RE
UR	542015030120	1.5	4	2.3	3	20	50	1	0.3
	542015030125	1.5	4	2.3	3	25	60	1	0.3
	542015030130	1.5	4	2.3	3	30	70	1	0.3
	542015030140	1.5	4	2.3	3	40	80	1	0.3
	542015030150	1.5	4	2.3	3	50	90	1	0.3
	542015030208	1.5	4	2.3	3	8	50	2	0.3
	542015030210	1.5	4	2.3	3	10	50	2	0.3
	542015030212	1.5	4	2.3	3	12	50	2	0.3
	542015030216	1.5	4	2.3	3	16	50	2	0.3
	542015030220	1.5	4	2.3	3	20	50	2	0.3
	542015030225	1.5	4	2.3	3	25	60	2	0.3
	542015030230	1.5	4	2.3	3	30	70	2	0.3
	542015030240	1.5	6	2.3	3	40	80	2	0.3
	542015030250	1.5	6	2.3	3	50	90	2	0.3
	542020010110	2	4	2	5	10	50	1	0.1
	542020010112	2	4	2	5	12	50	1	0.1
	542020010116	2	4	2	5	16	50	1	0.1
	542020010120	2	4	2	5	20	50	1	0.1
	542020010125	2	4	2	5	25	60	1	0.1
	542020010130	2	4	2	5	30	70	1	0.1
	542020010140	2	6	2	5	40	80	1	0.1
	542020010150	2	6	2	5	50	100	1	0.1
	542020010160	2	6	2	5	60	100	1	0.1
	542020010180	2	6	2	5	80	140	1	0.1
	542020010210	2	4	2	5	10	50	2	0.1
	542020010212	2	4	2	5	12	50	2	0.1
	542020010216	2	4	2	5	16	50	2	0.1
	542020010220	2	4	2	5	20	50	2	0.1
	542020010225	2	4	2	5	25	60	2	0.1
	542020010230	2	4	2	5	30	70	2	0.1
	542020010240	2	6	2	5	40	80	2	0.1
	542020010250	2	6	2	5	50	100	2	0.1
	542020010260	2	6	2	5	60	100	2	0.1

	Designation	DC	DCON-MS	APMX	LU	LUX	LF	BHTA(°)	RE
UR	542020010280	2	8	2	5	80	140	2	0.1
	542020020110	2	4	2	5	10	50	1	0.2
	542020020112	2	4	2	5	12	50	1	0.2
	542020020116	2	4	2	5	16	50	1	0.2
	542020020120	2	4	2	5	20	50	1	0.2
	542020020125	2	4	2	5	25	60	1	0.2
	542020020130	2	4	2	5	30	70	1	0.2
	542020020140	2	6	2	5	40	80	1	0.2
	542020020150	2	6	2	5	50	100	1	0.2
	542020020160	2	6	2	5	60	100	1	0.2
	542020020180	2	6	2	5	80	140	1	0.2
	542020020210	2	4	2	5	10	50	2	0.2
	542020020212	2	4	2	5	12	50	2	0.2
	542020020216	2	4	2	5	16	50	2	0.2
	542020020220	2	4	2	5	20	50	2	0.2
	542020020225	2	4	2	5	25	60	2	0.2
	542020020230	2	4	2	5	30	70	2	0.2
	542020020240	2	6	2	5	40	80	2	0.2
	542020020250	2	6	2	5	50	100	2	0.2
	542020020260	2	6	2	5	60	100	2	0.2
	542020020280	2	8	2	5	80	140	2	0.2
	542020030110	2	4	2	5	10	50	1	0.3
	542020030112	2	4	2	5	12	50	1	0.3
	542020030116	2	4	2	5	16	50	1	0.3
	542020030120	2	4	2	5	20	50	1	0.3
	542020030125	2	4	2	5	25	60	1	0.3
	542020030130	2	4	2	5	30	70	1	0.3
	542020030140	2	6	2	5	40	80	1	0.3
	542020030150	2	6	2	5	50	100	1	0.3
	542020030160	2	6	2	5	60	100	1	0.3
	542020030180	2	6	2	5	80	140	1	0.3
	542020030210	2	4	2	5	10	50	2	0.3
	542020030212	2	4	2	5	12	50	2	0.3

UR542

2 flutes tapered neck



CARBIDE

2

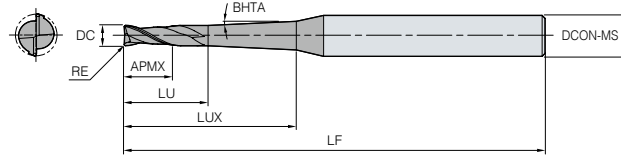
H-A
30°

Grade
PC315W

h5
shank

RE
±0.01

DC	Tolerance
Ø0.2 ~ Ø4	0.000 ~ -0.012



Designation	DC	DCON-MS	APMX	LU	LUX	LF	BHTA(°)	RE
UR 542020030216	2	4	2	5	16	50	2	0.3
542020030220	2	4	2	5	20	50	2	0.3
542020030225	2	4	2	5	25	60	2	0.3
542030050260	3	8	4.5	6	60	100	2	0.5
542030050270	3	8	4.5	6	70	120	2	0.5
542040020140	4	6	6	8	40	90	1	0.2
542040020150	4	6	6	8	50	100	1	0.2
542020030230	2	4	2	5	30	70	2	0.3
542020030240	2	6	2	5	40	80	2	0.3
542020030250	2	6	2	5	50	100	2	0.3
542020030260	2	6	2	5	60	100	2	0.3
542020030280	2	8	2	5	80	140	2	0.3
542020050110	2	4	2	5	10	50	1	0.5
542020050112	2	4	2	5	12	50	1	0.5
542020050116	2	4	2	5	16	50	1	0.5
542020050120	2	4	2	5	20	50	1	0.5
542020050125	2	4	2	5	25	60	1	0.5
542020050130	2	4	2	5	30	70	1	0.5
542020050140	2	6	2	5	40	80	1	0.5
542020050150	2	6	2	5	50	100	1	0.5
542020050160	2	6	2	5	60	100	1	0.5
542020050180	2	6	2	5	80	140	1	0.5
542020050210	2	4	2	5	10	50	2	0.5
542020050212	2	4	2	5	12	50	2	0.5
542020050216	2	4	2	5	16	50	2	0.5
542020050220	2	4	2	5	20	50	2	0.5
542020050225	2	4	2	5	25	60	2	0.5
542020050230	2	4	2	5	30	70	2	0.5
542020050240	2	6	2	5	40	80	2	0.5
542020050250	2	6	2	5	50	100	2	0.5
542020050260	2	6	2	5	60	100	2	0.5
542020050280	2	8	2	5	80	140	2	0.5
542030020116	3	6	4.5	6	16	60	1	0.2

Designation	DC	DCON-MS	APMX	LU	LUX	LF	BHTA(°)	RE
UR 542030020120	3	6	4.5	6	20	65	1	0.2
542030020130	3	6	4.5	6	30	70	1	0.2
542030020140	3	6	4.5	6	40	80	1	0.2
542030020150	3	6	4.5	6	50	90	1	0.2
542030020160	3	6	4.5	6	60	100	1	0.2
542030020216	3	6	4.5	6	16	60	2	0.2
542030020220	3	6	4.5	6	20	65	2	0.2
542030020230	3	6	4.5	6	30	70	2	0.2
542030020240	3	6	4.5	6	40	80	2	0.2
542030020250	3	8	4.5	6	50	90	2	0.2
542030020260	3	8	4.5	6	60	100	2	0.2
542030020270	3	8	4.5	6	70	120	2	0.2
542030030116	3	6	4.5	6	16	60	1	0.3
542030030120	3	6	4.5	6	20	65	1	0.3
542030030130	3	6	4.5	6	30	70	1	0.3
542030030140	3	6	4.5	6	40	80	1	0.3
542030030150	3	6	4.5	6	50	90	1	0.3
542030030160	3	6	4.5	6	60	100	1	0.3
542030030216	3	6	4.5	6	16	60	2	0.3
542030030220	3	6	4.5	6	20	65	2	0.3
542030030230	3	6	4.5	6	30	70	2	0.3
542030030240	3	6	4.5	6	40	80	2	0.3
542030030250	3	8	4.5	6	50	90	2	0.3
542030030260	3	8	4.5	6	60	100	2	0.3
542030030270	3	8	4.5	6	70	120	2	0.3
542030050116	3	6	4.5	6	16	60	1	0.5
542030050120	3	6	4.5	6	20	65	1	0.5
542030050130	3	6	4.5	6	30	70	1	0.5
542030050140	3	6	4.5	6	40	80	1	0.5
542030050150	3	6	4.5	6	50	90	1	0.5
542030050160	3	6	4.5	6	60	100	1	0.5
542030050216	3	6	4.5	6	16	60	2	0.5
542030050220	3	6	4.5	6	20	65	2	0.5

(mm)

UR542

2 flutes tapered neck



CARBIDE

2

H-A
30°

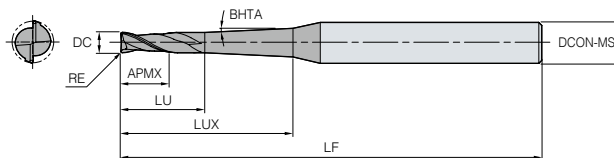
Grade
PC315W

h5
shank

RE
±0.01

DC
Ø0.2 ~ Ø4

Tolerance
0.000 ~ -0.012



(mm)

	Designation	DC	DCON-MS	APMX	LU	LUX	LF	BHTA(°)	RE
UR	542030050230	3	6	4.5	6	30	70	2	0.5
	542030050240	3	6	4.5	6	40	80	2	0.5
	542030050250	3	8	4.5	6	50	90	2	0.5
	542040020160	4	6	6	8	60	110	1	0.2
	542040020170	4	8	6	8	70	120	1	0.2
	542040020240	4	8	6	8	40	90	2	0.2
	542040020250	4	8	6	8	50	100	2	0.2
	542040020260	4	8	6	8	60	110	2	0.2
	542040020270	4	10	6	8	70	120	2	0.2
	542040030140	4	6	6	8	40	90	1	0.3
	542040030150	4	6	6	8	50	100	1	0.3
	542040030160	4	6	6	8	60	110	1	0.3
	542040030170	4	8	6	8	70	120	1	0.3

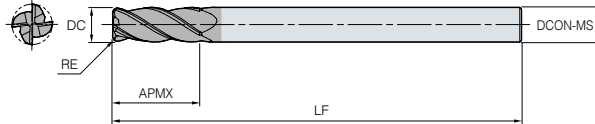
	Designation	DC	DCON-MS	APMX	LU	LUX	LF	BHTA(°)	RE
UR	542040030240	4	8	6	8	40	90	2	0.3
	542040030250	4	8	6	8	50	100	2	0.3
	542040030260	4	8	6	8	60	110	2	0.3
	542040030270	4	10	6	8	70	120	2	0.3
	542040050140	4	6	6	8	40	90	1	0.5
	542040050150	4	6	6	8	50	100	1	0.5
	542040050160	4	6	6	8	60	110	1	0.5
	542040050170	4	8	6	8	70	120	1	0.5
	542040050240	4	8	6	8	40	90	2	0.5
	542040050250	4	8	6	8	50	100	2	0.5
	542040050260	4	8	6	8	60	110	2	0.5
	542040050270	4	10	6	8	70	120	2	0.5

UR504

4 flutes radius



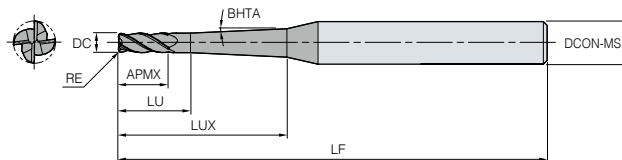
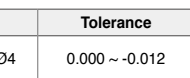
CARBIDE	4	H-A 40°	Grade PC315W	h5 shank	RE ±0.01	DC Ø3 ~ Ø20	Tolerance 0.000 ~ -0.030
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							(mm)						
Designation		DC	DCON-MS	APMX	LF	RE	Designation		DC	DCON-MS	APMX	LF	RE
UR	50403002	3	6	8	60	0.2	UR	50410003075	10	10	25	75	0.3
	50403002S4	3	4	8	60	0.2		50410005	10	10	25	100	0.5
	50403003	3	6	8	60	0.3		50410005075	10	10	25	75	0.5
	50403005	3	6	8	60	0.5		50410010	10	10	25	100	1
	50403005S4	3	4	8	60	0.5		50410010075	10	10	25	75	1
	50404002	4	6	10	70	0.2		50410015	10	10	25	100	1.5
	50404002S4	4	4	10	70	0.2		50410020	10	10	25	100	2
	50404003	4	6	10	70	0.3		50410025	10	10	25	100	2.5
	50404005	4	6	10	70	0.5		50412003	12	12	30	110	0.3
	50404005S4	4	4	10	70	0.5		50412003080	12	12	30	80	0.3
	50404010	4	6	10	70	1		50412005	12	12	30	110	0.5
	50404010S4	4	4	10	70	1		50412005080	12	12	30	80	0.5
	50405002	5	6	13	90	0.2		50412010	12	12	30	110	1
	50405003	5	6	13	90	0.3		50412010080	12	12	30	80	1
	50405003060	5	6	13	60	0.3		50412015	12	12	30	110	1.5
	50405005	5	6	13	90	0.5		50412020	12	12	30	110	2
	50405005060	5	6	13	60	0.5		50412025	12	12	30	110	2.5
	50405010	5	6	13	90	1		50412030	12	12	30	110	3
	50406002	6	6	13	90	0.2		50416005	16	16	32	150	0.5
	50406003	6	6	15	90	0.3		50416005100	16	16	32	100	0.5
	50406003060	6	6	15	60	0.3		50416010	16	16	32	150	1
	50406005	6	6	15	90	0.5		50416010100	16	16	32	100	1
	50406005060	6	6	15	60	0.5		50416015	16	16	32	150	1.5
	50406010	6	6	15	90	1		50416015100	16	16	32	100	1.5
	50406010060	6	6	15	60	1		50416020	16	16	32	150	2
	50408003	8	8	20	100	0.3		50416020100	16	16	32	100	2
	50408003070	8	8	20	70	0.3		50420005	20	20	38	150	0.5
	50408005	8	8	20	100	0.5		50420005100	20	20	38	100	0.5
	50408005070	8	8	20	70	0.5		50420010	20	20	38	150	1
	50408010	8	8	20	100	1		50420010100	20	20	38	100	1
	50408010070	8	8	20	70	1		50420015	20	20	38	150	1.5
	50408015	8	8	20	100	1.5		50420015100	20	20	38	100	1.5
	50408020	8	8	20	100	2		50420020	20	20	38	150	2
	50410003	10	10	25	100	0.3		50420020100	20	20	38	100	2

UR544

4 flutes tapered neck radius



(mm)

	Designation	DC	DCON-MS	APMX	LU	LUX	LF	BHTA(°)	RE
UR	544010010106	1	4	1.5	2.5	6	50	1	0.1
	544010010108	1	4	1.5	2.5	8	50	1	0.1
	544010010110	1	4	1.5	2.5	10	50	1	0.1
	544010010112	1	4	1.5	2.5	12	50	1	0.1
	544010010116	1	4	1.5	2.5	16	50	1	0.1
	544010010120	1	4	1.5	2.5	20	50	1	0.1
	544010010125	1	4	1.5	2.5	25	60	1	0.1
	544010010130	1	4	1.5	2.5	30	70	1	0.1
	544010010140	1	4	1.5	2.5	40	80	1	0.1
	544010010150	1	4	1.5	2.5	50	90	1	0.1
	544010010206	1	4	1.5	2.5	6	50	1	0.1
	544010010208	1	4	1.5	2.5	8	50	2	0.1
	544010010210	1	4	1.5	2.5	10	50	2	0.1
	544010010212	1	4	1.5	2.5	12	50	2	0.1
	544010010216	1	4	1.5	2.5	16	50	2	0.1
	544010010220	1	4	1.5	2.5	20	50	2	0.1
	544010010225	1	4	1.5	2.5	25	60	2	0.1
	544010010230	1	4	1.5	2.5	30	70	2	0.1
	544010010240	1	4	1.5	2.5	40	80	2	0.1
	544010010250	1	6	1.5	2.5	50	90	2	0.1
	544010020106	1	4	1.5	2.5	6	50	2	0.2
	544010020108	1	4	1.5	2.5	8	50	1	0.2
	544010020110	1	4	1.5	2.5	10	50	1	0.2
	544010020112	1	4	1.5	2.5	12	50	1	0.2
	544010020116	1	4	1.5	2.5	16	50	1	0.2
	544010020120	1	4	1.5	2.5	20	50	1	0.2
	544010020125	1	4	1.5	2.5	25	60	1	0.2
	544010020130	1	4	1.5	2.5	30	70	1	0.2
	544010020140	1	4	1.5	2.5	40	80	1	0.2
	544010020150	1	4	1.5	2.5	50	90	1	0.2
	544010020206	1	4	1.5	2.5	6	50	1	0.2
	544010020208	1	4	1.5	2.5	8	50	2	0.2
	544010020210	1	4	1.5	2.5	10	50	2	0.2
	544010020212	1	4	1.5	2.5	12	50	2	0.2
	544010020216	1	4	1.5	2.5	16	50	2	0.2
	544010020220	1	4	1.5	2.5	20	50	2	0.2

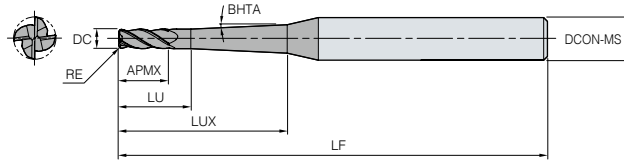
	Designation	DC	DCON-MS	APMX	LU	LUX	LF	BHTA(°)	RE
UR	544010020225	1	4	1.5	2.5	25	60	2	0.2
	544010020230	1	4	1.5	2.5	30	70	2	0.2
	544010020240	1	4	1.5	2.5	40	80	2	0.2
	544010020250	1	6	1.5	2.5	50	90	2	0.2
	544012010108	1.2	4	1.8	3	8	50	2	0.1
	544012010112	1.2	4	1.8	3	12	50	1	0.1
	544012010116	1.2	4	1.8	3	16	50	1	0.1
	544012010120	1.2	4	1.8	3	20	50	1	0.1
	544012010125	1.2	4	1.8	3	25	60	1	0.1
	544012010130	1.2	4	1.8	3	30	70	1	0.1
	544012010208	1.2	4	1.8	3	8	50	1	0.1
	544012010212	1.2	4	1.8	3	12	50	2	0.1
	544012010216	1.2	4	1.8	3	16	50	2	0.1
	544012010220	1.2	4	1.8	3	20	50	2	0.1
	544012010225	1.2	4	1.8	3	25	60	2	0.1
	544012010230	1.2	4	1.8	3	30	70	2	0.1
	544012020108	1.2	4	1.8	3	8	50	2	0.2
	544012020112	1.2	4	1.8	3	12	50	1	0.2
	544012020116	1.2	4	1.8	3	16	50	1	0.2
	544012020120	1.2	4	1.8	3	20	50	1	0.2
	544012020125	1.2	4	1.8	3	25	60	1	0.2
	544012020130	1.2	4	1.8	3	30	70	1	0.2
	544012020208	1.2	4	1.8	3	8	50	1	0.2
	544012020212	1.2	4	1.8	3	12	50	2	0.2
	544012020216	1.2	4	1.8	3	16	50	2	0.2
	544012020220	1.2	4	1.8	3	20	50	2	0.2
	544012020225	1.2	4	1.8	3	25	60	2	0.2
	544012020230	1.2	4	1.8	3	30	70	2	0.2
	544015010108	1.5	4	2.3	3	8	50	2	0.1
	544015010110	1.5	4	2.3	3	10	50	1	0.1
	544015010112	1.5	4	2.3	3	12	50	1	0.1
	544015010116	1.5	4	2.3	3	16	50	1	0.1
	544015010120	1.5	4	2.3	3	20	50	1	0.1
	544015010125	1.5	4	2.3	3	25	60	1	0.1
	544015010130	1.5	4	2.3	3	30	70	1	0.1
	544015010140	1.5	4	2.3	3	40	80	1	0.1

UR544

4 flutes tapered neck radius



CARBIDE	4	H-A 30°	Grade PC315W	h5 shank	RE ±0.01	DC Ø1 ~ Ø4	Tolerance 0.000 ~ -0.012
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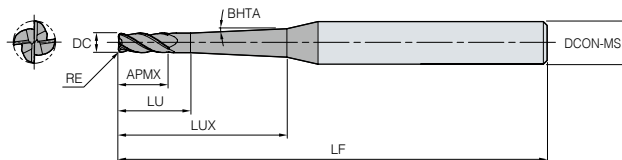
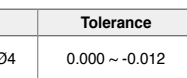


Designation	DC	DCON-MS	APMX	LU	LUX	LF	BHTA(°)	RE
UR 544015010150	1.5	4	2.3	3	50	90	1	0.1
544015010208	1.5	4	2.3	3	8	50	1	0.1
544015010210	1.5	4	2.3	3	10	50	2	0.1
544015010212	1.5	4	2.3	3	12	50	2	0.1
544015010216	1.5	4	2.3	3	16	50	2	0.1
544015010220	1.5	4	2.3	3	20	50	2	0.1
544015010225	1.5	4	2.3	3	25	60	2	0.1
544015010230	1.5	4	2.3	3	30	70	2	0.1
544015010240	1.5	6	2.3	3	40	80	2	0.1
544015010250	1.5	6	2.3	3	50	90	2	0.1
544015020108	1.5	4	2.3	3	8	50	2	0.2
544015020110	1.5	4	2.3	3	10	50	1	0.2
544015020112	1.5	4	2.3	3	12	50	1	0.2
544015020116	1.5	4	2.3	3	16	50	1	0.2
544015020120	1.5	4	2.3	3	20	50	1	0.2
544015020125	1.5	4	2.3	3	25	60	1	0.2
544015020130	1.5	4	2.3	3	30	70	1	0.2
544015020140	1.5	4	2.3	3	40	80	1	0.2
544015020150	1.5	4	2.3	3	50	90	1	0.2
544015020208	1.5	4	2.3	3	8	50	1	0.2
544015020210	1.5	4	2.3	3	10	50	2	0.2
544015020212	1.5	4	2.3	3	12	50	2	0.2
544015020216	1.5	4	2.3	3	16	50	2	0.2
544015020220	1.5	4	2.3	3	20	50	2	0.2
544015020225	1.5	4	2.3	3	25	60	2	0.2
544015020230	1.5	4	2.3	3	30	70	2	0.2
544015020240	1.5	6	2.3	3	40	80	2	0.2
544015020250	1.5	6	2.3	3	50	90	2	0.2
544015030108	1.5	4	2.3	3	8	50	2	0.3
544015030110	1.5	4	2.3	3	10	50	1	0.3
544015030112	1.5	4	2.3	3	12	50	1	0.3
544015030116	1.5	4	2.3	3	16	50	1	0.3
544015030120	1.5	4	2.3	3	20	50	1	0.3
544015030125	1.5	4	2.3	3	25	60	1	0.3
544015030130	1.5	4	2.3	3	30	70	1	0.3
544015030140	1.5	4	2.3	3	40	80	1	0.3

Designation	DC	DCON-MS	APMX	LU	LUX	LF	BHTA(°)	RE
UR 544015030150	1.5	4	2.3	3	50	90	1	0.3
544015030208	1.5	4	2.3	3	8	50	1	0.3
544015030210	1.5	4	2.3	3	10	50	2	0.3
544015030212	1.5	4	2.3	3	12	50	2	0.3
544015030216	1.5	4	2.3	3	16	50	2	0.3
544015030220	1.5	4	2.3	3	20	50	2	0.3
544015030225	1.5	4	2.3	3	25	60	2	0.3
544015030230	1.5	4	2.3	3	30	70	2	0.3
544015030240	1.5	6	2.3	3	40	80	2	0.3
544015030250	1.5	6	2.3	3	50	90	2	0.3
544020010110	2	4	2	5	10	50	2	0.1
544020010112	2	4	2	5	12	50	1	0.1
544020010116	2	4	2	5	16	50	1	0.1
544020010120	2	4	2	5	20	50	1	0.1
544020010125	2	4	2	5	25	60	1	0.1
544020010130	2	4	2	5	30	70	1	0.1
544020010140	2	6	2	5	40	80	1	0.1
544020010150	2	6	2	5	50	100	1	0.1
544020010160	2	6	2	5	60	100	1	0.1
544020010180	2	6	2	5	80	140	1	0.1
544020010210	2	4	2	5	10	50	1	0.1
544020010212	2	4	2	5	12	50	2	0.1
544020010216	2	4	2	5	16	50	2	0.1
544020010220	2	4	2	5	20	50	2	0.1
544020010225	2	4	2	5	25	60	2	0.1
544020010230	2	4	2	5	30	70	2	0.1
544020010240	2	6	2	5	40	80	2	0.1
544020010250	2	6	2	5	50	100	2	0.1
544020010260	2	6	2	5	60	100	2	0.1
544020010280	2	8	2	5	80	140	2	0.1
544020020110	2	4	2	5	10	50	2	0.2
544020020112	2	4	2	5	12	50	1	0.2
544020020116	2	4	2	5	16	50	1	0.2
544020020120	2	4	2	5	20	50	1	0.2
544020020125	2	4	2	5	25	60	1	0.2
544020020130	2	4	2	5	30	70	1	0.2

UR544

4 flutes tapered neck radius



(mm)

	Designation	DC	DCON-MS	APMX	LU	LUX	LF	BHTA(°)	RE
UR	544020020140	2	6	2	5	40	80	1	0.2
	544020020150	2	6	2	5	50	100	1	0.2
	544020020160	2	6	2	5	60	100	1	0.2
	544020020180	2	6	2	5	80	140	1	0.2
	544020020210	2	4	2	5	10	50	1	0.2
	544020020212	2	4	2	5	12	50	2	0.2
	544020020216	2	4	2	5	16	50	2	0.2
	544020020220	2	4	2	5	20	50	2	0.2
	544020020225	2	4	2	5	25	60	2	0.2
	544020020230	2	4	2	5	30	70	2	0.2
	544020020240	2	6	2	5	40	80	2	0.2
	544020020250	2	6	2	5	50	100	2	0.2
	544020020260	2	6	2	5	60	100	2	0.2
	544020020280	2	8	2	5	80	140	2	0.2
	544020030110	2	4	2	5	10	50	2	0.3
	544020030112	2	4	2	5	12	50	1	0.3
	544020030116	2	4	2	5	16	50	1	0.3
	544020030120	2	4	2	5	20	50	1	0.3
	544020030125	2	4	2	5	25	60	1	0.3
	544020030130	2	4	2	5	30	70	1	0.3
	544020030140	2	6	2	5	40	80	1	0.3
	544020030150	2	6	2	5	50	100	1	0.3
	544020030160	2	6	2	5	60	100	1	0.3
	544020030180	2	6	2	5	80	140	1	0.3
	544020030210	2	4	2	5	10	50	1	0.3
	544020030212	2	4	2	5	12	50	2	0.3
	544020030216	2	4	2	5	16	50	2	0.3
	544020030220	2	4	2	5	20	50	2	0.3
	544020030225	2	4	2	5	25	60	2	0.3
	544020030230	2	4	2	5	30	70	2	0.3
	544020030240	2	6	2	5	40	80	2	0.3
	544020030250	2	6	2	5	50	100	2	0.3
	544020030260	2	6	2	5	60	100	2	0.3
	544020030280	2	8	2	5	80	140	2	0.3
	544020050110	2	4	2	5	10	50	2	0.5
	544020050112	2	4	2	5	12	50	1	0.5

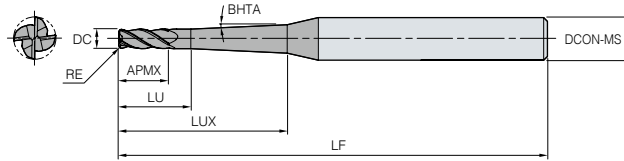
	Designation	DC	DCON-MS	APMX	LU	LUX	LF	BHTA(°)	RE
UR	544020050116	2	4	2	5	16	50	1	0.5
	544020050120	2	4	2	5	20	50	1	0.5
	544020050125	2	4	2	5	25	60	1	0.5
	544020050130	2	4	2	5	30	70	1	0.5
	544020050140	2	6	2	5	40	80	1	0.5
	544020050150	2	6	2	5	50	100	1	0.5
	544020050160	2	6	2	5	60	100	1	0.5
	544020050180	2	6	2	5	80	140	1	0.5
	544020050210	2	4	2	5	10	50	1	0.5
	544020050212	2	4	2	5	12	50	2	0.5
	544020050216	2	4	2	5	16	50	2	0.5
	544020050220	2	4	2	5	20	50	2	0.5
	544020050225	2	4	2	5	25	60	2	0.5
	544020050230	2	4	2	5	30	70	2	0.5
	544020050240	2	6	2	5	40	80	2	0.5
	544020050250	2	6	2	5	50	100	2	0.5
	544020050260	2	6	2	5	60	100	2	0.5
	544020050280	2	8	2	5	80	140	2	0.5
	544030020116	3	6	4.5	6	16	60	2	0.2
	544030020120	3	6	4.5	6	20	65	1	0.2
	544030020130	3	6	4.5	6	30	70	1	0.2
	544030020140	3	6	4.5	6	40	80	1	0.2
	544030020150	3	6	4.5	6	50	90	1	0.2
	544030020160	3	6	4.5	6	60	100	1	0.2
	544030020216	3	6	4.5	6	16	60	1	0.2
	544030020220	3	6	4.5	6	20	65	2	0.2
	544030020230	3	6	4.5	6	30	70	2	0.2
	544030020240	3	6	4.5	6	40	80	2	0.2
	544030020250	3	8	4.5	6	50	90	2	0.2
	544030020260	3	8	4.5	6	60	100	2	0.2
	544030020270	3	8	4.5	6	70	120	2	0.2
	544030030116	3	6	4.5	6	16	60	2	0.3
	544030030120	3	6	4.5	6	20	65	1	0.3
	544030030130	3	6	4.5	6	30	70	1	0.3
	544030030140	3	6	4.5	6	40	80	1	0.3
	544030030150	3	6	4.5	6	50	90	1	0.3

UR544

4 flutes tapered neck radius



CARBIDE	4	H-A 30°	Grade PC315W	h5 shank	RE ±0.01	DC Ø1 ~ Ø4	Tolerance 0.000 ~ -0.012
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Designation	DC	DCON-MS	APMX	LU	LUX	LF	BHTA(°)	RE
UR 544030030160	3	6	4.5	6	60	100	1	0.3
544030030216	3	6	4.5	6	16	60	1	0.3
544030030220	3	6	4.5	6	20	65	2	0.3
544030030230	3	6	4.5	6	30	70	2	0.3
544030030240	3	6	4.5	6	40	80	2	0.3
544030030250	3	8	4.5	6	50	90	2	0.3
544030030260	3	8	4.5	6	60	100	2	0.3
544030030270	3	8	4.5	6	70	120	2	0.3
544030050116	3	6	4.5	6	16	60	2	0.5
544030050120	3	6	4.5	6	20	65	1	0.5
544030050130	3	6	4.5	6	30	70	1	0.5
544030050140	3	6	4.5	6	40	80	1	0.5
544030050150	3	6	4.5	6	50	90	1	0.5
544030050160	3	6	4.5	6	60	100	1	0.5
544030050216	3	6	4.5	6	16	60	1	0.5
544030050220	3	6	4.5	6	20	65	2	0.5
544030050230	3	6	4.5	6	30	70	2	0.5
544030050240	3	6	4.5	6	40	80	2	0.5
544030050250	3	8	4.5	6	50	90	2	0.5
544030050260	3	8	4.5	6	60	100	2	0.5
544030050270	3	8	4.5	6	70	120	2	0.5
544040020140	4	6	6	8	40	90	2	0.2
544040020150	4	6	6	8	50	100	1	0.2

(mm)

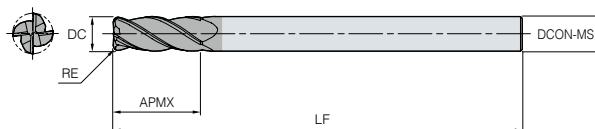
Designation	DC	DCON-MS	APMX	LU	LUX	LF	BHTA(°)	RE
UR 544040020160	4	6	6	8	60	110	1	0.2
544040020170	4	8	6	8	70	120	1	0.2
544040020240	4	8	6	8	40	90	1	0.2
544040020250	4	8	6	8	50	100	2	0.2
544040020260	4	8	6	8	60	110	2	0.2
544040020270	4	10	6	8	70	120	2	0.2
544040030140	4	6	6	8	40	90	2	0.3
544040030150	4	6	6	8	50	100	1	0.3
544040030160	4	6	6	8	60	110	1	0.3
544040030170	4	8	6	8	70	120	1	0.3
544040030240	4	8	6	8	40	90	1	0.3
544040030250	4	8	6	8	50	100	2	0.3
544040030260	4	8	6	8	60	110	2	0.3
544040030270	4	10	6	8	70	120	2	0.3
544040050140	4	6	6	8	40	90	2	0.5
544040050150	4	6	6	8	50	100	1	0.5
544040050160	4	6	6	8	60	110	1	0.5
544040050170	4	8	6	8	70	120	1	0.5
544040050240	4	8	6	8	40	90	1	0.5
544040050250	4	8	6	8	50	100	2	0.5
544040050260	4	8	6	8	60	110	2	0.5
544040050270	4	10	6	8	70	120	2	0.5

UXR504

4 flutes multi helix radius



CARBIDE	4	H-A 34°/36°	Grade PC315W	h5 shank	RE ±0.02	DC	Tolerance
						Ø1 ~ Ø20	0.000 ~ -0.030



(mm)

	Designation	DC	DCON-MS	APMX	LF	RE
UXR	504010005	1	6	2.5	50	0.05
	50401001	1	6	2.5	50	0.1
	50401002	1	6	2.5	50	0.2
	50401003	1	6	2.5	50	0.3
	504012005	1.2	6	3	50	0.05
	50401201	1.2	6	3	50	0.1
	50401202	1.2	6	3	50	0.2
	50401203	1.2	6	3	50	0.3
	504015005	1.5	6	4	50	0.05
	50401501	1.5	6	4	50	0.1
	50401502	1.5	6	4	50	0.2
	50401503	1.5	6	4	50	0.3
	50401505	1.5	6	4	50	0.5
	50402001	2	6	6	50	0.1
	50402002	2	6	6	50	0.2
	50402003	2	6	6	50	0.3
	50402005	2	6	6	50	0.5
	50402501	2.5	6	7	60	0.1
	50402502	2.5	6	7	60	0.2
	50402503	2.5	6	7	60	0.3
	50402505	2.5	6	7	60	0.5
	50403001	3	6	8	60	0.1
	50403002	3	6	8	60	0.2
	50403003	3	6	8	60	0.3
	50403005	3	6	8	60	0.5
	50403010	3	6	8	60	1
	50403501	3.5	6	10	70	0.1
	50403502	3.5	6	10	70	0.2
	50403503	3.5	6	10	70	0.3
	50403505	3.5	6	10	70	0.5
	50404001	4	6	10	70	0.1
	50404001100S4	4	4	10	100	0.1
	50404001S4	4	4	10	70	0.1
	50404002	4	6	10	70	0.2
	50404002100S4	4	4	10	100	0.2
	50404002S4	4	4	10	70	0.2

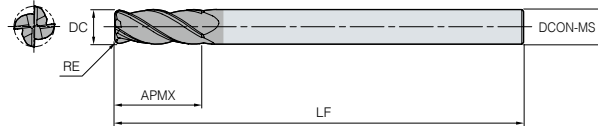
	Designation	DC	DCON-MS	APMX	LF	RE
UXR	50404003	4	6	10	70	0.3
	50404003100S4	4	4	10	100	0.3
	50404003S4	4	4	10	70	0.3
	50404005	4	6	10	70	0.5
	50404005100S4	4	4	10	100	0.5
	50404005S4	4	4	10	70	0.5
	50404010	4	6	10	70	1
	50404010100S4	4	4	10	100	1
	50404010S4	4	4	10	70	1
	50404501	4.5	6	11	80	0.1
	50404502	4.5	6	11	80	0.2
	50404503	4.5	6	11	80	0.3
	50404505	4.5	6	11	80	0.5
	50405001	5	6	13	90	0.1
	50405002	5	6	13	90	0.2
	50405003	5	6	13	90	0.3
	50405005	5	6	13	90	0.5
	50405010	5	6	13	90	1
	50405501	5.5	6	13	90	0.1
	50405502	5.5	6	13	90	0.2
	50405503	5.5	6	13	90	0.3
	50405505	5.5	6	13	90	0.5
	50405510	5.5	6	13	90	1
	50406001	6	6	15	90	0.1
	50406001060	6	6	15	60	0.1
	50406002	6	6	15	90	0.2
	50406002060	6	6	15	60	0.2
	50406003	6	6	15	90	0.3
	50406005	6	6	15	90	0.5
	50406005060	6	6	15	60	0.5
	50406005110	6	6	15	110	0.5
	50406005130	6	6	15	130	0.5
	50406010	6	6	15	90	1
	50406010060	6	6	15	60	1
	50406010110	6	6	15	110	1
	50406010130	6	6	15	130	1

UXR504

4 flutes multi helix radius



CARBIDE	4	H-A 34°/36°	Grade PC315W	h5 shank	RE ±0.02	DC Ø1 ~ Ø20	Tolerance 0.000 ~ -0.030
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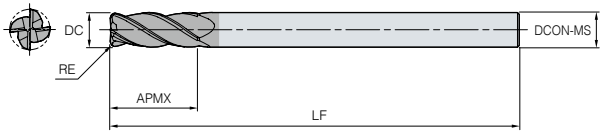
																					(mm)
Designation		DC	DCON-MS	APMX	LF	RE	Designation		DC	DCON-MS	APMX	LF	RE	Designation		DC	DCON-MS	APMX	LF	RE	
UXR	50406015	6	6	15	90	1.5	UXR	50410010130	10	10	22	130	1	UXR	50410010150	10	10	22	150	1	
	50406020	6	6	15	90	2		50410015	10	10	25	100	1.5		50410020	10	10	25	100	2	
	50407001	7	8	16	90	0.1		50410025	10	10	25	100	2.5		50410030	10	10	25	100	3	
	50407002	7	8	16	90	0.2		50410040	10	10	25	100	4		50411002	11	12	25	110	0.2	
	50407003	7	8	16	90	0.3		50411003	11	12	25	110	0.3		50411005	11	12	25	110	0.5	
	50407005	7	8	16	90	0.5		50411010	11	12	25	110	1		50411020	11	12	25	110	2	
	50407010	7	8	16	90	1		50412001	12	12	30	110	0.1		50412002	12	12	30	110	0.2	
	50407020	7	8	16	90	2		50412003	12	12	30	110	0.3		50412003080	12	12	30	80	0.3	
	50408001	8	8	20	100	0.1		50412005	12	12	30	110	0.5		50412005080	12	12	30	80	0.5	
	50408002	8	8	20	100	0.2		50412005130	12	12	30	130	0.5		50412005150	12	12	30	150	0.5	
	50408003	8	8	20	100	0.3		50412010	12	12	30	110	1		50412010080	12	12	30	80	1	
	50408003070	8	8	20	70	0.3		50412010130	12	12	30	130	1		50412010150	12	12	30	150	1	
	50408005	8	8	20	100	0.5		50412015	12	12	30	110	1.5		50412015080	12	12	30	80	1.5	
	50408005070	8	8	20	70	0.5		50412020	12	12	30	110	2		50412020080	12	12	30	80	2	
	50408005120	8	8	20	120	0.5		50412025	12	12	30	110	2.5		50412025080	12	12	30	80	2.5	
	50408005150	8	8	20	150	0.5		50412030	12	12	30	110	3		50412030080	12	12	30	80	3	
	50408010	8	8	20	100	1		50412040	12	12	30	110	4		50412050	12	12	30	110	5	
	50408010070	8	8	20	70	1		50414005	14	16	35	150	0.5		50414010	14	16	35	150	1	
	50408010120	8	8	20	120	1															
	50408010150	8	8	20	150	1															
	50408015	8	8	20	100	1.5															
	50408015070	8	8	20	70	1.5															
	50408020	8	8	20	100	2															
	50408020070	8	8	20	70	2															
	50408025	8	8	20	100	2.5															
	50408030	8	8	20	100	3															
	50410001	10	10	25	100	0.1															
	50410002	10	10	25	100	0.2															
	50410003	10	10	25	100	0.3															
	50410003075	10	10	25	75	0.3															
	50410005	10	10	25	100	0.5															
	50410005075	10	10	25	75	0.5															
	50410005130	10	10	22	130	0.5															
	50410005150	10	10	22	150	0.5															
	50410010	10	10	25	100	1															
	50410010075	10	10	25	75	1															

UXR504

4 flutes multi helix radius



CARBIDE	4	H-A 34°/36°	Grade PC315W	h5 shank	RE ±0.02	DC Ø1 ~ Ø20	Tolerance 0.000 ~ -0.030
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Designation		DC	DCON-MS	APMX	LF	RE
UXR	50414020	14	16	35	150	2
	50416005	16	16	40	100	0.5
	50416005150	16	16	40	150	0.5
	50416010	16	16	40	100	1
	50416010150	16	16	40	150	1
	50416015	16	16	40	100	1.5
	50416015150	16	16	40	150	1.5
	50416020	16	16	40	100	2
	50416020150	16	16	40	150	2
	50416030	16	16	40	100	3

(mm)

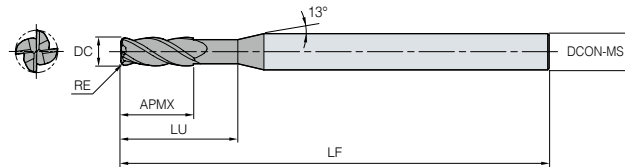
Designation		DC	DCON-MS	APMX	LF	RE
UXR	50420005	20	20	45	110	0.5
	50420005150	20	20	45	150	0.5
	50420010	20	20	45	110	1
	50420010150	20	20	45	150	1
	50420015	20	20	45	110	1.5
	50420015150	20	20	45	150	1.5
	50420020	20	20	45	110	2
	50420020150	20	20	45	150	2
	50420030	20	20	45	110	3

UXR514

4 flutes multi helix neck radius



CARBIDE	4	H-A 34°/36°	Grade PC315W	h5 shank	RE ±0.02	DC Ø1 ~ Ø20	Tolerance 0.000 ~ -0.030
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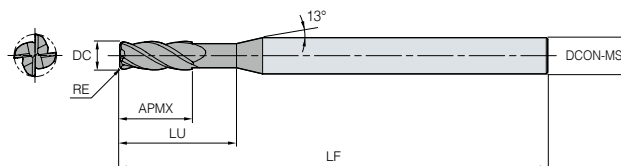
																(mm)
Designation		DC	DCON-MS	APMX	LU	LF	RE	Designation		DC	DCON-MS	APMX	LU	LF	RE	
UXR	51401000503	1	4	1.5	3	50	0.05	UXR	51401200503	1.2	4	1.8	3	50	0.05	
	51401000504	1	4	1.5	4	50	0.05		51401200504	1.2	4	1.8	4	50	0.05	
	51401000506	1	4	1.5	6	50	0.05		51401200506	1.2	4	1.8	6	50	0.05	
	51401000508	1	4	1.5	8	50	0.05		51401200508	1.2	4	1.8	8	50	0.05	
	51401000510	1	4	1.5	10	50	0.05		51401200510	1.2	4	1.8	10	50	0.05	
	51401000512	1	4	1.5	12	50	0.05		51401200512	1.2	4	1.8	12	50	0.05	
	51401000514	1	4	1.5	14	50	0.05		51401200516	1.2	4	1.8	16	50	0.05	
	51401000516	1	4	1.5	16	50	0.05		51401200520	1.2	4	1.8	20	50	0.05	
	51401000520	1	4	1.5	20	50	0.05		5140120103	1.2	4	1.8	3	50	0.1	
	5140100103	1	4	1.5	3	50	0.1		5140120104	1.2	4	1.8	4	50	0.1	
	5140100104	1	4	1.5	4	50	0.1		5140120106	1.2	4	1.8	6	50	0.1	
	5140100106	1	4	1.5	6	50	0.1		5140120108	1.2	4	1.8	8	50	0.1	
	5140100108	1	4	1.5	8	50	0.1		5140120110	1.2	4	1.8	10	50	0.1	
	5140100110	1	4	1.5	10	50	0.1		5140120112	1.2	4	1.8	12	50	0.1	
	5140100112	1	4	1.5	12	50	0.1		5140120116	1.2	4	1.8	16	50	0.1	
	5140100114	1	4	1.5	14	50	0.1		5140120120	1.2	4	1.8	20	50	0.1	
	5140100116	1	4	1.5	16	50	0.1		5140120203	1.2	4	1.8	3	50	0.2	
	5140100120	1	4	1.5	20	50	0.1		5140120204	1.2	4	1.8	4	50	0.2	
	5140100203	1	4	1.5	3	50	0.2		5140120206	1.2	4	1.8	6	50	0.2	
	5140100204	1	4	1.5	4	50	0.2		5140120208	1.2	4	1.8	8	50	0.2	
	5140100206	1	4	1.5	6	50	0.2		5140120210	1.2	4	1.8	10	50	0.2	
	5140100208	1	4	1.5	8	50	0.2		5140120212	1.2	4	1.8	12	50	0.2	
	5140100210	1	4	1.5	10	50	0.2		5140120216	1.2	4	1.8	16	50	0.2	
	5140100212	1	4	1.5	12	50	0.2		5140120220	1.2	4	1.8	20	50	0.2	
	5140100214	1	4	1.5	14	50	0.2		5140120303	1.2	4	1.8	3	50	0.3	
	5140100216	1	4	1.5	16	50	0.2		5140120304	1.2	4	1.8	4	50	0.3	
	5140100220	1	4	1.5	20	50	0.2		5140120306	1.2	4	1.8	6	50	0.3	
	5140100303	1	4	1.5	3	50	0.3		5140120308	1.2	4	1.8	8	50	0.3	
	5140100304	1	4	1.5	4	50	0.3		5140120310	1.2	4	1.8	10	50	0.3	
	5140100306	1	4	1.5	6	50	0.3		5140120312	1.2	4	1.8	12	50	0.3	
	5140100308	1	4	1.5	8	50	0.3		5140120316	1.2	4	1.8	16	50	0.3	
	5140100310	1	4	1.5	10	50	0.3		5140120320	1.2	4	1.8	20	50	0.3	
	5140100312	1	4	1.5	12	50	0.3		51401500504	1.5	4	2.3	4	50	0.05	
	5140100314	1	4	1.5	14	50	0.3		51401500506	1.5	4	2.3	6	50	0.05	
	5140100316	1	4	1.5	16	50	0.3		51401500508	1.5	4	2.3	8	50	0.05	
	5140100320	1	4	1.5	20	50	0.3		51401500510	1.5	4	2.3	10	50	0.05	

UXR514

4 flutes multi helix neck radius



CARBIDE	4	H-A 34°/36°	Grade PC315W	h5 shank	RE ±0.02	DC Ø1 ~ Ø20	Tolerance 0.000 ~ -0.030
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(mm)

	Designation	DC	DCON-MS	APMX	LU	LF	RE
UXR	51401500512	1.5	4	2.3	12	50	0.05
	51401500514	1.5	4	2.3	14	50	0.05
	51401500516	1.5	4	2.3	16	50	0.05
	51401500520	1.5	4	2.3	20	50	0.05
	51401500522	1.5	4	2.3	22	60	0.05
	51401500526	1.5	4	2.3	26	60	0.05
	5140150104	1.5	4	2.3	4	50	0.1
	5140150106	1.5	4	2.3	6	50	0.1
	5140150108	1.5	4	2.3	8	50	0.1
	5140150110	1.5	4	2.3	10	50	0.1
	5140150112	1.5	4	2.3	12	50	0.1
	5140150114	1.5	4	2.3	14	50	0.1
	5140150116	1.5	4	2.3	16	50	0.1
	5140150120	1.5	4	2.3	20	50	0.1
	5140150122	1.5	4	2.3	22	60	0.1
	5140150126	1.5	4	2.3	26	60	0.1
	5140150204	1.5	4	2.3	4	50	0.2
	5140150206	1.5	4	2.3	6	50	0.2
	5140150208	1.5	4	2.3	8	50	0.2
	5140150210	1.5	4	2.3	10	50	0.2
	5140150212	1.5	4	2.3	12	50	0.2
	5140150214	1.5	4	2.3	14	50	0.2
	5140150216	1.5	4	2.3	16	50	0.2
	5140150220	1.5	4	2.3	20	50	0.2
	5140150222	1.5	4	2.3	22	60	0.2
	5140150226	1.5	4	2.3	26	60	0.2
	5140150304	1.5	4	2.3	4	50	0.3
	5140150306	1.5	4	2.3	6	50	0.3
	5140150308	1.5	4	2.3	8	50	0.3
	5140150310	1.5	4	2.3	10	50	0.3
	5140150312	1.5	4	2.3	12	50	0.3
	5140150314	1.5	4	2.3	14	50	0.3
	5140150316	1.5	4	2.3	16	50	0.3
	5140150320	1.5	4	2.3	20	50	0.3
	5140150322	1.5	4	2.3	22	60	0.3
	5140150326	1.5	4	2.3	26	60	0.3

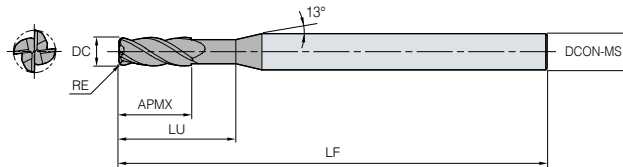
	Designation	DC	DCON-MS	APMX	LU	LF	RE
UXR	5140150504	1.5	4	2.3	4	50	0.5
	5140150506	1.5	4	2.3	6	50	0.5
	5140150508	1.5	4	2.3	8	50	0.5
	5140150510	1.5	4	2.3	10	50	0.5
	5140150512	1.5	4	2.3	12	50	0.5
	5140150514	1.5	4	2.3	14	50	0.5
	5140150516	1.5	4	2.3	16	50	0.5
	5140150520	1.5	4	2.3	20	50	0.5
	5140150522	1.5	4	2.3	22	60	0.5
	5140150526	1.5	4	2.3	26	60	0.5
	5140200106	2	4	3	6	50	0.1
	5140200108	2	4	3	8	50	0.1
	5140200110	2	4	3	10	50	0.1
	5140200112	2	4	3	12	50	0.1
	5140200114	2	4	3	14	50	0.1
	5140200116	2	4	3	16	50	0.1
	5140200120	2	4	3	20	50	0.1
	5140200122	2	4	3	22	60	0.1
	5140200126	2	4	3	26	60	0.1
	5140200130	2	4	3	30	70	0.1
	5140200206	2	4	3	6	50	0.2
	5140200208	2	4	3	8	50	0.2
	5140200210	2	4	3	10	50	0.2
	5140200212	2	4	3	12	50	0.2
	5140200214	2	4	3	14	50	0.2
	5140200216	2	4	3	16	50	0.2
	5140200220	2	4	3	20	50	0.2
	5140200222	2	4	3	22	60	0.2
	5140200226	2	4	3	26	60	0.2
	5140200230	2	4	3	30	70	0.2
	5140200306	2	4	3	6	50	0.3
	5140200308	2	4	3	8	50	0.3
	5140200310	2	4	3	10	50	0.3
	5140200312	2	4	3	12	50	0.3
	5140200314	2	4	3	14	50	0.3
	5140200316	2	4	3	16	50	0.3

UXR514

4 flutes multi helix neck radius



CARBIDE	4	H-A 34°/36°	Grade PC315W	h5 shank	RE ±0.02	DC Ø1 ~ Ø20	Tolerance 0.000 ~ -0.030
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Designation	DC	DCON-MS	APMX	LU	LF	RE
UXR 5140200320	2	4	3	20	50	0.3
5140200322	2	4	3	22	60	0.3
5140200326	2	4	3	26	60	0.3
5140200330	2	4	3	30	70	0.3
5140200506	2	4	3	6	50	0.5
5140200508	2	4	3	8	50	0.5
5140200510	2	4	3	10	50	0.5
5140200512	2	4	3	12	50	0.5
5140200514	2	4	3	14	50	0.5
5140200516	2	4	3	16	50	0.5
5140200520	2	4	3	20	50	0.5
5140200522	2	4	3	22	60	0.5
5140200526	2	4	3	26	60	0.5
5140200530	2	4	3	30	70	0.5
5140250108	2.5	4	4	8	50	0.1
5140250110	2.5	4	4	10	50	0.1
5140250112	2.5	4	4	12	50	0.1
5140250114	2.5	4	4	14	50	0.1
5140250116	2.5	4	4	16	50	0.1
5140250120	2.5	4	4	20	50	0.1
5140250126	2.5	4	4	26	60	0.1
5140250130	2.5	4	4	30	70	0.1
5140250208	2.5	4	4	8	50	0.2
5140250210	2.5	4	4	10	50	0.2
5140250212	2.5	4	4	12	50	0.2
5140250214	2.5	4	4	14	50	0.2
5140250216	2.5	4	4	16	50	0.2
5140250220	2.5	4	4	20	50	0.2
5140250226	2.5	4	4	26	60	0.2
5140250230	2.5	4	4	30	70	0.2
5140250308	2.5	4	4	8	50	0.3
5140250310	2.5	4	4	10	50	0.3
5140250312	2.5	4	4	12	50	0.3
5140250314	2.5	4	4	14	50	0.3
5140250316	2.5	4	4	16	50	0.3
5140250320	2.5	4	4	20	50	0.3

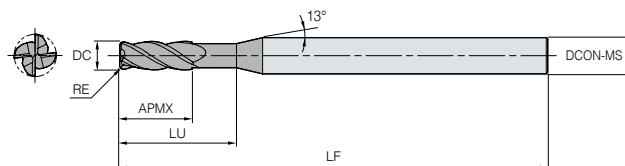
Designation	DC	DCON-MS	APMX	LU	LF	RE
UXR 5140250326	2.5	4	4	26	60	0.3
5140250330	2.5	4	4	30	70	0.3
5140250508	2.5	4	4	8	50	0.5
5140250510	2.5	4	4	10	50	0.5
5140250512	2.5	4	4	12	50	0.5
5140250514	2.5	4	4	14	50	0.5
5140250516	2.5	4	4	16	50	0.5
5140250520	2.5	4	4	20	50	0.5
5140250526	2.5	4	4	26	60	0.5
5140250530	2.5	4	4	30	70	0.5
5140300108	3	6	4.5	8	50	0.1
5140300110	3	6	4.5	10	50	0.1
5140300112	3	6	4.5	12	50	0.1
5140300114	3	6	4.5	14	60	0.1
5140300116	3	6	4.5	16	60	0.1
5140300120	3	6	4.5	20	60	0.1
5140300126	3	6	4.5	26	65	0.1
5140300130	3	6	4.5	30	70	0.1
5140300135	3	6	4.5	35	70	0.1
5140300140	3	6	4.5	40	80	0.1
5140300208	3	6	4.5	8	50	0.2
5140300210	3	6	4.5	10	50	0.2
5140300212	3	6	4.5	12	50	0.2
5140300214	3	6	4.5	14	60	0.2
5140300216	3	6	4.5	16	60	0.2
5140300220	3	6	4.5	20	60	0.2
5140300226	3	6	4.5	26	65	0.2
5140300230	3	6	4.5	30	70	0.2
5140300235	3	6	4.5	35	70	0.2
5140300240	3	6	4.5	40	80	0.2
5140300308	3	6	4.5	8	50	0.3
5140300310	3	6	4.5	10	50	0.3
5140300312	3	6	4.5	12	50	0.3
5140300314	3	6	4.5	14	60	0.3
5140300316	3	6	4.5	16	60	0.3
5140300320	3	6	4.5	20	60	0.3

UXR514

4 flutes multi helix neck radius



CARBIDE	4	H-A 34°/36°	Grade PC315W	h5 shank	RE ±0.02	DC Ø1 ~ Ø20	Tolerance 0.000 ~ -0.030
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(mm)

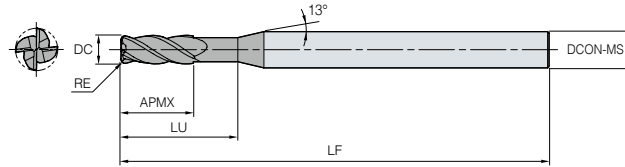
Designation	DC	DCON-MS	APMX	LU	LF	RE
UXR 5140300326	3	6	4.5	26	65	0.3
5140300330	3	6	4.5	30	70	0.3
5140300335	3	6	4.5	35	70	0.3
5140300340	3	6	4.5	40	80	0.3
5140300508	3	6	4.5	8	50	0.5
5140300510	3	6	4.5	10	50	0.5
5140300512	3	6	4.5	12	50	0.5
5140300514	3	6	4.5	14	60	0.5
5140300516	3	6	4.5	16	60	0.5
5140300520	3	6	4.5	20	60	0.5
5140300526	3	6	4.5	26	65	0.5
5140300530	3	6	4.5	30	70	0.5
5140300535	3	6	4.5	35	70	0.5
5140300540	3	6	4.5	40	80	0.5
5140301008	3	6	4.5	8	50	1
5140301010	3	6	4.5	10	50	1
5140301012	3	6	4.5	12	50	1
5140301014	3	6	4.5	14	60	1
5140301016	3	6	4.5	16	60	1
5140301020	3	6	4.5	20	60	1
5140301026	3	6	4.5	26	65	1
5140301030	3	6	4.5	30	70	1
5140301035	3	6	4.5	35	70	1
5140301040	3	6	4.5	40	80	1
5140400110	4	6	6	10	50	0.1
5140400112	4	6	6	12	50	0.1
5140400114	4	6	6	14	60	0.1
5140400116	4	6	6	16	60	0.1
5140400120	4	6	6	20	60	0.1
5140400126	4	6	6	26	65	0.1
5140400130	4	6	6	30	70	0.1
5140400135	4	6	6	35	70	0.1
5140400140	4	6	6	40	80	0.1
5140400145	4	6	6	45	90	0.1
5140400150	4	6	6	50	100	0.1
5140400210	4	6	6	10	50	0.2

Designation	DC	DCON-MS	APMX	LU	LF	RE
UXR 5140400212	4	6	6	12	50	0.2
5140400214	4	6	6	14	60	0.2
5140400216	4	6	6	16	60	0.2
5140400220	4	6	6	20	60	0.2
5140400226	4	6	6	26	65	0.2
5140400230	4	6	6	30	70	0.2
5140400235	4	6	6	35	70	0.2
5140400240	4	6	6	40	80	0.2
5140400245	4	6	6	45	90	0.2
5140400250	4	6	6	50	100	0.2
5140400310	4	6	6	10	50	0.3
5140400312	4	6	6	12	50	0.3
5140400314	4	6	6	14	60	0.3
5140400316	4	6	6	16	60	0.3
5140400320	4	6	6	20	60	0.3
5140400326	4	6	6	26	65	0.3
5140400330	4	6	6	30	70	0.3
5140400335	4	6	6	35	70	0.3
5140400340	4	6	6	40	80	0.3
5140400345	4	6	6	45	90	0.3
5140400350	4	6	6	50	100	0.3
5140400510	4	6	6	10	50	0.5
5140400512	4	6	6	12	50	0.5
5140400514	4	6	6	14	60	0.5
5140400516	4	6	6	16	60	0.5
5140400520	4	6	6	20	60	0.5
5140400526	4	6	6	26	65	0.5
5140400530	4	6	6	30	70	0.5
5140400535	4	6	6	35	70	0.5
5140400540	4	6	6	40	80	0.5
5140400545	4	6	6	45	90	0.5
5140400550	4	6	6	50	100	0.5
5140401010	4	6	6	10	50	1
5140401012	4	6	6	12	50	1
5140401014	4	6	6	14	60	1
5140401016	4	6	6	16	60	1

UXR514

4 flutes multi helix neck radius

CARBIDE	4	H-A 34°/36°	Grade PC315W	h5 shank	RE ±0.02	DC Ø1 ~ Ø20	Tolerance 0.000 ~ -0.030
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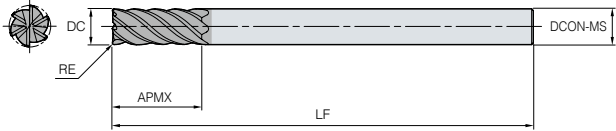
														(mm)	
Designation		DC	DCON-MS	APMX	LU	LF	RE	Designation		DC	DCON-MS	APMX	LU	LF	RE
UXR	5140401020	4	6	6	20	60	1	UXR	51408010100	8	8	20	35	100	1
	5140401026	4	6	6	26	65	1		51408015	8	8	12	25	70	1.5
	5140401030	4	6	6	30	70	1		51408020	8	8	12	25	70	2
	5140401035	4	6	6	35	70	1		51410001	10	10	15	30	75	0.1
	5140401040	4	6	6	40	80	1		51410002	10	10	15	30	75	0.2
	5140401045	4	6	6	45	90	1		51410003	10	10	15	30	75	0.3
	5140401050	4	6	6	50	100	1		51410003100	10	10	25	40	100	0.3
	51405001	5	6	8	15	60	0.1		51410005	10	10	15	30	75	0.5
	51405002	5	6	8	15	60	0.2		51410005100	10	10	25	40	100	0.5
	51405003	5	6	8	15	60	0.3		51410010	10	10	15	30	75	1
	51405005	5	6	8	15	60	0.5		51410010100	10	10	25	40	100	1
	51405010	5	6	8	15	60	1		51410015	10	10	15	30	75	1.5
	51405015	5	6	8	15	60	1.5		51410020	10	10	15	30	75	2
	51405020	5	6	8	15	60	2		51412002	12	12	18	32	80	0.2
	51406001	6	6	9	20	60	0.1		51412003	12	12	18	32	80	0.3
	51406002	6	6	9	20	60	0.2		51412003110	12	12	30	45	110	0.3
	51406003	6	6	9	20	60	0.3		51412005	12	12	18	32	80	0.5
	51406003090	6	6	15	30	90	0.3		51412005110	12	12	30	45	110	0.5
	51406005	6	6	9	20	60	0.5		51412010	12	12	18	32	80	1
	51406005090	6	6	15	30	90	0.5		51412010110	12	12	30	45	110	1
	51406010	6	6	9	20	60	1		51412015	12	12	18	32	80	1.5
	51406010090	6	6	15	30	90	1		51412020	12	12	18	32	80	2
	51406015	6	6	9	20	60	1.5		51416005	16	16	20	35	100	0.5
	51406020	6	6	9	20	60	2		51416005150	16	16	35	50	150	0.5
	51408001	8	8	12	25	70	0.1		51416010	16	16	20	35	100	1
	51408002	8	8	12	25	70	0.2		51416010150	16	16	35	50	150	1
	51408003	8	8	12	25	70	0.3		51420005	20	20	25	40	100	0.5
	51408003100	8	8	20	35	100	0.3		51420005150	20	20	40	55	150	0.5
	51408005	8	8	12	25	70	0.5		51420010	20	20	25	40	100	1
	51408005100	8	8	20	35	100	0.5		51420010150	20	20	40	55	150	1
	51408010	8	8	12	25	70	1								

UR506

6 flutes radius



CARBIDE	6	H-A 45°	Grade PC315W	h5 shank	RE ±0.015	DC Ø6 ~ Ø20	Tolerance 0.000 ~ -0.030
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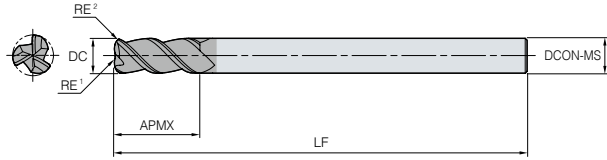
(mm)						
Designation		DC	DCON-MS	APMX	LF	RE
UR	50606003	6	6	15	90	0.3
	50606005	6	6	15	90	0.5
	50606010	6	6	15	90	1
	50608003	8	8	20	100	0.3
	50608005	8	8	20	100	0.5
	50608010	8	8	20	100	1
	50610003	10	10	25	100	0.3
	50610005	10	10	25	100	0.5
	50610010	10	10	25	100	1
	50612003	12	12	30	110	0.3
	50612005	12	12	30	110	0.5
	50612010	12	12	30	110	1
	50616005	16	16	32	150	0.5
	50616010	16	16	32	150	1
	50616015	16	16	32	150	1.5
	50616020	16	16	32	150	2
	50620005	20	20	38	150	0.5
	50620010	20	20	38	150	1
	50620015	20	20	38	150	1.5
	50620020	20	20	38	150	2

UDR503

3 flutes double radius



CARBIDE	3	H-A 45°	Grade PC315W	h5 shank	RE ±0.02	DC	Tolerance
						Ø6 ~ Ø20	0.000 ~ -0.020



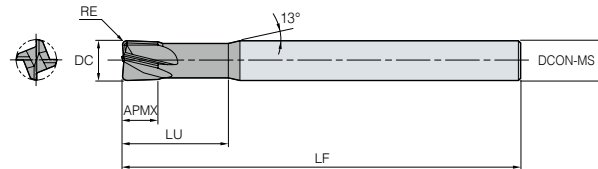
(mm)

Designation	DC	DCON-MS	APMX	LF	RE¹	RE²
UDR 50306005	6	6	10	90	0.5	6
50306010	6	6	10	90	1	6
50306020	6	6	10	90	2	6
50308005	8	8	16	100	0.5	8
50308010	8	8	16	100	1	8
50308020	8	8	16	100	2	8
50310005	10	10	20	100	0.5	10
50310010	10	10	20	100	1	10
50310020	10	10	20	100	2	10
50312005	12	12	24	110	0.5	12
50312010	12	12	24	110	1	12
50312020	12	12	24	110	2	12
50316005	16	16	32	150	0.5	16
50316010	16	16	32	150	1	16
50320005	20	20	40	150	0.5	20
50320010	20	20	40	150	1	20

USPM4

4 flutes radius Endmill for high speed machining

CARBIDE	4	H-A 10°	Grade PC315W	h5 shank	RE ±0.02	DC Ø1 ~ Ø20	Tolerance 0.000 ~ -0.030
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(mm)

	Designation	DC	DCON-MS	APMX	LU	LF	RE
USPM	4010-01	1	6	1	2.5	50	0.1
	4010-02	1	6	1	2.5	50	0.2
	4010-03	1	6	1	2.5	50	0.3
	4015-02	1.5	6	1.5	4	50	0.2
	4015-03	1.5	6	1.5	4	50	0.3
	4015-05	1.5	6	1.5	4	50	0.5
	4020-02	2	6	2	6	50	0.2
	4020-03	2	6	2	6	50	0.3
	4020-05	2	6	2	6	50	0.5
	4030-02	3	6	3	8	50	0.2
	4030-03	3	6	3	8	50	0.3
	4030-05	3	6	3	8	50	0.5
	4040-02	4	6	4	10	50	0.2
	4040-03	4	6	4	10	50	0.3
	4040-05	4	6	4	10	50	0.5
	4040-10	4	6	4	10	50	1
	4060-02	6	6	6	15	60	0.2
	4060-02L	6	6	6	15	90	0.2
	4060-03	6	6	6	15	60	0.3
	4060-03L	6	6	6	15	90	0.3
	4060-05	6	6	6	15	60	0.5
	4060-05L	6	6	6	15	90	0.5
	4060-10	6	6	6	15	60	1
	4060-10L	6	6	6	15	90	1
	4060-20	6	6	6	15	60	2
	4060-20L	6	6	6	15	90	2
	4080-02	8	8	8	20	70	0.2
	4080-02L	8	8	8	20	100	0.2
	4080-03	8	8	8	20	70	0.3
	4080-03L	8	8	8	20	100	0.3
	4080-05	8	8	8	20	70	0.5
	4080-05L	8	8	8	20	100	0.5
	4080-10	8	8	8	20	70	1
	4080-10L	8	8	8	20	100	1
	4080-20	8	8	8	20	70	2
	4080-20L	8	8	8	20	100	2

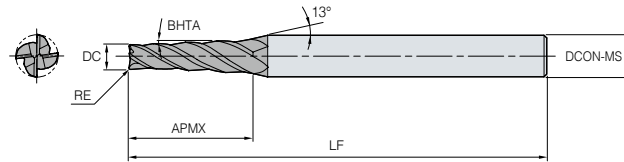
	Designation	DC	DCON-MS	APMX	LU	LF	RE
USPM	4100-02	10	10	10	25	75	0.2
	4100-02L	10	10	10	25	100	0.2
	4100-03	10	10	10	25	75	0.3
	4100-03L	10	10	10	25	100	0.3
	4100-05	10	10	10	25	75	0.5
	4100-05L	10	10	10	25	100	0.5
	4100-10	10	10	10	25	75	1
	4100-10L	10	10	10	25	100	1
	4100-15	10	10	10	25	75	1.5
	4100-15L	10	10	10	25	100	1.5
	4100-20	10	10	10	25	75	2
	4100-20L	10	10	10	25	100	2
	4120-03	12	12	12	30	80	0.3
	4120-03L	12	12	12	30	110	0.3
	4120-05	12	12	12	30	80	0.5
	4120-05L	12	12	12	30	110	0.5
	4120-10	12	12	12	30	80	1
	4120-10L	12	12	12	30	110	1
	4120-15	12	12	12	30	80	1.5
	4120-15L	12	12	12	30	110	1.5
	4120-20	12	12	12	30	80	2
	4120-20L	12	12	12	30	110	2
	4120-30	12	12	12	30	80	3
	4120-30L	12	12	12	30	110	3
	4160-05	16	16	16	35	100	0.5
	4160-05L	16	16	16	35	150	0.5
	4160-10	16	16	16	35	100	1
	4160-10L	16	16	16	35	150	1
	4160-20	16	16	16	35	100	2
	4160-20L	16	16	16	35	150	2
	4200-05	20	20	20	40	100	0.5
	4200-05L	20	20	20	40	150	0.5
	4200-10	20	20	20	40	100	1
	4200-10L	20	20	20	40	150	1
	4200-20	20	20	20	40	100	2
	4200-20L	20	20	20	40	150	2

UTR504

4 flutes tapered radius



CARBIDE	4	H-A 30°	Grade PC315W	h5 shank	RE ±0.01	DC Ø0.8 ~ Ø2.5	Tolerance 0.000 ~ -0.030
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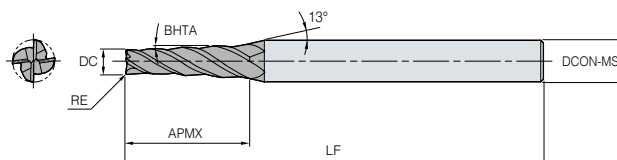
																(mm)
Designation		DC	DCON-MS	APMX	BHTA(°)	LF	RE	Designation		DC	DCON-MS	APMX	BHTA(°)	LF	RE	
UTR	504008010104	0.8	4	4	1	45	0.1	UTR	504010020112	1	4	12	1	50	0.2	
	504008010106	0.8	4	6	1	45	0.1		5040100201504	1	4	4	1.5	50	0.2	
	504008010108	0.8	4	8	1	45	0.1		5040100201506	1	4	6	1.5	50	0.2	
	5040080101504	0.8	4	4	1.5	45	0.1		5040100201508	1	4	8	1.5	50	0.2	
	5040080101506	0.8	4	6	1.5	45	0.1		5040100201510	1	4	10	1.5	50	0.2	
	5040080101508	0.8	4	8	1.5	45	0.1		5040100201512	1	4	12	1.5	50	0.2	
	504008020104	0.8	4	4	1	45	0.2		504010020204	1	4	4	2	50	0.2	
	504008020106	0.8	4	6	1	45	0.2		504010020206	1	4	6	2	50	0.2	
	504008020108	0.8	4	8	1	45	0.2		504010020208	1	4	8	2	50	0.2	
	5040080201504	0.8	4	4	1.5	45	0.2		504010020210	1	4	10	2	50	0.2	
	5040080201506	0.8	4	6	1.5	45	0.2		504010020212	1	4	12	2	50	0.2	
	5040080201508	0.8	4	8	1.5	45	0.2		504010020304	1	4	4	3	50	0.2	
	504010010104	1	4	4	1	50	0.1		504010020306	1	4	6	3	50	0.2	
	504010010106	1	4	6	1	50	0.1		504010020308	1	4	8	3	50	0.2	
	504010010108	1	4	8	1	50	0.1		504010020310	1	4	10	3	50	0.2	
	504010010110	1	4	10	1	50	0.1		504010020312	1	4	12	3	50	0.2	
	504010010112	1	4	12	1	50	0.1		504010030104	1	4	4	1	50	0.3	
	5040100101504	1	4	4	1.5	50	0.1		504010030106	1	4	6	1	50	0.3	
	5040100101506	1	4	6	1.5	50	0.1		504010030108	1	4	8	1	50	0.3	
	5040100101508	1	4	8	1.5	50	0.1		504010030110	1	4	10	1	50	0.3	
	5040100101510	1	4	10	1.5	50	0.1		504010030112	1	4	12	1	50	0.3	
	5040100101512	1	4	12	1.5	50	0.1		5040100301504	1	4	4	1.5	50	0.3	
	504010010204	1	4	4	2	50	0.1		5040100301506	1	4	6	1.5	50	0.3	
	504010010206	1	4	6	2	50	0.1		5040100301508	1	4	8	1.5	50	0.3	
	504010010208	1	4	8	2	50	0.1		5040100301510	1	4	10	1.5	50	0.3	
	504010010210	1	4	10	2	50	0.1		5040100301512	1	4	12	1.5	50	0.3	
	504010010212	1	4	12	2	50	0.1		504010030204	1	4	4	2	50	0.3	
	504010010304	1	4	4	3	50	0.1		504010030206	1	4	6	2	50	0.3	
	504010010306	1	4	6	3	50	0.1		504010030208	1	4	8	2	50	0.3	
	504010010308	1	4	8	3	50	0.1		504010030210	1	4	10	2	50	0.3	
	504010010310	1	4	10	3	50	0.1		504010030212	1	4	12	2	50	0.3	
	504010010312	1	4	12	3	50	0.1		504010030304	1	4	4	3	50	0.3	
	504010020104	1	4	4	1	50	0.2		504010030306	1	4	6	3	50	0.3	
	504010020106	1	4	6	1	50	0.2		504010030308	1	4	8	3	50	0.3	
	504010020108	1	4	8	1	50	0.2		504010030310	1	4	10	3	50	0.3	
	504010020110	1	4	10	1	50	0.2		504010030312	1	4	12	3	50	0.3	

UTR504

4 flutes tapered radius



CARBIDE	4	H-A 30°	Grade PC315W	h5 shank	RE ±0.01	DC Ø0.8 ~ Ø2.5	Tolerance 0.000 ~ -0.030
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(mm)

	Designation	DC	DCON-MS	APMX	BHTA(°)	LF	RE
UTR	504012010106	1.2	4	6	1	50	0.1
	504012010108	1.2	4	8	1	50	0.1
	504012010110	1.2	4	10	1	50	0.1
	504012010112	1.2	4	12	1	50	0.1
	504012010206	1.2	4	6	2	50	0.1
	504012010208	1.2	4	8	2	50	0.1
	504012010210	1.2	4	10	2	50	0.1
	504012010212	1.2	4	12	2	50	0.1
	504012010306	1.2	4	6	3	50	0.1
	504012010308	1.2	4	8	3	50	0.1
	504012010310	1.2	4	10	3	50	0.1
	504012010312	1.2	4	12	3	50	0.1
	504012020106	1.2	4	6	1	50	0.2
	504012020108	1.2	4	8	1	50	0.2
	504012020110	1.2	4	10	1	50	0.2
	504012020112	1.2	4	12	1	50	0.2
	504012020206	1.2	4	6	2	50	0.2
	504012020208	1.2	4	8	2	50	0.2
	504012020210	1.2	4	10	2	50	0.2
	504012020212	1.2	4	12	2	50	0.2
	504012020306	1.2	4	6	3	50	0.2
	504012020308	1.2	4	8	3	50	0.2
	504012020310	1.2	4	10	3	50	0.2
	504012020312	1.2	4	12	3	50	0.2
	504012030106	1.2	4	6	1	50	0.3
	504012030108	1.2	4	8	1	50	0.3
	504012030110	1.2	4	10	1	50	0.3
	504012030112	1.2	4	12	1	50	0.3
	504012030206	1.2	4	6	2	50	0.3
	504012030208	1.2	4	8	2	50	0.3
	504012030210	1.2	4	10	2	50	0.3
	504012030212	1.2	4	12	2	50	0.3
	504012030306	1.2	4	6	3	50	0.3
	504012030308	1.2	4	8	3	50	0.3
	504012030310	1.2	4	10	3	50	0.3
	504012030312	1.2	4	12	3	50	0.3

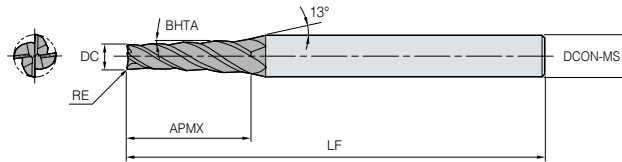
	Designation	DC	DCON-MS	APMX	BHTA(°)	LF	RE
UTR	504015010106	1.5	4	6	1	50	0.1
	504015010108	1.5	4	8	1	50	0.1
	504015010110	1.5	4	10	1	50	0.1
	504015010112	1.5	4	12	1	50	0.1
	504015010116	1.5	4	16	1	50	0.1
	504015010120	1.5	4	20	1	60	0.1
	504015010206	1.5	4	6	2	50	0.1
	504015010208	1.5	4	8	2	50	0.1
	504015010210	1.5	4	10	2	50	0.1
	504015010212	1.5	4	12	2	50	0.1
	504015010216	1.5	4	16	2	50	0.1
	504015010220	1.5	4	20	2	60	0.1
	504015010306	1.5	4	6	3	50	0.1
	504015010308	1.5	4	8	3	50	0.1
	504015010310	1.5	4	10	3	50	0.1
	504015010312	1.5	4	12	3	50	0.1
	504015010316	1.5	4	16	3	50	0.1
	504015010320	1.5	4	20	3	60	0.1
	504015020106	1.5	4	6	1	50	0.2
	504015020108	1.5	4	8	1	50	0.2
	504015020110	1.5	4	10	1	50	0.2
	504015020112	1.5	4	12	1	50	0.2
	504015020116	1.5	4	16	1	50	0.2
	504015020120	1.5	4	20	1	60	0.2
	504015020206	1.5	4	6	2	50	0.2
	504015020208	1.5	4	8	2	50	0.2
	504015020210	1.5	4	10	2	50	0.2
	504015020212	1.5	4	12	2	50	0.2
	504015020216	1.5	4	16	2	50	0.2
	504015020220	1.5	4	20	2	60	0.2
	504015020306	1.5	4	6	3	50	0.2
	504015020308	1.5	4	8	3	50	0.2
	504015020310	1.5	4	10	3	50	0.2
	504015020312	1.5	4	12	3	50	0.2
	504015020316	1.5	4	16	3	50	0.2
	504015020320	1.5	4	20	3	60	0.2

UTR504

4 flutes tapered radius



CARBIDE	4	H-A 30°	Grade PC315W	h5 shank	RE ±0.01	DC Ø0.8 ~ Ø2.5	Tolerance 0.000 ~ -0.030
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Designation	DC	DCON-MS	APMX	BHTA(°)	LF	RE
UTR 504015030106	1.5	4	6	1	50	0.3
504015030108	1.5	4	8	1	50	0.3
504015030110	1.5	4	10	1	50	0.3
504015030112	1.5	4	12	1	50	0.3
504015030116	1.5	4	16	1	50	0.3
504015030120	1.5	4	20	1	60	0.3
504015030206	1.5	4	6	2	50	0.3
504015030208	1.5	4	8	2	50	0.3
504015030210	1.5	4	10	2	50	0.3
504015030212	1.5	4	12	2	50	0.3
504015030216	1.5	4	16	2	50	0.3
504015030220	1.5	4	20	2	60	0.3
504015030306	1.5	4	6	3	50	0.3
504015030308	1.5	4	8	3	50	0.3
504015030310	1.5	4	10	3	50	0.3
504015030312	1.5	4	12	3	50	0.3
504015030316	1.5	4	16	3	50	0.3
504015030320	1.5	4	20	3	60	0.3
504020010108	2	4	8	1	50	0.1
504020010110	2	4	10	1	50	0.1
504020010112	2	4	12	1	50	0.1
504020010116	2	4	16	1	50	0.1
504020010120	2	4	20	1	60	0.1
504020010125	2	4	25	1	60	0.1
504020010208	2	4	8	2	50	0.1
504020010210	2	4	10	2	50	0.1
504020010212	2	4	12	2	50	0.1
504020010216	2	4	16	2	50	0.1
504020010220	2	4	20	2	60	0.1
504020010225	2	4	25	2	60	0.1
504020010308	2	4	8	3	50	0.1
504020010310	2	4	10	3	50	0.1
504020010312	2	4	12	3	50	0.1
504020010316	2	4	16	3	50	0.1
504020010320	2	6	20	3	60	0.1
504020010325	2	6	25	3	60	0.1

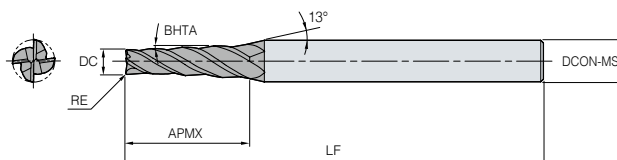
Designation	DC	DCON-MS	APMX	BHTA(°)	LF	RE
UTR 504020020108	2	4	8	1	50	0.2
504020020110	2	4	10	1	50	0.2
504020020112	2	4	12	1	50	0.2
504020020116	2	4	16	1	50	0.2
504020020120	2	4	20	1	60	0.2
504020020125	2	4	25	1	60	0.2
504020020208	2	4	8	2	50	0.2
504020020210	2	4	10	2	50	0.2
504020020212	2	4	12	2	50	0.2
504020020216	2	4	16	2	50	0.2
504020020220	2	4	20	2	60	0.2
504020020225	2	4	25	2	60	0.2
504020020308	2	4	8	3	50	0.2
504020020310	2	4	10	3	50	0.2
504020020312	2	4	12	3	50	0.2
504020020316	2	4	16	3	50	0.2
504020020320	2	6	20	3	60	0.2
504020020325	2	6	25	3	60	0.2
504020030108	2	4	8	1	50	0.3
504020030110	2	4	10	1	50	0.3
504020030112	2	4	12	1	50	0.3
504020030116	2	4	16	1	50	0.3
504020030120	2	4	20	1	60	0.3
504020030125	2	4	25	1	60	0.3
504020030208	2	4	8	2	50	0.3
504020030210	2	4	10	2	50	0.3
504020030212	2	4	12	2	50	0.3
504020030216	2	4	16	2	50	0.3
504020030220	2	4	20	2	60	0.3
504020030225	2	4	25	2	60	0.3
504020030308	2	4	8	3	50	0.3
504020030310	2	4	10	3	50	0.3
504020030312	2	4	12	3	50	0.3
504020030316	2	4	16	3	50	0.3
504020030320	2	6	20	3	60	0.3
504020030325	2	6	25	3	60	0.3

UTR504

4 flutes tapered radius



CARBIDE	4	H-A 30°	Grade PC315W	h5 shank	RE ±0.01	DC Ø0.8 ~ Ø2.5	Tolerance 0.000 ~ -0.030
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(mm)

	Designation	DC	DCON-MS	APMX	BHTA(°)	LF	RE
UTR	504025010110	2.5	4	10	1	50	0.1
	504025010112	2.5	4	12	1	50	0.1
	504025010116	2.5	4	16	1	50	0.1
	504025010120	2.5	4	20	1	60	0.1
	504025010125	2.5	4	25	1	60	0.1
	504025010130	2.5	4	30	1	60	0.1
	504025010210	2.5	4	10	2	50	0.1
	504025010212	2.5	4	12	2	50	0.1
	504025010216	2.5	4	16	2	50	0.1
	504025010220	2.5	4	20	2	60	0.1
	504025010225	2.5	6	25	2	60	0.1
	504025010230	2.5	6	30	2	60	0.1
	504025010310	2.5	4	10	3	50	0.1
	504025010312	2.5	4	12	3	50	0.1
	504025010316	2.5	6	16	3	50	0.1
	504025010320	2.5	6	20	3	60	0.1
	504025010325	2.5	6	25	3	60	0.1
	504025010330	2.5	6	30	3	60	0.1
	504025020110	2.5	4	10	1	50	0.2
	504025020112	2.5	4	12	1	50	0.2
	504025020116	2.5	4	16	1	50	0.2
	504025020120	2.5	4	20	1	60	0.2
	504025020125	2.5	4	25	1	60	0.2
	504025020130	2.5	4	30	1	60	0.2
	504025020210	2.5	4	10	2	50	0.2
	504025020212	2.5	4	12	2	50	0.2
	504025020216	2.5	4	16	2	50	0.2

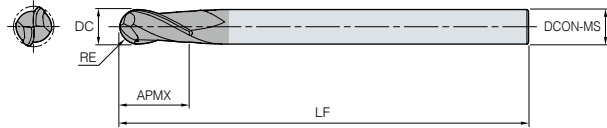
	Designation	DC	DCON-MS	APMX	BHTA(°)	LF	RE
UTR	504025020220	2.5	4	20	2	60	0.2
	504025020225	2.5	6	25	2	60	0.2
	504025020230	2.5	6	30	2	60	0.2
	504025020310	2.5	4	10	3	50	0.2
	504025020312	2.5	4	12	3	50	0.2
	504025020316	2.5	6	16	3	50	0.2
	504025020320	2.5	6	20	3	60	0.2
	504025020325	2.5	6	25	3	60	0.2
	504025020330	2.5	6	30	3	60	0.2
	504025030110	2.5	4	10	1	50	0.3
	504025030112	2.5	4	12	1	50	0.3
	504025030116	2.5	4	16	1	50	0.3
	504025030120	2.5	4	20	1	60	0.3
	504025030125	2.5	4	25	1	60	0.3
	504025030130	2.5	4	30	1	60	0.3
	504025030212	2.5	4	12	2	50	0.3
	504025030216	2.5	4	16	2	50	0.3
	504025030220	2.5	4	20	2	60	0.3
	504025030225	2.5	6	25	2	60	0.3
	504025030230	2.5	6	30	2	60	0.3
	504025030310	2.5	4	10	3	50	0.3
	504025030312	2.5	4	12	3	50	0.3
	504025030316	2.5	6	16	3	50	0.3
	504025030320	2.5	6	20	3	60	0.3
	504025030325	2.5	6	25	3	60	0.3
	504025030330	2.5	6	30	3	60	0.3

UB502

2 flutes ball



CARBIDE	2	H-A 30°	Grade PC303W	h5 shank	RE ±0.005	RE ±0.01	DC Ø0.1 ~ Ø6 Ø6.5 ~ Ø25	Tolerance 0.000 ~ -0.012 0.000 ~ -0.015
					DC6 or below	Above DC6		



Designation							DC	DCON-MS	APMX	LF	RE	
UB	502001S	0.1	4	0.1	40	0.05						
	502001	0.1	4	0.2	40	0.05						
	502001S3	0.1	3	0.2	40	0.05						
	5020015S	0.15	4	0.15	40	0.075						
	5020015	0.15	4	0.3	40	0.075						
	5020015S3	0.15	3	0.3	40	0.075						
	502002S	0.2	4	0.2	40	0.1						
	502002	0.2	4	0.4	40	0.1						
	502002S3	0.2	3	0.4	40	0.1						
	502003S	0.3	4	0.3	40	0.15						
	502003	0.3	4	0.6	40	0.15						
	502003S3	0.3	3	0.6	40	0.15						
	502004S	0.4	4	0.4	40	0.2						
	502004	0.4	4	0.8	40	0.2						
	502004S3	0.4	3	0.8	40	0.2						
	502005S	0.5	4	0.5	40	0.25						
	502005	0.5	4	1	40	0.25						
	502005S3	0.5	3	1	40	0.25						
	502006S	0.6	4	0.6	40	0.3						
	502006	0.6	4	1.2	40	0.3						
	502006S3	0.6	3	1.2	40	0.3						
	502007S	0.7	4	0.7	40	0.35						
	502007	0.7	4	1.4	40	0.35						
	502007S3	0.7	3	1.4	40	0.35						
	502008S	0.8	4	0.8	40	0.4						
	502008	0.8	4	1.6	40	0.4						
	502008S3	0.8	3	1.6	40	0.4						
	502009S	0.9	4	0.9	40	0.45						
	502009	0.9	4	1.8	40	0.45						
	502009S3	0.9	3	1.8	40	0.45						
	502010S	1	6	1.5	40	0.5						
	502010S3	1	3	2.5	50	0.5						
	502010S4	1	4	2.5	50	0.5						
	502010	1	6	2.5	50	0.5						
	502010070	1	6	2.5	70	0.5						
	502010100	1	6	2.5	100	0.5						
	502012S	1.2	6	2	40	0.6						
Designation							DC	DCON-MS	APMX	LF	RE	
UB	502012S3	1.2	3	3	50	0.6						
	502012S4	1.2	4	3	50	0.6						
	502012	1.2	6	3	50	0.6						
	502012070	1.2	6	3	70	0.6						
	502012100	1.2	6	3	100	0.6						
	502015S	1.5	6	2.5	40	0.75						
	502015S3	1.5	3	4	50	0.75						
	502015S4	1.5	4	4	50	0.75						
	502015	1.5	6	4	50	0.75						
	502015070	1.5	6	4	70	0.75						
	502015100	1.5	6	4	100	0.75						
	502020S	2	6	3	40	1						
	502020S3	2	3	5	50	1						
	502020S4	2	4	5	50	1						
	502020	2	6	5	50	1						
	502020080	2	6	5	80	1						
	502020100	2	6	5	100	1						
	502025S	2.5	6	4	40	1.25						
	502025S3	2.5	3	6	60	1.25						
	502025S4	2.5	4	6	60	1.25						
	502025	2.5	6	6	60	1.25						
	502025080	2.5	6	6	80	1.25						
	502025100	2.5	6	6	100	1.25						
	502030S	3	6	4.5	40	1.5						
	502030S3	3	3	6	60	1.5						
	502030S4	3	4	6	60	1.5						
	502030	3	6	6	60	1.5						
	502030080	3	6	6	80	1.5						
	502030100	3	6	6	100	1.5						
	502035	3.5	6	8	70	1.75						
	502040S	4	6	6	50	2						
	502040S4	4	4	8	70	2						
	502040	4	6	8	70	2						
	502040100S4	4	4	8	100	2						
	502040120S4	4	4	8	120	2						
	502040100	4	6	8	100	2						
	502040120	4	6	8	120	2						

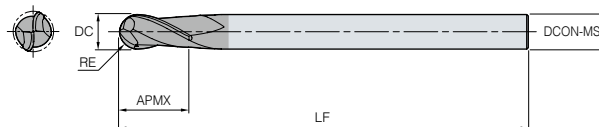
UB502

2 flutes ball



DC	Tolerance
Ø0.1 ~ Ø6	0.000 ~ -0.012
Ø6.5 ~ Ø25	0.000 ~ -0.015

DC6 or below Above DC6



(mm)

	Designation	DC	DCON-MS	APMX	LF	RE
UB	502045	4.5	6	9	80	2.25
	502050S	5	6	7.5	60	2.5
	502050	5	6	10	80	2.5
	502050S5	5	5	10	80	2.5
	502055	5.5	6	11	90	2.75
	502060S	6	6	9	50	3
	502060060	6	6	9	60	3
	502060080	6	6	9	80	3
	502060	6	6	12	90	3
	502060110	6	6	12	110	3
	502060130	6	6	12	130	3
	502060150	6	6	12	150	3
	502065	6.5	8	13	90	3.25
	502070	7	8	14	90	3.5
	502080S	8	8	12	50	4
	502080060	8	8	12	60	4
	502080080	8	8	12	80	4
	502080090	8	8	12	90	4
	502080	8	8	14	100	4
	502080130	8	8	14	130	4
	502080150	8	8	14	150	4
	502085	8.5	10	16	100	4.25
	502090	9	10	18	100	4.5
	502100S	10	10	15	50	5
	502100060	10	10	15	60	5
	502100080	10	10	15	80	5
	502100090	10	10	15	90	5
	502100	10	10	18	100	5
	502100130	10	10	18	130	5
	502100150	10	10	18	150	5

	Designation	DC	DCON-MS	APMX	LF	RE
UB	502100180	10	10	18	180	5
	502100200	10	10	18	200	5
	502110	11	12	20	100	5.5
	502120S	12	12	18	60	6
	502120080	12	12	18	80	6
	502120090	12	12	18	90	6
	502120100	12	12	18	100	6
	502120	12	12	24	110	6
	502120130	12	12	24	130	6
	502120150	12	12	24	150	6
	502120180	12	12	24	180	6
	502120200	12	12	24	200	6
	502130	13	12	24	100	6.5
	502140S12	14	12	26	100	7
	502140	14	14	26	100	7
	502140S16	14	16	26	100	7
	502150	15	16	28	140	7.5
	502160100	16	16	24	100	8
	502160130	16	16	24	130	8
	502160	16	16	30	150	8
	502160180	16	16	30	180	8
	502160200	16	16	30	200	8
	502180S16	18	16	34	150	9
	502180	18	18	34	150	9
	502200100	20	20	30	100	10
	502200130	20	20	30	130	10
	502200	20	20	38	150	10
	502200200	20	20	38	200	10
	502250120	25	25	50	120	12.5
	502250180	25	25	50	180	12.5

UB502P

2 flutes high precision ball



CARBIDE

2

H-A
30°

Grade
PC303W

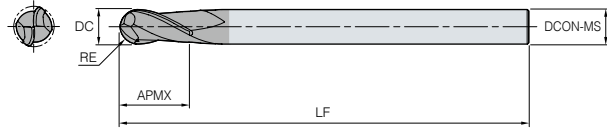
h5
shank

RE
±0.002

RE
±0.003

DC6 or below Above DC6

DC	Tolerance
Ø0.1 ~ Ø6	0.000 ~ -0.012
Ø8 ~ Ø12	0.000 ~ -0.015



(mm)

Designation	DC	DCON-MS	APMX	LF	RE
UB 502001P	0.1	4	0.2	40	0.05
5020015P	0.15	4	0.3	40	0.075
502002P	0.2	4	0.4	40	0.1
502003P	0.3	4	0.6	40	0.15
502004P	0.4	4	0.8	40	0.2
502005P	0.5	4	1	40	0.25
502006P	0.6	4	1.2	40	0.3
502007P	0.7	4	1.4	40	0.35
502008P	0.8	4	1.6	40	0.4
502009P	0.9	4	1.8	40	0.45
502010P	1	6	2.5	50	0.5
502012P	1.2	6	3	50	0.6
502015P	1.5	6	4	50	0.75
502020P	2	6	5	50	1
502025P	2.5	6	6	60	1.25
502030P	3	6	6	60	1.5
502040P	4	6	8	70	2
502050P	5	6	10	80	2.5
502060P	6	6	12	90	3
502080P	8	8	14	100	4
502100P	10	10	18	100	5
502120P	12	12	24	110	6

UB512

2 flutes long neck ball



CARBIDE

2

H-A
30°

Grade
PC303W

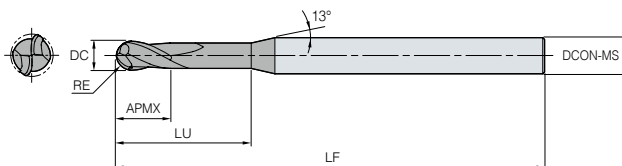
h5
shank

RE
±0.005

RE
±0.01

DC6 or below Above DC6

DC	Tolerance
Ø0.1 ~ Ø6	0.000 ~ -0.012
Ø8 ~ Ø12	0.000 ~ -0.015



(mm)

	Designation	DC	DCON-MS	APMX	LU	LF	RE
UB	512001002	0.1	4	0.1	0.2	40	0.05
	512001003	0.1	4	0.1	0.3	40	0.05
	512001005	0.1	4	0.1	0.5	40	0.05
	51200101	0.1	4	0.1	1	40	0.05
	512002005	0.2	4	0.2	0.5	40	0.1
	51200201	0.2	4	0.2	1	40	0.1
	512002015	0.2	4	0.2	1.5	40	0.1
	51200202	0.2	4	0.2	2	40	0.1
	51200203	0.2	4	0.2	3	40	0.1
	51200301	0.3	4	0.3	1	40	0.15
	512003015	0.3	4	0.3	1.5	40	0.15
	51200302	0.3	4	0.3	2	40	0.15
	512003025	0.3	4	0.3	2.5	40	0.15
	51200303	0.3	4	0.3	3	40	0.15
	51200304	0.3	4	0.3	4	40	0.15
	51200305	0.3	4	0.3	5	40	0.15
	51200401	0.4	4	0.4	1	40	0.2
	512004015	0.4	4	0.4	1.5	40	0.2
	51200402	0.4	4	0.4	2	40	0.2
	512004025	0.4	4	0.4	2.5	40	0.2
	51200403	0.4	4	0.4	3	40	0.2
	51200404	0.4	4	0.4	4	40	0.2
	51200405	0.4	4	0.4	5	40	0.2
	51200406	0.4	4	0.4	6	40	0.2
	51200408	0.4	4	0.4	8	40	0.2
	51200410	0.4	4	0.4	10	40	0.2
	51200501	0.5	4	0.5	1	45	0.25
	512005015	0.5	4	0.5	1.5	45	0.25
	51200502	0.5	4	0.5	2	45	0.25
	512005025	0.5	4	0.5	2.5	45	0.25
	51200503	0.5	4	0.5	3	45	0.25
	51200504	0.5	4	0.5	4	45	0.25
	51200505	0.5	4	0.5	5	45	0.25
	51200506	0.5	4	0.5	6	45	0.25

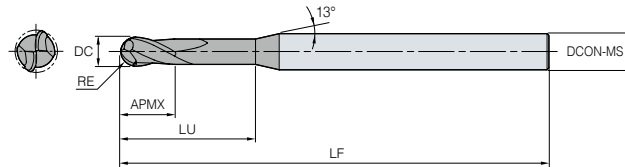
	Designation	DC	DCON-MS	APMX	LU	LF	RE
UB	51200508	0.5	4	0.5	8	45	0.25
	51200510	0.5	4	0.5	10	45	0.25
	51200512	0.5	4	0.5	12	45	0.25
	51200514	0.5	4	0.5	14	45	0.25
	51200516	0.5	4	0.5	16	45	0.25
	51200601	0.6	4	0.6	1	45	0.3
	51200602	0.6	4	0.6	2	45	0.3
	51200603	0.6	4	0.6	3	45	0.3
	51200604	0.6	4	0.6	4	45	0.3
	51200605	0.6	4	0.6	5	45	0.3
	51200606	0.6	4	0.6	6	45	0.3
	51200608	0.6	4	0.6	8	45	0.3
	51200610	0.6	4	0.6	10	45	0.3
	51200612	0.6	4	0.6	12	45	0.3
	51200614	0.6	4	0.6	14	45	0.3
	51200616	0.6	4	0.6	16	45	0.3
	51200702	0.7	4	0.7	2	45	0.35
	51200704	0.7	4	0.7	4	45	0.35
	51200706	0.7	4	0.7	6	45	0.35
	51200708	0.7	4	0.7	8	45	0.35
	51200710	0.7	4	0.7	10	45	0.35
	51200712	0.7	4	0.7	12	45	0.35
	51200801	0.8	4	0.8	1	45	0.4
	51200802	0.8	4	0.8	2	45	0.4
	51200803	0.8	4	0.8	3	45	0.4
	51200804	0.8	4	0.8	4	45	0.4
	51200805	0.8	4	0.8	5	45	0.4
	51200806	0.8	4	0.8	6	45	0.4
	51200807	0.8	4	0.8	7	45	0.4
	51200808	0.8	4	0.8	8	45	0.4
	51200810	0.8	4	0.8	10	45	0.4
	51200812	0.8	4	0.8	12	45	0.4
	51200814	0.8	4	0.8	14	45	0.4
	51200816	0.8	4	0.8	16	45	0.4

UB512

2 flutes long neck ball



CARBIDE	2	H-A 30°	Grade PC303W	h5 shank	RE ±0.005	RE ±0.01	DC Ø0.1 ~ Ø6 Ø8 ~ Ø12	Tolerance 0.000 ~ -0.012 0.000 ~ -0.015
					DC6 or below Above DC6			



Designation		DC	DCON-MS	APMX	LU	LF	RE
UB	51200820	0.8	4	0.8	20	45	0.4
	51200904	0.9	4	0.9	4	45	0.45
	51200906	0.9	4	0.9	6	45	0.45
	51200908	0.9	4	0.9	8	45	0.45
	51200910	0.9	4	0.9	10	45	0.45
	51201002	1	4	1	2	50	0.5
	512010025	1	4	1	2.5	50	0.5
	51201003	1	4	1	3	50	0.5
	51201004	1	4	1	4	50	0.5
	51201005	1	4	1	5	50	0.5
	51201006	1	4	1	6	50	0.5
	51201007	1	4	1	7	50	0.5
	51201008	1	4	1	8	50	0.5
	51201009	1	4	1	9	50	0.5
	51201010	1	4	1	10	50	0.5
	51201012	1	4	1	12	50	0.5
	51201014	1	4	1	14	50	0.5
	51201016	1	4	1	16	50	0.5
	51201018	1	4	1	18	50	0.5
	51201020	1	4	1	20	55	0.5
	51201022	1	4	1	22	60	0.5
	51201026	1	4	1	26	60	0.5
	51201030	1	4	1	30	70	0.5
	51201040	1	4	1	40	80	0.5
	51201050	1	4	1	50	100	0.5
	51201204	1.2	4	1.2	4	50	0.6
	51201206	1.2	4	1.2	6	50	0.6
	51201208	1.2	4	1.2	8	50	0.6
	51201210	1.2	4	1.2	10	50	0.6
	51201212	1.2	4	1.2	12	50	0.6
	51201214	1.2	4	1.2	14	50	0.6
	51201216	1.2	4	1.2	16	50	0.6
	51201220	1.2	4	1.2	20	55	0.6
	51201226	1.2	4	1.2	26	60	0.6

Designation		DC	DCON-MS	APMX	LU	LF	RE
UB	51201406	1.4	4	1.4	6	50	0.7
	51201408	1.4	4	1.4	8	50	0.7
	51201410	1.4	4	1.4	10	50	0.7
	51201412	1.4	4	1.4	12	50	0.7
	51201416	1.4	4	1.4	16	50	0.7
	51201503	1.5	4	1.5	3	50	0.75
	51201504	1.5	4	1.5	4	50	0.75
	51201505	1.5	4	1.5	5	50	0.75
	51201506	1.5	4	1.5	6	50	0.75
	51201507	1.5	4	1.5	7	50	0.75
	51201508	1.5	4	1.5	8	50	0.75
	51201510	1.5	4	1.5	10	50	0.75
	51201512	1.5	4	1.5	12	50	0.75
	51201514	1.5	4	1.5	14	50	0.75
	51201516	1.5	4	1.5	16	50	0.75
	51201518	1.5	4	1.5	18	50	0.75
	51201520	1.5	4	1.5	20	55	0.75
	51201522	1.5	4	1.5	22	60	0.75
	51201526	1.5	4	1.5	26	60	0.75
	51201530	1.5	4	1.5	30	70	0.75
	51201535	1.5	4	1.5	35	70	0.75
	51201540	1.5	4	1.5	40	80	0.75
	51201604	1.6	4	1.6	4	50	0.8
	51201606	1.6	4	1.6	6	50	0.8
	51201608	1.6	4	1.6	8	50	0.8
	51201610	1.6	4	1.6	10	50	0.8
	51201612	1.6	4	1.6	12	50	0.8
	51201616	1.6	4	1.6	16	50	0.8
	51201620	1.6	4	1.6	20	50	0.8
	51201804	1.8	4	1.8	4	50	0.9
	51201806	1.8	4	1.8	6	50	0.9
	51201808	1.8	4	1.8	8	50	0.9
	51201810	1.8	4	1.8	10	50	0.9
	51201812	1.8	4	1.8	12	50	0.9

UB512

2 flutes long neck ball



CARBIDE

2

H-A
30°

Grade
PC303W

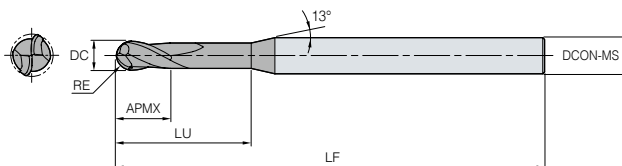
h5
shank

RE
±0.005

RE
±0.01

DC6 or below Above DC6

DC	Tolerance
Ø0.1 ~ Ø6	0.000 ~ -0.012
Ø8 ~ Ø12	0.000 ~ -0.015



(mm)

	Designation	DC	DCON-MS	APMX	LU	LF	RE
UB	51201816	1.8	4	1.8	16	50	0.9
	51201820	1.8	4	1.8	20	50	0.9
	51202004	2	4	2	4	50	1
	51202005	2	4	2	5	50	1
	51202006	2	4	2	6	50	1
	51202008	2	4	2	8	50	1
	51202010	2	4	2	10	50	1
	51202012	2	4	2	12	50	1
	51202014	2	4	2	14	50	1
	51202016	2	4	2	16	50	1
	51202018	2	4	2	18	55	1
	51202020	2	4	2	20	55	1
	51202022	2	4	2	22	60	1
	51202026	2	4	2	26	60	1
	51202030	2	4	2	30	70	1
	51202035	2	4	2	35	70	1
	51202040	2	4	2	40	80	1
	51202045	2	4	2	45	90	1
	51202050	2	4	2	50	100	1
	51202060	2	4	2	60	110	1
	51202508	2.5	4	2.5	8	50	1.25
	51202510	2.5	4	2.5	10	50	1.25
	51202512	2.5	4	2.5	12	50	1.25
	51202516	2.5	4	2.5	16	50	1.25
	51202520	2.5	4	2.5	20	50	1.25
	51202522	2.5	4	2.5	22	60	1.25
	51202526	2.5	4	2.5	26	60	1.25
	51202530	2.5	4	2.5	30	70	1.25
	51202535	2.5	4	2.5	35	70	1.25
	51202540	2.5	4	2.5	40	80	1.25
	51202545	2.5	4	2.5	45	90	1.25
	51202550	2.5	4	2.5	50	100	1.25
	51203006	3	6	3	6	50	1.5
	51203008	3	6	3	8	50	1.5

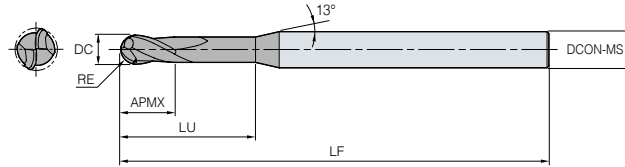
	Designation	DC	DCON-MS	APMX	LU	LF	RE
UB	51203010	3	6	3	10	50	1.5
	51203012	3	6	3	12	50	1.5
	51203014	3	6	3	14	60	1.5
	51203016	3	6	3	16	60	1.5
	51203018	3	6	3	18	60	1.5
	51203020	3	6	3	20	60	1.5
	51203022	3	6	3	22	65	1.5
	51203026	3	6	3	26	65	1.5
	51203030	3	6	3	30	70	1.5
	51203035	3	6	3	35	70	1.5
	51203040	3	6	3	40	80	1.5
	51203045	3	6	3	45	90	1.5
	51203050	3	6	3	50	100	1.5
	51203060	3	6	3	60	100	1.5
	51204008	4	6	4	8	50	2
	51204010	4	6	4	10	50	2
	51204012	4	6	4	12	50	2
	51204014	4	6	4	14	60	2
	51204016	4	6	4	16	60	2
	51204018	4	6	4	18	60	2
	51204020	4	6	4	20	60	2
	51204022	4	6	4	22	65	2
	51204026	4	6	4	26	65	2
	51204030	4	6	4	30	70	2
	51204035	4	6	4	35	70	2
	51204040	4	6	4	40	80	2
	51204045	4	6	4	45	90	2
	51204050	4	6	4	50	100	2
	51204055	4	6	4	55	100	2
	51204060	4	6	4	60	100	2
	51205015	5	6	6	15	60	2.5
	51205020	5	6	6	20	60	2.5
	51205026	5	6	6	26	65	2.5
	51205030	5	6	6	30	70	2.5

UB512

2 flutes long neck ball



CARBIDE	2	H-A 30°	Grade PC303W	h5 shank	RE ±0.005	RE ±0.01	DC Ø0.1 ~ Ø6 Ø8 ~ Ø12	Tolerance 0.000 ~ -0.012 0.000 ~ -0.015
					DC6 or below Above DC6			



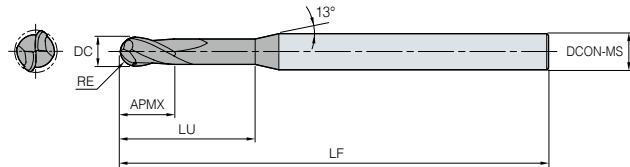
										(mm)
	Designation	DC	DCON-MS	APMX	LU	LF	RE			
UB	51205035	5	6	6	35	70	2.5			
	51205040	5	6	6	40	80	2.5			
	51205045	5	6	6	45	90	2.5			
	51205050	5	6	6	50	100	2.5			
	51205055	5	6	6	55	100	2.5			
	51205060	5	6	6	60	100	2.5			
	51206020	6	6	8	20	60	3			
	5120602090	6	6	12	20	90	3			
	51206030	6	6	8	30	60	3			
	5120603090	6	6	12	30	90	3			
	51208025	8	8	10	25	70	4			
	Designation	DC	DCON-MS	APMX	LU	LF	RE			
UB	51208025100	8	8	14	25	100	4			
	51208035	8	8	10	35	70	4			
	51208035100	8	8	14	35	100	4			
	51210030	10	10	12	30	75	5			
	51210030100	10	10	18	30	100	5			
	51210040	10	10	12	40	75	5			
	51210040100	10	10	18	40	100	5			
	51212032	12	12	14	32	80	6			
	51212032110	12	12	22	32	110	6			
	51212045	12	12	14	45	80	6			
	51212045110	12	12	22	45	110	6			

UB512S6

2 flutes long neck ball



CARBIDE	2	H-A 30°	Grade PC303W	h5 shank	RE ±0.005	DC Ø0.5 ~ Ø2	Tolerance 0.000 ~ -0.012
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(mm)

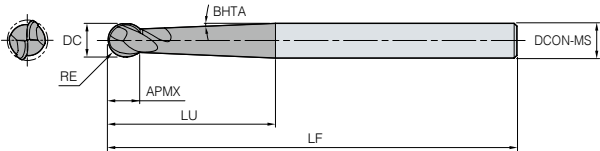
	Designation	DC	DCON-MS	APMX	LU	LF	RE
UB	51200501S6	0.5	6	0.5	1	45	0.25
	51200502S6	0.5	6	0.5	2	45	0.25
	51200504S6	0.5	6	0.5	4	45	0.25
	51200601S6	0.6	6	0.6	1	45	0.3
	51200602S6	0.6	6	0.6	2	45	0.3
	51200603S6	0.6	6	0.6	3	45	0.3
	51200604S6	0.6	6	0.6	4	45	0.3
	51200605S6	0.6	6	0.6	5	45	0.3
	51200606S6	0.6	6	0.6	6	45	0.3
	51200608S6	0.6	6	0.6	8	45	0.3
	51200610S6	0.6	6	0.6	10	45	0.3
	51200612S6	0.6	6	0.6	12	45	0.3
	51200614S6	0.6	6	0.6	14	45	0.3
	51200616S6	0.6	6	0.6	16	45	0.3
	51200801S6	0.8	6	0.8	1	45	0.4
	51200802S6	0.8	6	0.8	2	45	0.4
	51200803S6	0.8	6	0.8	3	45	0.4
	51200804S6	0.8	6	0.8	4	45	0.4
	51200805S6	0.8	6	0.8	5	45	0.4
	51200806S6	0.8	6	0.8	6	45	0.4
	51200808S6	0.8	6	0.8	8	45	0.4
	51200810S6	0.8	6	0.8	10	45	0.4
	51200812S6	0.8	6	0.8	12	45	0.4
	51200814S6	0.8	6	0.8	14	45	0.4
	51200816S6	0.8	6	0.8	16	45	0.4
	51200820S6	0.8	6	0.8	20	55	0.4
	51201002S6	1	6	1	2	50	0.5
	51201003S6	1	6	1	3	50	0.5
	51201004S6	1	6	1	4	50	0.5
	51201005S6	1	6	1	5	50	0.5
	51201006S6	1	6	1	6	50	0.5
	51201007S6	1	6	1	7	50	0.5
	51201008S6	1	6	1	8	50	0.5
	51201009S6	1	6	1	9	50	0.5
	51201010S6	1	6	1	10	50	0.5
	51201012S6	1	6	1	12	50	0.5
	51201014S6	1	6	1	14	50	0.5

	Designation	DC	DCON-MS	APMX	LU	LF	RE
UB	51201016S6	1	6	1	16	50	0.5
	51201018S6	1	6	1	18	50	0.5
	51201020S6	1	6	1	20	55	0.5
	51201022S6	1	6	1	22	60	0.5
	51201026S6	1	6	1	26	60	0.5
	51201030S6	1	6	1	30	70	0.5
	51201503S6	1.5	6	1.5	3	50	0.75
	51201504S6	1.5	6	1.5	4	50	0.75
	51201506S6	1.5	6	1.5	6	50	0.75
	51201508S6	1.5	6	1.5	8	50	0.75
	51201510S6	1.5	6	1.5	10	50	0.75
	51201512S6	1.5	6	1.5	12	50	0.75
	51201514S6	1.5	6	1.5	14	50	0.75
	51201516S6	1.5	6	1.5	16	50	0.75
	51201518S6	1.5	6	1.5	18	50	0.75
	51201520S6	1.5	6	1.5	20	55	0.75
	51201522S6	1.5	6	1.5	22	60	0.75
	51201526S6	1.5	6	1.5	26	60	0.75
	51201530S6	1.5	6	1.5	30	70	0.75
	51201535S6	1.5	6	1.5	35	70	0.75
	51201540S6	1.5	6	1.5	40	80	0.75
	51202004S6	2	6	2	4	50	1
	51202006S6	2	6	2	6	50	1
	51202008S6	2	6	2	8	50	1
	51202010S6	2	6	2	10	50	1
	51202012S6	2	6	2	12	50	1
	51202014S6	2	6	2	14	50	1
	51202016S6	2	6	2	16	50	1
	51202018S6	2	6	2	18	50	1
	51202020S6	2	6	2	20	50	1
	51202022S6	2	6	2	22	60	1
	51202026S6	2	6	2	26	60	1
	51202030S6	2	6	2	30	70	1
	51202035S6	2	6	2	35	70	1
	51202040S6	2	6	2	40	80	1
	51202045S6	2	6	2	45	90	1
	51202050S6	2	6	2	50	100	1

UB532

2 flutes lollipop style

CARBIDE	2	H-A 30°	Grade PC303W	h5 shank	RE ±0.01	DC Ø3 ~ Ø12	Tolerance 0.000 ~ -0.030
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(mm)

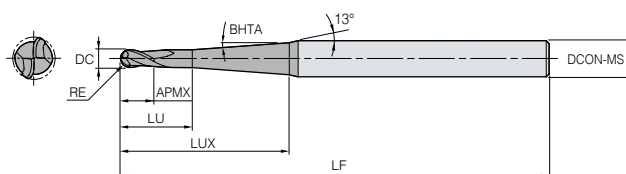
Designation	DC	DCON-MS	APMX	LU	LF	BHTA(°)	RE	
UB	532030	3	6	2.3	16	80	1.5	1.5
	532040	4	6	3.1	20	80	1.5	2
	532050	5	6	3.9	25	80	1.5	2.5
	532060	6	6	4.9	30	100	1.5	3
	532080	8	8	6.3	35	100	1.5	4
	532100	10	10	7.9	40	100	1.5	5
	532120	12	12	9.5	50	100	1.5	6

UB542

2 flutes tapered neck ball



CARBIDE	2	H-A 30°	Grade PC303W	h5 shank	RE ±0.005	RE ±0.01	DC	Tolerance
					DC6 or below	Above DC6	Ø0.1 ~ Ø6 Ø8 ~ Ø12	0.000 ~ -0.012 0.000 ~ -0.015



(mm)

Designation	DC	DCON-MS	APMX	LU	LUX	LF	BHTA(°)	RE
UB 54200105005	0.1	4	0.1	-	0.5	40	0.5	0.05
5420010501	0.1	4	0.1	-	1	40	0.5	0.05
54200110005	0.1	4	0.1	-	0.5	40	1	0.05
5420011001	0.1	4	0.1	-	1	40	1	0.05
54200115005	0.1	4	0.1	-	0.5	40	1.5	0.05
5420011501	0.1	4	0.1	-	1	40	1.5	0.05
54200120005	0.1	4	0.1	-	0.5	40	2	0.05
5420012001	0.1	4	0.1	-	1	40	2	0.05
54200130005	0.1	4	0.1	-	0.5	40	3	0.05
5420013001	0.1	4	0.1	-	1	40	3	0.05
5420020501	0.2	4	0.2	0.4	1	40	0.5	0.1
5420020502	0.2	4	0.2	0.4	2	40	0.5	0.1
5420020503	0.2	4	0.2	0.4	3	40	0.5	0.1
5420021001	0.2	4	0.2	0.4	1	40	1	0.1
5420021002	0.2	4	0.2	0.4	2	40	1	0.1
5420021003	0.2	4	0.2	0.4	3	40	1	0.1
5420021501	0.2	4	0.2	0.4	1	40	1.5	0.1
5420021502	0.2	4	0.2	0.4	2	40	1.5	0.1
5420021503	0.2	4	0.2	0.4	3	40	1.5	0.1
5420022001	0.2	4	0.2	0.4	1	40	2	0.1
5420022002	0.2	4	0.2	0.4	2	40	2	0.1
5420022003	0.2	4	0.2	0.4	3	40	2	0.1
5420023001	0.2	4	0.2	0.4	1	40	3	0.1
5420023002	0.2	4	0.2	0.4	2	40	3	0.1
5420023003	0.2	4	0.2	0.4	3	40	3	0.1
5420025002	0.2	4	0.2	0.4	2	40	5	0.1
5420025003	0.2	4	0.2	0.4	3	40	5	0.1
5420030502	0.3	4	0.3	0.6	2	40	0.5	0.15
5420030503	0.3	4	0.3	0.6	3	40	0.5	0.15
5420030504	0.3	4	0.3	0.6	4	40	0.5	0.15
5420030505	0.3	4	0.3	0.6	5	40	0.5	0.15
5420031002	0.3	4	0.3	0.6	2	40	1	0.15
5420031003	0.3	4	0.3	0.6	3	40	1	0.15
5420031004	0.3	4	0.3	0.6	4	40	1	0.15
5420031005	0.3	4	0.3	0.6	5	40	1	0.15
5420031502	0.3	4	0.3	0.6	2	40	1.5	0.15

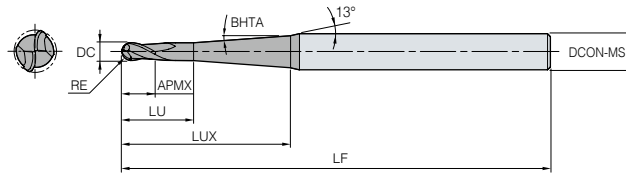
Designation	DC	DCON-MS	APMX	LU	LUX	LF	BHTA(°)	RE
UB 5420031503	0.3	4	0.3	0.6	3	40	1.5	0.15
5420031504	0.3	4	0.3	0.6	4	40	1.5	0.15
5420031505	0.3	4	0.3	0.6	5	40	1.5	0.15
5420032002	0.3	4	0.3	0.6	2	40	2	0.15
5420032003	0.3	4	0.3	0.6	3	40	2	0.15
5420032004	0.3	4	0.3	0.6	4	40	2	0.15
5420032005	0.3	4	0.3	0.6	5	40	2	0.15
5420033002	0.3	4	0.3	0.6	2	40	3	0.15
5420033003	0.3	4	0.3	0.6	3	40	3	0.15
5420033004	0.3	4	0.3	0.6	4	40	3	0.15
5420033005	0.3	4	0.3	0.6	5	40	3	0.15
5420035005	0.3	4	0.3	0.6	5	40	5	0.15
5420040502	0.4	4	0.4	0.8	2	50	0.5	0.2
5420040503	0.4	4	0.4	0.8	3	50	0.5	0.2
5420040504	0.4	4	0.4	0.8	4	50	0.5	0.2
5420040505	0.4	4	0.4	0.8	5	50	0.5	0.2
5420040506	0.4	4	0.4	0.8	6	50	0.5	0.2
5420041002	0.4	4	0.4	0.8	2	50	1	0.2
5420041003	0.4	4	0.4	0.8	3	50	1	0.2
5420041004	0.4	4	0.4	0.8	4	50	1	0.2
5420041005	0.4	4	0.4	0.8	5	50	1	0.2
5420041006	0.4	4	0.4	0.8	6	50	1	0.2
5420041502	0.4	4	0.4	0.8	2	50	1.5	0.2
5420041503	0.4	4	0.4	0.8	3	50	1.5	0.2
5420041504	0.4	4	0.4	0.8	4	50	1.5	0.2
5420041505	0.4	4	0.4	0.8	5	50	1.5	0.2
5420041506	0.4	4	0.4	0.8	6	50	1.5	0.2
5420042002	0.4	4	0.4	0.8	2	50	2	0.2
5420042003	0.4	4	0.4	0.8	3	50	2	0.2
5420042004	0.4	4	0.4	0.8	4	50	2	0.2
5420042005	0.4	4	0.4	0.8	5	50	2	0.2
5420042006	0.4	4	0.4	0.8	6	50	2	0.2
5420043002	0.4	4	0.4	0.8	2	50	3	0.2
5420043003	0.4	4	0.4	0.8	3	50	3	0.2
5420043004	0.4	4	0.4	0.8	4	50	3	0.2
5420043005	0.4	4	0.4	0.8	5	50	3	0.2

UB542

2 flutes tapered neck ball



CARBIDE	2	H-A 30°	Grade PC303W	h5 shank	RE ±0.005	RE ±0.01	DC Ø0.1 ~ Ø6 Ø8 ~ Ø12	Tolerance 0.000 ~ -0.012 0.000 ~ -0.015
					DC6 or below Above DC6			



										(mm)									
Designation		DC	DCON-MS	APMX	LU	LUX	LF	BHTA(°)	RE	Designation		DC	DCON-MS	APMX	LU	LUX	LF	BHTA(°)	RE
UB	5420043006	0.4	4	0.4	0.8	6	50	3	0.2	UB	5420061510	0.6	4	0.6	1.2	10	50	1.5	0.3
	5420045004	0.4	4	0.4	0.8	4	50	5	0.2		5420061512	0.6	4	0.6	1.2	12	50	1.5	0.3
	5420045006	0.4	4	0.4	0.8	6	50	5	0.2		5420062004	0.6	4	0.6	1.2	4	50	2	0.3
	5420050504	0.5	4	0.5	1	4	50	0.5	0.25		5420062006	0.6	4	0.6	1.2	6	50	2	0.3
	5420050506	0.5	4	0.5	1	6	50	0.5	0.25		5420062008	0.6	4	0.6	1.2	8	50	2	0.3
	5420050508	0.5	4	0.5	1	8	50	0.5	0.25		5420062010	0.6	4	0.6	1.2	10	50	2	0.3
	5420050510	0.5	4	0.5	1	10	50	0.5	0.25		5420062012	0.6	4	0.6	1.2	12	50	2	0.3
	5420051004	0.5	4	0.5	1	4	50	1	0.25		5420063004	0.6	4	0.6	1.2	4	50	3	0.3
	5420051006	0.5	4	0.5	1	6	50	1	0.25		5420063006	0.6	4	0.6	1.2	6	50	3	0.3
	5420051008	0.5	4	0.5	1	8	50	1	0.25		5420063008	0.6	4	0.6	1.2	8	50	3	0.3
	5420051010	0.5	4	0.5	1	10	50	1	0.25		5420063010	0.6	4	0.6	1.2	10	50	3	0.3
	5420051504	0.5	4	0.5	1	4	50	1.5	0.25		5420063012	0.6	4	0.6	1.2	12	50	3	0.3
	5420051506	0.5	4	0.5	1	6	50	1.5	0.25		5420080504	0.8	4	0.8	1.6	4	50	0.5	0.4
	5420051508	0.5	4	0.5	1	8	50	1.5	0.25		5420080506	0.8	4	0.8	1.6	6	50	0.5	0.4
	5420051510	0.5	4	0.5	1	10	50	1.5	0.25		5420080508	0.8	4	0.8	1.6	8	50	0.5	0.4
	5420052004	0.5	4	0.5	1	4	50	2	0.25		5420080510	0.8	4	0.8	1.6	10	50	0.5	0.4
	5420052006	0.5	4	0.5	1	6	50	2	0.25		5420080512	0.8	4	0.8	1.6	12	50	0.5	0.4
	5420052008	0.5	4	0.5	1	8	50	2	0.25		5420080516	0.8	4	0.8	1.6	16	50	0.5	0.4
	5420052010	0.5	4	0.5	1	10	50	2	0.25		5420081004	0.8	4	0.8	1.6	4	50	1	0.4
	5420053004	0.5	4	0.5	1	4	50	3	0.25		5420081006	0.8	4	0.8	1.6	6	50	1	0.4
	5420053006	0.5	4	0.5	1	6	50	3	0.25		5420081008	0.8	4	0.8	1.6	8	50	1	0.4
	5420053008	0.5	4	0.5	1	8	50	3	0.25		5420081010	0.8	4	0.8	1.6	10	50	1	0.4
	5420053010	0.5	4	0.5	1	10	50	3	0.25		5420081012	0.8	4	0.8	1.6	12	50	1	0.4
	5420060504	0.6	4	0.6	1.2	4	50	0.5	0.3		5420081016	0.8	4	0.8	1.6	16	50	1	0.4
	5420060506	0.6	4	0.6	1.2	6	50	0.5	0.3		5420081504	0.8	4	0.8	1.6	4	50	1.5	0.4
	5420060508	0.6	4	0.6	1.2	8	50	0.5	0.3		5420081506	0.8	4	0.8	1.6	6	50	1.5	0.4
	5420060510	0.6	4	0.6	1.2	10	50	0.5	0.3		5420081508	0.8	4	0.8	1.6	8	50	1.5	0.4
	5420060512	0.6	4	0.6	1.2	12	50	0.5	0.3		5420081510	0.8	4	0.8	1.6	10	50	1.5	0.4
	5420061004	0.6	4	0.6	1.2	4	50	1	0.3		5420081512	0.8	4	0.8	1.6	12	50	1.5	0.4
	5420061006	0.6	4	0.6	1.2	6	50	1	0.3		5420081516	0.8	4	0.8	1.6	16	50	1.5	0.4
	5420061008	0.6	4	0.6	1.2	8	50	1	0.3		5420082004	0.8	4	0.8	1.6	4	50	2	0.4
	5420061010	0.6	4	0.6	1.2	10	50	1	0.3		5420082006	0.8	4	0.8	1.6	6	50	2	0.4
	5420061012	0.6	4	0.6	1.2	12	50	1	0.3		5420082008	0.8	4	0.8	1.6	8	50	2	0.4
	5420061504	0.6	4	0.6	1.2	4	50	1.5	0.3		5420082010	0.8	4	0.8	1.6	10	50	2	0.4
	5420061506	0.6	4	0.6	1.2	6	50	1.5	0.3		5420082012	0.8	4	0.8	1.6	12	50	2	0.4
	5420061508	0.6	4	0.6	1.2	8	50	1.5	0.3		5420082016	0.8	4	0.8	1.6	16	50	2	0.4

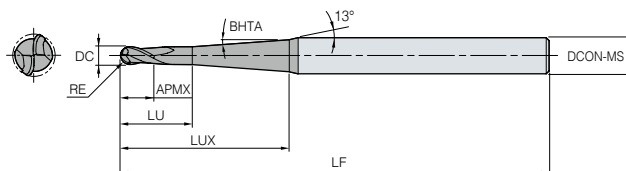
UB542

2 flutes tapered neck ball



DC6 or below Above DC6

DC	Tolerance
Ø0.1 ~ Ø6	0.000 ~ -0.012
Ø8 ~ Ø12	0.000 ~ -0.015



(mm)

Designation	DC	DCON-MS	APMX	LU	LUX	LF	BHTA(°)	RE
UB 5420083004	0.8	4	0.8	1.6	4	50	3	0.4
5420083006	0.8	4	0.8	1.6	6	50	3	0.4
5420083008	0.8	4	0.8	1.6	8	50	3	0.4
5420083010	0.8	4	0.8	1.6	10	50	3	0.4
5420083012	0.8	4	0.8	1.6	12	50	3	0.4
5420083016	0.8	4	0.8	1.6	16	50	3	0.4
5420100506	1	4	1	2.5	6	50	0.5	0.5
5420100508	1	4	1	2.5	8	50	0.5	0.5
5420100510	1	4	1	2.5	10	50	0.5	0.5
5420100512	1	4	1	2.5	12	50	0.5	0.5
5420100516	1	4	1	2.5	16	50	0.5	0.5
5420100520	1	4	1	2.5	20	50	0.5	0.5
5420100525	1	4	1	2.5	25	60	0.5	0.5
5420100530	1	4	1	2.5	30	70	0.5	0.5
5420100540	1	4	1	2.5	40	80	0.5	0.5
5420100550	1	4	1	2.5	50	90	0.5	0.5
5420101006	1	4	1	2.5	6	50	1	0.5
5420101008	1	4	1	2.5	8	50	1	0.5
5420101010	1	4	1	2.5	10	50	1	0.5
5420101012	1	4	1	2.5	12	50	1	0.5
5420101016	1	4	1	2.5	16	50	1	0.5
5420101020	1	4	1	2.5	20	50	1	0.5
5420101025	1	4	1	2.5	25	60	1	0.5
5420101030	1	4	1	2.5	30	70	1	0.5
5420101040	1	4	1	2.5	40	80	1	0.5
5420101050	1	4	1	2.5	50	90	1	0.5
5420101506	1	4	1	2.5	6	50	1.5	0.5
5420101508	1	4	1	2.5	8	50	1.5	0.5
5420101510	1	4	1	2.5	10	50	1.5	0.5
5420101512	1	4	1	2.5	12	50	1.5	0.5
5420101516	1	4	1	2.5	16	50	1.5	0.5
5420101520	1	4	1	2.5	20	50	1.5	0.5
5420101525	1	4	1	2.5	25	60	1.5	0.5
5420101530	1	4	1	2.5	30	70	1.5	0.5
5420101540	1	4	1	2.5	40	80	1.5	0.5
5420101550	1	4	1	2.5	50	90	1.5	0.5

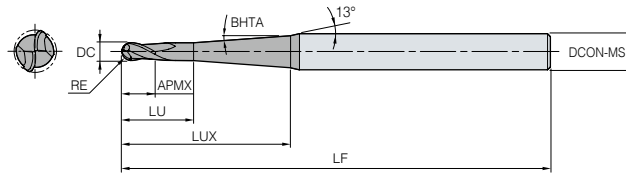
Designation	DC	DCON-MS	APMX	LU	LUX	LF	BHTA(°)	RE
UB 5420102006	1	4	1	2.5	6	50	2	0.5
5420102008	1	4	1	2.5	8	50	2	0.5
5420102010	1	4	1	2.5	10	50	2	0.5
5420102012	1	4	1	2.5	12	50	2	0.5
5420102016	1	4	1	2.5	16	50	2	0.5
5420102020	1	4	1	2.5	20	50	2	0.5
5420102025	1	4	1	2.5	25	60	2	0.5
5420102030	1	4	1	2.5	30	70	2	0.5
5420102040	1	4	1	2.5	40	80	2	0.5
5420102050	1	6	1	2.5	50	90	2	0.5
5420103006	1	4	1	2.5	6	50	3	0.5
5420103008	1	4	1	2.5	8	50	3	0.5
5420103010	1	4	1	2.5	10	50	3	0.5
5420103012	1	4	1	2.5	12	50	3	0.5
5420103016	1	4	1	2.5	16	50	3	0.5
5420103020	1	4	1	2.5	20	50	3	0.5
5420103025	1	4	1	2.5	25	60	3	0.5
5420103030	1	6	1	2.5	30	70	3	0.5
5420103040	1	6	1	2.5	40	80	3	0.5
5420103050	1	6	1	2.5	50	90	3	0.5
5420105030	1	6	1	2.5	30	70	5	0.5
5420120508	1.2	4	1.2	3	8	50	0.5	0.6
5420120512	1.2	4	1.2	3	12	50	0.5	0.6
5420120516	1.2	4	1.2	3	16	50	0.5	0.6
5420120520	1.2	4	1.2	3	20	50	0.5	0.6
5420120525	1.2	4	1.2	3	25	60	0.5	0.6
5420120530	1.2	4	1.2	3	30	70	0.5	0.6
5420121008	1.2	4	1.2	3	8	50	1	0.6
5420121012	1.2	4	1.2	3	12	50	1	0.6
5420121016	1.2	4	1.2	3	16	50	1	0.6
5420121020	1.2	4	1.2	3	20	50	1	0.6
5420121025	1.2	4	1.2	3	25	60	1	0.6
5420121030	1.2	4	1.2	3	30	70	1	0.6
5420121508	1.2	4	1.2	3	8	50	1.5	0.6
5420121512	1.2	4	1.2	3	12	50	1.5	0.6
5420121516	1.2	4	1.2	3	16	50	1.5	0.6

UB542

2 flutes tapered neck ball



CARBIDE	2	H-A 30°	Grade PC303W	h5 shank	RE ±0.005	RE ±0.01	DC Ø0.1 ~ Ø6 Ø8 ~ Ø12	Tolerance 0.000 ~ -0.012 0.000 ~ -0.015
					DC6 or below Above DC6			



										(mm)									
Designation		DC	DCON-MS	APMX	LU	LUX	LF	BHTA(°)	RE	Designation		DC	DCON-MS	APMX	LU	LUX	LF	BHTA(°)	RE
UB	5420121520	1.2	4	1.2	3	20	50	1.5	0.6	UB	5420151516	1.5	4	1.5	4	16	50	1.5	0.75
	5420121525	1.2	4	1.2	3	25	60	1.5	0.6		5420151520	1.5	4	1.5	4	20	50	1.5	0.75
	5420121530	1.2	4	1.2	3	30	70	1.5	0.6		5420151525	1.5	4	1.5	4	25	60	1.5	0.75
	5420122008	1.2	4	1.2	3	8	50	2	0.6		5420151530	1.5	4	1.5	4	30	70	1.5	0.75
	5420122012	1.2	4	1.2	3	12	50	2	0.6		5420151540	1.5	4	1.5	4	40	80	1.5	0.75
	5420122016	1.2	4	1.2	3	16	50	2	0.6		5420151550	1.5	4	1.5	4	50	90	1.5	0.75
	5420122020	1.2	4	1.2	3	20	50	2	0.6		5420121016	1.2	4	1.2	3	16	50	1	0.6
	5420122025	1.2	4	1.2	3	25	60	2	0.6		5420121020	1.2	4	1.2	3	20	50	1	0.6
	5420122030	1.2	4	1.2	3	30	70	2	0.6		5420121025	1.2	4	1.2	3	25	60	1	0.6
	5420123008	1.2	4	1.2	3	8	50	3	0.6		5420121030	1.2	4	1.2	3	30	70	1	0.6
	5420123012	1.2	4	1.2	3	12	50	3	0.6		5420121508	1.2	4	1.2	3	8	50	1.5	0.6
	5420123016	1.2	4	1.2	3	16	50	3	0.6		5420121512	1.2	4	1.2	3	12	50	1.5	0.6
	5420123020	1.2	4	1.2	3	20	50	3	0.6		5420121516	1.2	4	1.2	3	16	50	1.5	0.6
	5420123025	1.2	4	1.2	3	25	60	3	0.6		5420121520	1.2	4	1.2	3	20	50	1.5	0.6
	5420123030	1.2	6	1.2	3	30	70	5	0.6		5420121525	1.2	4	1.2	3	25	60	1.5	0.6
	5420150508	1.5	4	1.5	4	8	50	0.5	0.75		5420121530	1.2	4	1.2	3	30	70	1.5	0.6
	5420150510	1.5	4	1.5	4	10	50	0.5	0.75		5420122008	1.2	4	1.2	3	8	50	2	0.6
	5420150512	1.5	4	1.5	4	12	50	0.5	0.75		5420122012	1.2	4	1.2	3	12	50	2	0.6
	5420150516	1.5	4	1.5	4	16	50	0.5	0.75		5420122016	1.2	4	1.2	3	16	50	2	0.6
	5420150520	1.5	4	1.5	4	20	50	0.5	0.75		5420122020	1.2	4	1.2	3	20	50	2	0.6
	5420150525	1.5	4	1.5	4	25	60	0.5	0.75		5420122025	1.2	4	1.2	3	25	60	2	0.6
	5420150530	1.5	4	1.5	4	30	70	0.5	0.75		5420122030	1.2	4	1.2	3	30	70	2	0.6
	5420150540	1.5	4	1.5	4	40	80	0.5	0.75		5420123008	1.2	4	1.2	3	8	50	3	0.6
	5420150550	1.5	4	1.5	4	50	90	0.5	0.75		5420123012	1.2	4	1.2	3	12	50	3	0.6
	5420151008	1.5	4	1.5	4	8	50	1	0.75		5420123016	1.2	4	1.2	3	16	50	3	0.6
	5420151010	1.5	4	1.5	4	10	50	1	0.75		5420123020	1.2	4	1.2	3	20	50	3	0.6
	5420151012	1.5	4	1.5	4	12	50	1	0.75		5420123025	1.2	4	1.2	3	25	60	3	0.6
	5420151016	1.5	4	1.5	4	16	50	1	0.75		5420123030	1.2	6	1.2	3	30	70	5	0.6
	5420151020	1.5	4	1.5	4	20	50	1	0.75		5420150508	1.5	4	1.5	4	8	50	0.5	0.75
	5420151025	1.5	4	1.5	4	25	60	1	0.75		5420150510	1.5	4	1.5	4	10	50	0.5	0.75
	5420151030	1.5	4	1.5	4	30	70	1	0.75		5420150512	1.5	4	1.5	4	12	50	0.5	0.75
	5420151040	1.5	4	1.5	4	40	80	1	0.75		5420150516	1.5	4	1.5	4	16	50	0.5	0.75
	5420151050	1.5	4	1.5	4	50	90	1	0.75		5420150520	1.5	4	1.5	4	20	50	0.5	0.75
	5420151508	1.5	4	1.5	4	8	50	1.5	0.75		5420150525	1.5	4	1.5	4	25	60	0.5	0.75
	5420151510	1.5	4	1.5	4	10	50	1.5	0.75		5420150530	1.5	4	1.5	4	30	70	0.5	0.75
	5420151512	1.5	4	1.5	4	12	50	1.5	0.75		5420150540	1.5	4	1.5	4	40	80	0.5	0.75

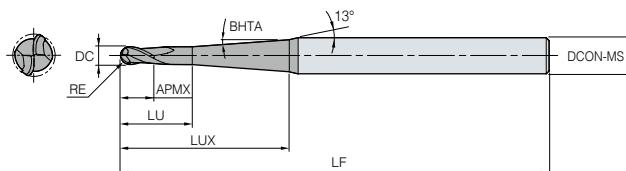
UB542

2 flutes tapered neck ball



DC6 or below Above DC6

DC	Tolerance
Ø0.1 ~ Ø6	0.000 ~ -0.012
Ø8 ~ Ø12	0.000 ~ -0.015



(mm)

Designation	DC	DCON-MS	APMX	LU	LUX	LF	BHTA(°)	RE
UB 5420150550	1.5	4	1.5	4	50	90	0.5	0.75
5420151008	1.5	4	1.5	4	8	50	1	0.75
5420151010	1.5	4	1.5	4	10	50	1	0.75
5420151012	1.5	4	1.5	4	12	50	1	0.75
5420151016	1.5	4	1.5	4	16	50	1	0.75
5420151020	1.5	4	1.5	4	20	50	1	0.75
5420151025	1.5	4	1.5	4	25	60	1	0.75
5420151030	1.5	4	1.5	4	30	70	1	0.75
5420151040	1.5	4	1.5	4	40	80	1	0.75
5420151050	1.5	4	1.5	4	50	90	1	0.75
5420151508	1.5	4	1.5	4	8	50	1.5	0.75
5420151510	1.5	4	1.5	4	10	50	1.5	0.75
5420151512	1.5	4	1.5	4	12	50	1.5	0.75
5420151516	1.5	4	1.5	4	16	50	1.5	0.75
5420151520	1.5	4	1.5	4	20	50	1.5	0.75
5420151525	1.5	4	1.5	4	25	60	1.5	0.75
5420151530	1.5	4	1.5	4	30	70	1.5	0.75
5420151540	1.5	4	1.5	4	40	80	1.5	0.75
5420151550	1.5	4	1.5	4	50	90	1.5	0.75
5420152008	1.5	4	1.5	4	8	50	2	0.75
5420152010	1.5	4	1.5	4	10	50	2	0.75
5420152012	1.5	4	1.5	4	12	50	2	0.75
5420152016	1.5	4	1.5	4	16	50	2	0.75
5420152020	1.5	4	1.5	4	20	50	2	0.75
5420152025	1.5	4	1.5	4	25	60	2	0.75
5420152030	1.5	4	1.5	4	30	70	2	0.75
5420152040	1.5	6	1.5	4	40	80	2	0.75
5420152050	1.5	6	1.5	4	50	90	2	0.75
5420153020	1.5	6	1.5	4	20	50	3	0.75
5420153030	1.5	6	1.5	4	30	70	3	0.75
5420153040	1.5	6	1.5	4	40	80	3	0.75
5420153050	1.5	8	1.5	4	50	90	3	0.75
5420155030	1.5	8	1.5	4	30	70	5	0.75
5420200510	2	4	2	5	10	50	0.5	1
5420200512	2	4	2	5	12	50	0.5	1
5420200516	2	4	2	5	16	50	0.5	1

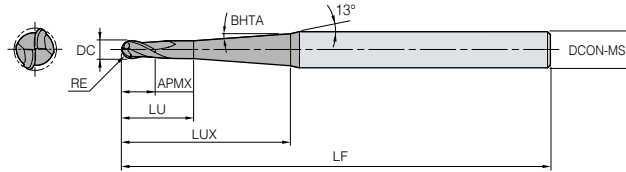
Designation	DC	DCON-MS	APMX	LU	LUX	LF	BHTA(°)	RE
UB 5420200520	2	4	2	5	20	50	0.5	1
5420200525	2	4	2	5	25	60	0.5	1
5420200530	2	4	2	5	30	70	0.5	1
5420200540	2	4	2	5	40	80	0.5	1
5420200550	2	6	2	5	50	100	0.5	1
5420200560	2	6	2	5	60	100	0.5	1
5420200580	2	6	2	5	80	140	0.5	1
5420201010	2	4	2	5	10	50	1	1
5420201012	2	4	2	5	12	50	1	1
5420201016	2	4	2	5	16	50	1	1
5420201020	2	4	2	5	20	50	1	1
5420201025	2	4	2	5	25	60	1	1
5420201030	2	4	2	5	30	70	1	1
5420201040	2	6	2	5	40	80	1	1
5420201050	2	6	2	5	50	100	1	1
5420201060	2	6	2	5	60	100	1	1
5420201080	2	6	2	5	80	140	1	1
5420201510	2	4	2	5	10	50	1.5	1
5420201512	2	4	2	5	12	50	1.5	1
5420201516	2	4	2	5	16	50	1.5	1
5420201520	2	4	2	5	20	50	1.5	1
5420201525	2	4	2	5	25	60	1.5	1
5420201530	2	6	2	5	30	70	1.5	1
5420201540	2	6	2	5	40	80	1.5	1
5420201550	2	6	2	5	50	100	1.5	1
5420201560	2	6	2	5	60	100	1.5	1
5420201580	2	6	2	5	80	140	1.5	1
5420202010	2	4	2	5	10	50	2	1
5420202012	2	4	2	5	12	50	2	1
5420202016	2	4	2	5	16	50	2	1
5420202020	2	4	2	5	20	55	2	1
5420202025	2	4	2	5	25	60	2	1
5420202030	2	4	2	5	30	70	2	1
5420202040	2	6	2	5	40	80	2	1
5420202050	2	6	2	5	50	90	2	1
5420202060	2	6	2	5	60	100	2	1

UB542

2 flutes tapered neck ball



CARBIDE	2	H-A 30°	Grade PC303W	h5 shank	RE ±0.005	RE ±0.01	DC Ø0.1 ~ Ø6 Ø8 ~ Ø12	Tolerance 0.000 ~ -0.012 0.000 ~ -0.015
					DC6 or below Above DC6			



										(mm)									
Designation		DC	DCON-MS	APMX	LU	LUX	LF	BHTA(°)	RE	Designation		DC	DCON-MS	APMX	LU	LUX	LF	BHTA(°)	RE
UB	5420202080	2	8	2	5	80	140	2	1	UB	5420400540	4	6	6	8	40	90	0.5	2
	5420203030	2	6	2	5	30	70	3	1		5420400550	4	6	6	8	50	100	0.5	2
	5420203040	2	6	2	5	40	80	3	1		5420400560	4	6	6	8	60	110	0.5	2
	5420203050	2	8	2	5	50	90	3	1		5420400570	4	6	6	8	70	120	0.5	2
	5420203060	2	8	2	5	60	100	3	1		5420401040	4	6	6	8	40	90	1	2
	5420203080	2	10	2	5	80	140	3	1		5420401050	4	6	6	8	50	100	1	2
	5420205030	2	8	2	5	30	70	5	1		5420401060	4	8	6	8	60	110	1	2
	5420205040	2	10	2	5	40	90	5	1		5420401070	4	8	6	8	70	120	1	2
	5420300516	3	6	4.5	6	16	60	0.5	1.5		5420401540	4	6	6	8	40	90	1.5	2
	5420300520	3	6	4.5	6	20	65	0.5	1.5		5420401550	4	8	6	8	50	100	1.5	2
	5420300530	3	6	4.5	6	30	70	0.5	1.5		5420401560	4	8	6	8	60	110	1.5	2
	5420300540	3	6	4.5	6	40	80	0.5	1.5		5420401570	4	8	6	8	70	120	1.5	2
	5420300550	3	6	4.5	6	50	90	0.5	1.5		5420403050	4	10	6	8	50	100	3	2
	5420300560	3	6	4.5	6	60	100	0.5	1.5		5420405050	4	12	6	8	50	100	5	2
	5420301016	3	6	4.5	6	16	60	1	1.5		5420501060	5	8	10	13	60	120	1	2.5
	5420301020	3	6	4.5	6	20	65	1	1.5		5420501560	5	8	10	13	60	120	1.5	2.5
	5420301030	3	6	4.5	6	30	70	1	1.5		5420503040	5	8	10	13	40	120	3	2.5
	5420301040	3	6	4.5	6	40	80	1	1.5		5420601060	6	8	12	15	60	120	1	3
	5420301050	3	6	4.5	6	50	90	1	1.5		5420601090	6	10	12	15	90	150	1	3
	5420301060	3	6	4.5	6	60	100	1	1.5		5420601560	6	10	12	15	60	120	1.5	3
	5420301070	3	6	4.5	6	70	120	1	1.5		5420601590	6	10	12	15	90	150	1.5	3
	5420301516	3	6	4.5	6	16	60	1.5	1.5		5420602060	6	10	12	15	60	120	2	3
	5420301520	3	6	4.5	6	20	65	1.5	1.5		5420602090	6	12	12	15	90	150	2	3
	5420301530	3	6	4.5	6	30	70	1.5	1.5		5420603060	6	12	12	15	60	120	3	3
	5420301540	3	6	4.5	6	40	80	1.5	1.5		5420603090	6	14	12	15	90	150	3	3
	5420301550	3	6	4.5	6	50	90	1.5	1.5		54208010100	8	12	14	18	100	150	1	4
	5420301560	3	6	4.5	6	60	100	1.5	1.5		5420801070	8	10	14	18	70	130	1	4
	5420302016	3	6	4.5	6	16	60	2	1.5		54208015100	8	14	14	18	100	150	1.5	4
	5420302020	3	6	4.5	6	20	65	2	1.5		5420801570	8	12	14	18	70	130	1.5	4
	5420302030	3	6	4.5	6	30	70	2	1.5		54208020100	8	14	14	18	100	150	2	4
	5420302040	3	6	4.5	6	40	80	2	1.5		5420802070	8	12	14	18	70	130	2	4
	5420302050	3	8	4.5	6	50	90	2	1.5		54208030100	8	18	14	18	100	150	3	4
	5420303030	3	6	4.5	6	30	70	3	1.5		5420803070	8	14	14	18	70	130	3	4
	5420303040	3	8	4.5	6	40	90	3	1.5		54210010100	10	14	18	22	100	200	1	5
	5420305030	3	8	4.5	6	30	70	5	1.5		5421001070	10	12	18	22	70	130	1	5
	5420305040	3	10	4.5	6	40	90	5	1.5		5421001080	10	14	18	22	80	150	1	5

UB542

2 flutes tapered neck ball



CARBIDE

2

H-A
30°

Grade
PC303W

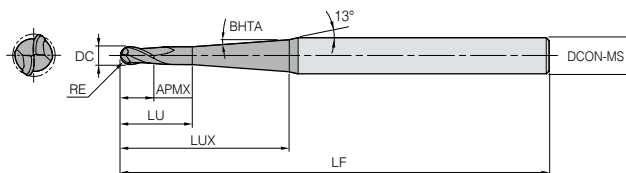
h5
shank

RE
±0.005

RE
±0.01

DC6 or below Above DC6

DC	Tolerance
Ø0.1 ~ Ø6	0.000 ~ -0.012
Ø8 ~ Ø12	0.000 ~ -0.015



(mm)

Designation	DC	DCON-MS	APMX	LU	LUX	LF	BHTA(°)	RE
UB 54210015100	10	16	18	22	100	200	1.5	5
5421001570	10	14	18	22	70	130	1.5	5
5421001580	10	14	18	22	80	150	1.5	5
54210020100	10	16	18	22	100	200	2	5
5421002070	10	14	18	22	70	130	2	5
5421002080	10	16	18	22	80	150	2	5
54210030100	10	20	18	22	100	200	3	5
5421003070	10	16	18	22	70	130	3	5
5421003080	10	18	18	22	80	150	3	5
54212010100	12	16	22	25	100	200	1	6
5421201060	12	14	22	25	60	130	1	6
5421201080	12	14	22	25	80	150	1	6
5421201090	12	16	22	25	90	180	1	6

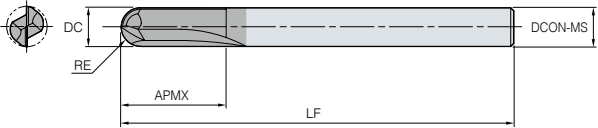
Designation	DC	DCON-MS	APMX	LU	LUX	LF	BHTA(°)	RE
UB 54212015100	12	16	22	25	100	200	1.5	6
5421201560	12	14	22	25	60	130	1.5	6
5421201580	12	16	22	25	80	150	1.5	6
5421201590	12	16	22	25	90	180	1.5	6
54212020100	12	18	22	25	100	200	2	6
5421202060	12	16	22	25	60	130	2	6
5421202080	12	16	22	25	80	150	2	6
5421202090	12	18	22	25	90	180	2	6
54212030100	12	20	22	25	100	200	3	6
5421203060	12	16	22	25	60	130	3	6
5421203080	12	18	22	25	80	150	3	6
5421203090	12	20	22	25	90	180	3	6

USB502

2 flutes straight ball



CARBIDE	2	H-A 0°	Grade PC303W	h5 shank	RE ±0.005	RE ±0.01	DC Ø3 ~ Ø6 Ø8 ~ Ø20	Tolerance 0.000 ~ -0.012 0.000 ~ -0.015
					DC6 or below	Above DC6		



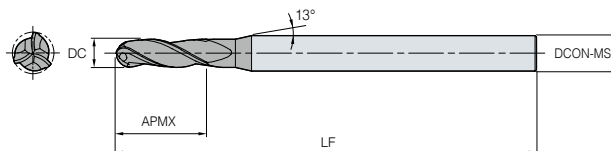
(mm)

Designation		DC	DCON-MS	APMX	LF	RE
USB	502030	3	6	10	70	1.5
	502040	4	6	12	70	2
	502050	5	6	18	90	2.5
	502060	6	6	20	90	3
	502080	8	8	25	100	4
	502100	10	10	30	100	5
	502120	12	12	32	110	6
	502160	16	16	35	150	8
	502200	20	20	40	150	10

UB503

3 flutes ball

CARBIDE	3	H-A 30°	Grade PC303W	h5 shank	RE ±0.005	RE ±0.01	DC	Tolerance
					DC6 or below	Above DC6	Ø1 ~ Ø12	0.000 ~ -0.020



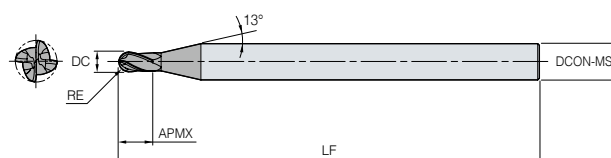
(mm)

Designation	DC	DCON-MS	APMX	LF	RE
UB 503010	1	6	1	50	0.5
503015	1.5	6	1.5	50	0.75
503020	2	6	2	50	1
503030	3	6	3	60	1.5
503040	4	6	4	70	2
503050	5	6	5	80	2.5
503060	6	6	6	90	3
503080	8	8	8	100	4
503100	10	10	10	100	5
503120	12	12	12	110	6

UB504

4 flutes ball

CARBIDE	4	H-A 30°	Grade PC303W	h5 shank	RE ±0.005	RE ±0.01	DC	Tolerance
					DC6 or below	Above DC6	Ø1 ~ Ø12	0.000 ~ -0.020



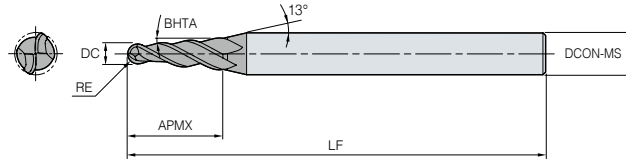
(mm)

Designation	DC	DCON-MS	APMX	LF	RE
UB 504010	1	6	1	50	0.5
504015	1.5	6	1.5	50	0.75
504020	2	6	2	50	1
504030	3	6	3	60	1.5
504040	4	6	4	70	2
504050	5	6	5	80	2.5
504060	6	6	6	90	3
504080	8	8	8	100	4
504100	10	10	10	100	5
504120	12	12	12	110	6

UTB502

2 flutes tapered ball

CARBIDE	2	H-A 30°	Grade PC303W	h5 shank	RE ±0.01	DC Ø0.3 ~ Ø2	Tolerance 0.000 ~ -0.030
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(mm)

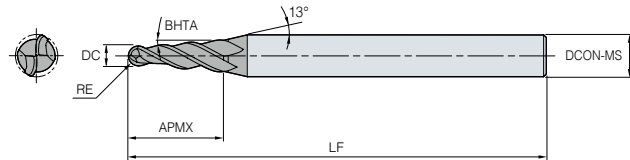
Designation	DC	DCON-MS	APMX	BHTA(°)	LF	RE
UTB 502003005	0.3	4	1.2	0.5	40	0.15
50200301	0.3	4	1.2	1	40	0.15
502003015	0.3	4	1.2	1.5	40	0.15
50200302	0.3	4	1.2	2	40	0.15
50200303	0.3	4	1.2	3	40	0.15
50200305	0.3	4	1.2	5	40	0.15
50200307	0.3	4	1.5	7	40	0.15
50200310	0.3	4	1.5	10	40	0.15
502004005	0.4	4	1.6	0.5	40	0.2
50200401	0.4	4	1.6	1	40	0.2
502004015	0.4	4	1.6	1.5	40	0.2
50200402	0.4	4	1.6	2	40	0.2
50200403	0.4	4	1.6	3	40	0.2
50200405	0.4	4	1.6	5	40	0.2
50200407	0.4	4	2	7	40	0.2
50200410	0.4	4	2	10	40	0.2
502005005	0.5	4	2	0.5	45	0.25
50200501	0.5	4	2	1	45	0.25
502005015	0.5	4	2	1.5	45	0.25
50200502	0.5	4	2	2	45	0.25
50200503	0.5	4	2	3	45	0.25
50200505	0.5	4	2	5	45	0.25
50200507	0.5	4	2.5	7	45	0.25
50200510	0.5	4	2.5	10	45	0.25
502006005	0.6	4	2	0.5	45	0.3
50200601	0.6	4	2	1	45	0.3
502006015	0.6	4	2	1.5	45	0.3
50200602	0.6	4	2	2	45	0.3
50200603	0.6	4	2	3	45	0.3
50200605	0.6	4	2	5	45	0.3
50200607	0.6	4	2.5	7	45	0.3
50200610	0.6	4	2.5	10	45	0.3
502007005	0.7	4	2.5	0.5	45	0.35
50200701	0.7	4	2.5	1	45	0.35
502007015	0.7	4	2.5	1.5	45	0.35
50200702	0.7	4	2.5	2	45	0.35

UTB502

2 flutes tapered ball



CARBIDE	2	H-A 30°	Grade PC303W	h5 shank	RE ±0.01	DC Ø0.3 ~ Ø2	Tolerance 0.000 ~ -0.030
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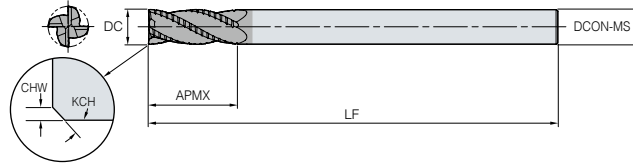
(mm)

Designation		DC	DCON-MS	APMX	BHTA(°)	LF	RE
UTB	50200703	0.7	4	2.5	3	45	0.35
	50200705	0.7	4	2.5	5	45	0.35
	50200707	0.7	4	3	7	45	0.35
	50200710	0.7	4	3	10	45	0.35
	502008005	0.8	4	3.2	0.5	45	0.4
	50200801	0.8	4	3.2	1	45	0.4
	502008015	0.8	4	3.2	1.5	45	0.4
	50200802	0.8	4	3.2	2	45	0.4
	50200803	0.8	4	3.2	3	45	0.4
	50200805	0.8	4	3.2	5	45	0.4
	50200807	0.8	4	3.2	7	45	0.4
	50200810	0.8	4	3.2	10	45	0.4
	502010005	1	4	4	0.5	50	0.5
	50201001	1	4	4	1	50	0.5
	502010015	1	4	4	1.5	50	0.5
	50201002	1	4	4	2	50	0.5
	50201003	1	4	4	3	50	0.5
	50201005	1	4	4	5	50	0.5
	50201007	1	4	4	7	50	0.5
	50201010	1	4	4	10	50	0.5
	502015005	1.5	4	6	0.5	50	0.75
	50201501	1.5	4	6	1	50	0.75
	502015015	1.5	4	6	1.5	50	0.75
	50201502	1.5	4	7	2	50	0.75
	50201503	1.5	4	8	3	50	0.75
	50201505	1.5	4	10	5	50	0.75
	50201507	1.5	4	10	7	50	0.75
	50201510	1.5	6	10	10	50	0.75
	502020005	2	4	6	0.5	50	1
	50202001	2	4	6	1	50	1
	502020015	2	4	6	1.5	50	1
	50202002	2	4	10	2	50	1
	50202003	2	4	10	3	50	1
	50202005	2	4	10	5	50	1
	50202007	2	6	10	7	50	1
	50202010	2	6	11	10	50	1

UF50

3~5 flutes chamfer pitch roughing

CARBIDE	3	4	5	H-A 29°/31°	Grade PC303W	CHAMFERED	h5 shank	RE ±0.025	DC Ø3 ~ Ø25	Tolerance 0.000 ~ -0.050
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(mm)

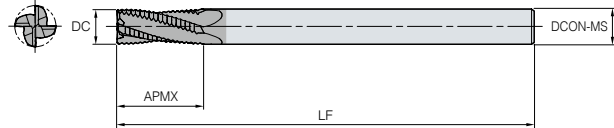
Designation		DC	DCON-MS	APMX	LF	KCH	CHW	NOF
UF	503030	3	6	8	50	45°	0.2	3
	503040	4	6	10	50	45°	0.2	3
	504050	5	6	13	50	45°	0.2	4
	504060	6	6	10	50	45°	0.2	4
	50406015	6	6	15	60	45°	0.2	4
	504070	7	8	18	70	45°	0.2	4
	504080	8	8	12	60	45°	0.2	4
	50408020	8	8	20	70	45°	0.2	4
	504090	9	10	22	75	45°	0.3	4
	504100	10	10	15	65	45°	0.3	4
	50410025	10	10	25	75	45°	0.3	4
	504110	11	12	27	80	45°	0.3	4
	504120	12	12	20	70	45°	0.3	4
	50412030	12	12	30	80	45°	0.3	4
	505130	13	12	35	100	45°	0.5	5
	505140S16	14	16	35	100	45°	0.5	5
	505140	14	14	35	100	45°	0.5	5
	505150	15	16	35	100	45°	0.5	5
	505160	16	16	25	80	45°	1	5
	50516040	16	16	40	100	45°	1	5
	505180S20	18	20	40	100	45°	1	5
	505180	18	18	40	100	45°	1	5
	505200	20	20	25	80	45°	1	5
	50520045	20	20	45	100	45°	1	5
	505250	25	25	45	100	45°	1	5

UF51

3~5 flutes fine pitch roughing



DC	Tolerance
Ø3	0.000 ~ -0.040
Ø4 ~ Ø6	0.000 ~ -0.048
Ø7 ~ Ø10	0.000 ~ -0.058
Ø11 ~ Ø18	0.000 ~ -0.070
Ø20 ~ Ø25	0.000 ~ -0.084



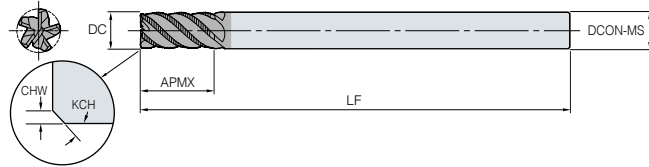
(mm)

Designation		DC	DCON-MS	APMX	LF	NOF
UF	513030	3	6	8	50	3
	513040	4	6	10	50	3
	513050	5	6	13	50	3
	513060	6	6	15	60	3
	51306020	6	6	20	60	3
	513070	7	8	18	70	3
	513080	8	8	20	70	3
	51308025	8	8	25	70	3
	514090	9	10	22	75	4
	514100	10	10	25	75	4
	51410030	10	10	30	75	4
	514110	11	12	27	80	4
	514120	12	12	30	80	4
	51412035	12	12	35	80	4
	514130	13	12	35	100	4
	514140	14	16	35	100	4
	514160	16	16	40	100	4
	514180	18	18	40	100	4
	514200	20	20	50	100	4
	515250	25	25	50	100	5

UF51H

3~5 flutes 45° helix fine pitch roughing

CARBIDE	3	4	5	H-A 45°	Grade PC303W	FINE	h5 shank	RE ±0.025	DC Ø3 ~ Ø25	Tolerance 0.000 ~ -0.050
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(mm)

Designation		DC	DCON-MS	APMX	LF	KCH	CHW	NOF
UF	513030H	3	6	8	50	45°	0.2	3
	513040H	4	6	10	50	45°	0.2	3
	514050H	5	6	13	50	45°	0.3	4
	514060H	6	6	10	50	45°	0.3	4
	51406015H	6	6	15	60	45°	0.3	4
	514070H	7	8	18	70	45°	0.4	4
	514080H	8	8	12	60	45°	0.4	4
	51408020H	8	8	20	70	45°	0.4	4
	514090H	9	10	22	75	45°	0.4	4
	514100H	10	10	15	65	45°	0.4	4
	51410025H	10	10	25	75	45°	0.4	4
	514110H	11	12	27	80	45°	0.5	4
	514120H	12	12	20	70	45°	0.5	4
	51412030H	12	12	30	80	45°	0.5	4
	515130H	13	12	35	100	45°	0.5	5
	515140S16H	14	16	35	100	45°	0.5	5
	515140H	14	14	35	100	45°	0.5	5
	515160H	16	16	25	80	45°	0.5	5
	51516040H	16	16	40	100	45°	0.5	5
	515180S20H	18	20	40	100	45°	0.5	5
	515180H	18	18	40	100	45°	0.5	5
	515200H	20	20	25	80	45°	0.5	5
	51520045H	20	20	45	100	45°	0.5	5
	515250H	25	25	45	100	45°	0.5	5

B Technical Information for G-Star Endmill

For low hardness

G-Star Endmill

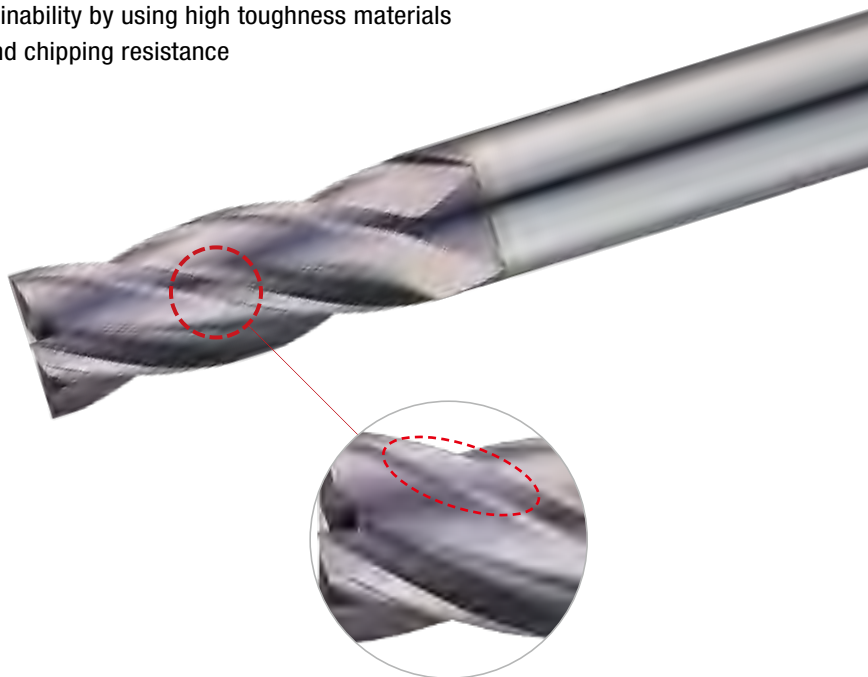
- Suitable for low hardness steel (HRC10 ~ 30); Alloy steel, Carbon steel, Pre-harden, Hardened steel etc.
- Suitable for general purpose machining from roughing to finishing; even on curved or sloped surfaces

Code system

Z	R	32	4H	08	02	S4
Type	Appearance	Length, Shank type	No. of flute	Cutting dia.	Corner R	Shank dia.
D: Dynamic Z: Zamus T: Thunder	B: Ball E: Flat R: Radius X: Flat S: Flat XB: Ball	30: Straight 31: Standard 32: Long Cutting Length 33: Long Shank 34: Tapered Neck	2: 2 flutes 4: 4 flutes 4H: 4 flutes (Helix 45°)	1 ~ 20	0.2 ~ 3 mm	4 ~ 20 mm

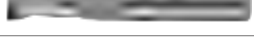
Features

- Excellent rake angle and cutting edge considered the characteristics of workpiece.
- Improved chipping resistance and enhanced machinability by using high toughness materials
- TiAIN coating for enhanced oxidation resistance and chipping resistance



- Excellent workpiece finishes by 45°high helix angle

Line-up

Designation	Appearance	Type	Range	Page
DB312		2 flutes ball Endmill	Ø1.0 ~ 20.0	B177
DB342		2 flutes tapered neck type ball Endmill	Ø1.0 ~ 12.0	B178
TX202		2 flutes short shank flat Endmill	Ø1.0 ~ 20.0	B179
TX204		4 flutes short shank flat Endmill	Ø1.0 ~ 20.0	B180
TX222		2 flutes long flat Endmill	Ø3.0 ~ 20.0	B181
TX224		4 flutes long flat Endmill	Ø3.0 ~ 20.0	B182
TX302		2 flutes flat Endmill	Ø1.0 ~ 20.0	B183
TX304		4 flutes flat Endmill	Ø1.0 ~ 20.0	B184
TX304H		4 flutes 45° helix flat Endmill	Ø3.0 ~ 20.0	B185
TXB202		2 flutes short shank ball Endmill	Ø1.0 ~ 20.0	B186
TXB204		4 flutes short shank ball Endmill	Ø2.0 ~ 20.0	B187
TXB222		2 flutes long ball Endmill	Ø3.0 ~ 20.0	B188
TXB232		2 flutes long shank ball Endmill	Ø3.0 ~ 20.0	B189
TXB302		2 flutes ball Endmill	Ø1.0 ~ 20.0	B190
TXB304		4 flutes ball Endmill	Ø1.0 ~ 20.0	B191
ZE302P		2 flutes flat Endmill	Ø1.0 ~ 20.0	B192
ZE304P		4 flutes flat Endmill	Ø1.0 ~ 20.0	B193
ZE322		2 flutes extra long flat Endmill	Ø3.0 ~ 20.0	B194
ZE324		4 flutes extra long flat Endmill	Ø3.0 ~ 20.0	B195
ZR304H		4 flutes 45° helix radius Endmill	Ø3.0 ~ 12.0	B196
ZR322		2 flutes long shank radius Endmill	Ø3.0 ~ 12.0	B197
ZR324		4 flutes long shank radius Endmill	Ø3.0 ~ 12.0	B198
ZR324H		4 flutes 45° helix radius Endmill	Ø6.0 ~ 12.0	B199

DB312

2 flutes ball

FINE
GRAIN

2

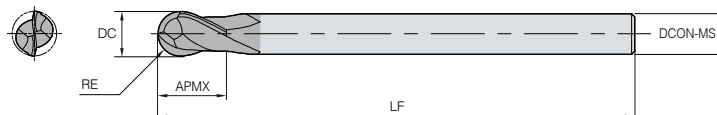
H-A
30°

Grade
AiTiN

h6.5
shank

RE
±0.01

DC	Tolerance
All sizes	0 ~ -0.02



(mm)

Designation	DC	DCON-MS	APMX	LF	RE
DB 312010S4	1	4	2.5	50	0.5
312010	1	6	2.5	50	0.5
312012	1.2	6	3	50	0.6
312015	1.5	6	4	50	0.75
312020S4	2	4	5	50	1
312020	2	6	5	50	1
312025	2.5	6	6	60	1.25
312030S3	3	3	8	60	1.5
312030S4	3	4	8	60	1.5
312030	3	6	8	60	1.5
312035	3.5	6	8	70	1.75
312040S4	4	4	8	70	2
312040	4	6	8	70	2
312045	4.5	6	8	70	2.25
312050	5	6	10	80	2.5
312055	5.5	6	10	80	2.75
312060S	6	6	12	60	3
312060	6	6	12	90	3
312065	6.5	8	12	90	3.25
312070	7	8	14	90	3.5
312080S	8	8	14	60	4
312080	8	8	14	100	4
312090	9	10	18	100	4.5
312100S	10	10	18	60	5
312100	10	10	18	100	5
312120	12	12	22	110	6
312140	14	14	26	110	7
312160	16	16	30	140	8
312180	18	18	34	140	9
312200	20	20	38	160	10

※ The above specifications are subject to change without prior notice for product quality improvement.

• Applicable Workpiece

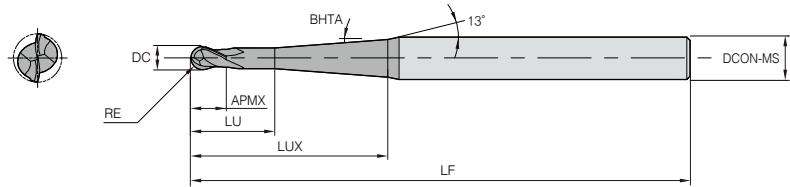
Carbon steel ~ HB225	Alloy steel HB225~325	Pre-hardened steel HrC30~50	Hardened steel		Copper	Graphite	Cast iron ~ FCD500	Aluminum	Stainless steel
			SKD61~HrC55	SKD11 HrC55~					
◎	◎	○							

◎: EXCELLENT ○: GOOD

DB342

2 flutes tapered neck type ball

FINE GRAIN	2	H-A 30°	Grade AiTiN	h6 shank	RE ±0.01	DC All sizes	Tolerance 0 ~ -0.02
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(mm)

Designation	DC	DCON-MS	APMX	LU	LUX	LF	BHTA(°)	RE
DB 34201015	1	6	2	4	23	60	1°30′	0.5
34201050	1	6	2	4	23	60	5°	0.5
34201030	1	6	2	4	42	80	3°	0.5
34202015	2	6	4	6	23	60	1°30′	1
34202050	2	6	4	6	23	60	5°	1
34202030	2	6	4	6	41	80	3°	1
34203030	3	6	6	8	32	70	3°	1.5
34203015	3	6	6	8	52	90	1°30′	1.5
34204030	4	6	8	10	28	70	3°	2
34204015	4	6	8	10	49	90	1°30′	2
34205030	5	8	10	12	41	90	3°	2.5
34205015	5	8	10	12	61	110	1°30′	2.5
34206030	6	8	12	15	34	90	3°	3
34206015	6	8	12	15	53	110	1°30′	3
34208030	8	10	14	17	36	100	3°	4
34208015	8	10	14	17	55	120	1°30′	4
34210030	10	12	18	21	40	110	3°	5
34210015	10	12	18	21	59	130	1°30′	5
34212030	12	16	22	25	63	140	3°	6
34212015	12	16	22	25	83	160	1°30′	6

※ The above specifications are subject to change without prior notice for product quality improvement.

• Applicable Workpiece

Carbon steel ~ HB225	Alloy steel HB225~325	Pre-hardened steel HrC30~50	Hardened steel		Copper	Graphite	Cast iron ~FCD500	Aluminum	Stainless steel
			SKD61~HrC55	SKD11 HrC55~					
◎	◎	○							

◎: EXCELLENT ○: GOOD

TX202

2 flutes short shank flat



FINE
GRAIN

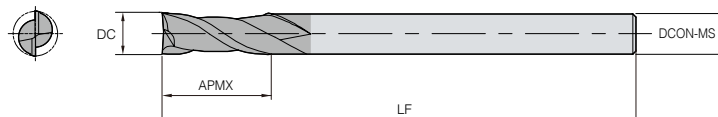
2

H-A
30°

Grade
AiTiN

h6
shank

DC	Tolerance
Ø1 ~ Ø3	-0.014 ~ -0.028
Ø4 ~ Ø6	-0.02 ~ -0.038
Ø7 ~ Ø10	-0.025 ~ -0.047
Ø12 ~ Ø18	-0.032 ~ -0.059
Ø20 ~	-0.04 ~ -0.073



(mm)

Designation	DC	DCON-MS	APMX	LF
TX 202010	1	3	3	39
202015	1.5	3	5	39
202020	2	3	7	39
202025	2.5	3	8	39
202030	3	3	10	39
202040	4	4	14	51
202050	5	5	16	51
202060	6	6	19	64
202080	8	8	21	64
202100	10	10	25	70
202120	12	12	25	76
202160	16	16	32	89
202200	20	20	38	102

※ The above specifications are subject to change without prior notice for product quality improvement.

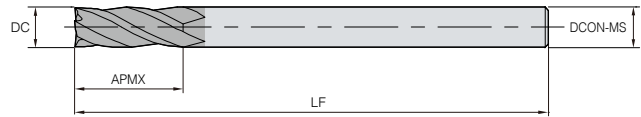
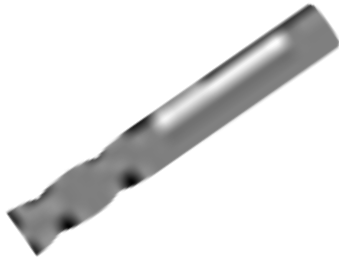
• Applicable Workpiece

Carbon steel ~ HB225	Alloy steel HB225~325	Pre-hardened steel HrC30~50	Hardened steel		Copper	Graphite	Cast iron ~ FCD500	Aluminum	Stainless steel
			SKD61~HrC55	SKD11 HrC55~					
◎	◎	○							

◎: EXCELLENT ○: GOOD

TX204

4 flutes short shank flat



FINE
GRAIN

4

H-A
30°

Grade
AiTiN

h6
shank

DC	Tolerance
Ø1 ~ Ø3	-0.014 ~ -0.028
Ø4 ~ Ø6	-0.02 ~ -0.038
Ø7 ~ Ø10	-0.025 ~ -0.047
Ø12 ~ Ø18	-0.032 ~ -0.059
Ø20 ~	-0.04 ~ -0.073

(mm)

Designation	DC	DCON-MS	APMX	LF
TX 204010	1	3	3	39
204015	1.5	3	5	39
204020	2	3	7	39
204025	2.5	3	8	39
204030	3	3	10	39
204040	4	4	14	51
204050	5	5	16	51
204060	6	6	19	64
204080	8	8	21	64
204100	10	10	25	70
204120	12	12	25	76
204160	16	16	32	89
204200	20	20	38	102

※ The above specifications are subject to change without prior notice for product quality improvement.

• Applicable Workpiece

Carbon steel ~ HB225	Alloy steel HB225~325	Pre-hardened steel HrC30~50	Hardened steel		Copper	Graphite	Cast iron ~FCD500	Aluminum	Stainless steel
			SKD61~HrC55	SKD11 HrC55~					
◎	◎	○							

◎: EXCELLENT ○: GOOD

TX222

2 flutes long flat



FINE
GRAIN

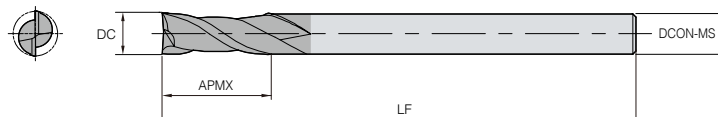
2

H-A
30°

Grade
AiTiN

h6
shank

DC	Tolerance
Ø1 ~ Ø3	-0.014 ~ -0.028
Ø4 ~ Ø6	-0.02 ~ -0.038
Ø7 ~ Ø10	-0.025 ~ -0.047
Ø12 ~ Ø18	-0.032 ~ -0.059
Ø20 ~	-0.04 ~ -0.073



(mm)

Designation	DC	DCON-MS	APMX	LF
TX 222030	3	3	20	60
222040	4	4	20	60
222050	5	5	25	75
222060	6	6	30	75
222080	8	8	30	75
222100	10	10	40	100
222120	12	12	45	100
222140	14	14	45	100
222160	16	16	45	100
222180	18	18	45	100
222200	20	20	45	100

※ The above specifications are subject to change without prior notice for product quality improvement.

• Applicable Workpiece

Carbon steel ~ HB225	Alloy steel HB225~325	Pre-hardened steel HrC30~50	Hardened steel		Copper	Graphite	Cast iron ~ FCD500	Aluminum	Stainless steel
			SKD61~HrC55	SKD11 HrC55~					
◎	◎	○							

◎: EXCELLENT ○: GOOD

TX224

4 flutes long flat

FINE
GRAIN

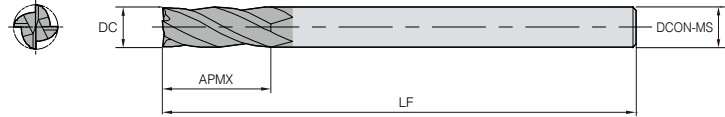
4

H-A
30°

Grade
AiTiN

h6
shank

DC	Tolerance
Ø1 ~ Ø3	-0.014 ~ -0.028
Ø4 ~ Ø6	-0.02 ~ -0.038
Ø7 ~ Ø10	-0.025 ~ -0.047
Ø12 ~ Ø18	-0.032 ~ -0.059
Ø20 ~	-0.04 ~ -0.073



(mm)

Designation	DC	DCON-MS	APMX	LF
TX 224030	3	3	20	60
224040	4	4	20	60
224050	5	5	25	75
224060	6	6	30	75
224080	8	8	30	75
224081	8	8	30	100
224100	10	10	40	100
224120	12	12	45	100
224140	14	14	45	100
224160	16	16	45	100
224180	18	18	45	100
224200	20	20	45	100

※ The above specifications are subject to change without prior notice for product quality improvement.

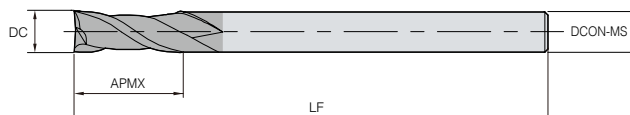
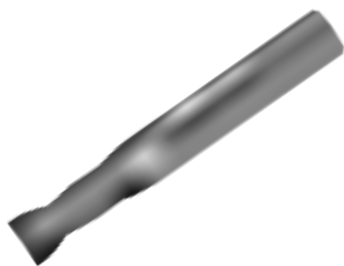
• Applicable Workpiece

Carbon steel ~ HB225	Alloy steel HB225~325	Pre-hardened steel HrC30~50	Hardened steel		Copper	Graphite	Cast iron ~FCD500	Aluminum	Stainless steel
			SKD61~HrC55	SKD11 HrC55~					
◎	◎	○							

◎: EXCELLENT ○: GOOD

TX302

2 flutes flat



FINE
GRAIN

2

H-A
30°

Grade
AiTiN

h6
shank

DC	Tolerance
Ø1 ~ Ø3	-0.014 ~ -0.028
Ø4 ~ Ø6	-0.02 ~ -0.038
Ø7 ~ Ø10	-0.025 ~ -0.047
Ø12 ~ Ø18	-0.032 ~ -0.059
Ø20 ~	-0.04 ~ -0.073

(mm)

Designation	DC	DCON-MS	APMX	LF
TX 302010	1	4	3	50
302015	1.5	4	4	50
302020	2	4	6	50
302025	2.5	4	8	50
302030	3	4	9	50
302040	4	4	11	50
302050	5	6	13	50
302060	6	6	16	50
302070	7	8	16	60
302080	8	8	19	60
302090	9	10	19	60
302100	10	10	25	75
302120	12	12	30	75
302140	14	14	32	75
302160	16	16	32	100
302180	18	18	32	100
302200	20	20	38	100

※ The above specifications are subject to change without prior notice for product quality improvement.

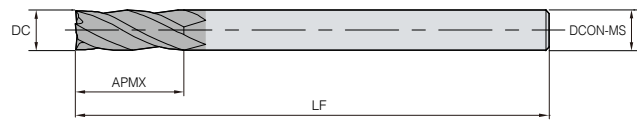
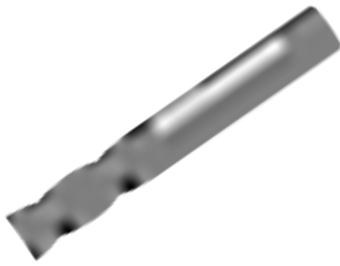
• Applicable Workpiece

Carbon steel ~ HB225	Alloy steel HB225~325	Pre-hardened steel HrC30~50	Hardened steel		Copper	Graphite	Cast iron ~ FCD500	Aluminum	Stainless steel
			SKD61~HrC55	SKD11 HrC55~					
◎	◎	○							

◎: EXCELLENT ○: GOOD

TX304

4 flutes flat


FINE
GRAIN

4

H-A
30°

Grade
AiTiN

h6
shank

DC	Tolerance
Ø1 ~ Ø3	-0.014 ~ -0.028
Ø4 ~ Ø6	-0.02 ~ -0.038
Ø7 ~ Ø10	-0.025 ~ -0.047
Ø12 ~ Ø18	-0.032 ~ -0.059
Ø20 ~	-0.04 ~ -0.073

(mm)					
Designation		DC	DCON-MS	APMX	LF
TX	304010	1	4	3	50
	304015	1.5	4	4	50
	304020	2	4	6	50
	304025	2.5	4	8	50
	304030	3	4	9	50
	304040	4	4	11	50
	304050	5	6	13	50
	304060	6	6	16	50
	304070	7	8	16	60
	304080	8	8	19	60
	304090	9	10	19	60
	304100	10	10	25	75
	304120	12	12	30	75
	304140	14	14	32	75
	304160	16	16	32	100
	304180	18	18	32	100
	304200	20	20	38	100

※ The above specifications are subject to change without prior notice for product quality improvement.

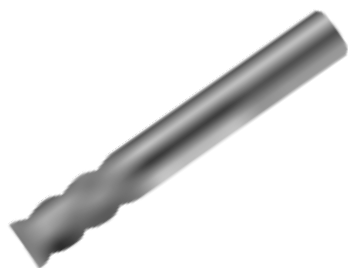
• Applicable Workpiece

Carbon steel ~ HB225	Alloy steel HB225~325	Pre-hardened steel HrC30~50	Hardened steel		Copper	Graphite	Cast iron ~FCD500	Aluminum	Stainless steel
			SKD61~HrC55	SKD11 HrC55~					
◎	◎	○							

◎: EXCELLENT ○: GOOD

TX304H

4 flutes 45° helix flat



FINE
GRAIN

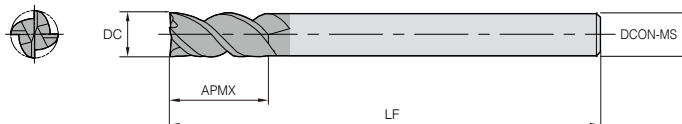
4

H-A
45°

Grade
AiTiN

h6
shank

DC	Tolerance
Ø1 ~ Ø3	-0.014 ~ -0.028
Ø4 ~ Ø6	-0.02 ~ -0.038
Ø7 ~ Ø10	-0.025 ~ -0.047
Ø12 ~ Ø18	-0.032 ~ -0.059
Ø20 ~	-0.04 ~ -0.073



(mm)

Designation	DC	DCON-MS	APMX	LF
TX 304H030	3	6	8	50
304H030S3	3	3	8	50
304H030S4	3	4	8	50
304H040	4	6	11	50
304H040S4	4	4	11	50
304H050	5	6	13	50
304H060	6	6	13	50
304H080	8	8	19	60
304H100	10	10	22	70
304H120	12	12	26	75
304H130	13	12	26	80
304H140	14	14	26	80
304H160	16	16	32	90
304H180	18	18	32	100
304H200	20	20	38	100

※ The above specifications are subject to change without prior notice for product quality improvement.

• Applicable Workpiece

Carbon steel ~ HB225	Alloy steel HB225~325	Pre-hardened steel HrC30~50	Hardened steel		Copper	Graphite	Cast iron ~FCD500	Aluminum	Stainless steel
			SKD61~HrC55	SKD11 HrC55~					
◎	◎	○							

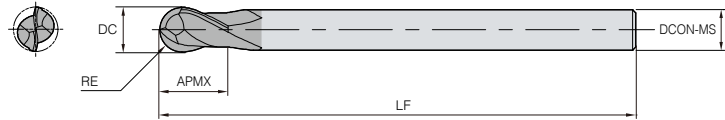
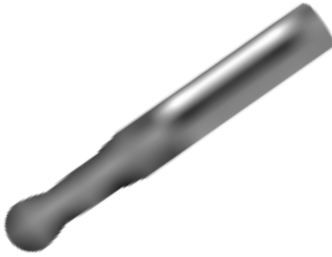
◎: EXCELLENT ○: GOOD

TXB202**2 flutes short shank ball****FINE
GRAIN****2****H-A
30°****Grade
AiTiN****h6
shank****RE
±0.02****DC**

All sizes

Tolerance

0 ~ -0.04



(mm)

Designation		DC	DCON-MS	APMX	LF	RE
TXB	202010	1	3	3	39	0.5
	202015	1.5	3	5	39	0.75
	202020	2	3	7	39	1
	202025	2.5	3	8	39	1.25
	202030	3	3	10	39	1.5
	202040	4	4	14	51	2
	202050	5	5	16	51	2.5
	202060	6	6	19	64	3
	202080	8	8	21	64	4
	202100	10	10	25	70	5
	202120	12	12	25	76	6
	202160	16	16	32	89	8
	202200	20	20	38	100	10

※ The above specifications are subject to change without prior notice for product quality improvement.

• Applicable Workpiece

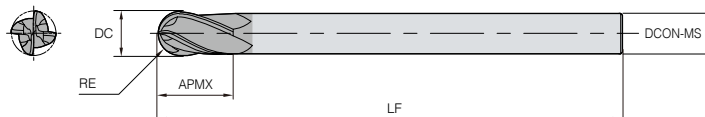
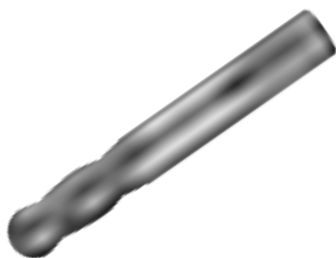
Carbon steel ~ HB225	Alloy steel HB225~325	Pre-hardened steel HrC30~50	Hardened steel		Copper	Graphite	Cast iron ~FCD500	Aluminum	Stainless steel
			SKD61~HrC55	SKD11 HrC55~					
◎	◎	○							

◎: EXCELLENT ○: GOOD

TXB204

4 flutes short shank ball

FINE GRAIN	4	H-A 30°	Grade AiTiN	h6 shank	RE ±0.01	DC All sizes	Tolerance 0 ~ -0.04
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(mm)

	Designation	DC	DCON-MS	APMX	LF	RE
TXB	204020	2	3	7	39	1
	204030	3	3	10	39	1.5
	204040	4	4	14	51	2
	204050	5	5	16	51	2.5
	204060	6	6	19	64	3
	204080	8	8	21	64	4
	204100	10	10	25	70	5
	204120	12	12	25	76	6
	204160	16	16	32	89	8
	204200	20	20	38	100	10

※ The above specifications are subject to change without prior notice for product quality improvement.

• Applicable Workpiece

Carbon steel ~ HB225	Alloy steel HB225~325	Pre-hardened steel HrC30~50	Hardened steel		Copper	Graphite	Cast iron ~ FCD500	Aluminum	Stainless steel
			SKD61~HrC55	SKD11 HrC55~					
◎	◎	○							

◎: EXCELLENT ○: GOOD

TXB222

2 flutes long ball

FINE
GRAIN

2

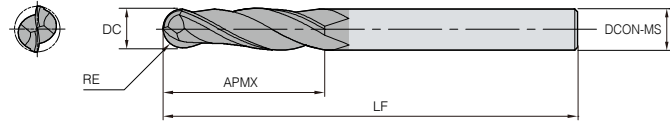
H-A
30°

Grade
AiTiN

h6
shank

RE
±0.02

DC
All sizes

Tolerance
0 ~ -0.04


(mm)

Designation	DC	DCON-MS	APMX	LF	RE
TXB 222030	3	3	20	60	1.5
222040	4	4	20	60	2
222050	5	5	25	75	2.5
222060	6	6	30	75	3
222080	8	8	30	100	4
222100	10	10	40	100	5
222120	12	12	45	100	6
222140	14	14	45	100	7
222160	16	16	45	100	8
222180	18	18	45	100	9
222200	20	20	45	100	10

※ The above specifications are subject to change without prior notice for product quality improvement.

• Applicable Workpiece

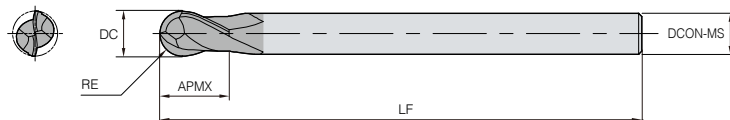
Carbon steel ~ HB225	Alloy steel HB225~325	Pre-hardened steel HrC30~50	Hardened steel		Copper	Graphite	Cast iron ~FCD500	Aluminum	Stainless steel
			SKD61~HrC55	SKD11 HrC55~					
◎	◎	○							

◎: EXCELLENT ○: GOOD

TXB232

2 flutes long shank ball

FINE GRAIN	2	H-A 30°	Grade AiTiN	h6 shank	RE ±0.02	DC All sizes	Tolerance 0 ~ -0.04
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(mm)

Designation		DC	DCON-MS	APMX	LF	RE
TXB	232030	3	3	5	75	1.5
	232040	4	4	8	75	2
	232050	5	5	9	75	2.5
	232060	6	6	10	100	3
	232060-75	6	6	10	75	3
	232080	8	8	12	100	4
	232080-75	8	8	12	75	4
	232100	10	10	14	100	5
	232100L	10	10	14	150	5
	232120	12	12	16	100	6
	232120L	12	12	16	150	6
	232140	14	14	18	100	7
	232160	16	16	22	150	8
	232200	20	20	26	150	10

※ The above specifications are subject to change without prior notice for product quality improvement.

• Applicable Workpiece

Carbon steel ~ HB225	Alloy steel HB225~325	Pre-hardened steel HrC30~50	Hardened steel		Copper	Graphite	Cast iron ~ FCD500	Aluminum	Stainless steel
			SKD61~HrC55	SKD11 HrC55~					
◎	◎	○							

◎: EXCELLENT ○: GOOD

TXB302

2 flutes ball

FINE
GRAIN

2

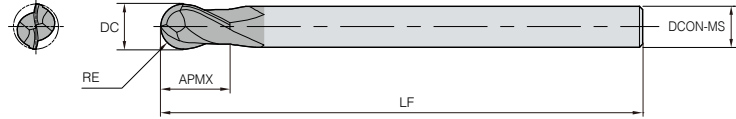
H-A
30°

Grade
AiTiN

h6
shank

RE
±0.02

DC
All sizes

Tolerance
0 ~ -0.04


(mm)

Designation	DC	DCON-MS	APMX	LF	RE
TXB 302010	1	4	2	50	0.5
302015	1.5	4	3	50	0.75
302020	2	4	4	50	1
302025	2.5	4	6	50	1.25
302030	3	4	6	50	1.5
302040	4	4	8	50	2
302050	5	6	10	50	2.5
302060	6	6	12	50	3
302080	8	8	14	60	4
302100	10	10	18	75	5
302120	12	12	22	75	6
302140	14	14	32	75	7
302160	16	16	32	100	8
302180	18	18	32	100	9
302200	20	20	38	100	10

※ The above specifications are subject to change without prior notice for product quality improvement.

• Applicable Workpiece

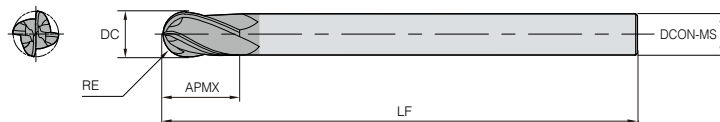
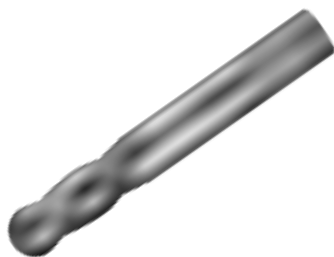
Carbon steel ~ HB225	Alloy steel HB225~325	Pre-hardened steel HrC30~50	Hardened steel		Copper	Graphite	Cast iron ~FCD500	Aluminum	Stainless steel
			SKD61~HrC55	SKD11 HrC55~					
◎	◎	○							

◎: EXCELLENT ○: GOOD

TXB304

4 flutes ball

FINE GRAIN	4	H-A 30°	Grade AiTiN	h6 shank	RE ±0.02	DC All sizes	Tolerance 0 ~ -0.04
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(mm)

	Designation	DC	DCON-MS	APMX	LF	RE
TXB	304010	1	4	2	50	0.5
	304015	1.5	4	3	50	0.75
	304020	2	4	4	50	1
	304030	3	4	6	50	1.5
	304040	4	4	8	50	2
	304050	5	6	10	50	2.5
	304060	6	6	12	50	3
	304080	8	8	14	60	4
	304100	10	10	18	75	5
	304120	12	12	22	75	6
	304140	14	14	32	75	7
	304160	16	16	32	100	8
	304180	18	18	32	100	9
	304200	20	20	38	100	10

※ The above specifications are subject to change without prior notice for product quality improvement.

• Applicable Workpiece

Carbon steel ~ HB225	Alloy steel HB225~325	Pre-hardened steel HrC30~50	Hardened steel		Copper	Graphite	Cast iron ~ FCD500	Aluminum	Stainless steel
			SKD61~HrC55	SKD11 HrC55~					
◎	◎	○							

◎: EXCELLENT ○: GOOD

ZE302P

2 flutes flat

FINE
GRAIN

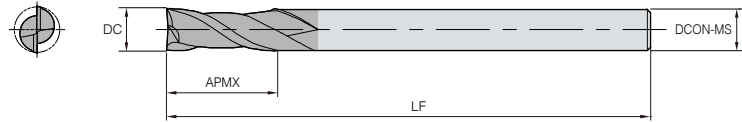
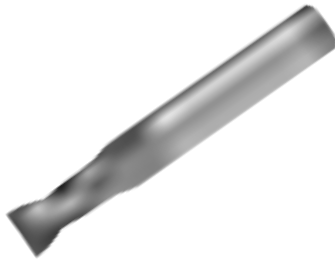
2

H-A
30°

Grade
AiTiN

h6
shank

DC	Tolerance
All sizes	0 ~ -0.02



(mm)

Designation	DC	DCON-MS	APMX	LF
ZE 302010P	1	6	2.5	50
302015P	1.5	6	4	50
302020P	2	6	6	50
302025P	2.5	6	8	50
302030P	3	6	10	50
302035P	3.5	6	10	50
302040P	4	6	12	50
302045P	4.5	6	14	50
302050P	5	6	15	60
302055P	5.5	6	15	60
302060P	6	6	15	60
302065P	6.5	8	18	60
302070P	7	8	20	65
302075P	7.5	8	20	65
302080P	8	8	20	65
302085P	8.5	10	22	70
302090P	9	10	22	70
302095P	9.5	10	24	70
302100P	10	10	25	70
302105P	10.5	12	26	80
302110P	11	12	30	80
302115P	11.5	12	30	80
302120P	12	12	30	80
302130P	13	12	35	90
302140P	14	14	35	100
302150P	15	16	40	100
302160P	16	16	40	100
302180P	18	18	45	100
302200P	20	20	45	100

※ The above specifications are subject to change without prior notice for product quality improvement.

• Applicable Workpiece

Carbon steel ~ HB225	Alloy steel HB225~325	Pre-hardened steel HrC30~50	Hardened steel		Copper	Graphite	Cast iron ~FCD500	Aluminum	Stainless steel
			SKD61~HrC55	SKD11 HrC55~					
◎	◎	○							

◎: EXCELLENT ○: GOOD

ZE304P

4 flutes flat

FINE
GRAIN

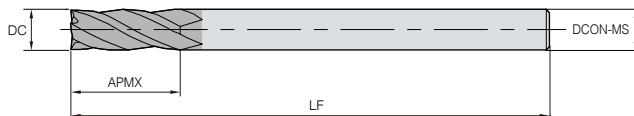
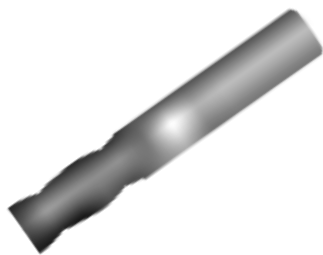
4

H-A
30°

Grade
AiTiN

h6
shank

DC	Tolerance
All sizes	0 ~ -0.02



(mm)

Designation	DC	DCON-MS	APMX	LF
ZE 304010P	1	6	2.5	50
304015P	1.5	6	4	50
304020P	2	6	6	50
304025P	2.5	6	8	50
304030P	3	6	10	50
304035P	3.5	6	10	50
304040P	4	6	12	50
304045P	4.5	6	14	50
304050P	5	6	15	60
304055P	5.5	6	15	60
304060P	6	6	15	60
304065P	6.5	8	18	60
304070P	7	8	20	65
304075P	7.5	8	20	65
304080P	8	8	20	65
304085P	8.5	10	22	70
304090P	9	10	22	70
304095P	9.5	10	24	70
304100P	10	10	25	70
304105P	10.5	12	26	80
304110P	11	12	30	80
304115P	11.5	12	30	80
304120P	12	12	30	80
304130P	13	12	35	90
304140P	14	14	35	100
304150P	15	16	40	100
304160P	16	16	40	100
304180P	18	18	45	100
304200P	20	20	45	100

※ The above specifications are subject to change without prior notice for product quality improvement.

• Applicable Workpiece

Carbon steel ~ HB225	Alloy steel HB225~325	Pre-hardened steel HrC30~50	Hardened steel		Copper	Graphite	Cast iron ~ FCD500	Aluminum	Stainless steel
			SKD61~HrC55	SKD11 HrC55~					
◎	◎	○							

◎: EXCELLENT ○: GOOD

ZE322

2 flutes extra long flat

FINE
GRAIN

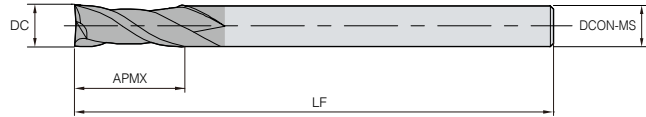
2

H-A
30°

Grade
AiTiN

h6
shank

DC	Tolerance
All sizes	0 ~ -0.03



(mm)

Designation	DC	DCON-MS	APMX	LF
ZE 322030	3	6	15	60
322031	3	6	20	70
322030S	3	3	20	100
322040	4	6	15	60
322041	4	6	20	70
322040S	4	4	20	100
322050	5	6	20	60
322051	5	6	20	80
322052	5	6	25	100
322060	6	6	20	80
322061	6	6	30	100
322062	6	6	40	150
322080	8	8	30	90
322081	8	8	35	100
322082	8	8	40	150
322100	10	10	30	90
322101	10	10	35	100
322102	10	10	45	150
322103	10	10	55	180
322120	12	12	30	90
322121	12	12	40	110
322122	12	12	50	150
322123	12	12	60	200
322140	14	14	40	120
322141	14	14	60	150
322160	16	16	50	140
322161	16	16	70	160
322162	16	16	80	200
322180	18	18	50	140
322200	20	20	60	150
322201	20	20	100	200
322202	20	20	130	250

* The above specifications are subject to change without prior notice for product quality improvement.

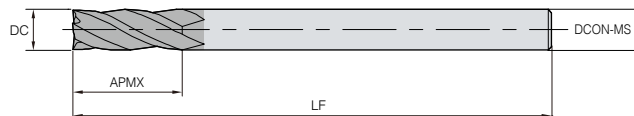
• Applicable Workpiece

Carbon steel ~ HB225	Alloy steel HB225~325	Pre-hardened steel HrC30~50	Hardened steel		Copper	Graphite	Cast iron ~FCD500	Aluminum	Stainless steel
			SKD61~HrC55	SKD11 HrC55~					
◎	◎	○							

◎: EXCELLENT ○: GOOD

ZE324

4 flutes extra long flat



FINE
GRAIN

4

H-A
30°

Grade
AiTiN

h6
shank

DC	Tolerance
All sizes	0 ~ -0.03

(mm)

Designation	DC	DCON-MS	APMX	LF
ZE 324030	3	6	15	60
324031	3	6	20	70
324030S	3	3	20	100
324040	4	6	15	60
324041	4	6	20	70
324040S	4	4	20	100
324050	5	6	20	60
324051	5	6	20	80
324052	5	6	25	100
324060	6	6	20	80
324061	6	6	30	100
324062	6	6	40	150
324080	8	8	30	90
324081	8	8	35	100
324082	8	8	40	150
324100	10	10	30	90
324101	10	10	35	100
324102	10	10	45	150
324103	10	10	55	180
324120	12	12	30	90
324121	12	12	40	110
324122	12	12	50	150
324123	12	12	60	200
324140	14	14	40	120
324141	14	14	60	150
324160	16	16	50	140
324161	16	16	70	160
324162	16	16	80	200
324180	18	18	50	140
324200	20	20	60	150
324201	20	20	100	200
324202	20	20	130	250

※ The above specifications are subject to change without prior notice for product quality improvement.

• Applicable Workpiece

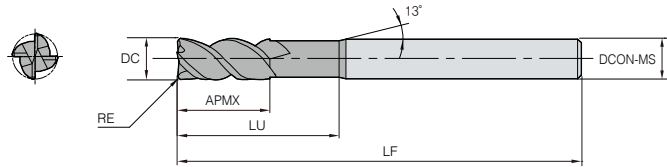
Carbon steel ~ HB225	Alloy steel HB225~325	Pre-hardened steel HrC30~50	Hardened steel		Copper	Graphite	Cast iron ~ FCD500	Aluminum	Stainless steel
			SKD61 ~HrC55	SKD11 HrC55~					
◎	◎	○							

◎: EXCELLENT ○: GOOD

ZR304H

4 flutes 45° helix radius

FINE GRAIN	4	H-A 45°	Grade AiTiN	h6 shank	RE ±0.02	DC All sizes	Tolerance 0 ~ -0.03
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(mm)

Designation	DC	DCON-MS	APMX	LU	LF	RE
ZR 304H0303	3	6	4	12	55	0.3
304H0302S3	3	3	4	12	55	0.2
304H0303S4	3	4	4	12	55	0.3
304H0305	3	6	4	12	55	0.5
304H0305S3	3	3	4	12	55	0.5
304H0305S4	3	4	4	12	55	0.5
304H0402S4	4	4	5	16	55	0.2
304H0403	4	6	5	16	55	0.3
304H0403S4	4	4	5	16	55	0.3
304H0405	4	6	5	16	55	0.5
304H0405S4	4	4	5	16	55	0.5
304H0605	6	6	7	20	60	0.5
304H0610	6	6	7	20	60	1
304H0805	8	8	10	25	65	0.5
304H0810	8	8	10	25	65	1
304H1005	10	10	12	30	70	0.5
304H1010	10	10	12	30	70	1
304H1015	10	10	12	30	70	1.5
304H1020	10	10	12	30	70	2
304H1205	12	12	15	30	80	0.5
304H1210	12	12	15	30	80	1
304H1215	12	12	15	30	80	1.5
304H1220	12	12	15	30	80	2

※ The above specifications are subject to change without prior notice for product quality improvement.

• Applicable Workpiece

Carbon steel ~ HB225	Alloy steel HB225~325	Pre-hardened steel HrC30~50	Hardened steel		Copper	Graphite	Cast iron ~FCD500	Aluminum	Stainless steel
			SKD61~HrC55	SKD11 HrC55~					
◎	◎	○							

◎: EXCELLENT ○: GOOD

ZR322

2 flutes long shank radius

FINE
GRAIN

2

H-A
30°

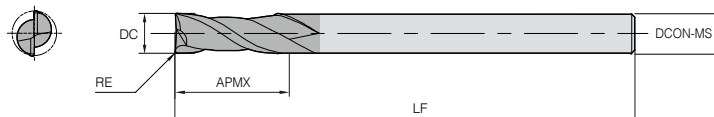
Grade
AiTiN

h6
shank

RE
±0.02

DC
All sizes

Tolerance
0 ~ -0.03



(mm)

Designation	DC	DCON-MS	APMX	LF	RE
ZR 3220302S4	3	4	8	60	0.2
3220302	3	6	8	60	0.2
3220303	3	6	8	60	0.3
3220305S4	3	4	8	60	0.5
3220305	3	6	8	60	0.5
3220402S4	4	4	11	70	0.2
3220402	4	6	11	70	0.2
3220403	4	6	11	70	0.3
3220405S4	4	4	11	70	0.5
3220405	4	6	11	70	0.5
3220410S4	4	4	11	70	1
3220410	4	6	11	70	1
3220502	5	6	13	80	0.2
3220503	5	6	13	80	0.3
3220505	5	6	13	80	0.5
3220510	5	6	13	80	1
3220602	6	6	13	90	0.2
3220603	6	6	13	90	0.3
3220605	6	6	13	90	0.5
3220610	6	6	13	90	1
3220803	8	8	19	100	0.3
3220805	8	8	19	100	0.5
3220810	8	8	19	100	1
3220815	8	8	19	100	1.5
3220820	8	8	19	100	2
3221003	10	10	22	100	0.3
3221005	10	10	22	100	0.5
3221010	10	10	22	100	1
3221015	10	10	22	100	1.5
3221020	10	10	22	100	2
3221025	10	10	22	100	2.5
3221205	12	12	26	110	0.5
3221210	12	12	26	110	1
3221215	12	12	26	110	1.5
3221220	12	12	26	110	2
3221225	12	12	26	110	2.5
3221230	12	12	26	110	3

※ The above specifications are subject to change without prior notice for product quality improvement.

• Applicable Workpiece

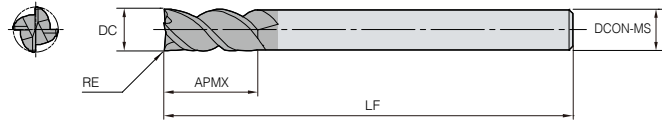
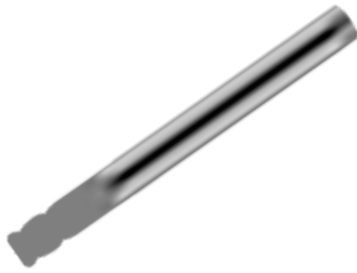
Carbon steel ~ HB225	Alloy steel HB225~325	Pre-hardened steel HrC30~50	Hardened steel		Copper	Graphite	Cast iron ~ FCD500	Aluminum	Stainless steel
			SKD61~HrC55	SKD11 HrC55~					
◎	◎	○							

◎: EXCELLENT ○: GOOD

ZR324

4 flutes long shank radius

FINE GRAIN	4	H-A 30°	Grade AiTiN	h6 shank	RE ±0.02	DC All sizes	Tolerance 0 ~ -0.03
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(mm)

Designation	DC	DCON-MS	APMX	LF	RE
ZR					
3240302S4	3	4	8	60	0.2
3240302	3	6	8	60	0.2
3240303	3	6	8	60	0.3
3240305S4	3	4	8	60	0.5
3240305	3	6	8	60	0.5
3240402S4	4	4	11	70	0.2
3240402	4	6	11	70	0.2
3240403	4	6	11	70	0.3
3240405S4	4	4	11	70	0.5
3240405	4	6	11	70	0.5
3240410S4	4	4	11	70	1
3240410	4	6	11	70	1
3240502	5	6	13	80	0.2
3240503	5	6	13	80	0.3
3240505	5	6	13	80	0.5
3240510	5	6	13	80	1
3240602	6	6	13	90	0.2
3240603	6	6	13	90	0.3
3240605	6	6	13	90	0.5
3240610	6	6	13	90	1
3240803	8	8	19	100	0.3
3240805	8	8	19	100	0.5
3240810	8	8	19	100	1
3240815	8	8	19	100	1.5
3240820	8	8	19	100	2
3241003	10	10	22	100	0.3
3241005	10	10	22	100	0.5
3241010	10	10	22	100	1
3241015	10	10	22	100	1.5
3241020	10	10	22	100	2
3241025	10	10	22	100	2.5
3241205	12	12	26	110	0.5
3241210	12	12	26	110	1
3241215	12	12	26	110	1.5
3241220	12	12	26	110	2
3241225	12	12	26	110	2.5
3241230	12	12	26	110	3

※ The above specifications are subject to change without prior notice for product quality improvement.

• Applicable Workpiece

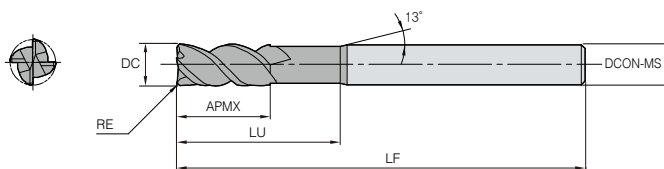
Carbon steel ~ HB225	Alloy steel HB225~325	Pre-hardened steel HrC30~50	Hardened steel		Copper	Graphite	Cast iron ~FCD500	Aluminum	Stainless steel
			SKD61~HrC55	SKD11 HrC55~					
◎	◎	○							

◎: EXCELLENT ○: GOOD

ZR324H

4 flutes 45° helix radius

FINE GRAIN	4	H-A 45°	Grade AiTiN	h6 shank	RE ±0.02	DC All sizes	Tolerance 0 ~ -0.03
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(mm)

Designation		DC	DCON-MS	APMX	LU	LF	RE
ZR	324H0605	6	6	9	20	90	0.5
	324H0610	6	6	9	20	90	1
	324H0805	8	8	12	25	100	0.5
	324H0810	8	8	12	25	100	1
	324H1005	10	10	15	32	100	0.5
	324H1010	10	10	15	32	100	1
	324H1015	10	10	15	32	100	1.5
	324H1020	10	10	15	32	100	2
	324H1205	12	12	18	38	110	0.5
	324H1210	12	12	18	38	110	1
	324H1215	12	12	18	38	110	1.5
	324H1220	12	12	18	38	110	2

※ The above specifications are subject to change without prior notice for product quality improvement.

• Applicable Workpiece

Carbon steel ~ HB225	Alloy steel HB225~325	Pre-hardened steel HrC30~50	Hardened steel		Copper	Graphite	Cast iron ~ FCD500	Aluminum	Stainless steel
			SKD61~HrC55	SKD11 HrC55~					
◎	◎	○							

◎: EXCELLENT ○: GOOD

High efficient roughing Endmill

R⁺ Endmill

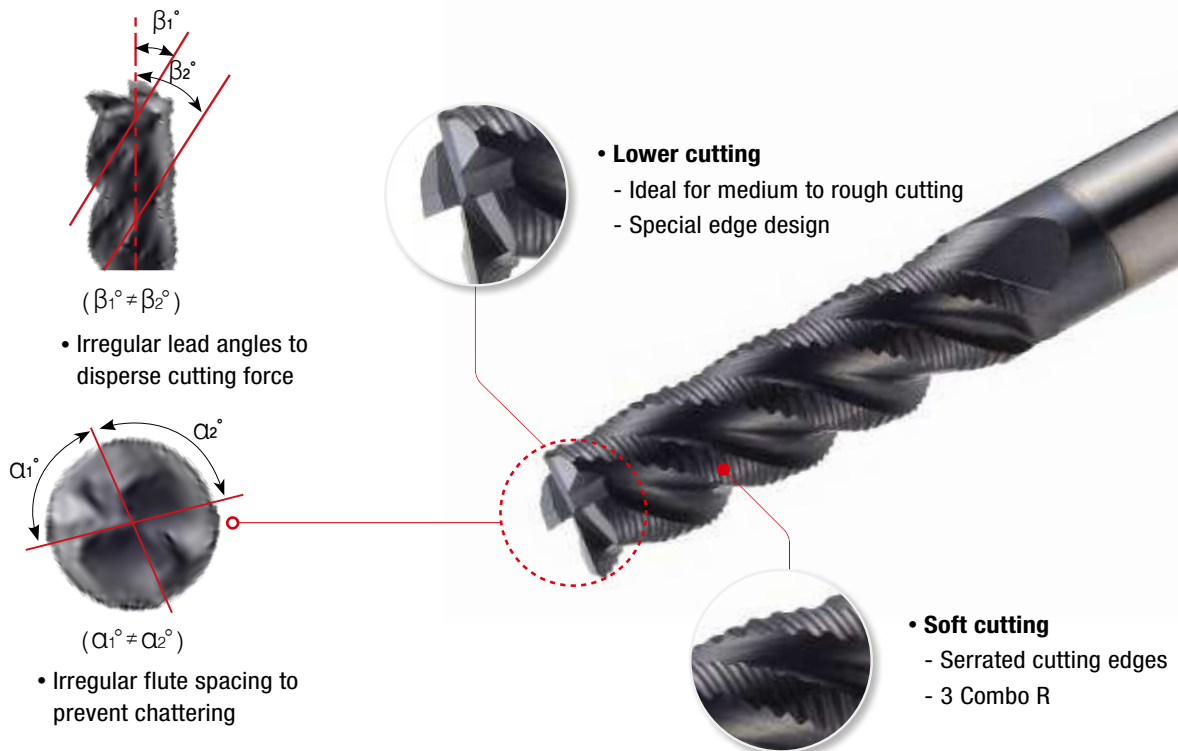
- Cost-effective cutting-edge design for rough machining
- Specifically designed corners as irregular flute spacing and lead angle

Code system

RP(A)E	4	080	-	075	-	RG	(-H)
Roughing Plus (Aluminum) Endmill	Tooth	Drill dia. 080: Ø8		Overall length 075: 75mm		Flute information	Helix angle H: High Helix L: Low Helix
						<Cutting-edge> F: Fine Pitch M: Medium Pitch R: Roughing Pitch X: 2-tooth finishing, 2-tooth roughing	<Flute> P: Irregular Flutes/Leads F: Irregular Flutes L: Irregular Leads G: General

Features

- Excellent machining efficiency - Special design for medium to rough cutting
- Longer cutting life - Extended tool cost thanks to newly applied grades
- Higher cutting performance - Blade design ideal for roughing

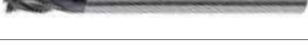


Grade system

Carbide Roughing		HSS Roughing	
FN30T	Carbide, uncoated	HN30T	HSS PM, uncoated
PC10T	Carbide, TiCN coated	HN20T	HSS, uncoated
PC20T	Carbide, TiN coated	HC10T	HSS, TiCN Coating
PC30T	Carbide, TiAlN coated	HC20T	HSS, TiN Coating
PC40T	Carbide, TiAlCrN coated	HC30T	HSS PM, TiAlN coated

B Technical Information for R⁺ Endmill

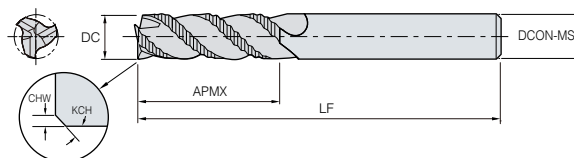
Line-up

Designation	Appearance	Type	Range	Page
RPAE		Wave roughing Endmill for Al	Ø6.0 ~ 25.0	B202
RPE-FP-H		Fine pitch standard type roughing Endmill	Ø5.0 ~ 20.0	B202
RPLE-FP-H		Fine pitch long type roughing Endmill	Ø5.0 ~ 20.0	B203
RPE-XG		Endmill for finishing and roughing	Ø5.0 ~ 20.0	B203
RPE-FP-L		Roughing Endmill for fine pitches	Ø5.0 ~ 20.0	B204
RPE-RG		Standard roughing Endmill	Ø5.0 ~ 20.0	B204
RPE-RG		4 flutes roughing Endmill	Ø6.0 ~ 20.0	B205
RPE-FF		Roughing Endmill for fine pitches	Ø6.0 ~ 20.0	B205
RPE-FP		Roughing Endmill for fine pitches	Ø6.0 ~ 20.0	B206
RPE-RG		Standard roughing Endmill	Ø6.0 ~ 50.0	B206

RPAE (Carbide)

Wave roughing Endmill for Al

3	H-A 45°	h6 shank	Grade PC30T	DC	Tolerance
				~ Ø6 ~ Ø25	0.000 ~ -0.050



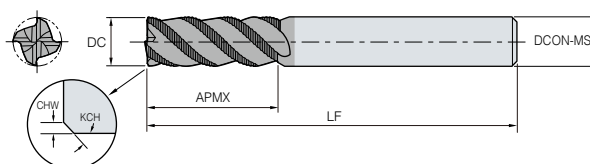
(mm)

Designation	DC	DCON-MS	APMX	LF	CHW	KCH
RPAE 3060-063	6	6	18	63	0.3	45°
3070-063	7	8	23	63	0.3	45°
3080-063	8	8	23	63	0.3	45°
3090-080	9	10	30	80	0.3	45°
3100-080	10	10	30	80	0.3	45°
3110-080	11	12	32	80	0.5	45°
3120-080	12	12	32	80	0.5	45°
3140-080	14	14	32	80	0.5	45°
3160-105	16	16	48	105	0.5	45°
3180-105	18	18	48	105	0.5	45°
3200-105	20	20	50	105	0.5	45°
3250-105	25	25	50	105	0.5	45°

RPE-FP-H (Carbide, High helix angle, irregular flute spacing and lead)

Fine pitch standard type roughing

4	H-A 40°/42°	h6 shank	Grade PC30T	DC	Tolerance
				Ø5 ~ Ø20	0.000 ~ -0.050



(mm)

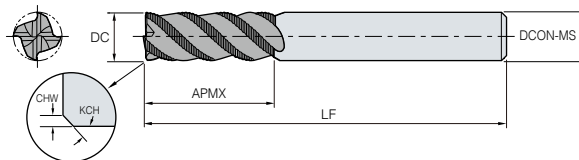
Designation	DC	DCON-MS	APMX	LF	CHW	KCH
RPE 4050-057-FP-H	5	6	13	57	0.3	45°
4060-057-FP-H	6	6	13	57	0.5	45°
4080-063-FP-H	8	8	19	63	0.5	45°
4100-072-FP-H	10	10	22	72	0.5	45°
4120-082-FP-H	12	12	26	82	0.5	45°
4140-082-FP-H	14	16	26	82	0.6	45°
4160-092-FP-H	16	16	32	92	0.6	45°
4180-092-FP-H	18	20	32	92	0.6	45°
4200-0104-FP-H	20	20	38	104	0.6	45°

RPLE-FP-H (Carbide, High helix angle, irregular flute spacing and lead)

Fine pitch long type roughing



DC	Tolerance
Ø5 ~ Ø20	0.000 ~ -0.050



(mm)

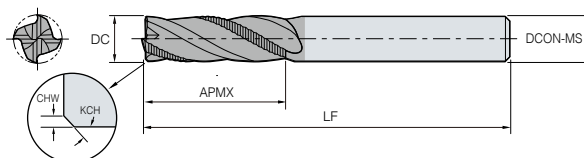
Designation	DC	DCON-MS	APMX	LF	CHW	KCH
RPLE 4050-063-FP-H	5	6	19	63	0.3	45°
4060-063-FP-H	6	8	19	63	0.5	45°
4080-072-FP-H	8	8	28	72	0.5	45°
4100-082-FP-H	10	10	34	82	0.5	45°
4120-097-FP-H	12	12	40	97	0.5	45°
4140-097-FP-H	14	16	40	97	0.6	45°
4160-108-FP-H	16	16	48	108	0.6	45°
4180-108-FP-H	18	20	48	108	0.6	45°
4200-122-FP-H	20	20	56	122	0.6	45°

RPE-XG (Carbide)

Endmill for finishing and roughing



DC	Tolerance
Ø5 ~ Ø20	0.000 ~ -0.050



(mm)

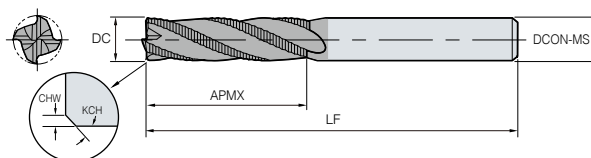
Designation	DC	DCON-MS	APMX	LF	CHW	KCH
RPE 4060-052-XG	6	6	14	52	0.25	45°
4070-063-XG	7	8	18	63	0.3	45°
4080-063-XG	8	8	18	63	0.3	45°
4090-080-XG	9	10	22	80	0.3	45°
4100-080-XG	10	10	22	80	0.3	45°
4110-080-XG	11	12	26	80	0.4	45°
4120-080-XG	12	12	26	80	0.4	45°
4140-080-XG	14	14	30	80	0.4	45°
4160-105-XG	16	16	34	105	0.6	45°
4180-105-XG	18	18	38	105	0.6	45°
4200-105-XG	20	20	42	105	0.6	45°

RPE-FP-L (Carbide, High helix angle, irregular flute spacing and lead)

Fine pitch long type roughing



DC	Tolerance
Ø5 ~ Ø20	0.000 ~ -0.050



(mm)

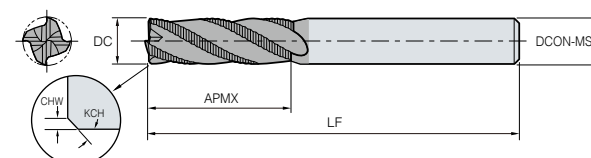
Designation	DC	DCON-MS	APMX	LF	CHW	KCH
RPE 4050-060-FP-L	5	6	13	60	0.3	45°
4060-080-FP-L	6	8	13	80	0.5	45°
4080-080-FP-L	8	8	19	80	0.5	45°
4100-080-FP-L	10	10	22	80	0.5	45°
4120-080-FP-L	12	12	26	80	0.5	45°
4140-085-FP-L	14	16	26	85	0.6	45°
4160-100-FP-L	16	16	32	100	0.6	45°
4180-100-FP-L	18	20	32	100	0.6	45°
4200-105-FP-L	20	20	38	105	0.6	45°

RPE-RG (Carbide)

Endmill for finishing and roughing



DC	Tolerance
Ø5 ~ Ø20	0.000 ~ -0.050



(mm)

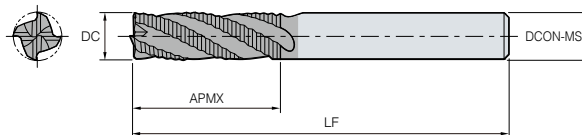
Designation	DC	DCON-MS	APMX	LF	CHW	KCH
RPE 4050-050-RG	5	6	13	50	0.3	45°
4060-050-RG	6	6	16	50	0.3	45°
4080-060-RG	8	8	20	60	0.3	45°
4100-075-RG	10	10	25	75	0.3	45°
4120-080-RG	12	12	30	80	0.4	45°
4140-100-RG	14	16	35	100	0.6	45°
4160-100-RG	16	16	40	100	0.6	45°
4180-110-RG	18	20	40	110	0.6	45°
4200-110-RG	20	20	45	110	0.6	45°

RPE-RG (HSS PM)

4 flutes roughing



DC	Tolerance
Ø6 ~ Ø20	±0.1



(mm)

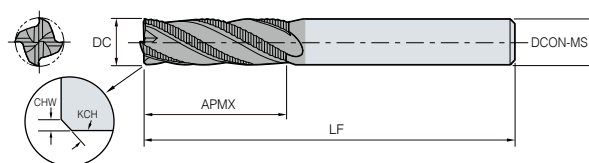
Designation	DC	DCON-MS	APMX	LF
RPE 4060-060-RG	6	6	20	60
4070-070-RG	7	10	20	70
4080-075-RG	8	10	25	75
4090-075-RG	9	10	30	75
4100-085-RG	10	10	35	85
4120-100-RG	12	12	40	100
4140-100-RG	14	16	40	100
4160-110-RG	16	16	50	110
4180-110-RG	18	20	50	110
4200-125-RG	20	20	60	125

RPE-FF (HSS PM, irregular flute spacing and lead)

Roughing Endmill for fine pitches



DC	Tolerance
Ø6 ~ Ø20	±0.1



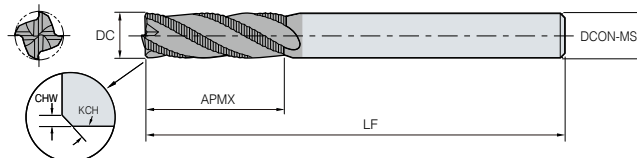
(mm)

Designation	DC	DCON-MS	APMX	LF	CHW	KCH
RPE 4060-060-FF	6	6	20	60	0.5	45°
4070-070-FF	7	10	20	70	0.5	45°
4080-075-FF	8	10	25	75	0.5	45°
4090-075-FF	9	10	30	75	0.5	45°
4100-085-FF	10	10	35	85	0.5	45°
4120-100-FF	12	12	40	100	0.6	45°
4140-100-FF	14	12	40	100	0.6	45°
4160-110-FF	16	16	50	110	0.6	45°
4180-110-FF	18	16	50	110	0.6	45°
4200-125-FF	20	20	60	125	0.6	45°

RPE-FP (HSS PM, irregular flute spacing and lead)

Roughing Endmill for fine pitches

4	H-A 25°/30°	HSS PM	h6 shank	Grade HC30T	DC Ø6 ~ Ø12 Ø12.1 ~ Ø20	Tolerance 0.000 ~ -0.050 0.000 ~ -0.100
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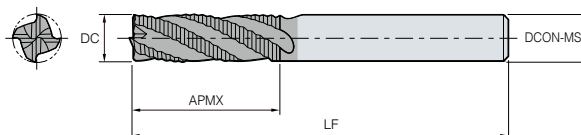
Designation	DC	DCON-MS	APMX	LF	CHW	KCH
RPE 4060-080-FP	6	6	13	80	0.5	45°
4070-080-FP	7	10	16	80	0.5	45°
4080-085-FP	8	10	19	85	0.5	45°
4090-095-FP	9	10	19	95	0.5	45°
4100-100-FP	10	10	22	100	0.5	45°

Designation	DC	DCON-MS	APMX	LF	CHW	KCH
RPE 4120-110-FP	12	12	26	110	0.6	45°
4140-110-FP	14	12	26	110	0.6	45°
4160-125-FP	16	16	32	125	0.6	45°
4180-125-FP	18	16	32	125	0.6	45°
4200-140-FP	20	20	38	140	0.6	45°

RPE-RG (HSS)

Roughing

4	H-A 30°	HSS Co 8%	h6 shank	Grade HN20T HC20T HC10T	DC Ø6 ~ Ø50	Tolerance ±0.1
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Designation	DC	DCON-MS	APMX	LF
RPE 4060-060-RG	6	6	15	60
4070-065-RG	7	8	20	65
4080-065-RG	8	8	20	65
4090-075-RG	9	10	25	75
4100-075-RG	10	10	25	75
4110-080-RG	11	12	30	80
4120-080-RG	12	12	30	80
4130-090-RG	13	12	35	90
4140-090-RG	14	12	35	90
4150-095-RG	15	12	40	95
4160-095-RG	16	16	40	95
4170-095-RG	17	16	40	95
4180-105-RG	18	16	40	105
4190-110-RG	19	16	45	110
4200-110-RG	20	20	45	110
4210-110-RG	21	20	45	110
4220-110-RG	22	20	45	110
4230-110-RG	23	20	45	110

Designation	DC	DCON-MS	APMX	LF
RPE 4240-120-RG	24	25	50	120
4250-120-RG	25	25	50	120
4260-120-RG	26	25	50	120
4270-125-RG	27	25	55	125
4280-125-RG	28	25	55	125
4300-125-RG	30	25	55	125
4320-145-RG	32	32	60	145
4340-145-RG	34	32	60	145
4350-145-RG	35	32	60	145
4360-145-RG	36	32	60	145
4380-150-RG	38	32	65	150
4400-150-RG	40	32	65	150
4420-155-RG	42	42	65	155
4440-155-RG	44	42	65	155
4450-160-RG	45	42	70	160
4460-160-RG	46	42	70	160
4500-160-RG	50	42	70	160

B Technical Information for S-Star Endmill

Endmill for Stainless steel machining

S-Star Endmill

- Stable high speed processing with minimum vibration, unequal index and optimal rake angle
- High machinability and low vibration with the application of unequal index on the cutting edge
- Minimum vibration through optimized helix angle and R gash application which also enhanced chip emission and strength improvement
- Reduced friction resistance and improved chip emission by applying new coatings with high surface hardness oxidation resistance
- Newly strengthened flute with enhanced chipping resistance, and deposition resistance

Code system

SP	FE	4	120	-	075	(-VOON00)	-	R10	(-N)
S-Star Endmill		No. of flute 4 : 4flutes	Tool dia 120 : Ø 2.0mm		Length 075 : 75mm	Flute length V10: 10mm		Corner radius R10: 1.0mm C05: 0.5mm	N: Nick type
	Type FE: Flat Endmill RE: Radius Endmill BE: Ball Endmill XE: Roughing Endmill				Neck length N20: 20mm				

Features



Applying high toughness substrate

- Stable cutting is ensured with better chipping resistance by applying high toughness substrate

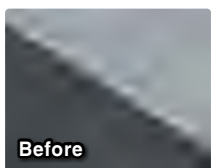


Applied differential AlCrN coating layer depth per tool size

- Applied multi coating layers
- Increased lubrication due to containing Cr
- Enhanced stability against frictional heat
- Improved wear resistance from thicker coating layers

Cutting edge treatment

- Improved chipping resistance in the beginning of cutting
- Better wear resistance and stable cutting
- High quality of product from cutting edge treatment stabilization

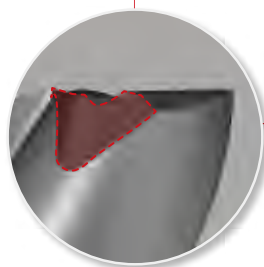


Additionally finished edge

- Enhanced surface finish due to increased 1st O.D grinding roughness
- High quality cutting edge and good welding resistance

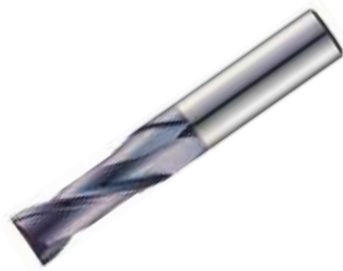
Uneven flute spacing / R gash

- High chip evacuation through R gash shape
- Stability in shouldering machining

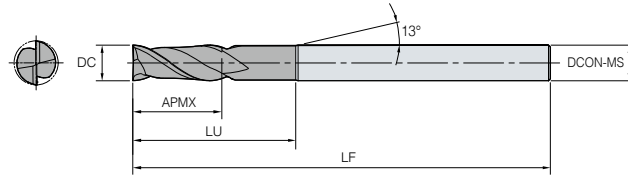


SPFE2000

2 flutes flat



DC	Tolerance
Ø1 ~ Ø5	0.000 ~ -0.015
Ø6	0.000 ~ -0.020
Ø8 ~ Ø20	0.000 ~ -0.030



(mm)

Designation	DC	DCON-MS	APMX	LU	LF
SPFE 2010-050	1	6	2.5	7.5	50
2012-050	1.2	6	3	8	50
2015-050	1.5	6	4	9	50
2020-050	2	6	6	11	50
2025-050	2.5	6	7	12	50
2030-055	3	6	8	13	55
2040-055	4	6	10	15	55
2050-055	5	6	15	20	55
2060-060	6	6	15	22	60
2080-070	8	8	20	32	70
2100-075	10	10	25	33	75
2120-080	12	12	30	35	80
2140-090	14	16	35	42	90
2160-100	16	16	42	52	100
2180-100	18	16	45	-	100
2200-100	20	20	48	50	100

• Applicable Workpiece

◎: EXCELLENT ○: GOOD

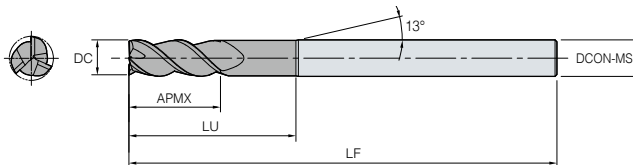
CARBON STEEL (~ HB225)	ALLOY STEEL (HB225 ~ 325)	PRE-HARDENED STEEL (HRC30 ~ 50)	HARDENED STEEL		CAST IRON ~ FCD500	ALUMINUM	STAINLESS STEEL	Ti-Alloy	Ni-Alloy
			SKD61 (~ HRC55)	SKD11 (HRC55 ~)			◎	○	○

SPFE3000

3 flutes flat



DC	Tolerance
Ø1 ~ Ø5	0.000 ~ -0.015
Ø6	0.000 ~ -0.020
Ø8 ~ Ø20	0.000 ~ -0.030



(mm)

Designation	DC	DCON-MS	APMX	LU	LF
SPFE					
3010-050	1	6	2.5	7.5	50
3012-050	1.2	6	3	8	50
3015-050	1.5	6	4	9	50
3020-050	2	6	6	11	50
3025-050	2.5	6	7	12	50
3030-055	3	6	8	13	55
3030-060-V10	3	6	10	15	60
3040-055	4	6	10	15	55
3040-060-V12	4	6	12	17	60
3050-055	5	6	13	18	55
3060-060	6	6	15	22	60
3060-065-V20	6	6	20	27	65
3080-070	8	8	20	32	70
3080-080-V30	8	8	30	42	80
3100-075	10	10	25	33	75
3100-085-V35	10	10	35	43	85
3120-080	12	12	30	35	80
3120-090-V40	12	12	40	45	90
3140-090	14	16	35	42	90
3160-100	16	16	42	52	100
3180-100	18	16	45	-	100
3200-100	20	20	48	50	100

• Applicable Workpiece

◎: EXCELLENT ○: GOOD

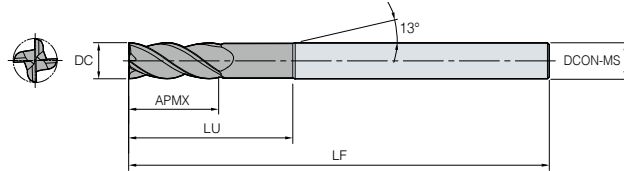
CARBON STEEL (~ HB225)	ALLOY STEEL (HB225 ~ 325)	PRE-HARDENED STEEL (HRC30 ~ 50)	HARDENED STEEL		CAST IRON ~ FCD500	ALUMINUM	STAINLESS STEEL	Ti-Alloy	Ni-Alloy
			SKD61 (~ HRC55)	SKD11 (HRC55 ~)			◎	○	○

SPFE4000

4 flutes flat



DC	Tolerance
Ø1 ~ Ø5.5	0.000 ~ -0.015
Ø6 ~ Ø7	0.000 ~ -0.020
Ø8 ~ Ø20	0.000 ~ -0.030



(mm)

Designation	DC	DCON-MS	APMX	LU	LF
SPFE					
4010-050	1	6	2.5	7.5	50
4012-050	1.2	6	3	8	50
4015-050	1.5	6	4	9	50
4020-050	2	6	6	11	50
4025-050	2.5	6	7	12	50
4030-055	3	6	8	13	55
4030-060-V10	3	6	10	15	60
4035-055	3.5	6	10	15	55
4040-055	4	6	10	15	55
4040-060-V12	4	6	12	17	60
4045-055	4.5	6	12	17	55
4050-055	5	6	15	20	55
4055-060	5.5	6	15	20	60
4060-060	6	6	15	22	60
4060-065-V20	6	6	20	27	65
4065-060	6.5	8	15	22	60
4070-080	7	8	20	42	80
4080-070	8	8	20	32	70
4080-070-V25	8	8	25	32	70
4080-080-V30	8	8	30	42	80
4085-070	8.5	10	20	28	70
4090-080	9	10	25	38	80
4100-075	10	10	25	33	75
4100-085-V35	10	10	35	43	85
4120-080	12	12	30	35	80
4120-090-V40	12	12	40	45	90
4140-090	14	16	35	42	90
4160-100	16	16	42	52	100
4180-100	18	16	45	-	100
4200-100	20	20	48	50	100

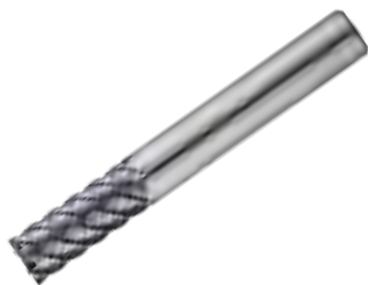
• Applicable Workpiece

◎: EXCELLENT ○: GOOD

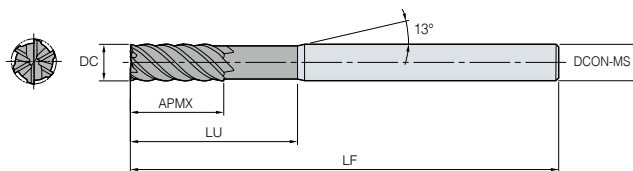
CARBON STEEL (~ HB225)	ALLOY STEEL (HB225 ~ 325)	PRE-HARDENED STEEL (HRC30 ~ 50)	HARDENED STEEL		CAST IRON ~ FCD500	ALUMINUM	STAINLESS STEEL	Ti-Alloy	Ni-Alloy
			SKD61 (~ HRC55)	SKD11 (HRC55 ~)			◎	○	○

SPFE6000

6 flutes flat



6	H-A 45°	h5 shank	Grade PC325	DC	Tolerance
				Ø6 Ø8 ~ Ø20	0.000 ~ -0.020 0.000 ~ -0.030



(mm)

Designation	DC	DCON-MS	APMX	LU	LF
SPFE					
6060-060	6	6	15	22	60
6080-070	8	8	20	32	70
6100-075	10	10	25	33	75
6120-080	12	12	30	35	80
6160-100	16	16	42	52	100
6200-100	20	20	48	50	100

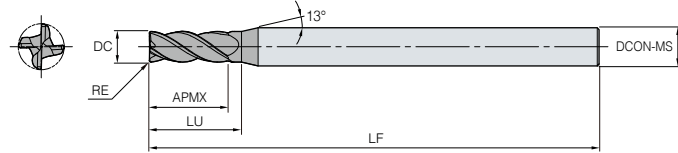
• Applicable Workpiece

◎: EXCELLENT ○: GOOD

CARBON STEEL (~ HB225)	ALLOY STEEL (HB225 ~ 325)	PRE-HARDENED STEEL (HRC30 ~ 50)	HARDENED STEEL		CAST IRON ~ FCD500	ALUMINUM	STAINLESS STEEL	Ti-Alloy	Ni-Alloy
			SKD61 (~ HRC55)	SKD11 (HRC55 ~)					
○	○	○					◎	○	○

SPRE4000**4 flutes radius**

DC	Tolerance
Ø1 ~ Ø5.5	0.000 ~ -0.015
Ø6 ~ Ø7	0.000 ~ -0.020
Ø8 ~ Ø20	0.000 ~ -0.030



(mm)

Designation	DC	DCON-MS	APMX	LU	LF	RE
SPRE						
4010-050-R01	1	6	2.5	7.5	50	0.1
4010-050-R02	1	6	2.5	7.5	50	0.2
4012-050-R01	1.2	6	3	8	50	0.1
4015-050-R01	1.5	6	4	9	50	0.1
4015-050-R02	1.5	6	4	9	50	0.2
4020-050-R01	2	6	6	11	50	0.1
4020-050-R02	2	6	6	11	50	0.2
4025-050-R02	2.5	6	7	12	50	0.2
4030-055-R02	3	6	8	13	55	0.2
4030-055-R03	3	6	8	13	55	0.3
4030-055-R05	3	6	8	13	55	0.5
4040-055-R02	4	6	10	15	55	0.2
4040-055-R03	4	6	10	15	55	0.3
4040-055-R05	4	6	10	15	55	0.5
4050-055-R02	5	6	15	20	55	0.2
4050-055-R03	5	6	15	20	55	0.3
4050-055-R05	5	6	15	20	55	0.5
4060-060-R03	6	6	15	22	60	0.3
4060-060-R05	6	6	15	22	60	0.5
4060-060-R10	6	6	15	22	60	1
4070-060-R03	7	8	15	22	60	0.3
4080-070-R02	8	8	20	32	70	0.2
4080-070-R03	8	8	20	32	70	0.3
4080-070-R05	8	8	20	32	70	0.5
4080-070-R10	8	8	20	32	70	1
4100-075-R03	10	10	25	33	75	0.3
4100-075-R05	10	10	25	33	75	0.5
4100-075-R10	10	10	25	33	75	1
4100-075-R15	10	10	25	33	75	1.5
4100-075-R20	10	10	25	33	75	2
4100-075-R30	10	10	25	33	75	3
4120-080-R03	12	12	30	35	80	0.3
4120-080-R05	12	12	30	80	12	0.5

• Applicable Workpiece

◎: EXCELLENT ○: GOOD

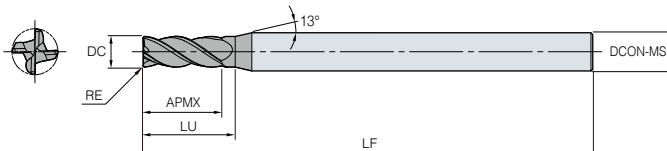
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			SKD61 (~ HRC55)	SKD11 (HRC55 ~)			◎	○	○

SPRE4000

4 flutes radius



DC	Tolerance
Ø1 ~ Ø5.5	0.000 ~ -0.015
Ø6 ~ Ø7	0.000 ~ -0.020
Ø8 ~ Ø20	0.000 ~ -0.030



(mm)

Designation		DC	DCON-MS	APMX	LU	LF	RE
SPRE	4120-080-R10	12	12	30	35	80	1
	4120-080-R15	12	12	30	35	80	1.5
	4120-080-R20	12	12	30	35	80	2
	4120-080-R30	12	12	30	35	80	3
	4120-080-R40	12	12	30	35	80	4
	4140-090-R05	14	16	35	42	90	0.5
	4140-090-R10	14	16	35	42	90	1
	4140-090-R20	14	16	35	42	90	2
	4140-090-R30	14	16	35	42	90	3
	4140-090-R40	14	16	35	42	90	4
	4160-100-R05	16	16	42	52	100	0.5
	4160-100-R10	16	16	42	52	100	1
	4160-100-R20	16	16	42	52	100	2
	4160-100-R30	16	16	42	52	100	3
	4160-100-R40	16	16	42	52	100	4
	4160-100-R50	16	16	42	52	100	5
	4180-100-R05	18	16	45	-	100	0.5
	4200-100-R05	20	20	48	50	100	0.5
	4200-100-R10	20	20	48	50	100	1
	4200-100-R20	20	20	48	50	100	2
	4200-100-R30	20	20	48	50	100	3
	4200-100-R40	20	20	48	50	100	4
	4200-100-R50	20	20	48	50	100	5

• Applicable Workpiece

◎: EXCELLENT ○: GOOD

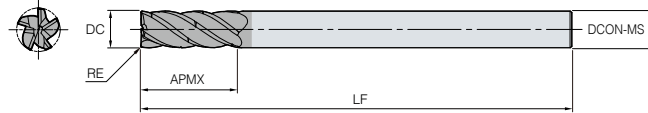
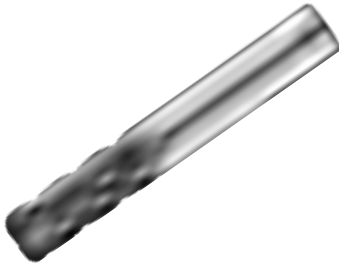
CARBON STEEL (~ HB225)	ALLOY STEEL (HB225 ~ 325)	PRE-HARDENED STEEL (HRC30 ~ 50)	HARDENED STEEL		CAST IRON ~ FCD500	ALUMINUM	STAINLESS STEEL	Ti-Alloy	Ni-Alloy
			SKD61 (~ HRC55)	SKD11 (HRC55 ~)					
○	○	○					◎	○	○

SPRE5000

5 flutes neck type radius



DC	Tolerance
Ø6	0.000 ~ -0.020
Ø8 ~ Ø20	0.000 ~ -0.030



(mm)

Designation	DC	DCON-MS	APMX	LF	RE
SPRE					
5060-060-R05-N	6	6	15	60	0.5
5060-070-R05-N	6	6	24	70	0.5
5080-070-R05-N	8	8	20	70	0.5
5080-090-R05-N	8	8	32	90	0.5
5100-075-R05-N	10	10	25	75	0.5
5100-100-R05-N	10	10	40	100	0.5
5120-080-R05-N	12	12	30	80	0.5
5120-080-R10-N	12	12	30	80	1
5120-080-R20-N	12	12	30	80	2
5120-080-R30-N	12	12	30	80	3
5120-110-R05-N	12	12	48	110	0.5
5160-100-R05-N	16	16	42	100	0.5
5160-100-R10-N	16	16	42	100	1
5160-100-R20-N	16	16	42	100	2
5160-100-R30-N	16	16	42	100	3
5160-100-R40-N	16	16	42	100	4
5160-130-R05-N	16	16	65	130	0.5
5200-110-R05-N	20	20	48	110	0.5
5200-130-R05-N	20	20	65	130	0.5

• Applicable Workpiece

◎: EXCELLENT ○: GOOD

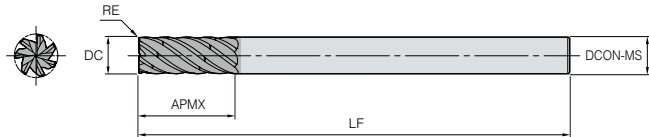
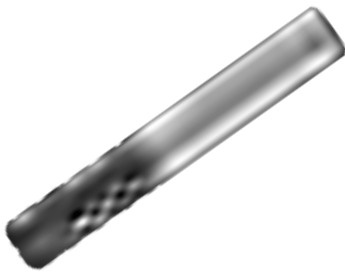
CARBON STEEL (~ HB225)	ALLOY STEEL (HB225 ~ 325)	PRE-HARDENED STEEL (HRC30 ~ 50)	HARDENED STEEL	CAST IRON ~ FCD500	ALUMINUM	STAINLESS STEEL	Ti-Alloy	Ni-Alloy
			SKD61 (~ HRC55) SKD11 (HRC55 ~)					
○	○	○				◎	○	○

SPRE7000

7 flutes neck type radius



DC	Tolerance
Ø6	0.000 ~ -0.020
Ø8 ~ Ø20	0.000 ~ -0.030



(mm)

Designation	DC	DCON-MS	APMX	LF	RE
SPRE					
7060-060-R05-N	6	6	15	60	0.5
7060-070-R05-N	6	6	24	70	0.5
7080-070-R05-N	8	8	20	70	0.5
7080-090-R05-N	8	8	32	90	0.5
7100-075-R05-N	10	10	25	75	0.5
7100-100-R05-N	10	10	40	100	0.5
7120-080-R05-N	12	12	30	80	0.5
7120-080-R10-N	12	12	30	80	1
7120-080-R20-N	12	12	30	80	2
7120-080-R30-N	12	12	30	80	3
7120-080-R40-N	12	12	30	80	4
7120-110-R05-N	12	12	48	110	0.5
7160-100-R05-N	16	16	42	100	0.5
7160-100-R10-N	16	16	42	100	1
7160-100-R20-N	16	16	42	100	2
7160-100-R30-N	16	16	42	100	3
7160-100-R40-N	16	16	42	100	4
7160-100-R50-N	16	16	42	100	5
7160-130-R05-N	16	16	65	130	0.5
7200-110-R05-N	20	20	48	110	0.5
7200-130-R05-N	20	20	65	130	0.5

• Applicable Workpiece

◎: EXCELLENT ○: GOOD

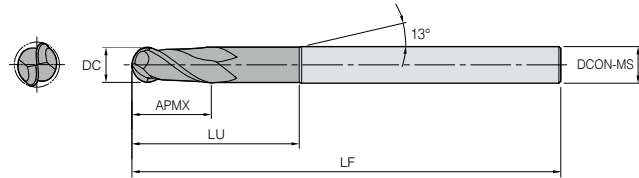
CARBON STEEL (~ HB225)	ALLOY STEEL (HB225 ~ 325)	PRE-HARDENED STEEL (HRC30 ~ 50)	HARDENED STEEL		CAST IRON ~ FCD500	ALUMINUM	STAINLESS STEEL	Ti-Alloy	Ni-Alloy
			SKD61 (~ HRC55)	SKD11 (HRC55 ~)			◎	○	○

SPBE2000

2 flutes ball



2	H-A 38°	h5 shank	Grade PC325	DC	Tolerance
				Ø1 ~ Ø5 Ø6 Ø8 ~ Ø12	0.000 ~ -0.015 0.000 ~ -0.020 0.000 ~ -0.030



		(mm)				
Designation		DC	DCON-MS	APMX	LU	LF
SPBE	2010-050	1	6	3	8	50
	2020-050	2	6	6	11	50
	2030-050	3	6	8	13	50
	2030-070	3	6	8	13	70
	2040-050	4	6	10	15	50
	2040-070	4	6	10	15	70
	2050-050	5	6	13	18	50
	2050-080	5	6	13	18	80
	2060-050	6	6	13	20	50
	2060-090	6	6	13	52	90
	2080-060	8	8	19	24	60
	2080-100	8	8	19	62	100
	2100-070	10	10	22	30	70
	2100-100	10	10	22	58	100
	2120-075	12	12	26	30	75
	2120-110	12	12	26	63	110

• Applicable Workpiece

◎: EXCELLENT ○: GOOD

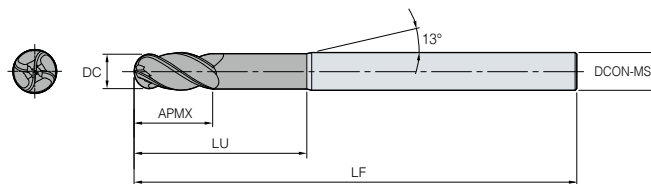
CARBON STEEL (~ HB225)	ALLOY STEEL (HB225 ~ 325)	PRE-HARDENED STEEL (HRC30 ~ 50)	HARDENED STEEL		CAST IRON ~ FCD500	ALUMINUM	STAINLESS STEEL	Ti-Alloy	Ni-Alloy
			SKD61 (~ HRC55)	SKD11 (HRC55 ~)					
○	○	○					◎	○	○

SPBE4000

4 flutes ball



DC	Tolerance
Ø3 ~ Ø5	0.000 ~ -0.015
Ø6	0.000 ~ -0.020
Ø8 ~ Ø20	0.000 ~ -0.030



(mm)

Designation	DC	DCON-MS	APMX	LU	LF
SPBE 4030-060	3	6	8	13	60
4040-070	4	6	8	13	70
4050-080	5	6	12	17	80
4060-090	6	6	12	52	90
4080-100	8	8	16	62	100
4100-100	10	10	20	58	100
4120-100	12	12	25	53	100
4160-100	16	16	30	50	100
4200-100	20	20	38	48	100

• Applicable Workpiece

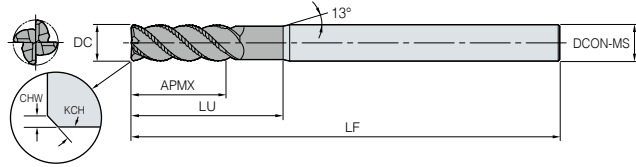
◎: EXCELLENT ○: GOOD

CARBON STEEL (~ HB225)	ALLOY STEEL (HB225 ~ 325)	PRE-HARDENED STEEL (HRC30 ~ 50)	HARDENED STEEL		CAST IRON ~ FCD500	ALUMINUM	STAINLESS STEEL	Ti-Alloy	Ni-Alloy
			SKD61 (~ HRC55)	SKD11 (HRC55 ~)			◎	○	○

SPXE3000/4000/5000

3~5 flutes roughing

3	4	5	H-A 45°	h5 shank	Grade PC325	DC	Tolerance
						Ø3 ~ Ø20	0.000 ~ -0.050



(mm)

Designation	DC	DCON-MS	APMX	LU	LF	CHW	KCH
SPXE							
3030-050-C02	3	6	8	13	50	0.2	45°
3040-050-C02	4	6	10	15	50	0.2	45°
4050-050-C03	5	6	13	18	50	0.3	45°
4060-060-C03	6	6	13	22	60	0.3	45°
4060-060-V10N20S6-C03	6	6	10	20	60	0.3	45°
4070-070-C04	7	8	18	32	70	0.4	45°
4080-070-C04	8	8	19	32	70	0.4	45°
4080-070-V12N25S8-C04	8	8	12	25	70	0.4	45°
4090-070-C04	9	10	20	28	70	0.4	45°
4100-075-C04	10	10	22	33	75	0.4	45°
4100-075-V15N30S10-C04	10	10	15	30	75	0.4	45°
4110-080-C05	11	12	25	33	80	0.5	45°
4120-080-C05	12	12	26	35	80	0.5	45°
4120-080-V20N35S12-C05	12	12	20	35	80	0.5	45°
5060-060-C05	6	6	13	22	60	0.5	45°
5080-065-C05	8	8	19	27	65	0.5	45°
5100-070-C05	10	10	22	30	70	0.5	45°
5120-080-C05	12	12	26	35	80	0.5	45°
5140-090-C05	14	16	28	40	90	0.5	45°
5160-100-C05	16	16	32	50	100	0.5	45°
5160-100-V42-C05	16	16	42	52	100	0.5	45°
5200-100-C05	20	20	38	48	100	0.5	45°
5200-100-V45-C05	20	20	45	50	100	0.5	45°

• Applicable Workpiece

◎: EXCELLENT ○: GOOD

CARBON STEEL (~ HB225)	ALLOY STEEL (HB225 ~ 325)	PRE-HARDENED STEEL (HRC30 ~ 50)	HARDENED STEEL		CAST IRON ~ FCD500	ALUMINUM	STAINLESS STEEL	Ti-Alloy	Ni-Alloy
			SKD61 (~ HRC55)	SKD11 (HRC55 ~)					
○	○	○					◎	○	○

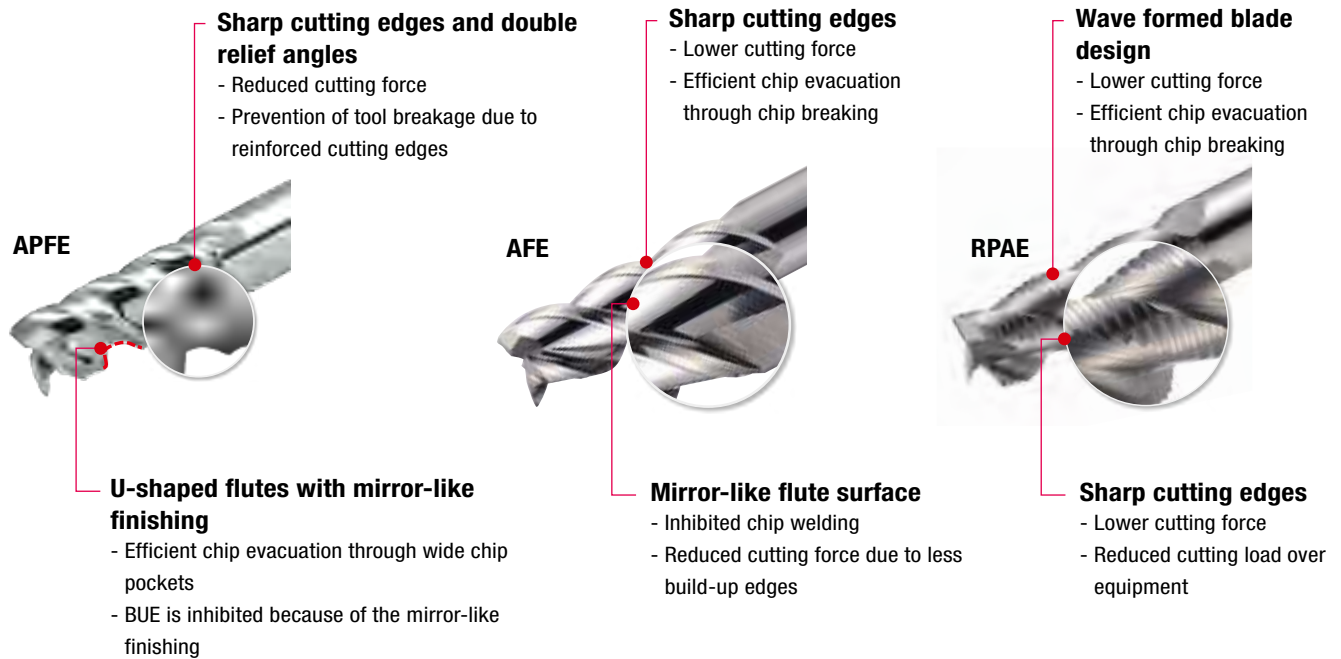
B Technical Information for A-Star Endmill

Endmill series for aluminum machining

A-Star Endmill

- Endmill series for Aluminum machining
- Optimized Solutions for each application - The wide selection of tools are available for various machining processes
- Higher Machining Efficiency - Advanced flute design and cutting edge technology are applied

Features



Line-up

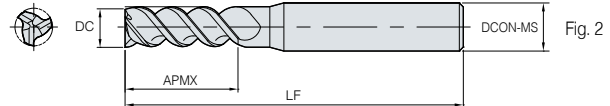
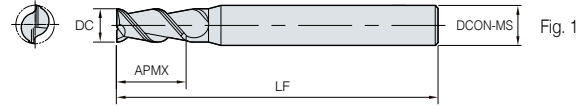
Designation	Appearance	Product name	Size(Ø)	Page
APFE		2~3 flutes flat Endmill	Ø2.0 ~ 20.0	B220
APMFE		2~3 flutes middle flat Endmill	Ø3.0 ~ 20.0	B221
APLFE		2~3 flutes long flat Endmill	Ø3.0 ~ 20.0	B222
APBE		2 flutes ball Endmill	Ø1.0 ~ 12.0	B223
AFE		3 flutes short flat Endmill	Ø1.0 ~ 20.0	B224
AFE		3 flutes flat Endmill	Ø1.0 ~ 20.0	B225
AFE		3 flutes long flat Endmill	Ø1.0 ~ 20.0	B226
APRE		3 flutes roughing Endmill	Ø4.0 ~ 25.0	B228
RPAE		3 flutes wave roughing Endmill	Ø6.0 ~ 25.0	B229

APFE2000/3000

2~3 flutes flat



DC	Tolerance
Ø1 ~ Ø5	0.000 ~ -0.020
Ø6 ~ Ø8	0.000 ~ -0.025
Ø10 ~ Ø20	0.000 ~ -0.030



(mm)

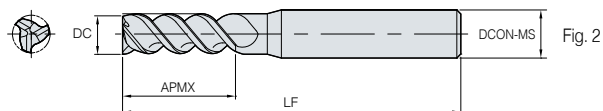
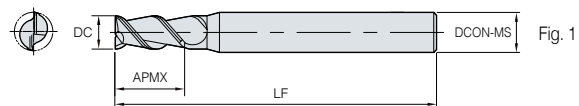
Designation		DC	DCON-MS	APMX	LF	Fig.
APFE	2020-050-S4	2	4	6	50	1
	2025-050	2.5	6	8	50	1
	2030-050	3	6	9	50	1
	2040-050	4	6	12	50	1
	2050-050	5	6	15	50	1
	2060-050	6	6	18	50	1
	2080-060	8	8	20	60	1
	2100-075	10	10	30	75	1
	2120-075	12	12	32	75	1
	2160-100	16	16	45	100	1
	2200-100	20	20	45	100	1
	3020-050-S4	2	4	6	50	2
	3025-050	2.5	6	8	50	2
	3030-050	3	6	9	50	2
	3040-050	4	6	12	50	2
	3050-050	5	6	15	50	2
	3060-050	6	6	18	50	2
	3080-060	8	8	20	60	2
	3100-075	10	10	30	75	2
	3120-075	12	12	32	75	2
	3160-100	16	16	45	100	2
	3200-100	20	20	45	100	2

APMFE2000/3000

2~3 middle flutes flat



DC	Tolerance
Ø1 ~ Ø5	0.000 ~ -0.020
Ø6 ~ Ø8	0.000 ~ -0.025
Ø10 ~ Ø20	0.000 ~ -0.030



(mm)

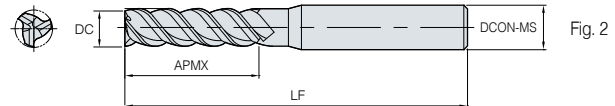
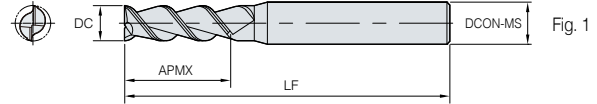
Designation		DC	DCON-MS	APMX	LF	Fig.
APMFE	2030-060	3	6	11	60	1
	2040-060	4	6	14	60	1
	2050-060	5	6	17	60	1
	2060-065	6	6	22	65	1
	2080-065	8	8	25	65	1
	2100-080	10	10	37	80	1
	2120-080	12	12	40	80	1
	2160-110	16	16	55	110	1
	2200-125	20	20	60	125	1
	3030-060	3	6	11	60	2
	3040-060	4	6	14	60	2
	3050-060	5	6	17	60	2
	3060-065	6	6	22	65	2
	3080-065	8	8	25	65	2
	3100-080	10	10	37	80	2
	3120-080	12	12	40	80	2
	3160-110	16	16	55	110	2
	3200-125	20	20	60	125	2

APLFE2000/3000

2~3 flutes long flat



DC	Tolerance
Ø1 ~ Ø5	0.000 ~ -0.020
Ø6 ~ Ø8	0.000 ~ -0.025
Ø10 ~ Ø20	0.000 ~ -0.030



(mm)

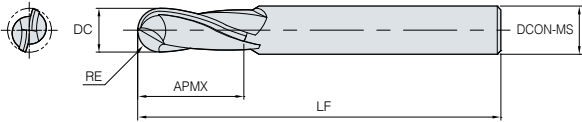
Designation		DC	DCON-MS	APMX	LF	Fig.
APLFE	2030-060	3	6	12	60	1
	2040-060	4	6	16	60	1
	2050-060	5	6	20	60	1
	2060-075	6	6	25	75	1
	2080-075	8	8	32	75	1
	2100-100	10	10	45	100	1
	2120-100	12	12	45	100	1
	2160-150	16	16	65	150	1
	2200-150	20	20	75	150	1
	3030-060	3	6	12	60	2
	3040-060	4	6	16	60	2
	3050-060	5	6	20	60	2
	3060-075	6	6	25	75	2
	3080-075	8	8	32	75	2
	3100-100	10	10	45	100	2
	3120-100	12	12	45	100	2
	3160-150	16	16	65	150	2
	3200-150	20	20	75	150	2

APBE2000

2 flutes ball



DC	Tolerance
Ø1 ~ Ø5	0.000 ~ -0.020
Ø6 ~ Ø8	0.000 ~ -0.025
Ø10 ~ Ø20	0.000 ~ -0.030



(mm)

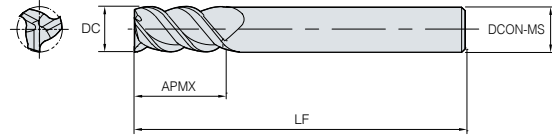
Designation		DC	DCON-MS	APMX	LF	RE
APBE	2010-050	1	4	2	50	0.5
	2015-050	1.5	4	3	50	0.75
	2020-050	2	4	4	50	1
	2025-050	2.5	4	5	50	1.25
	2030-050	3	4	6	50	1.5
	2035-050	3.5	4	7	50	1.75
	2040-050	4	4	8	50	2
	2045-050	4.5	6	9	50	2.25
	2050-050	5	6	10	50	2.5
	2055-050	5.5	6	11	50	2.75
	2060-050	6	6	12	50	3
	2080-060	8	8	16	60	4
	2100-075	10	10	20	75	5
	2120-075	12	12	24	75	6

AFE3000

3 flutes short flat



3	H-A 45°	h6 shank	Grade H05S	DC	Tolerance
				Ø1 ~ Ø12 Ø16 ~ Ø20	0.000 ~ -0.020 0.000 ~ -0.030



(mm)

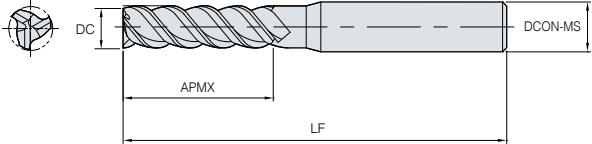
	Designation	DC	DCON-MS	APMX	LF
AFE	3010-040-V2S6	1	6	2	40
	3010-040-V2.5S6	1	6	2.5	40
	3015-040-V3S6	1.5	6	3	40
	3020-040-V3S6	2	6	3	40
	3030-045-V4S6	3	6	4	45
	3030-045-V8S6	3	6	8	45
	3040-045-V5S6	4	6	5	45
	3040-045-V8S6	4	6	8	45
	3040-045-V11S6	4	6	11	45
	3050-045-V6S6	5	6	6	45
	3060-050-V7S6	6	6	7	50
	3060-050-V13S6	6	6	13	50
	3080-060-V9S8	8	8	9	60
	3080-060-V19S8	8	8	19	60
	3100-065-V11S10	10	10	11	65
	3100-065-V22S10	10	10	22	65
	3120-070-V13S12	12	12	13	70
	3120-070-V26S12	12	12	26	70
	3160-090-V18S16	16	16	18	90
	3160-090-V32S16	16	16	32	90
	3200-090-V22S20	20	20	22	90
	3200-090-V38S20	20	20	38	90

AFE3000

3 flutes flat



			Grade H05S	DC	Tolerance
				Ø1 ~ Ø12	0.000 ~ -0.020
				Ø16 ~ Ø20	0.000 ~ -0.030



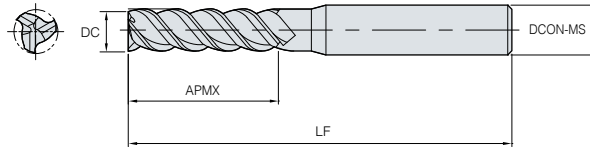
					(mm)
	Designation	DC	DCON-MS	APMX	LF
AFE	3010-050-V3S6	1	6	3	50
	3015-050-V5S6	1.5	6	5	50
	3020-050-V6S6	2	6	6	50
	3030-055-V11S6	3	6	11	55
	3040-055-V13S6	4	6	13	55
	3050-055-V17S6	5	6	17	55
	3060-060-V17S6	6	6	17	60
	3080-070-V22S8	8	8	22	70
	3100-075-V27S10	10	10	27	75
	3120-080-V32S12	12	12	32	80
	3160-100-V42S16	16	16	42	100
	3200-100-V48S20	20	20	48	100

AFE3000

3 flutes long flat



DC	Tolerance
Ø1 ~ Ø12	0.000 ~ -0.020
Ø16 ~ Ø20	0.000 ~ -0.030



(mm)

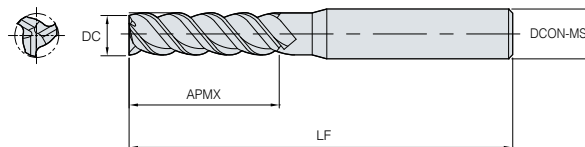
	Designation	DC	DCON-MS	APMX	LF
AFE	3010-060-V4S6	1	6	4	60
	3010-060-V6S6	1	6	6	60
	3015-060-V6S6	1.5	6	6	60
	3015-060-V8S6	1.5	6	8	60
	3015-060-V10S6	1.5	6	10	60
	3020-060-V8S6	2	6	8	60
	3020-060-V10S6	2	6	10	60
	3020-060-V12S6	2	6	12	60
	3030-065-V15S6	3	6	15	65
	3030-070-V20S6	3	6	20	70
	3030-075-V25S6	3	6	25	75
	3030-080-V30S6	3	6	30	80
	3040-065-V16S6	4	6	16	65
	3040-070-V20S6	4	6	20	70
	3040-075-V26S6	4	6	26	75
	3040-080-V30S6	4	6	30	80
	3060-060-V22S6	6	6	22	60
	3060-070-V25S6	6	6	25	70
	3060-075-V30S6	6	6	30	75
	3060-080-V35S6	6	6	35	80
	3060-090-V42S6	6	6	42	90
	3060-100-V50S6	6	6	50	100
	3080-080-V28S8	8	8	28	80
	3080-080-V30S8	8	8	30	80
	3080-085-V35S8	8	8	35	85
	3080-090-V40S8	8	8	40	90
	3080-095-V45S8	8	8	45	95
	3080-100-V50S8	8	8	50	100
	3080-105-V55S8	8	8	55	105
	3080-110-V65S8	8	8	65	110
	3100-090-V32S10	10	10	32	90
	3100-090-V35S10	10	10	35	90
	3100-090-V40S10	10	10	40	90
	3100-100-V45S10	10	10	45	100
	3100-100-V50S10	10	10	50	100

AFE3000

3 flutes long flat



				DC	Tolerance
				Ø1 ~ Ø12	0.000 ~ -0.020
				Ø16 ~ Ø20	0.000 ~ -0.030



(mm)

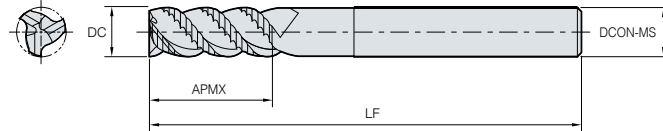
	Designation	DC	DCON-MS	APMX	LF
AFE	3100-110-V55S10	10	10	55	110
	3100-110-V60S10	10	10	60	110
	3100-120-V65S10	10	10	65	120
	3120-095-V40S12	12	12	40	95
	3120-100-V45S12	12	12	45	100
	3120-100-V50S12	12	12	50	100
	3120-110-V55S12	12	12	55	110
	3120-110-V60S12	12	12	60	110
	3120-120-V65S12	12	12	65	120
	3120-120-V70S12	12	12	70	120
	3120-135-V75S12	12	12	75	135
	3160-105-V52S16	16	16	52	105
	3160-110-V55S16	16	16	55	110
	3160-130-V65S16	16	16	65	130
	3160-150-V75S16	16	16	75	150
	3160-160-V85S16	16	16	85	160
	3160-180-V95S16	16	16	95	180
	3160-190-V105S16	16	16	105	190
	3160-200-V115S16	16	16	115	200
	3200-110-V55S20	20	20	55	110
	3200-130-V65S20	20	20	65	130
	3200-150-V75S20	20	20	75	150
	3200-160-V85S20	20	20	85	160
	3200-180-V95S20	20	20	95	180
	3200-190-V105S20	20	20	105	190
	3200-200-V115S20	20	20	115	200
	3200-220-V125S20	20	20	125	220

APRE3000

3 flutes roughing



3	H-A 45°	h6 shank	Grade H05S	DC	Tolerance
				Ø1 ~ Ø8 Ø8.5 ~ Ø25	0.000 ~ -0.070 0.000 ~ -0.010



(mm)

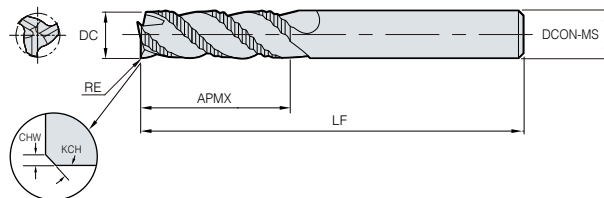
	Designation	DC	DCON-MS	APMX	LF
APRE	3040-050	4	6	8	50
	3050-050	5	6	13	50
	3060-050	6	6	15	50
	3065-060	6.5	8	16	60
	3070-060	7	8	16	60
	3075-060	7.5	8	20	60
	3080-060	8	8	20	60
	3085-075	8.5	10	20	75
	3090-075	9	10	20	75
	3095-075	9.5	10	22	75
	3100-075	10	10	25	75
	3110-075	11	12	30	75
	3120-075	12	12	30	75
	3130-075	13	14	30	75
	3140-075	14	16	32	75
	3150-075	15	16	32	75
	3160-100	16	16	35	100
	3170-100	17	20	35	100
	3180-100	18	20	35	100
	3200-100	20	20	45	100
	3250-105	25	25	50	105

RPAE3000

3 flutes wave roughing



DC	Tolerance
Ø6 ~ Ø10	0.000 ~ -0.058
Ø11 ~ Ø18	0.000 ~ -0.070
Ø20 ~ Ø25	0.000 ~ -0.084



(mm)

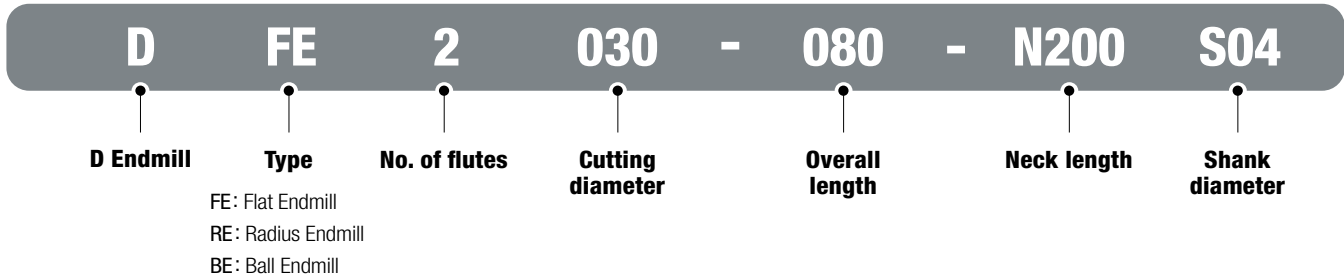
Designation		DC	DCON-MS	APMX	LF	KCH	CHW
RPAE	3060-063	6	6	18	63	45°	0.3
	3070-063	7	8	23	63	45°	0.3
	3080-063	8	8	23	63	45°	0.3
	3090-080	9	10	30	80	45°	0.3
	3100-080	10	10	30	80	45°	0.3
	3110-080	11	12	32	80	45°	0.5
	3120-080	12	12	32	80	45°	0.5
	3140-080	14	14	32	80	45°	0.5
	3160-105	16	16	48	105	45°	0.5
	3180-105	18	18	48	105	45°	0.5
	3200-105	20	20	50	105	45°	0.5
	3250-105	25	25	50	105	45°	0.5

Diamond coated Endmill

D Endmill

- Tangential cutting-edge geometries for excellent surface finish
- Excellent wear resistance due to high hardness and high purity diamond coating
- Advanced surface finish and cutting performance thanks to sharp edges and tangential tool geometries

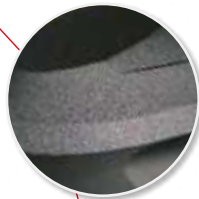
Code system



Features

• Tangential cutting-edge geometries

- One-Pass grinding system
- Prevents stepped cone on the machined surface
- Applied on 2 flutes and 4 flutes ball types



• Center-matched ball shape (4-flutes)

- Ball point shape for high feed machining
- Improved rigidity and excellent surface finish

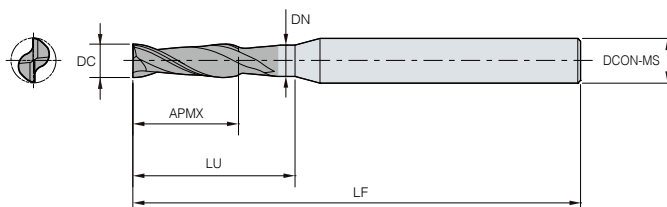


DFE2000

Flat



2	H-A 30°	h6 shank	Grade ND3000	DC	Tolerance
				Ø0.2 ~ Ø5	0.000 ~ -0.020
				Ø6 ~ Ø12	0.000 ~ -0.030



(mm)

Designation	DC	DCON-MS	APMX	LU	LF	DN
DFE 2002-045-N004S04	0.2	4	0.4	-	45	-
2003-045-N006S04	0.3	4	0.6	-	45	-
2004-045-N008S04	0.4	4	0.8	-	45	-
2005-045-N010S04	0.5	4	1	-	45	-
2006-045-N012S04	0.6	4	1.2	-	45	-
2007-045-N015S04	0.7	4	1.5	-	45	-
2008-045-N020S04	0.8	4	2	-	45	-
2009-045-N025S04	0.9	4	2.5	-	45	-
2010-045-N030S04	1	4	3	-	45	0.95
2010-045-N050S04	1	4	3	5	45	0.95
2010-060-N050S04	1	4	3	5	60	0.95
2010-060-N100S04	1	4	3	10	60	0.95
2010-060-N150S04	1	4	3	15	60	0.95
2010-060-N200S04	1	4	3	20	60	0.95
2010-060-N250S04	1	4	3	25	60	0.95
2015-060-N050S04	1.5	4	4	5	60	1.425
2015-060-N060S04	1.5	4	4	6	60	1.425
2015-060-N100S04	1.5	4	4	10	60	1.425
2015-060-N150S04	1.5	4	4	15	60	1.425
2015-060-N200S04	1.5	4	4	20	60	1.425
2015-060-N250S04	1.5	4	4	25	60	1.425
2020-045-N060S04	2	4	6	-	45	1.9
2020-045-N080S04	2	4	6	8	45	1.9
2020-080-N080S04	2	4	6	8	80	1.9
2020-080-N100S04	2	4	6	10	80	1.9
2020-080-N120S04	2	4	6	12	80	1.9
2020-080-N150S04	2	4	6	15	80	1.9
2020-080-N200S04	2	4	6	20	80	1.9

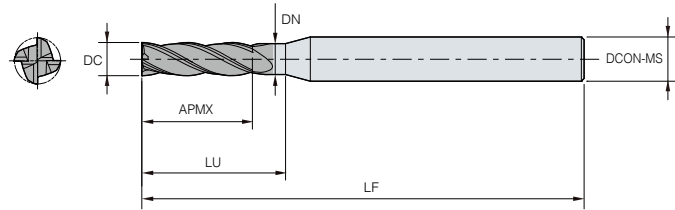
Designation	DC	DCON-MS	APMX	LU	LF	DN
DFE 2020-080-N250S04	2	4	6	25	80	1.9
2020-080-N300S04	2	4	6	30	80	1.9
2020-080-N400S04	2	4	6	40	80	1.9
2030-050-N100S06	3	6	9	10	50	2.85
2030-080-N100S04	3	4	9	10	80	2.85
2030-080-N120S04	3	4	9	12	80	2.85
2030-080-N200S04	3	4	9	20	80	2.85
2030-080-N250S04	3	4	9	25	80	2.85
2030-080-N300S04	3	4	9	30	80	2.85
2030-080-N400S04	3	4	9	40	80	2.85
2040-050-N120S06	4	6	12	-	50	2.85
2040-050-N160S06	4	6	12	16	50	3.8
2040-080-N160S04	4	4	12	16	80	3.8
2050-060-N150S06	5	6	15	-	60	4.75
2050-060-N200S06	5	6	15	20	60	4.75
2050-110-N200S06	5	6	15	20	110	4.75
2060-060-N180S06	6	6	18	-	60	6
2060-110-N250S06	6	6	18	25	110	5.7
2060-150-N250S06	6	6	18	25	150	5.7
2080-070-N250S08	8	8	25	-	70	8
2080-150-N400S08	8	8	25	40	150	7.6
2100-080-N300S10	10	10	30	-	80	10
2100-150-N500S10	10	10	30	50	150	9.5
2100-160-N500S10	10	10	30	50	160	9.5
2120-080-N250S12	12	12	25	-	80	12
2120-080-N350S12	12	12	35	-	80	12
2120-150-N600S12	12	12	35	60	150	11.4
2120-160-N600S12	12	12	35	60	160	11.4

DFE4000

Flat



DC	Tolerance
Ø2 - Ø4	0.000 ~ -0.020
Ø6 - Ø12	0.000 ~ -0.030



(mm)

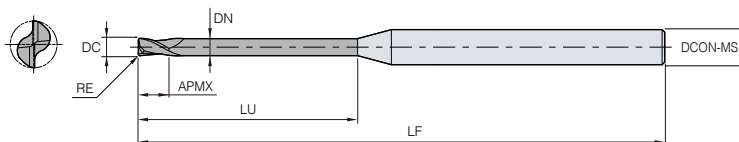
	Designation	DC	DCON-MS	APMX	LU	LF	DN
DFE	4020-045-N060S04	2	4	6	8	45	1.9
	4020-060-N100S04	2	4	10	12	60	1.9
	4030-050-N100S06	3	6	10	12	50	2.85
	4030-060-N150S03	3	3	15	-	60	2.85
	4030-060-N150S04	3	4	15	18	60	2.85
	4040-050-N120S06	4	6	12	-	50	2.85
	4040-050-N150S06	4	6	15	18	50	3.8
	4040-080-N200S04	4	4	20	-	80	4
	4060-060-N180S06	6	6	18	-	60	6
	4060-110-N300S06	6	6	30	-	110	6
	4060-150-N300S06	6	6	30	-	150	6
	4080-070-N250S08	8	8	25	-	70	8
	4080-110-N400S08	8	8	40	-	110	8
	4080-150-N400S08	8	8	40	-	150	8
	4100-080-N250S10	10	10	25	-	80	10
	4100-110-N400S10	10	10	40	-	110	10
	4100-150-N500S10	10	10	50	-	150	10
	4120-080-N300S12	12	12	30	-	80	12
	4120-110-N400S12	12	12	40	-	110	12
	4120-110-N450S12	12	12	45	-	110	12
	4120-150-N500S12	12	12	50	-	150	12
	4120-150-N550S12	12	12	55	-	150	12

DRE2000

Radius



				DC	Tolerance
				ALL	0.000 ~ -0.020



(mm)

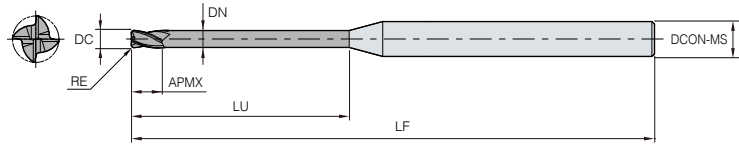
	Designation	DC	DCON-MS	APMX	LU	LF	RE	DN
DRE	2005-060-N050S04-R005	0.5	4	1	5	60	0.05	-
	2005-060-N080S04-R005	0.5	4	1	8	60	0.05	-
	2005-060-N100S04-R005	0.5	4	1	10	60	0.05	-
	2006-060-N060S04-R005	0.6	4	1.2	6	60	0.05	-
	2006-060-N090S04-R005	0.6	4	1.2	9	60	0.05	-
	2006-060-N120S04-R005	0.6	4	1.2	12	60	0.05	-
	2008-060-N040S04-R005	0.8	4	1.6	4	60	0.05	-
	2008-060-N080S04-R005	0.8	4	1.6	8	60	0.05	-
	2008-060-N100S04-R005	0.8	4	1.6	10	60	0.05	-
	2010-060-N050S04-R005	1	4	2	5	60	0.05	-
	2010-060-N100S04-R005	1	4	2	10	60	0.05	-
	2010-060-N100S04-R02	1	4	2	10	60	0.2	-
	2010-060-N150S04-R02	1	4	2	15	60	0.2	-
	2010-060-N200S04-R005	1	4	2	20	60	0.05	-
	2010-060-N200S04-R02	1	4	2	20	60	0.2	-
	2015-060-N050S04-R005	1.5	4	3	5	60	0.05	-
	2015-060-N100S04-R005	1.5	4	3	10	60	0.05	-
	2015-060-N150S04-R005	1.5	4	3	15	60	0.05	-
	2015-060-N150S04-R02	1.5	4	3	15	60	0.2	-
	2015-060-N200S04-R005	1.5	4	3	20	60	0.05	-
	2015-060-N200S04-R02	1.5	4	3	20	60	0.2	-
	2020-060-N120S04-R02	2	4	3.5	12	60	0.2	-
	2020-060-N120S04-R03	2	4	3.5	12	60	0.3	-
	2020-060-N180S04-R03	2	4	3.5	18	60	0.3	-
	2020-060-N250S04-R02	2	4	3.5	25	60	0.2	-
	2020-060-N250S04-R03	2	4	3.5	25	60	0.3	-
	2020-060-N300S04-R02	2	4	3.5	30	60	0.2	-
	2020-060-N300S04-R03	2	4	3.5	30	60	0.3	-
	2030-080-N100S04-R02	3	4	4	10	80	0.2	-
	2030-080-N200S04-R02	3	4	4	20	80	0.2	-
	2030-080-N200S04-R03	3	4	4	20	80	0.3	-
	2030-080-N300S04-R02	3	4	4	30	80	0.2	-
	2030-080-N400S04-R02	3	4	4	40	80	0.2	-
	2030-080-N400S04-R03	3	4	4	40	80	0.3	-

DRE4000

Radius



DC	Tolerance
Ø0.2 ~ Ø4	0.000 ~ -0.020
Ø6 ~ Ø12	0.000 ~ -0.030



(mm)

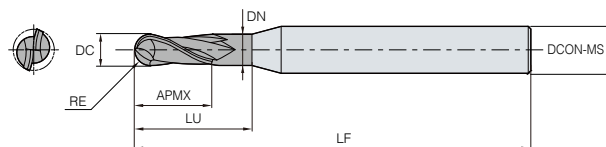
	Designation	DC	DCON-MS	APMX	LU	LF	RE	DN
DRE	4020-060-N120S04-R03	2	4	3.5	12	60	0.3	-
	4020-060-N180S04-R02	2	4	3.5	18	60	0.2	-
	4020-060-N180S04-R03	2	4	3.5	18	60	0.3	-
	4020-060-N250S04-R02	2	4	3.5	25	60	0.2	-
	4020-060-N250S04-R03	2	4	3.5	25	60	0.3	-
	4020-060-N300S04-R02	2	4	3.5	30	60	0.2	-
	4020-060-N300S04-R03	2	4	3.5	30	60	0.3	-
	4030-080-N100S04-R05	3	4	4	10	80	0.5	-
	4030-080-N200S04-R03	3	4	4	20	80	0.3	-
	4030-080-N200S04-R05	3	4	4	20	80	0.5	-
	4030-080-N300S04-R03	3	4	4	30	80	0.3	-
	4030-080-N300S04-R05	3	4	4	30	80	0.5	-
	4030-080-N400S04-R03	3	4	4	40	80	0.3	-
	4030-080-N400S04-R05	3	4	4	40	80	0.5	-
	4040-100-N200S04-R03	4	4	6	20	100	0.3	-
	4060-110-N250S06-R03	6	6	9	25	110	0.3	-
	4060-110-N250S06-R05	6	6	9	25	110	0.5	-
	4060-150-N300S06-R05	6	6	9	30	150	0.5	-
	4080-110-N300S08-R03	8	8	12	30	110	0.3	-
	4080-110-N300S08-R05	8	8	12	30	110	0.5	-
	4080-150-N400S08-R05	8	8	12	40	150	0.5	-
	4100-110-N350S10-R05	10	10	15	35	110	0.5	-
	4100-160-N450S10-R05	10	10	15	45	160	0.5	-
	4120-110-N400S12-R05	12	12	18	40	110	0.5	-
	4120-160-N450S12-R05	12	12	18	45	160	0.5	-

DBE2000

Ball



2	H-A 30°	h6 shank	Grade ND3000	DC	Tolerance
				Ø0.3 ~ Ø5	0.000 ~ -0.020
				Ø6 ~ Ø12	0.000 ~ -0.030



(mm)

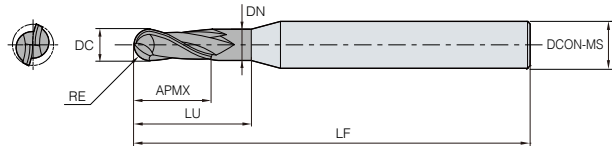
	Designation	DC	DCON-MS	APMX	LU	LF	RE	DN
DBE	2003-045-N012S04	0.3	4	1.2	-	45	0.15	-
	2003-045-N020S04	0.3	4	1.2	2	45	0.15	-
	2004-045-N015S04	0.4	4	1.5	-	45	0.2	-
	2004-045-N020S04	0.4	4	1.5	2	45	0.2	-
	2004-045-N030S04	0.4	4	1.5	3	45	0.2	-
	2004-045-N040S04	0.4	4	1.5	4	45	0.2	-
	2004-045-N050S04	0.4	4	1.5	5	45	0.2	-
	2005-045-N020S04	0.5	4	2	-	45	0.25	-
	2005-045-N030S04	0.5	4	2	3	45	0.25	-
	2005-045-N040S04	0.5	4	2	4	45	0.25	-
	2005-045-N050S04	0.5	4	2	5	45	0.25	-
	2005-045-N060S04	0.5	4	2	6	45	0.25	-
	2006-045-N020S04	0.6	4	2	2	45	0.3	0.6
	2006-045-N030S04	0.6	4	2	3	45	0.3	0.6
	2006-045-N040S04	0.6	4	2	4	45	0.3	0.57
	2006-045-N050S04	0.6	4	2	5	45	0.3	0.57
	2006-045-N060S04	0.6	4	2	6	45	0.3	0.57
	2006-045-N080S04	0.6	4	2	8	45	0.3	0.57
	2006-045-N100S04	0.6	4	2	10	45	0.3	0.57
	2008-045-N030S04	0.8	4	2.5	3	45	0.4	0.76
	2008-045-N040S04	0.8	4	2.5	4	45	0.4	0.76
	2008-045-N050S04	0.8	4	2.5	5	45	0.4	0.76
	2008-045-N060S04	0.8	4	2.5	6	45	0.4	0.76
	2008-045-N080S04	0.8	4	2.5	8	45	0.4	0.76
	2008-045-N100S04	0.8	4	2.5	10	45	0.4	0.76
	2010-060-N030S04	1	4	3	3	60	0.5	0.95
	2010-060-N040S04	1	4	3	4	60	0.5	0.95
	2010-060-N050S04	1	4	3	5	60	0.5	0.95
	2010-060-N060S04	1	4	3	6	60	0.5	0.95
	2010-060-N080S04	1	4	3	8	60	0.5	0.95
	2010-060-N100S04	1	4	3	10	60	0.5	0.95
	2010-060-N120S04	1	4	3	12	60	0.5	0.95
	2010-060-N150S04	1	4	3	15	60	0.5	0.95
	2010-060-N200S04	1	4	3	20	60	0.5	0.95
	2010-080-N250S04	1	4	3	25	80	0.5	0.95
	2010-080-N300S04	1	4	3	30	80	0.5	0.95

DBE2000

Ball



				DC	Tolerance
				Ø0.3 ~ Ø5 Ø6 ~ Ø12	0.000 ~ -0.020 0.000 ~ -0.030



(mm)

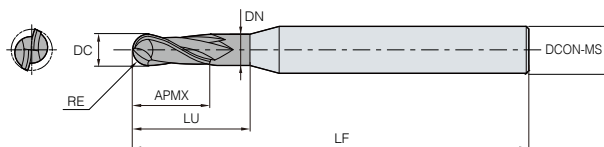
	Designation	DC	DCON-MS	APMX	LU	LF	RE	DN
DBE	2010-080-N350S04	1	4	3	35	80	0.5	0.95
	2010-080-N400S04	1	4	3	40	80	0.5	0.95
	2015-060-N050S04	1.5	4	4	5	60	0.75	1.425
	2015-060-N080S04	1.5	4	4	8	60	0.75	1.425
	2015-080-N100S04	1.5	4	4	10	80	0.75	1.425
	2015-080-N120S04	1.5	4	4	12	80	0.75	1.425
	2015-080-N150S04	1.5	4	4	15	80	0.75	1.425
	2015-080-N180S04	1.5	4	4	18	80	0.75	1.425
	2015-080-N200S04	1.5	4	4	20	80	0.75	1.425
	2015-080-N250S04	1.5	4	4	25	80	0.75	1.425
	2015-080-N300S04	1.5	4	4	30	80	0.75	1.425
	2015-080-N350S04	1.5	4	4	35	80	0.75	1.425
	2015-080-N400S04	1.5	4	4	40	80	0.75	1.425
	2020-060-N060S04	2	4	6	-	60	1	1.9
	2020-060-N080S04	2	4	6	8	60	1	1.9
	2020-080-N100S04	2	4	6	10	80	1	1.9
	2020-080-N150S04	2	4	6	15	80	1	1.9
	2020-080-N200S04	2	4	6	20	80	1	1.9
	2020-080-N250S04	2	4	6	25	80	1	1.9
	2020-080-N300S04	2	4	6	30	80	1	1.9
	2020-080-N350S04	2	4	6	35	80	1	1.9
	2020-100-N400S04	2	4	6	40	100	1	1.9
	2020-100-N450S04	2	4	6	45	100	1	1.9
	2020-100-N500S04	2	4	6	50	100	1	1.9
	2020-100-N600S04	2	4	6	60	100	1	2.85
	2030-060-N100S04	3	4	9	10	60	1.5	2.85
	2030-100-N150S04	3	4	9	15	100	1.5	2.85
	2030-100-N150S06	3	6	9	15	100	1.5	2.85
	2030-100-N200S04	3	4	9	20	100	1.5	2.85
	2030-100-N250S04	3	4	9	25	100	1.5	2.85
	2030-100-N300S04	3	4	9	30	100	1.5	2.85
	2030-100-N350S04	3	4	9	35	100	1.5	2.85
	2030-100-N400S04	3	4	9	40	100	1.5	2.85
	2030-100-N500S04	3	4	9	50	100	1.5	2.85
	2030-100-N600S04	3	4	9	60	100	1.5	2.85
	2040-060-N160S04	4	4	12	16	60	2	3.8

DBE2000

Ball



				DC Ø0.3 ~ Ø5 Ø6 ~ Ø12	Tolerance 0.000 ~ -0.020 0.000 ~ -0.030
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(mm)

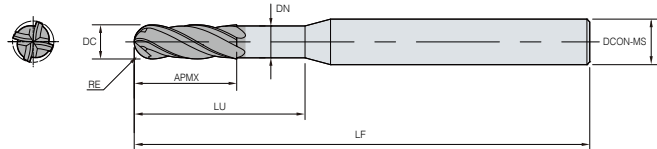
	Designation	DC	DCON-MS	APMX	LU	LF	RE	DN
DBE	2040-080-N160S04	4	4	12	16	80	2	3.8
	2040-080-N300S04	4	4	12	30	80	2	3.8
	2040-100-N160S04	4	4	12	16	100	2	3.8
	2040-100-N400S04	4	4	12	40	100	2	3.8
	2040-130-N160S04	4	4	12	16	130	2	3.8
	2040-130-N400S04	4	4	12	40	130	2	3.8
	2040-150-N160S04	4	4	12	16	150	2	3.8
	2040-150-N500S04	4	4	12	50	150	2	3.8
	2050-110-N200S06	5	6	15	20	110	2.5	4.75
	2060-080-N250S06	6	6	20	25	80	3	5.7
	2060-110-N250S06	6	6	20	25	110	3	5.7
	2060-110-N400S06	6	6	20	40	110	3	5.7
	2060-150-N300S06	6	6	20	30	150	3	5.7
	2060-150-N500S06	6	6	20	50	150	3	5.7
	2080-080-N300S08	8	8	25	30	80	4	7.6
	2080-110-N300S08	8	8	25	30	110	4	7.6
	2080-110-N400S08	8	8	25	40	110	4	7.6
	2080-150-N500S08	8	8	25	50	150	4	7.6
	2080-200-N400S08	8	8	25	40	200	4	7.6
	2100-080-N400S10	10	10	30	40	80	5	9.5
	2100-110-N400S10	10	10	30	40	110	5	9.5
	2100-110-N500S10	10	10	30	50	110	5	9.5
	2100-150-N600S10	10	10	30	60	150	5	9.5
	2100-160-N600S10	10	10	30	60	160	5	9.5
	2100-200-N500S10	10	10	30	50	200	5	9.5
	2120-110-N500S12	12	12	35	50	110	6	11.4
	2120-150-N500S12	12	12	35	50	150	6	11.4
	2120-160-N500S12	12	12	35	50	160	6	11.4
	2120-200-N600S12	12	12	35	60	200	6	11.4

DBE4000

Ball



4	H-A 30°	h6 shank	Grade ND3000	DC	Tolerance
				Ø1 - Ø4 Ø6 - Ø12	0.000 ~ -0.020 0.000 ~ -0.030



(mm)

	Designation	DC	DCON-MS	APMX	LU	LF	RE	DN
DBE	4010-060-N030S04	1	4	3	-	60	0.5	-
	4010-060-N050S04	1	4	3	5	60	0.5	-
	4010-060-N100S04	1	4	3	10	60	0.5	-
	4010-060-N150S04	1	4	3	15	60	0.5	-
	4010-060-N200S04	1	4	3	20	60	0.5	-
	4010-060-N250S04	1	4	3	25	60	0.5	-
	4015-045-N040S04	1.5	4	4	-	45	0.75	-
	4015-050-N040S04	1.5	4	4	-	50	0.75	-
	4015-060-N100S04	1.5	4	4	10	60	0.75	-
	4015-060-N150S04	1.5	4	4	15	60	0.75	-
	4015-060-N200S04	1.5	4	4	20	60	0.75	-
	4015-060-N250S04	1.5	4	4	25	60	0.75	-
	4020-060-N080S04	2	4	6	8	60	1	1.9
	4020-080-N100S04	2	4	6	10	80	1	1.9
	4020-080-N200S04	2	4	6	20	80	1	1.9
	4020-080-N300S04	2	4	6	30	80	1	1.9
	4020-080-N400S04	2	4	6	40	80	1	1.9
	4030-060-N100S04	3	4	9	10	60	1.5	2.85
	4030-100-N150S04	3	4	9	15	100	1.5	2.85
	4030-100-N200S04	3	4	9	20	100	1.5	2.85
	4030-100-N300S04	3	4	9	30	100	1.5	2.85
	4030-100-N400S04	3	4	9	40	100	1.5	2.85
	4030-100-N500S04	3	4	9	50	100	1.5	2.85
	4040-060-N160S04	4	4	12	16	60	2	-
	4040-080-N160S04	4	4	12	16	80	2	-
	4040-100-N160S04	4	4	12	16	100	2	-
	4040-130-N160S04	4	4	12	16	130	2	-
	4060-080-N250S06	6	6	20	25	80	3	-
	4060-110-N250S06	6	6	20	25	110	3	-
	4060-150-N300S06	6	6	20	30	150	3	-
	4080-080-N300S08	8	8	25	30	80	4	-
	4080-110-N300S08	8	8	25	30	110	4	-
	4080-150-N350S08	8	8	25	35	150	4	-
	4080-200-N400S08	8	8	25	40	200	4	-
	4100-080-N350S10	10	10	30	35	80	5	-
	4100-110-N350S10	10	10	30	35	110	5	-
	4100-150-N400S10	10	10	30	40	150	5	-
	4100-160-N400S10	10	10	30	40	160	5	-
	4100-200-N500S10	10	10	30	50	200	5	-
	4120-110-N500S12	12	12	35	50	110	6	-
	4120-150-N500S12	12	12	35	50	150	6	-
	4120-200-N600S12	12	12	35	60	200	6	-

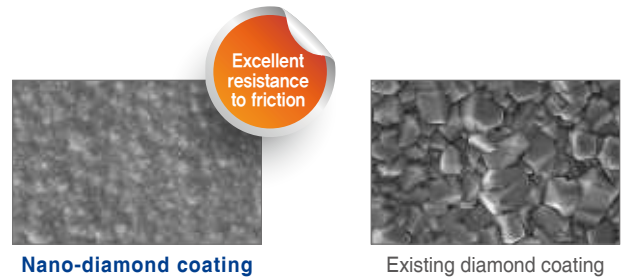
Router Endmill for machining composite materials

Composite Router Endmill

- Router Endmills optimized for machining composite materials (CFRP/GFRP)
- Excellent tool life thanks to nano-crystal diamond coating
- Blade design for reducing flaking and burrs
- Improved productivity through high efficiency machining

Features

- Diamond-coated grade ND2100 for machining composite materials
- High hardness diamond coating (over Hv 8,000)
- Nano-diamond coating with excellent resistance to friction and welding
- Improved resistance to flaking thanks by applying



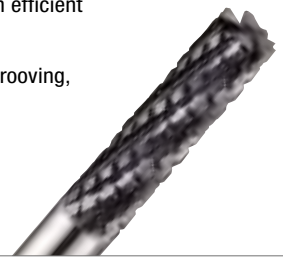
CCDR (Dual Helix Router Endmill)

- Dual helix design to inhibit flaking on upper and lower faces of workpieces
- Endmill for finishing, profiling, and grooving



CCHR (High-performance Router Endmill)

- Multi flute neck shaped for high efficient machining
- Endmill for shape contouring, grooving, roughing



CCR (Router Endmill)

- Down cut design for low vibrations and cutting force
- Endmill for roughing, profiling, and grooving



CCLR (Low Helix Router Endmill)

- Fewer burrs thanks to the low axial cutting force
- Endmill for finishing, profiling, and blind groove making



CCRR (Reverse Helix Router Endmill)

- Reverse helix design to inhibit a drift in the workpiece's course
- Endmill for finishing, profiling, and through groove making

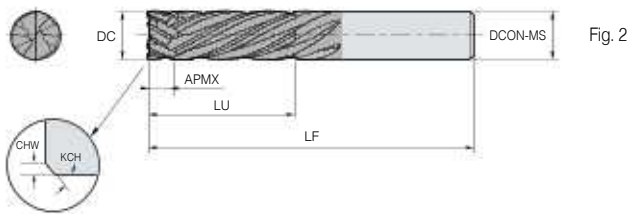
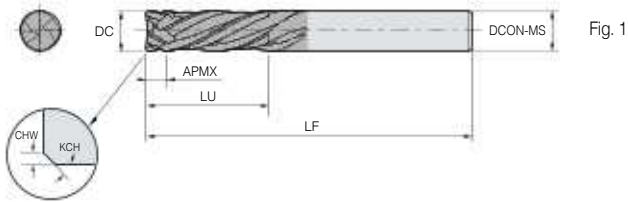


CCDR4000/6000

Flat



4	6	H-A 30°/30°	h5 shank	Grade ND2100	DC	Tolerance
					Ø6 - Ø12	0.000 ~ -0.030



(mm)

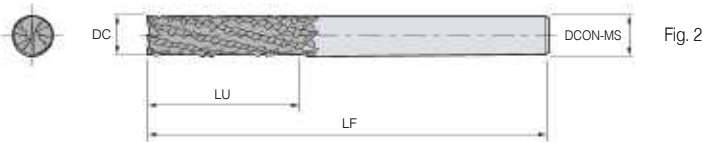
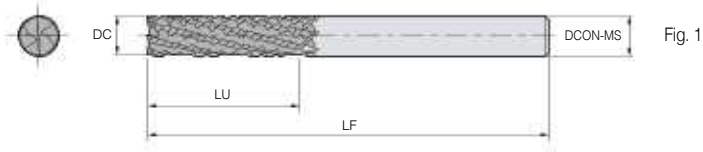
	Designation	DC	DCON-MS	APMX	LU	LF	KCH	CHW	Fig.
CCDR	4060-065	6	6	3	18	65	45°	0.1	1
	4080-075	8	8	4	24	75	45°	0.2	1
	6100-085	10	10	5	30	85	45°	0.2	2
	6120-100	12	12	6	36	100	45°	0.2	2

CCHR4000/6000

Flat



4	6	H-A 15°	h5 shank	Grade ND2100	DC	Tolerance
					Ø6 - Ø12	0.000 ~ -0.050



(mm)

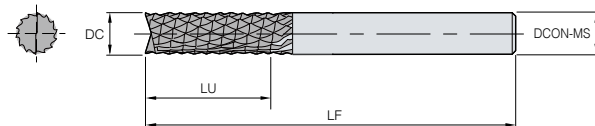
	Designation	DC	DCON-MS	APMX	LF	Fig.
CCHR	4060-065	6	6	18	65	1
	4080-075	8	8	24	75	1
	6100-085	10	10	30	85	2
	6120-100	12	12	36	100	2

CCR2000

Flat



				DC	Tolerance
				Ø4 ~ Ø12	-0.020 ~ -0.080



(mm)

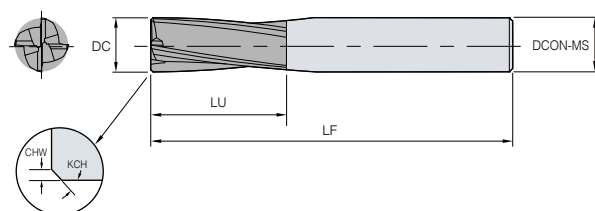
	Designation	DC	DCON-MS	APMX	LF
CCR	2040-050	4	4	12	50
	2050-050	5	5	15	50
	2060-065	6	6	18	65
	2080-075	8	8	24	75
	2100-085	10	10	30	85
	2120-100	12	12	36	100

CCLR4000

Flat



				DC	Tolerance
				Ø4 ~ Ø12	0.000 ~ -0.030



(mm)

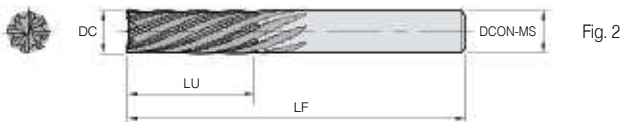
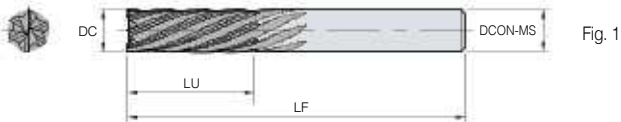
	Designation	DC	DCON-MS	APMX	LF	KCH	CHW
CCLR	4040-050	4	4	12	50	45°	0.1
	4050-050	5	5	15	50	45°	0.1
	4060-065	6	6	18	65	45°	0.1
	4080-075	8	8	24	75	45°	0.2
	4100-085	10	10	30	85	45°	0.2
	4120-100	12	12	36	100	45°	0.2

CCRR6000/8000

Flat



6	8	H-A -25°	h5 shank	Grade ND2100	DC	Tolerance
					Ø6 ~ Ø12	0.000 ~ -0.030



(mm)

Designation		DC	DCON-MS	APMX	LF	Fig.
CCRR	6060-065	6	6	18	65	1
	6080-075	8	8	24	75	1
	8100-085	10	10	30	85	2
	8120-100	12	12	36	100	2

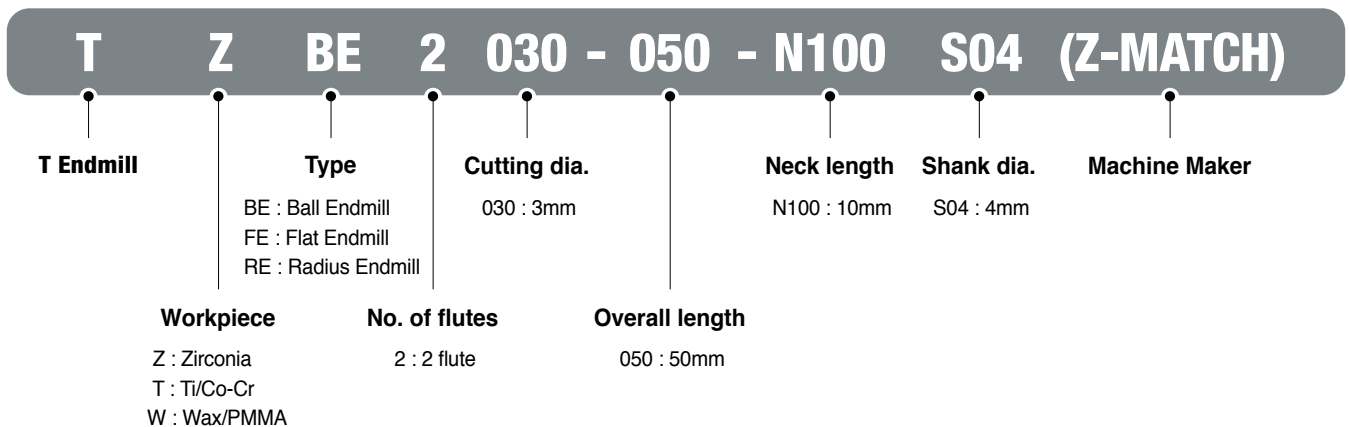
B Technical Information for T Endmill

Endmill for machining dental prostheses

T Endmill

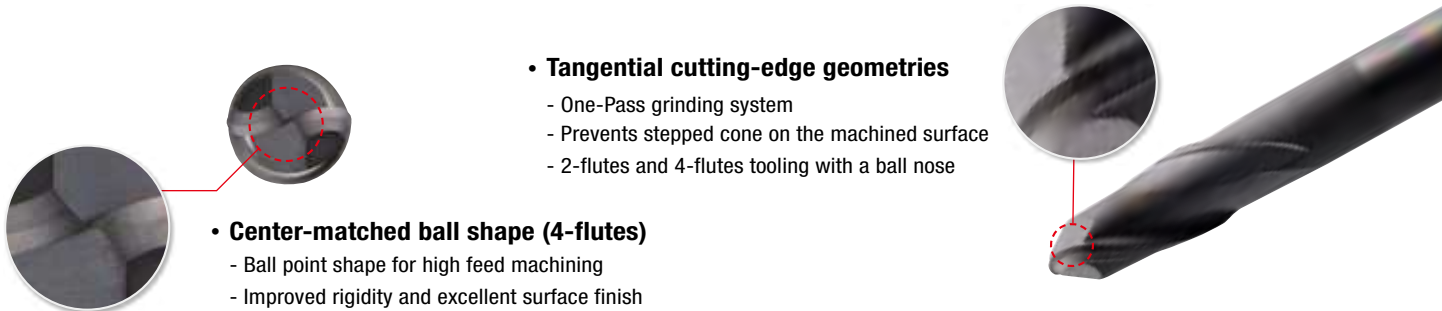
- For machining dental prostheses made of zirconia, titanium, Co-Cr, wax, PMMA, etc
- Optimized cutting performance by matching a proper grade with each type of materials
- Inhibited unevenness and excellent finish in machined surfaces due to the optimized cutting-edge design
- Specialized tool shape for each machine type

Code system



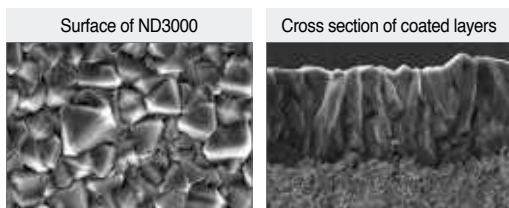
Features

- **A dedicated tool for each machine** - Meets marketplace demands
- **A specialized grade for each workpiece** - Provides optimized performance for various materials of implants
- **Optimized cutting-edge design** - Enables excellent machinability



Grade solution for zirconia

- **Development of ND3000 (Diamond-coated grade)**
 - High hardness diamond coating that is excellent in machining graphite and ceramic
 - Optimized for high speed and medium duty cutting thanks to its excellent grip to coated layers

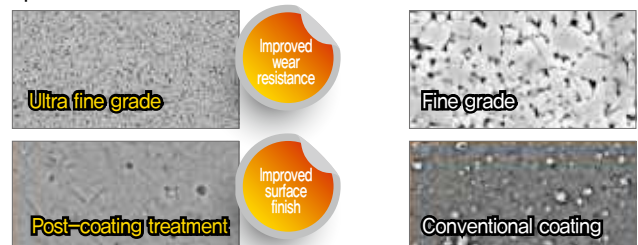


Surface of ND3000
High hardness diamond coating (Hv 10,000) provides excellent wear resistance

Cross section of coated layers
Specialized grade for Zirconia provides excellent adhesion

Grade solution for titanium

- **Development of PC2510 (Coated grade for high hardened steel)**
 - Post-coating treatment was applied to improve surface finish
 - A grade optimized for interrupted machining of high hardness steels and wet treatment accompanying high thermal shock.
- Its ultrafine substrate features high toughness which allows stable performance

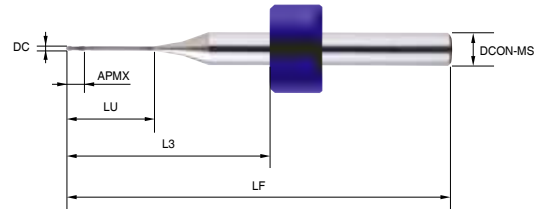


Special T Endmill order form

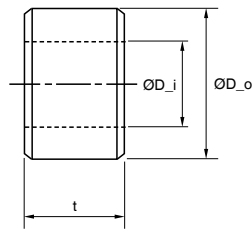
- Stop rings and other tool resources can be made to order

[Data Sheet]

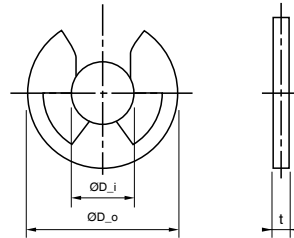
Type of machine	
Workpiece	
Dental material	
Cutting diameter (DC)	
Shank diameter (DCON-MS)	
Cutting length (APMX)	
Neck length (LU)	
Stop ring position (L3)	
Overall length (LF)	
Stop ring shape	



[Stop ring specification]



< Plastic ring >



< E type ring >

Type	Stop ring			Shank diameter		
	ØD_o	ØD_i	t	Ø3	Ø4	~ Ø6
Plastic ring	Ø7.55	Ø3	4.45	●		
	Ø7.7	Ø4	5.0		●	
	Ø10.5	~ Ø6	6.5			●
E type ring	~ Ø6.0	Ø2.5	0.4	●		

(mm)

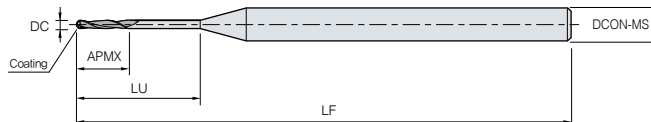
※ Stop ring can be made to order when specified sizes are send to an adjacent KORLOY sales office

ROLAND Type

(DWX-30, DWX-50, DWX-51D, DWX-52D)



2	H-A 30°	h5 shank	Grade PC2510 PC2010 PVD	Grade ND3000 DIA	DC Ø0.3 ~ Ø1 Ø2	Tolerance 0.000 ~ -0.015 0.000 ~ -0.020
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(mm)

	Designation	Coating	DC	DCON-MS	APMX	LU	LF	RE
TZBE	2003-050-N015S04(ROLAND)	-	0.3	4	1.5	20	50	DC/2
	2006-050-N12(ROLAND)	PVD	0.6	4	1.5	12	50	DC/2
	2010-050-N16(ROLAND)	DIA	1	4	2	16	50	DC/2
	2020-050-N20(ROLAND)	DIA	2	4	3	20	50	DC/2
TWBE	2010-050-N16(ROLAND)	-	1	4	2.2	16	50	DC/2
	2020-050-N20(ROLAND)	-	2	4	3.2	20	50	DC/2
ETRE	006050-S4M325-R	E.D DIA	0.6	4	10	10	50	DC/2
	010050-S4M170-R	E.D DIA	1	4	12	12	50	DC/2
	020050-S4M120-R	E.D DIA	2	4	17	17	50	DC/2

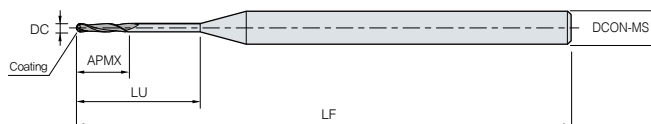
※ E.D DIA: Electro deposition DIA

IMES-ICORE Type

(250i, 340i, 450i)



2	H-A 30°	h5 shank	Grade PC2510 PC2010 PVD	Grade ND3000 DIA	DC Ø0.3 ~ Ø1 Ø2 ~ Ø13	Tolerance 0.000 ~ -0.015 0.000 ~ -0.020
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(mm)

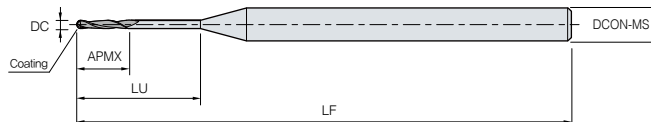
	Designation	Coating	DC	DCON-MS	APMX	LU	LF	RE
TZBE	2003-048-N015S03(450I)	PVD	0.3	3	1.8	3	48	DC/2
	2006-048-N080S03(450I)	PVD	0.6	3	1.5	8	48	DC/2
	2010-048-N14(450I)	DIA	1	3	4.5	14	48	DC/2
	2025-048-N20(450I)	DIA	2.5	3	6	20	48	DC/2
	2006-053-N090S06(450IS6)	PVD	0.6	6	3	9	53	DC/2
	2010-053-N142S06(450IS6)	DIA	1	6	4.5	14.2	53	DC/2
	2025-053-N201S06(450IS6)	DIA	2.5	6	7.5	20.1	53	DC/2
TWBE	2010-048-N140S03(250I340I450I)	-	1	3	4.5	14	48	DC/2
	2025-048-N200S03(250I340I450I)	-	2.5	3	6	20	48	DC/2
TTBE	2010-040-N090S03(450i)	PVD	1	3	2	9	40	DC/2
	2020-040-N120S03(450i)	PVD	2	3	4	12	40	DC/2
	2030-0385-N120S03(450i)	PVD	3	3	4	12	38.5	DC/2
ETND	006039-S3M325-T5(450i)	E.D DIA	10.3	3	10.3	10.3	39.8	DC/2
ETRE	010039-S3M230(450i)	E.D DIA	10.2	3	10.2	10.2	39.8	DC/2
	025039-S3M120(450i)	E.D DIA	13	3	13	13	39.8	DC/2

※ E.D DIA: Electro deposition DIA

WIELAND SELECT Type



DC	Tolerance
Ø1.0	0.000 ~ -0.015
Ø1.1 ~ Ø3.0	0.000 ~ -0.020



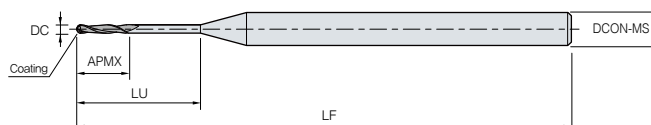
(mm)

Designation		Coating	DC	DCON-MS	APMX	LU	LF	RE
TZBE	2007-040-N145S03(WIELAND)	PVD	0.7	3	2.5	14.5	40	DC/2
	2010-040-N145S03(WIELAND)	DIA	1	3	4.5	14.5	40	DC/2
	2025-040-N200S03(WIELAND)	DIA	2.5	3	7.5	20	40	DC/2
TWBE	2010-040-N145S03(WIELAND)	-	1	3	4.5	14.5	40	DC/2
	2025-040-N200S03(WIELAND)	-	2.5	3	7.5	20	40	DC/2

WIELAND MINI Type



DC	Tolerance
Ø1.0	0.000 ~ -0.015
Ø1.1 ~ Ø3.0	0.000 ~ -0.020



(mm)

Designation		Coating	DC	DCON-MS	APMX	LU	LF	RE
TZBE	2010-035-N140S03(WIELAND MINI)	-	1	3	4.5	14	35	DC/2
	2025-035-N200S03(WIELAND MINI)	-	2.5	3	6	20	35	DC/2

AMANN GIRRBACH Type

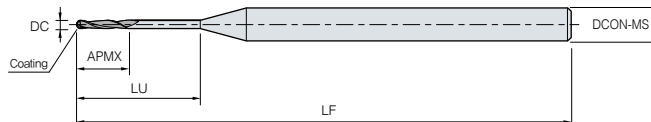
(CERA-MILL)



DC	Tolerance
Ø1.0 Ø1.1 ~ Ø3.0	0.000 ~ -0.015 0.000 ~ -0.020

PVD

DIA



(mm)

Designation	Coating	DC	DCON-MS	APMX	LU	LF	RE
TZBE	2006-047-N135S03(CERAMILL)	PVD	0.6	3	2.4	13.5	DC/2
	2010-047-N165S03(CERAMILL)	DIA	1	3	2	16.5	DC/2
	2025-047-N180S03(CERAMILL)	DIA	2.5	3	4	18	DC/2
TWBE	2010-047-N165S03(CERAMILL)	-	1	3	2	16.5	DC/2
	2025-047-N180S03(CERAMILL)	-	2.5	3	4	18	DC/2

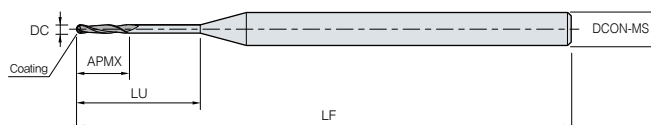
VHF-S1 Type



DC	Tolerance
Ø1.0 Ø1.1 ~ Ø3.0	0.000 ~ -0.015 0.000 ~ -0.020

PVD

DIA



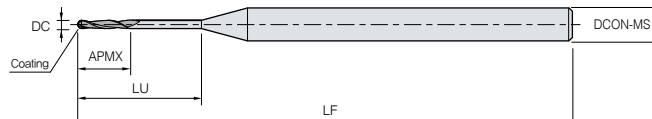
(mm)

Designation		Coating	DC	DCON-MS	APMX	LU	LF	RE
TZBE	2006-035-N030S03(VHF S1)	PVD	0.6	3	1.5	3	35	DC/2
	2010-040-N145S03(VHF S1)	DIA	1	3	5	14.5	40	DC/2
	2020-040-N160S03(VHF S1)	DIA	2	3	4	16	40	DC/2
TWBE	2010-040-N145S03(VHF S1)	-	1	3	5	14.5	40	DC/2
	2020-040-N160S03(VHF S1)	-	2	3	4	16	40	DC/2

VHF-K4/CM Type



DC	Tolerance
Ø1.0	0.000 ~ -0.015
Ø1.1 ~ Ø3.0	0.000 ~ -0.020



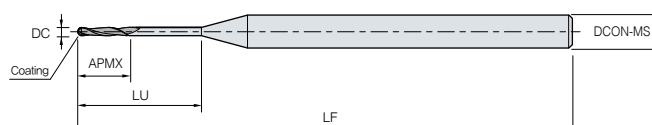
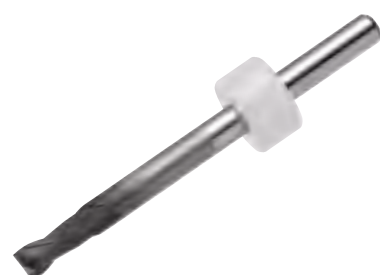
(mm)

Designation		Coating	DC	DCON-MS	APMX	LU	LF	RE
TWBE	2006-035-N030S03(VHF K4)	-	0.6	3	1	3	35	DC/2
	2010-035-N160S03(VHF K4)	-	1	3	2	16	35	DC/2
	3020-035-N160S03(VHF K4)	-	2	3	4	16	35	DC/2

CHARLY Type



DC	Tolerance
Ø1.0	0.000 ~ -0.015
Ø1.1 ~ Ø3.0	0.000 ~ -0.020



(mm)

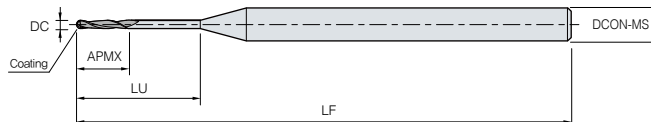
Designation		Coating	DC	DCON-MS	APMX	LU	LF	RE
TZBE	2005-0385-N100S03(CHARLY)	PVD	0.5	3	1.5	10	38.5	DC/2
	2010-0385-N120S03(CHARLY)	DIA	1	3	2	12	38.5	DC/2
	2015-0385-N120S03(CHARLY)	DIA	1.5	3	3	12	38.5	DC/2
TZFE	2030-0385-N060S03(CHARLY)-DY	DIA	3	3	6	-	38.5	DC/2
TZBE	2010-0385-N120S03(CHARLY)-N	-	1	3	2	12	38.5	DC/2
	2015-0385-N120S03(CHARLY)-N	-	1.5	3	3	12	38.5	DC/2
	2030-0385-N060S03(CHARLY)-N	-	3	3	6	-	38.5	DC/2

ZIRKONZAHN M5 Type



DC	Tolerance
Ø1.0	0.000 ~ -0.015
Ø1.1 ~ Ø3.0	0.000 ~ -0.020

DIA



(mm)

Designation		Coating	DC	DCON-MS	APMX	LU	LF	RE
TZBE	2010-057-N120S03(ZIRKON)-DYD	DIA	1	3	6	12	57	DC/2
	2020-057-N180S03(ZIRKON)-DYD	DIA	2	3	10	18	57	DC/2
	2003-057-N060S03(ZIRKONZAHN)-DY	-	0.3	3	3	6	57	DC/2
	2005-057-N100S03(ZIRKONZHAN)-DY	-	0.5	3	3	10	57	DC/2
	2010-057-N120S03(ZIRKONZHAN)-DY	-	1	3	6	12	57	DC/2
	2020-057-N180S03(ZIRKONZHAN)-DY	-	2	3	10	18	57	DC/2

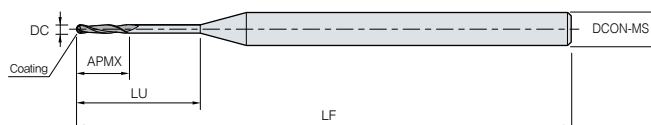
ZIRKONZAHN M1/M4 Type



DC	Tolerance
Ø1.0	0.000 ~ -0.015
Ø1.1 ~ Ø3.0	0.000 ~ -0.020

PVD

DIA

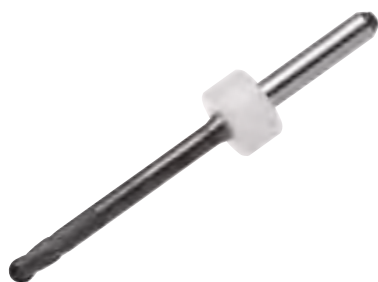


(mm)

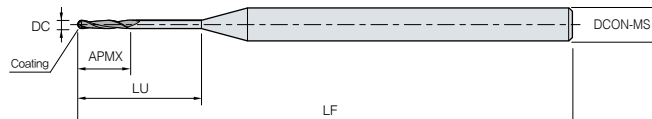
Designation		Coating	DC	DCON-MS	APMX	LU	LF	RE
TZBE	2005-050-N108S06(ZIRKON M1)	PVD	0.5	6	3.5	10.8	50	DC/2
	2010-050-N120S06(ZIRKON M1)	DIA	1	6	6	12	50	DC/2
	2020-050-N180S06(ZIRKON M1)	DIA	2	6	10	18	50	DC/2
	2005-050-N095S06(ZIRKON M1)	-	0.5	6	3	9.5	50	DC/2
	2010-050-N120S06(ZIRKON M1)-N	-	1	6	6	12	50	DC/2
	2020-050-N180S06(ZIRKON M1)-N	-	2	6	10	18	50	DC/2
TTBE	2010-050-N120S06(ZIRKON M1)	PVD	1	6	1	12	50	DC/2
	2020-050-N120S06(ZIRKON M1)	PVD	2	6	3	12	50	DC/2
	2030-050-N180S06(ZIRKON M1)	PVD	3	6	4	18	50	DC/2

SIRONA Type

(MC-X5)



2	H-A 30°	h5 shank	Grade PC2510 PC2010 PVD	Grade ND3000 DIA	DC Ø1.0 Ø1.1 ~ Ø3.0	Tolerance 0.000 ~ -0.015 0.000 ~ -0.020
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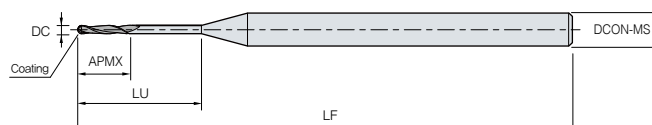


(mm)

	Designation	Coating	DC	DCON-MS	APMX	LU	LF	RE
TZBE	2005-042-N060S03(MC-X5)	PVD	0.5	3	1	6	42	DC/2
	2010-043-N170S03(MC-X5)	DIA	1	3	3	17	43	DC/2
	4025-044-N240S03(MC-X5)	DIA	2.5	3	5	24	44	DC/2
TWBE	2010-043-N170S03(MC-X5)	-	1	3	3	17	43	DC/2
	2025-044-N240S3(MC-X5)	-	2.5	3	5	24	44	DC/2

YENA Type

2	H-A 30°	h5 shank	Grade PC2510 PC2010 PVD	Grade ND3000 DIA	DC Ø1.0 Ø1.1 ~ Ø3.0	Tolerance 0.000 ~ -0.015 0.000 ~ -0.020
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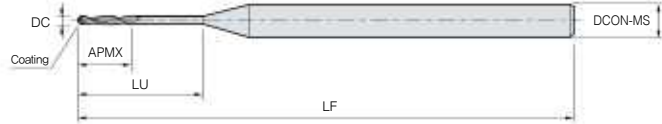


(mm)

	Designation	Coating	DC	DCON-MS	APMX	LU	LF	RE
TZBE	2010-045-N160S04(YENA)-DY	DIA	1	4	3	16	45	DC/2
	2020-045-N160S04(YENA)-DY	DIA	2	4	3	16	45	DC/2
TWBE	2010-045-N160S04(YENA)-DY	-	1	4	3	16	45	DC/2
	2020-045-N160S04(YENA)-DY	-	2	4	3	16	45	DC/2
TTBE	2020-045-N122S04(YENA)	PVD	2	4	3.6	12.2	45	DC/2
	2030-045-N125S04(YENA)	PVD	3	4	5.5	12.5	45	DC/2

LEZIRTH Type

			Grade PC2510 PC2010 PVD	DC Ø1.0 Ø1.1 ~ Ø3.0	Tolerance 0.000 ~ -0.015 0.000 ~ -0.020
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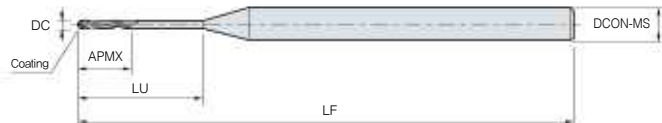
(mm)

	Designation	Coating	DC	DCON-MS	APMX	LU	LF	RE
TTBE	2010-050-V1.2N8S6	PVD	1	6	1.2	8	50	DC/2
	2015-045-V1.7N8.5S6	PVD	1.5	6	1.7	8.5	45	DC/2
TTE	2020-050-V2.2N12S6	PVD	2	6	2.2	12	50	DC/2
	2030-060-V3.5N16S6	PVD	3	6	3.5	16	60	DC/2

ARUM Type

(4X/5X-100)

			Grade PC2510 PC2010 PVD	Grade ND3000 DIA	DC Ø1.0 Ø1.1 ~ Ø3.0	Tolerance 0.000 ~ -0.015 0.000 ~ -0.020
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(mm)

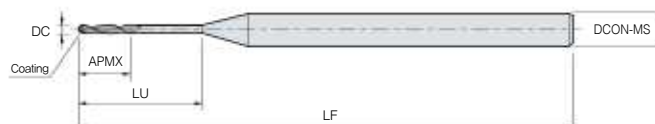
	Designation	Coating	DC	DCON-MS	APMX	LU	LF	RE
TZBE	2006-063-N120S06(4X-100)	PVD	0.6	6	1.5	12	63	DC/2
	2010-063-N160S06(4X-100)	DIA	1	6	2.5	16	63	DC/2
	2020-063-N200S06(4X-100)	DIA	2	6	6	20	63	DC/2
TWBE	2010-063-N160S06(4X-100)	-	1	6	2.5	16	63	DC/2
	2020-063-N200S06(4X-100)	-	2	6	6	20	63	DC/2

ARUM Type

(5X-150)



2	H-A 30°	h5 shank	Grade PC2510 PC2010 PVD	Grade ND3000 DIA	DC	Tolerance
					Ø1.0 Ø1.1 ~ Ø3.0	0.000 ~ -0.015 0.000 ~ -0.020



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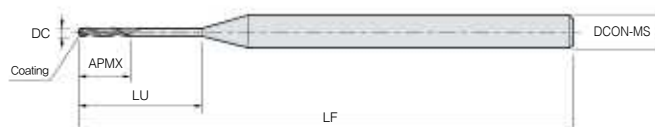
* E.D DIA: Electro deposition DIA

ARUM Type

(5X-200)



2	H-A 30°	h5 shank	Grade PC2510 PC2010 PVD	Grade ND3000 DIA	DC	Tolerance
					Ø1.0 Ø1.1 ~ Ø3.0	0.000 ~ -0.015 0.000 ~ -0.020





									(mm)
Designation		Coating	DC	DCON-MS	APMX	LU	LF	RE	
TZBE	2006-050-N080S06(5X-200)	PVD	0.6	6	1.6	8	50	DC/2	
	2010-053-N140S06(5X-200)	DIA	1	6	2	14	53	DC/2	
	2020-055-N200S06(5X-200)	DIA	2	6	6	20	55	DC/2	
TWBE	2010-053-N140S06(5X-200)	-	1	6	2	14	53	DC/2	
	2020-055-N200S06(5X-200)	-	2	6	6	20	55	DC/2	
ETND	006050-H6M270-T7	E.D DIA	0.6	6	8	8	50	DC/2	
ETRE	010050-H6M200	E.D DIA	1	6	10	10	50	DC/2	
	015050-H6M170	E.D DIA	1.5	6	10	10	50	DC/2	
	025050-H6M140	E.D DIA	2.5	6	12	12	50	DC/2	

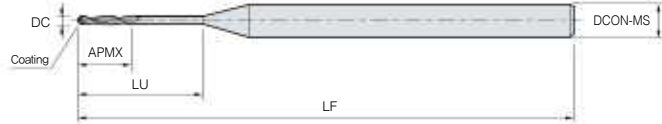
* E.D DIA: Electro deposition DIA

ARUM Type

(5X-300)



2	H-A 30°	h5 shank	Grade PC2510 PC2010 PVD	Grade ND3000 DIA	DC Ø1.0 Ø1.1 ~ Ø3.0	Tolerance 0.000 ~ -0.015 0.000 ~ -0.020
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(mm)

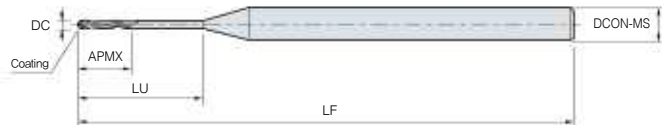
	Designation	Coating	DC	DCON-MS	APMX	LU	LF	RE
TZBE	2003-045-N120S04(5X-300)	PVD	0.3	4	0.6	12	45	DC/2
	2006-045-N100S04(5X-300)	PVD	0.6	4	1.2	10	45	DC/2
	2010-050-N160S04(5X-300)	DIA	1	4	2	16	50	DC/2
	2020-050-N180S04(5X-300)	DIA	2	4	6	18	50	DC/2
TWBE	2010-050-N160S04(5X-300)	-	1	4	2	16	50	DC/2
	2020-050-N180S04(5X-300)	-	2	4	6	18	50	DC/2

ARUM Type

(Metal Tool)



2	H-A 30°	h5 shank	Grade PC2510 PC2010 PVD	DC Ø1.0 Ø1.1 ~ Ø3.0	Tolerance 0.000 ~ -0.015 0.000 ~ -0.020
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(mm)

	Designation	Coating	DC	DCON-MS	APMX	LU	LF	RE
TTBE	2010-050-N100S06(ARUM)-DY	PVD	1	6	2	10	50	DC/2
	2015-050-N100S06(ARUM)-DY	PVD	1.5	6	3	10	50	DC/2
	2020-050-N120S06(ARUM)-DY	PVD	2	6	4	12	50	DC/2
	2025-050-N140S06(ARUM)-DY	PVD	2.5	6	8	14	50	DC/2
	2030-050-N120S06(ARUM)-DY	PVD	3	6	6	12	50	DC/2

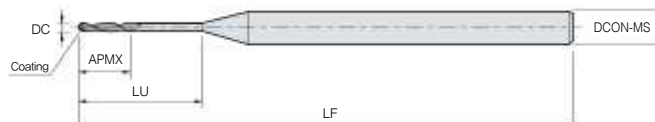
Z-MATCH/CAMELEON Type



DC	Tolerance
Ø1.0	0.000 ~ -0.015
Ø1.1 ~ Ø3.0	0.000 ~ -0.020

PVD

DIA



(mm)

Designation		Coating	DC	DCON-MS	APMX	LU	LF	RE
TZBE	2005-055-N8(Z-MATCH)	PVD	0.5	6	1.5	8	55	DC/2
	2010-055-N16(Z-MATCH)	DIA	1	6	3	16	55	DC/2
	2020-055-N20(Z-MATCH)	DIA	2	6	6	20	55	DC/2
TWBE	2010-055-N160S06(Z-MATCH)	-	1	6	3	16	55	DC/2
	2020-055-N200S06(Z-MATCH)	-	2	6	6	20	55	DC/2

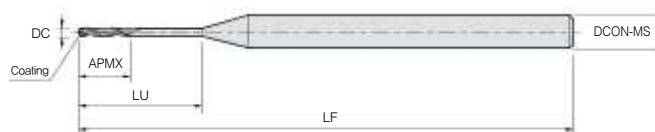
CAMELEON CS Type



DC	Tolerance
Ø1.0	0.000 ~ -0.015
Ø1.1 ~ Ø3.0	0.000 ~ -0.020

PVD

DIA



(mm)

Designation		Coating	DC	DCON-MS	APMX	LU	LF	RE
TZBE	2005-040-N080S03(CAMELEON CS)	PVD	0.5	3	1	8	40	DC/2
	2010-040-N160S03(CAMELEON CS)	DIA	1	3	2	16	40	DC/2
	2020-040-N200S03(CAMELEON CS)	DIA	2	3	4	20	40	DC/2
TWBE	2010-040-N160S03(CAMELEON CS)	-	1	3	2	16	40	DC/2
	2020-040-N200S03(CAMELEON CS)	-	2	3	4	20	40	DC/2

CAMELEON Type

(Metal Tool)



2	H-A 30°	h5 shank	Grade PC2510 PC2010 PVD	DC Ø1.0 Ø1.1 ~ Ø3.0	Tolerance 0.000 ~ -0.015 0.000 ~ -0.020
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(mm)

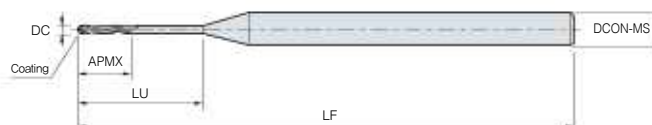
Designation	Coating	DC	DCON-MS	APMX	LU	LF	RE
TTBE 2015-045-N7(Z-MATCH-M)	PVD	1.5	6	3	7	45	DC/2
2020-045-N7(Z-MATCH-M)	PVD	2	6	3	7	45	DC/2
2030-045-N10(Z-MATCH-M)	PVD	3	6	4	10	45	DC/2
2015-055-N8(CAMELEON-M)	PVD	1.5	6	1.5	8	55	DC/2
2020-055-N8(CAMELEON-M)	PVD	2	6	2	8	55	DC/2
2030-055-N10(CAMELEON-M)	PVD	3	6	3	10	55	DC/2

RND Type

(DS200-5Z, DM-100, DS-4WA)



2	H-A 30°	h5 shank	Grade PC2510 PC2010 PVD	Grade ND3000 DIA	DC Ø1.0 Ø1.1 ~ Ø3.0	Tolerance 0.000 ~ -0.015 0.000 ~ -0.020
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(mm)

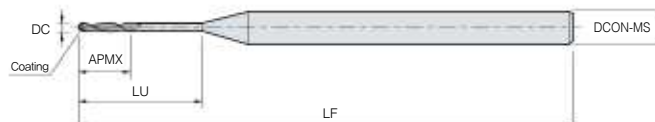
Designation	Coating	DC	DCON-MS	APMX	LU	LF	RE
TZBE 2006-045-N120S03(DS200-5Z)	PVD	0.6	3	1.2	12	45	DC/2
2010-045-N180S03(DS200-5Z)	DIA	1	3	3	18	45	DC/2
2020-045-N200S03(DS200-5Z)	DIA	2	3	4	20	45	DC/2
TWBE 2010-045-N180S03(DS200-5Z)	-	1	3	3	18	45	DC/2
2020-045-N200S03(DS200-5Z)	-	2	3	4	20	45	DC/2
TTBE 2015-050-N100S06(RND)-DY	PVD	1.5	6	3	10	50	DC/2
2020-050-N130S06(RND)-DY	PVD	2	6	4	13	50	DC/2
2030-050-N120S06(RND)-DY	PVD	3	6	6	12	50	DC/2

MANIX Type

(Metal Tool)



2	H-A 30°	h5 shank	Grade PC2510 PC2010 PVD	Grade ND3000 DIA	DC Ø1.0 Ø1.1 ~ Ø3.0	Tolerance 0.000 ~ -0.015 0.000 ~ -0.020
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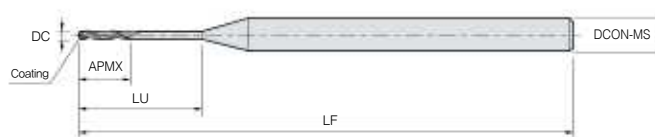


(mm)

	Designation	Coating	DC	DCON-MS	APMX	LU	LF	DC
TZBE	2006-045-N145S04(ZX-5SD)	PVD	0.6	4	1	14.5	45	DC/2
	2010-050-N160S04(ZX-5SD)	DIA	1	4	3	16	50	DC/2
	2020-050-N200S04(ZX-5SD)	DIA	2	4	6	20	50	DC/2
TWBE	2010-050-N165S04(MANIX)	-	1	4	2.8	16.5	50	DC/2
	2020-050-N200S04(MANIX)	-	2	4	5	20	50	DC/2
TTBE	2015-045-N105S06(MANIX)	PVD	1.5	6	2	10.5	45	DC/2
	2020-045-N125S06(MANIX)	PVD	2	6	4.8	12.5	45	DC/2
	2025-045-N120S06(MANIX)	PVD	2.5	6	3	12	45	DC/2
	2030-045-N125S06(MANIX)	PVD	3	6	4	12.5	45	DC/2

SEUNGWON DI Type

2	H-A 30°	h5 shank	Grade PC2510 PC2010 PVD	DC Ø1.0 Ø1.1 ~ Ø3.0	Tolerance 0.000 ~ -0.015 0.000 ~ -0.020
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(mm)

	Designation	Coating	DC	DCON-MS	APMX	LU	LF	RE
TTBE	2015-050-N100S06(SW DI-M)	PVD	1.5	6	8	10	50	DC/2
	2020-050-N120S06(SW DI-M)	PVD	2	6	8	12	50	DC/2

PROTECH INNOTION Type

(Proden, Prodia, Nexus)



2	H-A 30°	h5 shank	Grade PC2510 PC2010 PVD	Grade ND3000 DIA	DC Ø1.0 Ø1.1 ~ Ø3.0	Tolerance 0.000 ~ -0.015 0.000 ~ -0.020
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(mm)

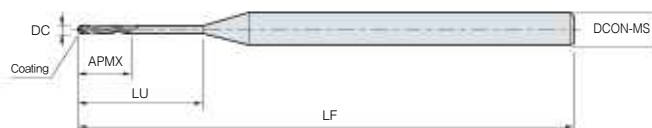
	Designation	Coating	DC	DCON-MS	APMX	LU	LF	RE
TTBE	2006-060-N115S06(PRODEN)	PVD	0.6	6	2.2	11.5	60	DC/2
	2010-060-N180S06(PRODEN)	DIA	1	6	3	18	60	DC/2
	2020-060-N200S06(PRODEN)	DIA	2	6	5	20	60	DC/2
TWBE	2010-060-N180S06(PRODEN)	-	1	6	3	18	60	DC/2
	2020-060-N200S06(PRODEN)	-	2	6	5	20	60	DC/2

PROTECH INNOTION Type

(Metal Tool-Monster, Nexus, Protech)



2	H-A 30°	h5 shank	Grade PC2510 PC2010 PVD	DC Ø1.0 Ø1.1 ~ Ø3.0	Tolerance 0.000 ~ -0.015 0.000 ~ -0.020
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(mm)

	Designation	Coating	DC	DCON-MS	APMX	LU	LF	RE
TTBE	2010-051-N100S06(NEXUS)	PVD	1	6	2	10	51	DC/2
	2015-051-N100S06(NEXUS)	PVD	1.5	6	2	10	51	DC/2
	2020-051-N120S06(NEXUS)	PVD	2	6	3	12	51	DC/2
	2030-051-N120S06(NEXUS)	PVD	3	6	4.5	12	51	DC/2

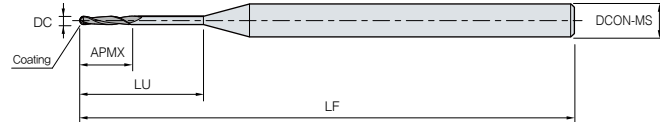
CERACUBE/TRION-Z Type



DC	Tolerance
Ø1.0	0.000 ~ -0.015
Ø1.1 ~ Ø3.0	0.000 ~ -0.020

PVD

DIA



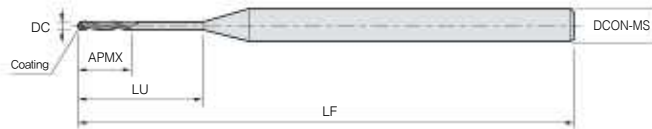
(mm)

Designation		Coating	DC	DCON-MS	APMX	LU	LF	RE
TZBE	2005-050-N22(TRION-Z)	PVD	0.5	4	2	22	50	DC/2
	2010-050-N18(TRION-Z)	DIA	1	4	3	18	50	DC/2
	2020-050-N20(TRION-Z)	DIA	2	4	7	20	50	DC/2
TWBE	2010-050-N180S04(TRION-Z)	-	1	4	3	18	50	DC/2
	2020-050-N200S04(TRION-Z)	-	2	4	7	20	50	DC/2

DENTIUM Type

(Rainbow, Rainbow-Mill, Rainbow-Mill Zr 2nd)

			Grade PC2510 PC2010 PVD	Grade ND3000 DIA	DC Ø1.0 Ø1.1 ~ Ø3.0	Tolerance 0.000 ~ -0.015 0.000 ~ -0.020
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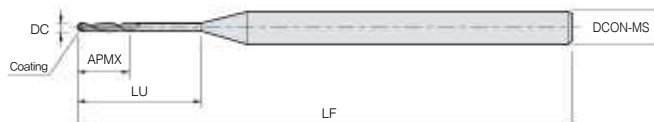
(mm)

	Designation	Coating	DC	DCON-MS	APMX	LU	LF	RE
TZBE	2005-050-N080S06(DENTIUM)	DIA	0.5	6	2	8	50	DC/2
	2010-050-N160S06(DENTIUM)	DIA	1	6	3	16	50	DC/2
	2020-050-N180S06(DENTIUM)	DIA	2	6	6	18	50	DC/2
	2005-043-N080S03(DENTIUM)	DIA	0.5	3	2	8	43	DC/2
	2010-050-N160S03(DENTIUM)	DIA	1	3	3	16	50	DC/2
	2020-050-N180S03(DENTIUM)	DIA	2	3	6	18	50	DC/2
	2005-045-N080S04(DENTIUM)	DIA	0.5	4	2	8	45	DC/2
	2010-045-N160S04(DENTIUM)	DIA	1	4	3	16	45	DC/2
	2025-045-N180S04(DENTIUM)	DIA	2.5	4	6	18	45	DC/2
TWBE	2005-050-N080S06(DENTIUM)	-	0.5	6	2	8	50	DC/2
	2010-050-N160S06(DENTIUM)	-	1	6	3	16	50	DC/2
	2020-050-N180S06(DENTIUM)	-	2	6	6	18	50	DC/2
	2010-050-N160S03(DENTIUM)	-	1	3	3	16	50	DC/2
	2020-050-N180S03(DENTIUM)	-	2	3	6	18	50	DC/2
	2010-045-N160S04(DENTIUM)	-	1	4	3	16	45	DC/2
	2025-045-N180S04(DENTIUM)	-	2.5	4	6	18	45	DC/2
TTBE	2010-050-N100S06(DENTIUM)	PVD	1	6	2	10	50	DC/2
	2015-050-N100S06(DENTIUM)	PVD	1.5	6	3	10	50	DC/2
	2020-050-N120S06(DENTIUM)	PVD	2	6	4	12	50	DC/2
	2030-050-N120S06(DENTIUM)	PVD	3	6	6	12	50	DC/2
ETRE	006050-H6M325-D(DENTIUM)	E.D DIA	0.6	6	6	6	50	DC/2
	010050-H6M170-D(DENTIUM)	E.D DIA	1	6	10	10	50	DC/2
	020050-H6M120-D(DENTIUM)	E.D DIA	2	6	12	12	50	DC/2
	006040-H3M325-D(DENTIUM)	E.D DIA	0.6	3	6	6	40	DC/2
	010045-H3M170-D(DENTIUM)	E.D DIA	1	3	10	10	45	DC/2
	020045-H3M120-D(DENTIUM)	E.D DIA	2	3	12	12	45	DC/2
	006045-H4M325-D(DENTIUM)	E.D DIA	0.6	4	6	6	45	DC/2
	010045-H4M170-D(DENTIUM)	E.D DIA	1	4	10	10	45	DC/2
	020045-H4M120-D(DENTIUM)	E.D DIA	2	4	12	12	45	DC/2

※ E.D DIA: Electro deposition DIA

DOF Type

				DC Ø1.0 Ø1.1 ~ Ø3.0	Tolerance 0.000 ~ -0.015 0.000 ~ -0.020
DIA					

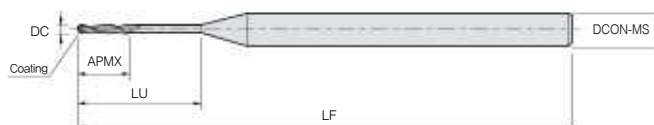


(mm)

Designation		Coating	DC	DCON-MS	APMX	LU	LF	RE
TZBE	2010-044-N19(DOF)	DIA	1	3	6	19	44	DC/2
	2020-044-N20(DOF)	DIA	2	3	8	20	44	DC/2
TWBE	2010-044-N19(DOF)	-	1	3	6	19	44	DC/2
	2020-044-N20(DOF)	-	2	3	8	20	44	DC/2

ELBEN Type

					DC Ø1.0 Ø1.1 ~ Ø3.0	Tolerance 0.000 ~ -0.015 0.000 ~ -0.020
PVD				DIA		



(mm)

Designation		Coating	DC	DCON-MS	APMX	LU	LF	RE
TZBE	2006-043-N080S06(ELBEN)	PVD	0.6	6	1.2	8	43	DC/2
	2010-050-N160S06(ELBEN)	DIA	1	6	3	16	50	DC/2
	2020-050-N200S06(ELBEN)	DIA	2	6	6	20	50	DC/2
TWBE	2010-050-N160S06(ELBEN)	-	1	6	3	16	50	DC/2
	2020-050-N200S06(ELBEN)	-	2	6	6	20	50	DC/2
TTBE	2010-045-N050S06(ELBEN)	PVD	1	6	2	5	45	DC/2
	2020-045-N120S06(ELBEN)	PVD	2	6	4	12	45	DC/2
	2030-045-N150S06(ELBEN)	PVD	3	6	6	15	45	DC/2

CEREC Type

(MC-XL)



2	H-A 30°	h5 shank	Grade ND3000	DC	Tolerance
			DIA	Ø1.0 Ø1.1 ~ Ø3.0	0.000 ~ -0.015 0.000 ~ -0.020



(mm)

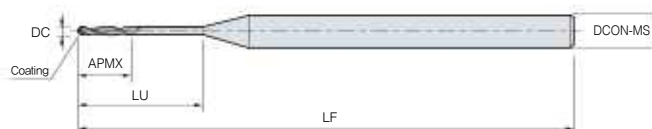
	Designation	Coating	DC	DCON-MS	APMX	LU	LF	RE
ETRE	018038-T35M230-12	DIA	1.73	3.5	8	12	38.1	DC/2
	009038-T35M230-12S	DIA	0.9	3.5	4	12	38.1	DC/2

UGINT Type

(D-100, D-200)



2	H-A 30°	h5 shank	Grade PC2510 PC2010	Grade ND3000	DC	Tolerance
			PVD	DIA	Ø1.0 Ø1.1 ~ Ø3.0	0.000 ~ -0.015 0.000 ~ -0.020



(mm)

	Designation	Coating	DC	DCON-MS	APMX	LU	LF	RE
TTBE	2007-040-N145S03(D-100)	PVD	0.7	3	2.5	14.5	40	DC/2
	2010-040-N145S03(D-100)	DIA	1	3	5	14.5	40	DC/2
	2020-040-N160S03(D-100)	DIA	2	3	4	16	40	DC/2
	2006-043-N080S06(D-200)	PVD	0.6	6	1.2	8	43	DC/2
	2010-050-N160S06(D-200)	DIA	1	6	3	16	50	DC/2
	2020-050-N200S06(D-200)	DIA	2	6	6	20	50	DC/2
TWBE	2010-040-N145S03(D-100)	-	1	3	5	14.5	40	DC/2
	2020-040-N160S03(D-100)	-	2	3	4	16	40	DC/2
	2010-050-N160S06(D-200)	-	1	6	3	16	50	DC/2
	2020-050-N200S06(D-200)	-	2	6	6	20	50	DC/2
TTBE	2015-050-N100S06(D-200)	PVD	1.5	6	7.4	10.2	50	DC/2
	2020-050-N120S06(D-200)	PVD	2	6	7.6	12.1	50	DC/2

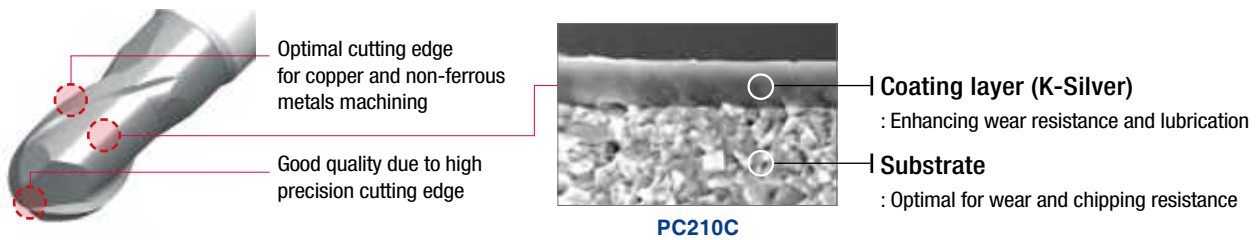
Long tool life and good surface roughness for electrode machining

C-Max

Copper

- Superior lubricity, wear resistance & chipping resistance due to the K-Silver coating layer and optimal substrate
- Optimal for copper and non-ferrous metal machining
- Wide line up (ball, flat, radius & long neck type)
- Long tool life and good surface roughness for electrode machining

Features

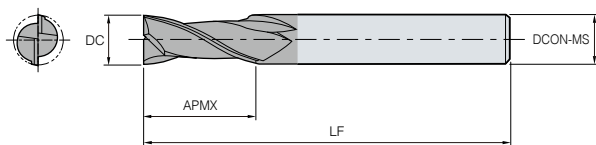


CFE2000

Flat



2	H-A 30°	Grade	DC	Tolerance
		PC210C	Ø0.5 ~ Ø6 Ø8 ~ Ø12	0.00 ~ 0.01 0.00 ~ 0.02



(mm)

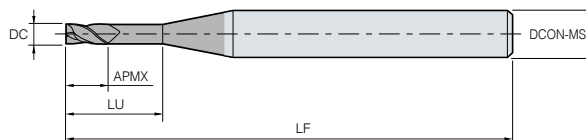
Designation	DC	DCON-MS	APMX	LF
CFE 2010-040	1	4	2.5	40
2015-040	1.5	4	4	40
2020-045	2	4	5	45
2030-045	3	6	8	45
2040-050	4	6	11	50
2050-060	5	6	13	60
2060-060	6	6	13	60
2080-060	8	8	19	60
2100-070	10	10	22	70
2120-075	12	12	26	75

CFNE2000

Long neck type flat



DC	Tolerance
Ø0.5 ~ Ø6	0.00 ~ 0.01
Ø8 ~ Ø12	0.00 ~ 0.02



(mm)

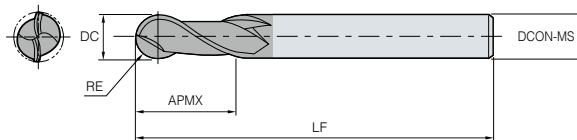
Designation		DC	DCON-MS	APMX	LU	LF
CFNE	2005-045-N2	0.5	4	0.8	2	45
	2005-045-N4	0.5	4	0.8	4	45
	2005-045-N6	0.5	4	0.8	6	45
	2005-050-N8	0.5	4	0.8	8	50
	2010-045-N4	1	4	1.5	4	45
	2010-045-N6	1	4	1.5	6	45
	2010-050-N8	1	4	1.5	8	50
	2010-050-N10	1	4	1.5	10	50
	2015-045-N6	1.5	4	2.3	6	45
	2015-050-N8	1.5	4	2.3	8	50
	2015-050-N10	1.5	4	2.3	10	50
	2015-050-N12	1.5	4	2.3	12	50
	2020-045-N6	2	4	3	6	45
	2020-050-N8	2	4	3	8	50
	2020-050-N10	2	4	3	10	50
	2020-055-N12	2	4	3	12	50
	2030-050-N10	3	4	4.5	10	50
	2030-050-N12	3	4	4.5	12	50
	2030-060-N14	3	4	4.5	14	60
	2030-060-N16	3	4	4.5	16	60
	2040-050-N12	4	6	6	12	50
	2040-050-N16	4	6	6	16	50
	2040-060-N20	4	6	6	20	60

CBE2000

Ball



2	H-A 30°	Grade PC210C	RE ±0.005	DC Ø0.5 ~ Ø6 Ø8 ~ Ø12	Tolerance 0.00 ~ 0.01 0.00 ~ 0.02
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(mm)

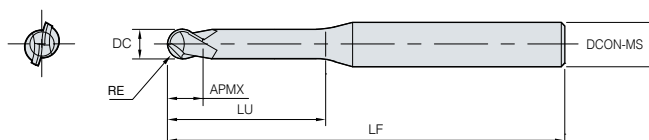
Designation	DC	DCON-MS	APMX	LF	RE
CBE 2010-050	1	4	2.5	50	0.5
2015-050	1.5	4	4	50	0.75
2020-050	2	4	5	50	1
2030-060	3	6	8	60	1.2
2040-070	4	6	8	70	2
2050-080	5	6	10	80	2.5
2060-080	6	6	12	80	3
2080-090	8	8	14	90	4
2100-100	10	10	18	100	5
2120-110	12	12	22	110	6

CBNE2000

Long neck type ball



2	H-A 30°	Grade PC210C	DC Ø0.5 ~ Ø6 Ø8 ~ Ø12	Tolerance 0.00 ~ 0.01 0.00 ~ 0.02
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(mm)

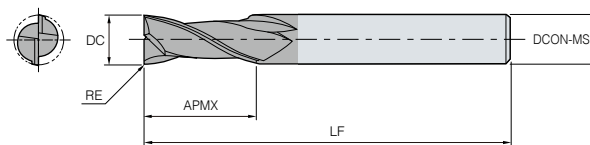
Designation	DC	DCON-MS	APMX	LU	LF	RE
CBNE 2005-045-N2	0.5	4	0.5	2	45	0.25
2005-045-N4	0.5	4	0.5	4	45	0.25
2005-045-N6	0.5	4	0.5	6	45	0.25
2005-050-N8	0.5	4	0.5	8	50	0.25
2010-045-N4	1	4	1	4	45	0.5
2010-045-N6	1	4	1	6	45	0.5
2010-050-N8	1	4	1	8	50	0.5
2010-050-N10	1	4	1	10	50	0.5
2015-050-N8	1.5	4	1.5	8	50	0.75
2015-050-N10	1.5	4	1.5	10	50	0.75
2015-050-N12	1.5	4	1.5	12	50	0.75
2015-055-N14	1.5	4	1.5	14	55	0.75
2020-050-N8	2	4	2	8	50	1
2020-050-N10	2	4	2	10	50	1
2020-050-N12	2	4	2	12	50	1
2020-055-N14	2	4	2	14	55	1
2030-050-N10	3	4	3	10	50	1.5
2030-050-N12	3	4	3	12	50	1.5
2030-055-N14	3	4	3	14	55	1.5
2030-055-N16	3	4	3	16	60	1.5
2040-060-N16	4	6	4	16	60	2
2040-060-N20	4	6	4	20	60	2
2040-070-N25	4	6	4	25	70	2
2040-070-N30	4	6	4	30	70	2

CRE2000

Radius



DC	Tolerance
Ø0.5 ~ Ø6	0.00 ~ 0.01
Ø8 ~ Ø12	0.00 ~ 0.02



(mm)

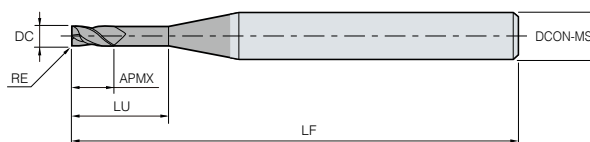
Designation	DC	DCON-MS	APMX	LF	RE
CRE 2020-045-R05	2	4	5	45	0.5
2030-045-R05	3	6	8	45	0.5
2040-050-R05	4	6	11	50	0.5
2050-060-R05	5	6	13	60	0.5
2060-060-R05	6	6	13	60	0.5
2080-060-R10	8	8	19	60	1
2100-070-R10	10	10	22	70	1
2120-075-R10	12	12	26	75	1

CRNE2000

Long neck type radius



DC	Tolerance
Ø0.5 ~ Ø6	0.00 ~ 0.01
Ø8 ~ Ø12	0.00 ~ 0.02



(mm)

Designation	DC	DCON-MS	APMX	LU	LF	RE
CRNE 2010-045-R02N4	1	4	1.5	4	45	0.2
2010-045-R02N6	1	4	1.5	6	45	0.2
2010-050-R02N8	1	4	1.5	8	50	0.2
2010-050-R02N10	1	4	1.5	10	50	0.2
2015-045-R02N6	1.5	4	2.3	6	45	0.2
2015-050-R02N8	1.5	4	2.3	8	50	0.2
2015-050-R02N10	1.5	4	2.3	10	50	0.2
2015-050-R02N12	1.5	4	2.3	12	50	0.2
2020-045-R05N6	2	4	3	6	45	0.5
2020-050-R05N8	2	4	3	8	50	0.5
2020-050-R05N10	2	4	3	10	50	0.5
2020-055-R05N12	2	4	3	12	50	0.5
2030-050-R05N10	3	4	4.5	10	50	0.5
2030-050-R05N12	3	4	4.5	12	50	0.5
2030-060-R05N14	3	4	4.5	14	60	0.5
2030-060-R05N16	3	4	4.5	16	60	0.5
2040-050-R05N12	4	6	6	12	50	0.5
2040-050-R05N16	4	6	6	16	50	0.5
2040-060-R05N20	4	6	6	20	60	0.5

Longer tool life and good surface finishes

PCD Endmill

- Longer tool life and good surface roughness
- Reducing burrs at non-ferrous metals machining
- 1000 type: Ultra finishing for non-ferrous metals
- 2000 type: Optimal for aluminum alloy, carbon steel, graphite and reinforced Plastic machining

PDE1000/2000

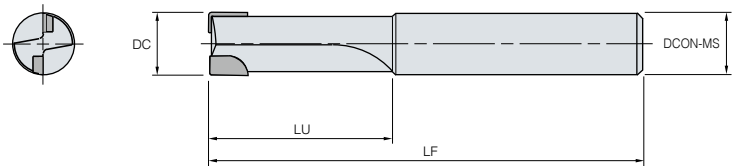
Flat

1

2

H-A
0°

Grade
DP200



		(mm)			
Designation		DC	DCON-MS	LU	LF
PDE	1040	4	6	15	45
	1050	5	6	15	50
	1060	6	6	20	60
	2060	6	8	20	60
	2070	7	8	20	60
	2080	8	8	20	60
	2090	9	10	25	70
	2100	10	10	25	70
	2120	12	12	25	75

Special Order Form for Endmills

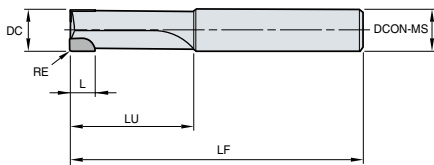


Fig 1

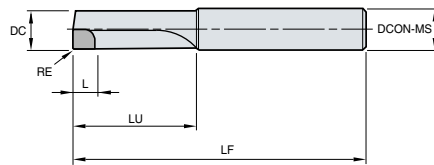
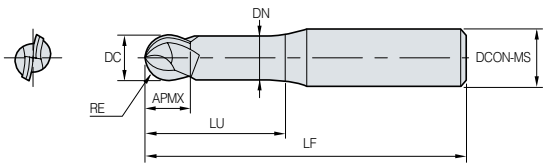


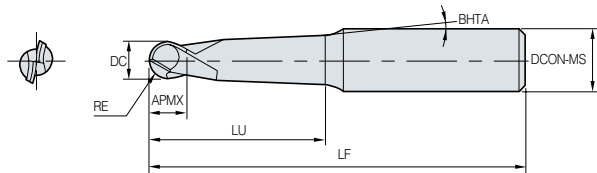
Fig 2

Designation	Fig	NOF	mm					
			DC	DCON-MS	RE	L	LU	LF
PDES								

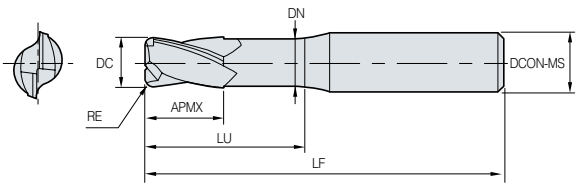
※ Depending on customer requests, we can make special Endmill



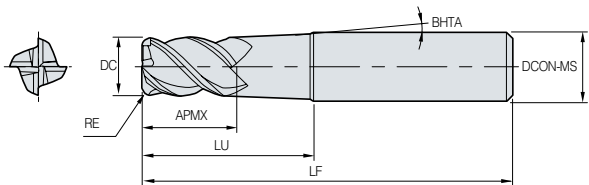
Designation	NOF	RE	DC	DCON-MS	DN	APMX	LU	LF



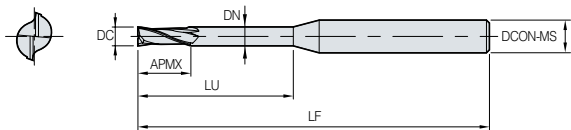
Designation	NOF	RE	DC	DCON-MS	APMX	LU	LF	BHTA(°)



Designation	NOF	DC	DCON-MS	DN	RE	APMX	LU	LF



Designation	NOF	RE	DC	DCON-MS	APMX	LU	LF	BHTA(°)



Designation	NOF	DC	DCON-MS	DN	APMX	LU	LF